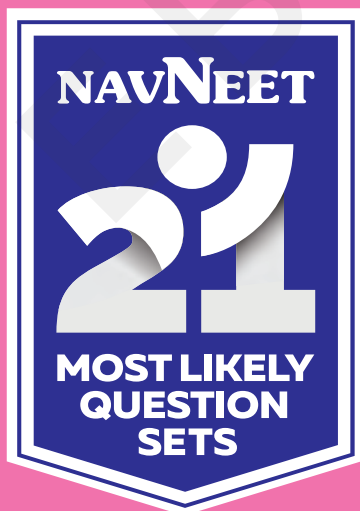


For
2026
Examinations



MATHEMATICS (PART – II)

Board's
Question Paper
of July 2025
Included

Solution on
QR Code

STD. X



GET FREE
SMART
DigiBook
Your Smart Book Platform





SMART
DigiBook

SMART DigiBook platform offers a digital version of your favourite Navneet books; available in PDF format and bundled with digital resources.

The interactive tools on SMART DigiBook will enhance your learning experience resulting in a better understanding of concepts.

Make learning fun and interactive with **SMART features** on **SMART DigiBook**



3 Simple Steps to SMART Learning

Step 1 : Scan the QR Code, download the SMART DigiBook App & click on Register.

Step 2 : Login using your credentials. (Registered users can login directly.)

Step 3 : Click the Redeem Access Code button and enter the Access Code mentioned. 



Earn NavCoins

- Earn NavCoins when you redeem this book.
- Earn more when you learn on the App.
- Use your NavCoins to win exciting rewards.



- Select Books/Digital Resources are available on purchase.
- Digital Assets may vary from book-to-book.
- Mobile version supports Android 10.0 (and above) & iOS 12.0 (and above).
- Access valid for 365 days from the date of redemption. Term & Condition apply.
- For any queries please contact : 1800 266 6676 (Monday to Saturday, between 9 am to 6 pm).

Access Code :



2026

**An excellent book for the
Preparation of the Board's
Examination**

MATHEMATICS

(PART – II)

STANDARD X

This book contains ...

1. Format (Outline) of the Question Paper at the beginning of the book
2. Study Tips for Students
3. **Section 1** : Board's July 2024 Question Paper with full Solution
4. **Section 2** : Question Sets for chapter-wise study and Revision :
 - (a) Guidance on how to attempt the question
 - (b) Model answers with solution of each question type
 - (c) Questions for Practice
5. Separate question set based on important theorems
6. Creative type question (open ended questions, heuristic approach, approximation, etc.) and challenging questions included in each set.
7. **Section 3** : Board's July 2025 Question Paper for Practice (with Solution on QR Code)

With 21 MLQ... Success awaits you!

By
NAVNEET



NAVNEET EDUCATION LIMITED

D0060

Outline of Question Paper

MATHEMATICS (PART – I) AND MATHEMATICS (PART – II)

Q. No.	Types of Questions	Marks	Marks with options
Q. 1	(A) Solve Multiple choice questions <i>(1 mark each)</i>	4	4
	(B) Solve 4 out of 4 <i>(1 mark each)</i>	4	4
Q. 2	(A) Complete 2 activities out of 3 <i>(2 marks each)</i>	4	6
	(B) Solve any 4 out of 5 subquestions <i>(2 marks each)</i>	8	10
Q. 3	(A) Complete 1 activity out of 2 <i>(3 marks each)</i>	3	6
	(B) Solve any 2 out of 4 subquestions <i>(3 marks each)</i>	6	12
Q. 4	Solve any 2 out of 3 subquestions <i>(4 marks each)</i>	8	12
Q. 5	Solve any 1 out of 2 subquestions <i>(3 marks each)</i>	3	6
	Total Marks	40	60

Q. 3 (B) One of the subquestions should be asked on prove the given statement.

Q. 4 These questions should be out of the textbooks but based on prescribed syllabus and also challenging.

Q. 5 These questions should be creative for which the responses should be as follows :

For Example :

- ☛ Expecting the student to express his/her own views
- ☛ To draw a figure as per the given information
- ☛ To complete an incomplete construction
- ☛ To complete the given flow chart
- ☛ To construct a problem from given information or situation
- ☛ Open ended questions
- ☛ Estimation and Approximation
- ☛ Comprehension of mathematical passage.

Study Tips for Students

Dear Students,

Your Board examination is approaching near. The coming days are for self-study and to make an effective study plan. Navneet 21 Most Likely Question Sets provide a deeper understanding of important topics and how questions are asked from these topics.

Navneet 21 Most Likely Question Sets are prepared according to the latest syllabus and the format of the Question Paper.

Important questions and answers from the examination point of view have been incorporated in this book.

1. The questions are based on the syllabus of Mathematics : Part-II
2. MCQ in **Q. 1 (A)** is objective type. This is based on your knowledge and understanding of the subject. **Q. 1 (B)** Short answer questions.
3. The sub-questions in **Q. 2 (A)** are activity based questions. To solve this, you have to fill in the blanks/boxes. These are easy questions.
4. The sub-questions in **Q. 2 (B)** are of **2 marks** and based on knowledge and understanding. They usually would be from the textbook.
5. The sub-questions in **Q. 3 (A)** is activity based questions of 3 marks and **Q. 3 (B)** of **3 marks** each based on knowledge, understanding and application.
6. **Q. 4** of **4 marks** is very important. It is a challenging question based on the syllabus but the question will not be from the textbook.
7. **Q. 5** is of constructive nature. This question is to evaluate your capacity to express your views regarding the given question. The question may be of various types based on the syllabus.

In this book, each set is divided into 1 mark, 2 marks and 3 marks questions with solutions. Moreover challenging question carrying 4 marks are also included in each set. Assignments in each case have been given for practice.



NAVNEET EDUCATION LIMITED

★ Bhavani Shankar Road, Dadar, **Mumbai – 400 028**. (Tel. 6662 6565)

www.navneet.com • e-mail : publications@navneet.com

Branches :

Pune • Nagpur • Nashik

Contents

	Page No.
Section 1 : BOARD'S QUESTION PAPER : JULY 2024 <i>(With Full Solution and Marking Scheme)</i> ...	5
Section 2 : MOST LIKELY QUESTION SETS	
1. Theorems ...	26
2. Similarity ...	40
3. Pythagoras Theorem ...	72
4. Circle ...	102
5. Geometric Constructions ...	138
6. Coordinate Geometry ...	154
7. Trigonometry ...	181
8. Mensuration ...	205
Section 3 : BOARD'S QUESTION PAPER : JULY 2025 (FOR PRACTICE) ...	234

© All rights reserved. No part of this publication or its associated digital content, including but not limited to content accessible via QR codes, online platforms, or other electronic means, may be copied, adapted, abridged, translated, stored in any retrieval system, or transmitted in any form or by any means graphic, electronic, mechanical, photographic, or otherwise without the prior written permission of the copyright holder, M/s. Navneet Education Limited. Applications for permission should be addressed to the publisher. Breach of this condition will entail legal action and prosecution without further notice.

Every effort has been made to avoid errors or omissions in this publication. However, errors or omissions may have occurred. Any such discrepancies brought to our notice will be corrected in the next edition. The publisher, author, or seller will not be held liable for any loss or damage arising from the use of this publication.

In case of binding mistake, misprints, or missing pages, etc., the publisher's entire liability and your exclusive remedy are replacement of the book within one month of purchase by a similar edition/reprint of the book.

Published by Navneet Education Limited, Dantali, Gujarat.
Printed by Navneet Education Limited, Dantali, Gujarat.

2626
CTP
(18/09/25)

Section 1**BOARD'S QUESTION PAPER : JULY 2024****(With Full Solution and Marking Scheme)****Time : 2 Hours]****[Max. Marks : 40**

- Note :**
- (i) *All questions are compulsory.*
 - (ii) *Use of calculator is not allowed.*
 - (iii) *The numbers to the right of the questions indicate full marks.*
 - (iv) *In case of MCQs [Q. No. ① (A)], only the first attempt will be evaluated and will be given credit.*
 - (v) *For every MCQ, the correct alternative (A), (B), (C) or (D) with subquestion number is to be written as an answer.*
 - (vi) *Draw the proper figures for answers wherever necessary.*
 - (vii) *The marks of construction should be clear. Do not erase them.*
 - (viii) *Diagram is essential for writing the proof of the theorem.*

In Q. ① (A), 4 multiple choice subquestions each of 1 mark will be asked. Four alternatives for each subquestion will be given. Only one of them will be correct. Write the letter of the alternative against the question number.

Note :

- (1) In Q. ① (A), all the subquestions are compulsory.
- (2) Method of solution is not expected. Write the correct alternative letter.
- (3) Hints have been given to make the students understand them. There is no need to write them in the examination.
- (4) Assessment will be done only to the first attempt of the answer.

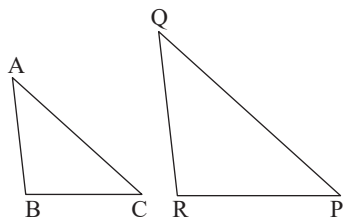
Q. 1 (A) For each of the following subquestions four alternative answers are given. Choose the correct alternative and write its alphabet :

4

- (i) In $\triangle ABC$ and $\triangle PQR$, in a one to one correspondence of vertices,

$$\frac{AB}{QR} = \frac{BC}{PR} = \frac{CA}{PQ},$$

then statement is true.



- (A) $\triangle PQR \sim \triangle ABC$ (B) $\triangle PQR \sim \triangle CAB$
 (C) $\triangle CBA \sim \triangle PQR$ (D) $\triangle BCA \sim \triangle PQR$
- (ii) Two circles of radii 5.5 cm and 3.3 cm respectively touch each other externally. Then the distance between their centres is
 (A) 4.4 cm (B) 8.8 cm (C) 2.2 cm (D) 8.3 cm
- (iii) Seg AB is parallel to Y-axis and coordinates of point A are (1, 3) then the coordinates of point B are
 (A) (3, 1) (B) (5, 3) (C) (3, 0) (D) (1, -3)
- (iv) The volume of a cube of side 10 cm is
 (A) 1000 cm^3 (B) 100 cm^3
 (C) $10,000 \text{ cm}^3$ (D) 10 cm^3

Answers

- (i) (B) (ii) (B) (iii) (D) (iv) (A). (1 mark each)

Explanations : Students are not expected to write the explanation.

Here, it is given for the better understanding of the students.

- (i) $P \leftrightarrow C, Q \leftrightarrow A, R \leftrightarrow B$.
 (ii) The distance between the centres of circles touching externally is equal to the sum of their radii.
 (iii) The line parallel to Y-axis will have the same x-coordinates for the points A and B.
 (iv) The volume of a cube $= l^3$.

In Q. 1 (B), 4 subquestions each of 1 mark will be asked. You have to solve all of them.

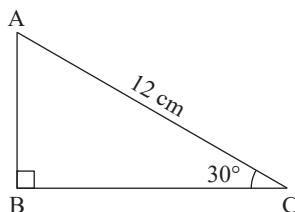
Note :

- (1) These questions will be based on X Standard (Part II) textbook.
- (2) Answers with solution are expected.

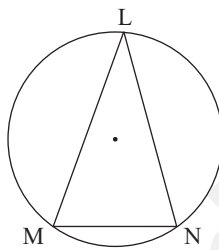
Q. 1 (B) Solve the following subquestions :

4

- (i) In $\triangle ABC$, $\angle B = 90^\circ$, $\angle C = 30^\circ$,
 $AC = 12$ cm, then find AB .



- (ii) In the given figure,
 $m(\text{arc } MN) = 70^\circ$,
 find $\angle MLN$.



- (iii) Find the value of $\sin \theta \times \operatorname{cosec} \theta$.
- (iv) If the radius of a circle is 4 cm and the length of an arc is 10 cm, then find the area of the sector.

Solutions

(i) **Ans.**

$$\angle B = 90^\circ, \angle C = 30^\circ.$$

$$\therefore \angle A = 60^\circ$$

... (The remaining angle of $\triangle ABC$)

$\therefore \triangle ABC$ is a $30^\circ - 60^\circ - 90^\circ$ triangle.

$$AB = \frac{1}{2} AC$$

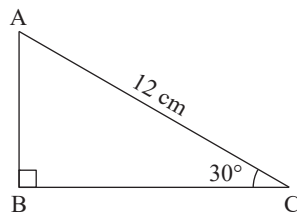
... (Side opposite to 30°) ($\frac{1}{2}$ mark)

$$\therefore AB = \frac{1}{2} \times 12 \text{ cm}$$

$$\therefore AB = 6 \text{ cm}$$

($\frac{1}{2}$ mark)

$$AB = 6 \text{ cm.}$$



(ii) Ans.

$$\angle MLN = \frac{1}{2} m(\text{minor arc MN})$$

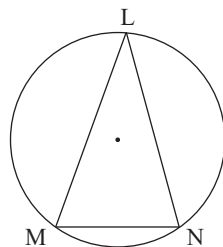
... (Inscribed angle theorem) ($\frac{1}{2}$ mark)

$$= \frac{1}{2} \times 70^\circ$$

... [Given : $m(\text{arc MN}) = 70^\circ$]

$$= 35^\circ$$

$$\angle MLN = 35^\circ.$$



($\frac{1}{2}$ mark)

(iii) Ans.

$$\operatorname{cosec} \theta = \frac{1}{\sin \theta} \quad \dots \text{ (The relation between trigonometric ratios)}$$

($\frac{1}{2}$ mark)

$$\sin \theta \times \operatorname{cosec} \theta$$

$$= \sin \theta \times \frac{1}{\sin \theta}$$

$$= 1$$

($\frac{1}{2}$ mark)

The value of $\sin \theta \times \operatorname{cosec} \theta$ is **1**.

(iv) Ans.

$r = 4$ cm, the length (l) of the arc = 10 cm, $A(\text{sector}) = ?$

$$\text{The area of the sector} = \frac{r}{2} \times l \quad (\frac{1}{2} \text{ mark})$$

$$= \frac{4}{2} \times 10 = 20 \quad (\frac{1}{2} \text{ mark})$$

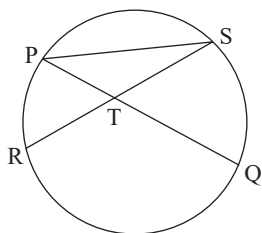
The area of the sector is **20 cm²**.

In Q. 2 (A), 3 activities each of 2 marks will be given. You have to attempt any two. Various types of activities, e.g. (i) find the length, angle (ii) write proof with the help of fill in the blanks/boxes, etc. can be asked.

Q. 2 (A) Complete the following activities and rewrite them :

(Any two) 4

- (i) In the given figure, chord PQ and chord RS intersect at point T. Then complete the following activity to prove,



$$\angle STQ = \frac{1}{2} [m(\text{arc SQ}) + m(\text{arc PR})]$$

Activity :

$$\angle STQ = \angle SPQ + \boxed{}$$

... (Remote interior angles theorem of a triangle)

$$= \frac{1}{2} m \left[(\text{arc SQ}) + \boxed{} \right]$$

... (Inscribed angle theorem)

$$= \frac{1}{2} \left[\boxed{} + \boxed{} \right]$$

- (ii) If $\sec \theta = \frac{25}{7}$, complete the following activity to find the value of $\tan \theta$.

Activity :

$$\text{We have, } 1 + \tan^2 \theta = \sec^2 \theta$$

$$\therefore 1 + \tan^2 \theta = \boxed{}$$

$$\therefore \tan^2 \theta = \frac{625}{49} - \boxed{}$$

$$\therefore \tan^2 \theta = \frac{625 - 49}{49}$$

$$\therefore \tan^2 \theta = \frac{\boxed{}}{49}$$

$$\therefore \tan \theta = \boxed{}$$

- (iii) Complete the following activity to find the volume of a cone, if the radius of its base is 7 cm and its perpendicular height is 6 cm.

Activity :

Here, radius (r) = 7 cm, height (h) = 6 cm

The volume of a cone = ... (Formula)

$$= \frac{1}{3} \times \frac{22}{7} \times \text{} \times 6$$

$$= \frac{1}{3} \times \frac{22}{7} \times \text{} \times 6$$

$\therefore$ the volume of the cone = cm^3 .

Solutions

(i) Activity :

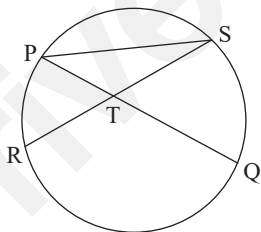
$$\angle \text{STQ} = \angle \text{SPQ} + \angle \text{PSR}$$

... (Remote interior angles theorem of a triangle)

$$= \frac{1}{2} m(\text{arc SQ}) + \frac{1}{2} m(\text{arc PR})$$

... (Inscribed angle theorem)

$$= \frac{1}{2} \left[m(\text{arc SQ}) + m(\text{arc PR}) \right]$$



(ii) Activity :

We have, $1 + \tan^2 \theta = \sec^2 \theta$

$$\therefore 1 + \tan^2 \theta = \left(\frac{25}{7} \right)^2$$

$$\therefore \tan^2 \theta = \frac{625}{49} - \text{}$$

$$\therefore \tan^2 \theta = \frac{625 - 49}{49}$$

$$\therefore \tan^2 \theta = \frac{\text{>}}{49}$$

$$\therefore \tan \theta = \frac{\text{>}}{7}$$

(iii) Activity :

Here, radius (r) = 7 cm, height (h) = 6 cm

The volume of a cone = $\frac{1}{3}\pi r^2 h$... (Formula)

$$= \frac{1}{3} \times \frac{22}{7} \times (7)^2 \times 6 = \frac{1}{3} \times \frac{22}{7} \times 49 \times 6$$

$\therefore$ the volume of the cone = 308 cm^3 .

(Marking scheme : $\frac{1}{2}$ mark for each correct answer written in each box.)

In Q. 2 (B), 5 subquestions each of 2 marks will be asked. You have to solve any four of them.

Note : Answers with solution are expected.

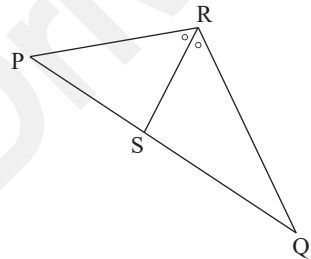
Q. 2 (B) Solve the following subquestions : (Any four) 8

(i) Construct a tangent to a circle with centre P and radius 3.2 cm at any point M on it.

(ii) In $\triangle PQR$, seg RS bisects $\angle PRQ$.

If $PR = 15$, $RQ = 20$,

$PS = 12$, then find SQ .

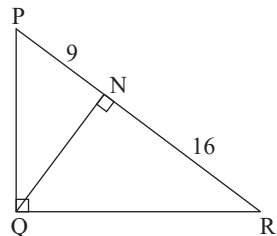


(iii) Find the surface area of a sphere of diameter 14 cm.

(iv) In the given figure, $\angle PQR = 90^\circ$,

seg $QN \perp$ seg PR ,

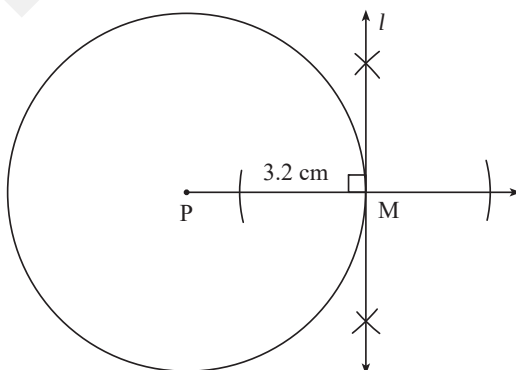
$PN = 9$, $NR = 16$, find QN .



(v) Find the slope of the line passing through the points A(3, 3) and B(5, 7).

Solutions

(i)



Line l is a tangent to the circle at the point M .

[**Marking scheme :**

(a) To draw a circle of radius 3.2 cm and mark point M on it. ($\frac{1}{2}$ mark)

(b) To draw ray PM . ($\frac{1}{2}$ mark)

(c) To draw tangent at point M . (1 mark)]

(ii) In $\triangle RPQ$, seg RS bisects $\angle PRQ$.

$\therefore$ by the theorem of an angle bisector of a triangle,

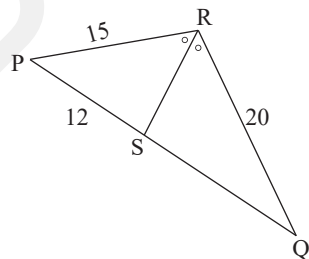
$$\frac{RP}{RQ} = \frac{PS}{SQ} \quad (\frac{1}{2} \text{ mark})$$

$$\therefore \frac{15}{20} = \frac{12}{SQ} \quad \dots \text{ (Substituting the given values) } \quad (\frac{1}{2} \text{ mark})$$

$$\therefore SQ = \frac{12 \times 20}{15} \quad (\frac{1}{2} \text{ mark})$$

$$\therefore SQ = 16 \quad (\frac{1}{2} \text{ mark})$$

Ans. $SQ = 16$.



(iii) The diameter = 14 cm.

$\therefore$ radius (r) = 7 cm. ($\frac{1}{2}$ mark)

The surface area of a sphere = $4\pi r^2$... (Formula) ($\frac{1}{2}$ mark)

$$= 4 \times \frac{22}{7} \times (7)^2 \quad \text{($\frac{1}{2}$ mark)}$$

$$= 4 \times \frac{22}{7} \times 7 \times 7 = 88 \times 7$$

$$= 616 \quad \text{($\frac{1}{2}$ mark)}$$

Ans. The surface area of the sphere is **616 cm²**.

(iv) In $\triangle PQR$, $\angle PQR = 90^\circ$.

seg $QN \perp$ hypotenuse PR .

$\therefore$ by the theorem of geometric mean,

$$QN^2 = PN \times NR \quad \text{($\frac{1}{2}$ mark)}$$

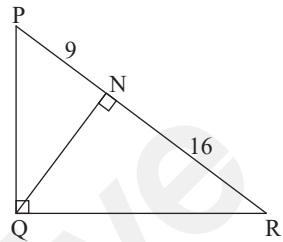
$$= 9 \times 16 \quad \text{($\frac{1}{2}$ mark)}$$

$\therefore QN = 3 \times 4$... (Taking square roots of both the sides)

($\frac{1}{2}$ mark)

$\therefore QN = 12$ ($\frac{1}{2}$ mark)

Ans. $QN = 12$ units.



(v) Let $A(3, 3) \equiv (x_1, y_1)$; $B(5, 7) \equiv (x_2, y_2)$

The slope of a line = $\frac{y_2 - y_1}{x_2 - x_1}$ ($\frac{1}{2}$ mark)

$\therefore$ the slope of the line passing through the points A and B

$$= \frac{7 - 3}{5 - 3} \quad \text{($\frac{1}{2}$ mark)}$$

$$= \frac{4}{2} \quad \text{($\frac{1}{2}$ mark)}$$

$$= 2 \quad \text{($\frac{1}{2}$ mark)}$$

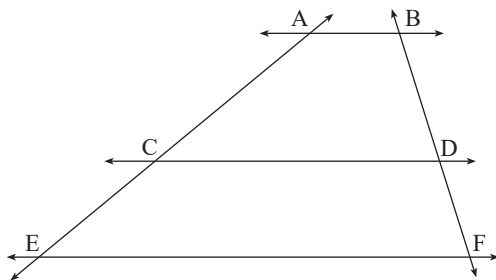
Ans. The slope of the line AB is **2**.

In Q. 3 (A), 2 activities each of 3 marks will be given. You have to attempt any one. Various types of activities, e.g. (i) to find the length of a segment, measure of an angle, (ii) write proof with the help of fill in the blanks/boxes can be asked.

Q. 3 (A) Complete the following activities and rewrite them :

(Any one) 3

(i)



In the given figure, $AB \parallel CD \parallel EF$. If $AC = 12$, $CE = 9$, $BD = 8$, then complete the following activity to find DF .

Activity : $AB \parallel CD \parallel EF$

$$\therefore \frac{AC}{\boxed{}} = \frac{\boxed{}}{DF}$$

... (Property of three parallel lines and their transversals)

$$\therefore \frac{12}{\boxed{}} = \frac{\boxed{}}{DF}$$

$$\therefore DF = \frac{8 \times 9}{\boxed{}}$$

$$\therefore DF = \boxed{}.$$

(ii) A cylinder is having radius 7 cm and height 21 cm. Complete the following activity to calculate its volume and circumference of its base.

Activity :

Radius (r) = 7 cm, Height (h) = 21 cm.

The volume of a cylinder = ... (Formula)

$$= \frac{22}{7} \times 7 \times 7 \times \text{$$

$$= 154 \times 21$$

$$= \text{ cm}^3$$

The circumference of the base = ... (Formula)

$$= 2 \times \frac{22}{7} \times \text{$$

$$= \text{ cm.}$$

Solutions

(i) Activity :

$AB \parallel CD \parallel EF$

$$\therefore \frac{AC}{CE} = \frac{BD}{DF}$$

... (Property of three parallel lines and their transversals)

$$\therefore \frac{12}{9} = \frac{8}{DF}$$

$$\therefore DF = \frac{8 \times 9}{12} \quad \therefore DF = \text{}.$$

(ii) Activity :

Radius (r) = 7 cm, Height (h) = 21 cm.

The volume of a cylinder = $\pi r^2 h$... (Formula)

$$= \frac{22}{7} \times 7 \times 7 \times \text{$$

$$= 154 \times 21$$

$$= \text{ cm}^3$$

The circumference of the base = $2\pi r$... (Formula)

$$= 2 \times \frac{22}{7} \times 7$$

$$= 44 \text{ cm.}$$

(Marking scheme : $\frac{1}{2}$ mark for each correct answer written in each box.)

In Q. 3 (B), four subquestions each of 3 marks will be asked. You have to solve any two of them. In this question, one of the subquestions would be one of the theorems from the textbook.

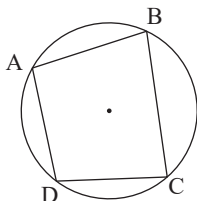
Note : Answers with solution are expected.

Q. 3 (B) Solve the following subquestions : (Any two) 6

- (i) Prove that 'Opposite angles of a cyclic quadrilateral are supplementary'.
- (ii) Draw a circle with centre P and radius 3.5 cm. Take point Q at a distance 8 cm from the centre. Construct tangents to the circle from point Q.
- (iii) Show that points P(-2, 3), Q(1, 2), R(4, 1) are collinear.
- (iv) If $\triangle PQR \sim \triangle LMN$, $9 \times A(\triangle PQR) = 16 \times A(\triangle LMN)$ and $QR = 20$, then find MN.

Solutions

- (i) **Given :** $\square ABCD$ is a cyclic quadrilateral. }
To prove : $\angle DAB + \angle DCB = 180^\circ$ } ($\frac{1}{2}$ mark)
 $\angle ABC + \angle ADC = 180^\circ$ }



($\frac{1}{2}$ mark for figure)

Proof : $\angle DAB$ is inscribed in arc DCB and intercepts arc DCB.

$$\therefore \angle DAB = \frac{1}{2} m(\text{arc DCB})$$

... (Inscribed angle theorem) ... (1)

$\angle DCB$ is inscribed in arc DAB and intercepts arc DAB.

$$\therefore \angle DCB = \frac{1}{2} m(\text{arc DAB})$$

... (Inscribed angle theorem) ... (2)

($\frac{1}{2}$ mark)

Adding (1) and (2),

$$\angle DAB + \angle DCB = \frac{1}{2} m(\text{arc DCB}) + \frac{1}{2} m(\text{arc DAB}) \quad (\frac{1}{2} \text{ mark})$$

$$\therefore \angle DAB + \angle DCB = \frac{1}{2} [m(\text{arc DCB}) + m(\text{arc DAB})]$$

Arc DCB and arc DAB constitute a complete circle.

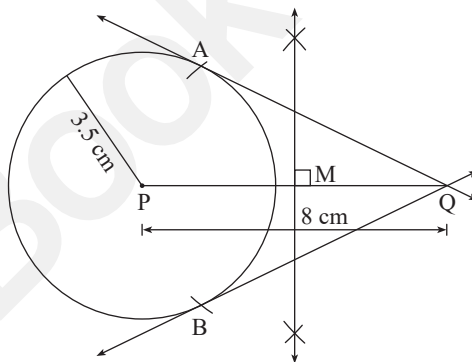
$$\therefore \angle DAB + \angle DCB = \frac{1}{2} \times 360^\circ$$

... (Measure of a circle is 360°) ($\frac{1}{2}$ mark)

$$\therefore \angle DAB + \angle DCB = 180^\circ \quad (\frac{1}{2} \text{ mark})$$

Similarly, we can prove, $\angle ABC + \angle ADC = 180^\circ$.

(ii) Ans.



Ray QA and ray QB are the tangents to the circle from the point Q.

[**Marking scheme :**

- (1) To draw a circle of radius 3.5 cm. ($\frac{1}{2}$ mark)
(2) To mark Q such that $PQ = 8$ cm. ($\frac{1}{2}$ mark)
(3) To draw perpendicular bisector of PQ. ($\frac{1}{2}$ mark)
(4) With M as centre and radius = MQ locating points A and B on the circle. ($\frac{1}{2}$ mark)
(5) To draw tangents QA and QB. ($\frac{1}{2}$ mark + $\frac{1}{2}$ mark)]
-

(iii) **Proof :**

Let $P(-2, 3) \equiv (x_1, y_1)$, $Q(1, 2) \equiv (x_2, y_2)$ and $R(4, 1) \equiv (x_3, y_3)$

$$\text{Slope of line PQ} = \frac{y_2 - y_1}{x_2 - x_1} \quad (\frac{1}{2} \text{ mark})$$

$$= \frac{2 - 3}{1 - (-2)} \quad (\frac{1}{2} \text{ mark})$$

$$= \frac{-1}{1 + 2} = -\frac{1}{3} \quad \dots (1) \quad (\frac{1}{2} \text{ mark})$$

$$\begin{aligned} \text{Slope of line QR} &= \frac{y_3 - y_2}{x_3 - x_2} \\ &= \frac{1 - 2}{4 - 1} = -\frac{1}{3} \quad \dots (2) \quad (\frac{1}{2} \text{ mark}) \end{aligned}$$

From (1) and (2),

the slope of the PQ = the slope of line QR and point Q lies on both the lines. ($\frac{1}{2}$ mark)

$\therefore$ points P, Q and R are collinear. ($\frac{1}{2}$ mark)

(iv) **Solution :**

$$9 \times A(\triangle PQR) = 16 \times A(\triangle LMN)$$

$$\therefore \frac{A(\triangle PQR)}{A(\triangle LMN)} = \frac{16}{9} \quad \dots (1) \quad (\frac{1}{2} \text{ mark})$$

$$\triangle PQR \sim \triangle LMN \quad \dots (\text{Given})$$

$$\therefore \frac{A(\triangle PQR)}{A(\triangle LMN)} = \frac{QR^2}{MN^2} \quad \dots (\text{Theorem of areas of similar triangles})$$

($\frac{1}{2}$ mark)



$$\therefore \frac{16}{9} = \frac{(20)^2}{MN^2} \quad \dots \text{ [From (1) and given]} \quad (\frac{1}{2} \text{ mark})$$

$$\therefore \frac{4}{3} = \frac{20}{MN} \quad \dots \text{ (Taking square roots of both the sides)} \quad (\frac{1}{2} \text{ mark})$$

$$\therefore MN = \frac{20 \times 3}{4} \quad (\frac{1}{2} \text{ mark})$$

$$\therefore MN = 15 \quad (\frac{1}{2} \text{ mark})$$

Ans. MN = 15.

In Q. 4, 3 subquestions each of 4 marks will be asked. You have to solve any two of them. The questions asked here will be out of the textbook based on the prescribed syllabus and also challenging.

Note : Answers with solution are expected.

Q. 4 Solve the following subquestions : (Any two) 8

(i) $\triangle ABC$ is an equilateral triangle. Point D is on seg BC such that

$$BD = \frac{1}{5} BC. \text{ Then prove that } \frac{AD^2}{AB^2} = \frac{21}{25}.$$

(ii) $\triangle LMN \sim \triangle LQP$. In $\triangle LMN$, $LM = 3.6$ cm, $\angle L = 50^\circ$,

$$LN = 4.2 \text{ cm and } \frac{LM}{LQ} = \frac{4}{7}. \text{ Construct } \triangle LQP.$$

(iii) To find the width of the river, a man observes the top of a tree, on the opposite bank making an angle of elevation of 60° . When he moves 24 metres backward from bank and observes the same top of the tree, his line of vision makes an angle of elevation of 30° . Find the height of the tree and width of the river. ($\sqrt{3} = 1.73$)

Solutions

(i) **Construction :** Draw $AP \perp$ side BC. ($\frac{1}{2}$ mark)

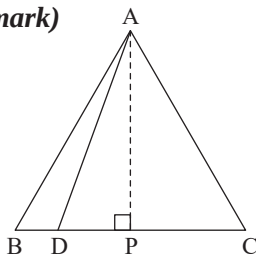
Proof : $\triangle ABC$ is an equilateral triangle.

$$\therefore \angle B = 60^\circ \text{ and } AB = AC = BC.$$

$$\angle APB = 90^\circ \quad \dots \text{ (Construction)}$$

$$\therefore \angle BAP = 30^\circ$$

$\dots$ (The remaining angle of $\triangle ABP$.)



$\therefore \triangle ABP$ is a $30^\circ - 60^\circ - 90^\circ$ triangle.

$$\therefore AP = \frac{\sqrt{3}}{2} AB \dots \text{(The side opposite to } 60^\circ) \dots (1) \text{ (}\frac{1}{2}\text{ mark)}$$

$$\text{and } BP = \frac{1}{2} AB = \frac{1}{2} BC \dots \text{(The side opposite to } 30^\circ) \dots (2)$$

$$DP = BP - BD \dots (B - D - P) \text{ (}\frac{1}{2}\text{ mark)}$$

$$= \frac{1}{2} BC - \frac{1}{5} BC \dots \text{[From (2) and given]}$$

$$= \frac{5BC - 2BC}{10} = \frac{3}{10} BC. \dots (3) \text{ (}\frac{1}{2}\text{ mark)}$$

In $\triangle ADP$, by Pythagoras theorem,

$$AD^2 = AP^2 + DP^2 \text{ (}\frac{1}{2}\text{ mark)}$$

$$= \left(\frac{\sqrt{3}}{2} AB \right)^2 + \left(\frac{3}{10} BC \right)^2 \dots \text{[From (1) and (3)] (}\frac{1}{2}\text{ mark)}$$

$$= \frac{3}{4} AB^2 + \frac{9}{100} AB^2 \dots (BC = AB)$$

$$= \frac{75AB^2 + 9AB^2}{100}$$

$$\therefore AD^2 = \frac{84AB^2}{100} \text{ (}\frac{1}{2}\text{ mark)}$$

$$\therefore AD^2 = \frac{21AB^2}{25}$$

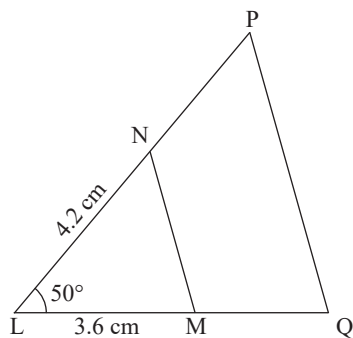
$$\therefore \frac{AD^2}{AB^2} = \frac{21}{25}. \text{ (}\frac{1}{2}\text{ mark)}$$

(ii) Steps of construction :

- (1) Construct $\triangle LMN$ such that $LM = 3.6$ cm, $\angle L = 50^\circ$ and $LN = 4.2$ cm.
- (2) Draw ray LX making a suitable angle with LM .
- (3) Take equal parts $LL_1, L_1L_2, L_2L_3, L_3L_4, L_4L_5, L_5L_6, L_6L_7$ on ray LX .



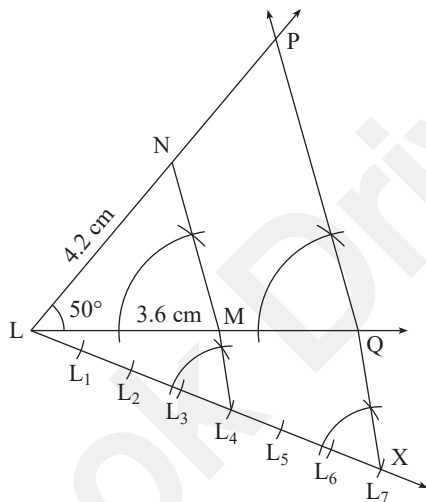
- (4) Join L_4M .
- (5) Draw L_7Q parallel to L_4M and intersecting ray LM in point Q .
- (6) Draw QP parallel to MN and intersecting ray LN in point P .
- (7) $\triangle LQP$ is the required triangle.



Rough figure

[**Note :** Here the steps of construction are given to understand the construction. It is not necessary to write them in the examination.]

Ans.



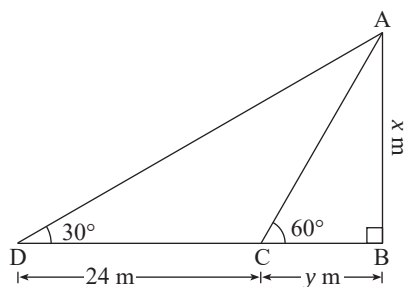
[**Marking scheme :**

- (1) To construct $\triangle LMN$. (1 mark)
- (2) To mark points $L_1, L_2, L_3, L_4, L_5, L_6, L_7$ on ray LX . (1 mark)
- (3) To divide ray LM in the ratio $4 : 7$ to locate point Q . (1 mark)
- (4) To draw $QP \parallel MN$ and obtain $\triangle LQP$. (1 mark)]

- (iii) Let AB represent the tree on the opposite bank of the river.

CB represents the width of the river.

C is the initial position of the observer.



($\frac{1}{2}$ mark for figure)

D is the final position of the observer.

$$DC = 24 \text{ m.}$$

Let AB be x m and BC be y m.

$$\angle BCA = 60^\circ \text{ and } \angle BDA = 30^\circ.$$

In $\triangle ABC$,

$$\tan \angle BCA = \tan 60^\circ = \frac{AB}{BC} \quad (\frac{1}{2} \text{ mark})$$

$$\therefore \sqrt{3} = \frac{x}{y}$$

$$\therefore x = \sqrt{3}y \quad \dots (1) \quad (\frac{1}{2} \text{ mark})$$

In $\triangle ABD$,

$$\tan \angle ADB = \tan 30^\circ = \frac{AB}{BD} \quad (\frac{1}{2} \text{ mark})$$

$$\therefore \frac{1}{\sqrt{3}} = \frac{x}{24+y}$$

$$\therefore x = \frac{24+y}{\sqrt{3}} \quad \dots (2) \quad (\frac{1}{2} \text{ mark})$$

From (1) and (2),

$$\sqrt{3}y = \frac{24+y}{\sqrt{3}} \quad (\frac{1}{2} \text{ mark})$$

$$\therefore \sqrt{3}y \times \sqrt{3} = 24 + y$$

$$\therefore 3y = 24 + y$$

$$\therefore 3y - y = 24$$

$$\therefore 2y = 24$$

$$\therefore y = 12 \quad \dots \text{(Width of the river)} \quad \dots (3) \quad (\frac{1}{2} \text{ mark})$$

The height of the tree = x m

$$= \sqrt{3} \times 12 \text{ m} \quad \dots [\text{From (1) and (3)}]$$

$$= 1.73 \times 12 \text{ m}$$

$$= 20.76 \text{ m} \quad (\frac{1}{2} \text{ mark})$$

Ans The height of the tree is **20.76 m** and
the width of the river is **12 m**.

In **Q. 5**, 2 subquestions each of 3 marks will be asked. You have to solve any one of them. The questions may be open ended or constructive in nature. Mostly the questions will not be from the textbook, but based on syllabus.

Note : Answers with solution are expected.

Q. 5 Solve the following subquestions : (Any one) 3

- (i) The diameter of a circular garden is 13 metres. The distance between two gates of the garden is 13 metres. An electric pole is to be erected on the circumference of the garden such that the difference between the distance of the pole from the gates is 7 metres. Can such a pole be erected? If yes, find the distance of the pole from each gate.
- (ii) The line $x - 6y + 11 = 0$ bisects the segment joining the points $(8, -1)$ and $(0, k)$, then find the value of k .

Solutions

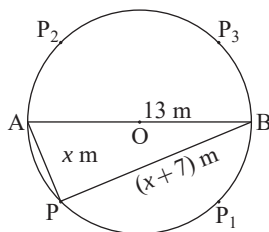
(i) Solution :

AB is the diameter of the garden.

AB = 13 m.

Let A and B represent the gates of the garden.

Let P be the position of the electric pole on the circumference of the garden.



($\frac{1}{2}$ mark for figure)

Let AP be x m.

Then from the given condition,

$$BP = (x + 7) \text{ m.}$$

$$\angle APB = 90^\circ \quad \dots \text{ (Angle inscribed in a semicircle)} \quad (\frac{1}{2} \text{ mark})$$

$\therefore \triangle APB$ is a right angled triangle.

By Pythagoras theorem,

$$AB^2 = AP^2 + BP^2 \quad (\frac{1}{2} \text{ mark})$$

$$\therefore (13)^2 = (x)^2 + (x + 7)^2$$

$$\therefore x^2 + x^2 + 14x + 49 = 169 \quad \dots \text{ (Transposing the sides)}$$

$$\therefore 2x^2 + 14x + 49 - 169 = 0$$

$$\therefore 2x^2 + 14x - 120 = 0$$

$$\therefore x^2 + 7x - 60 = 0 \quad \dots \text{ (Dividing by 2)} \quad (\frac{1}{2} \text{ mark})$$

$$\therefore x^2 + 12x - 5x - 60 = 0$$

$$\therefore x(x + 12) - 5(x + 12) = 0$$

$$\therefore (x + 12)(x - 5) = 0$$

$$\therefore x + 12 = 0 \quad \text{or} \quad x - 5 = 0$$

$$\therefore x = -12 \quad \text{or} \quad x = 5 \quad (\frac{1}{2} \text{ mark})$$

But the distance cannot be negative.

$$\therefore x = -12 \text{ is unacceptable.}$$

$$\therefore x = 5$$

$$x + 7 = 5 + 7 = 12$$

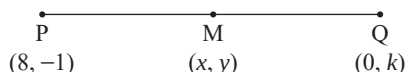
$$\therefore AP = 5 \text{ m, and } BP = 12 \text{ m} \quad (\frac{1}{2} \text{ mark})$$

Ans. A pole can be erected on the circumference of the garden.

The pole will be at a distance of **5 m** from one gate and at a distance of **12 m** from the other gate.

[There are 4 such positions for the pole. 2 poles (P and P_1) on the lower side of diameter and 2 poles (P_2 and P_3) on the upper side of the diameter.]

(ii) Solution :



(Fig. just for understanding)



Let PQ be the segment joining the points P and Q.

Let $P(8, -1) \equiv (x_1, y_1)$ and $Q(0, k) \equiv (x_2, y_2)$.

Let line $x - 6y + 11 = 0$ bisect the segment PQ at the point M.

$\therefore$ point M is the midpoint of seg PQ.

By midpoint formula,

$$x = \frac{x_1 + x_2}{2} \qquad y = \frac{y_1 + y_2}{2} \qquad \left(\frac{1}{2} \text{ mark}\right)$$

$$= \frac{8 + 0}{2} \qquad = \frac{-1 + k}{2} \qquad \left(\frac{1}{2} \text{ mark}\right)$$

$$= \frac{8}{2}$$

$$= 4$$

$$\therefore M\left(4, \frac{-1 + k}{2}\right) \qquad \left(\frac{1}{2} \text{ mark}\right)$$

Now, point M lies on the line $x - 6y + 11 = 0$.

$\therefore$ the coordinates of the point M satisfy the equation of the line.

Substituting $x = 4$ and $y = \frac{-1 + k}{2}$ in the equation of the line

$x - 6y + 11 = 0$, we get,

$$4 - 6\left(\frac{-1 + k}{2}\right) + 11 = 0 \qquad \left(\frac{1}{2} \text{ mark}\right)$$

$$\therefore 4 + 3 - 3k + 11 = 0$$

$$\therefore -3k = -4 - 3 - 11$$

$$\therefore -3k = -18 \qquad \left(\frac{1}{2} \text{ mark}\right)$$

$$\therefore k = \frac{-18}{-3}$$

$$\therefore k = 6 \qquad \left(\frac{1}{2} \text{ mark}\right)$$

Ans. The value of k is **6**.

Question Set 1

THEOREMS

CHAPTER 1

SIMILARITY

Theorem 1.1 : Basic Proportionality Theorem

If a line parallel to a side of a triangle intersects the remaining sides in two distinct points, then the line divides the sides in the same proportion.

(March '24) (3 marks)

Given : In $\triangle XYZ$,

- (i) Line $AB \parallel$ side YZ .
- (ii) Line AB intersects side XY and side XZ in points A and B respectively such that $X-A-Y$ and $X-B-Z$.

To prove : $\frac{XA}{AY} = \frac{XB}{BZ}$.

Construction : Draw seg BY and seg AZ .

Proof : $\triangle XAB$ and $\triangle BAY$ have a common vertex B and their bases XA and AY lie on the same line XY .

$\therefore$ they have equal heights.

$\therefore$ their areas are proportional to their corresponding bases.

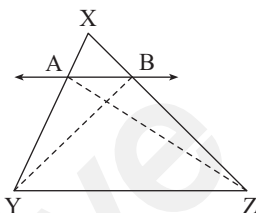
$$\therefore \frac{A(\triangle XAB)}{A(\triangle BAY)} = \frac{XA}{AY} \quad \dots (1)$$

$\triangle XAB$ and $\triangle ABZ$ have a common vertex A and their bases XB and BZ lie on the same line XZ .

$\therefore$ they have equal heights.

$\therefore$ their areas are proportional to their corresponding bases.

$$\therefore \frac{A(\triangle XAB)}{A(\triangle ABZ)} = \frac{XB}{BZ} \quad \dots (2)$$



$\triangle BAY$ and $\triangle ABZ$ lie between the same two parallel lines AB and YZ.

$\therefore$ they have equal heights, also they have same base AB.

$$\therefore A(\triangle BAY) = A(\triangle ABZ) \quad \dots (3)$$

$\therefore$ from (1), (2) and (3), we get,

$$\frac{A(\triangle XAB)}{A(\triangle BAY)} = \frac{A(\triangle XAB)}{A(\triangle ABZ)} \quad \dots (4)$$

$\therefore$ from (1), (2) and (4), we get,

$$\frac{XA}{AY} = \frac{XB}{BZ}.$$

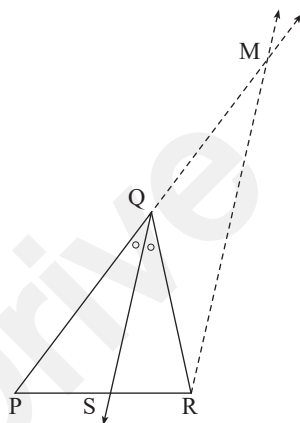
Theorem 1.2 : Theorem of an Angle Bisector of a Triangle

The bisector of an angle of a triangle divides the side opposite to the angle in the ratio of the remaining sides.

Given : In $\triangle PQR$, bisector of $\angle PQR$ intersects side PR in point S such that P-S-R.

To prove : $\frac{PQ}{QR} = \frac{PS}{SR}.$

Construction : Draw a ray RM parallel to ray QS through point R, intersecting PQ extended in the point M such that P-Q-M.



Proof : In $\triangle PMR$,

Ray QS $\parallel$ side MR ... (Construction)

$\therefore$ by basic proportionality theorem,

$$\frac{PQ}{QM} = \frac{PS}{SR} \quad \dots (1)$$

Ray QS $\parallel$ side MR and line PM is the transversal.

$$\therefore \angle PQS \cong \angle QMR \quad \dots \text{ (Corresponding angles) } \dots (2)$$

Ray QS $\parallel$ side MR and line QR is the transversal.

$$\therefore \angle SQR \cong \angle QRM \quad \dots \text{ [Alternate angles] } \dots (3)$$

$$\angle PQS \cong \angle SQR \quad \dots \text{ (Ray QS is the bisector of } \angle PQR) \dots (4)$$

In $\triangle QRM$,

$$\angle QMR \cong \angle QRM \quad \dots \text{ [From (2), (3) and (4)]}$$

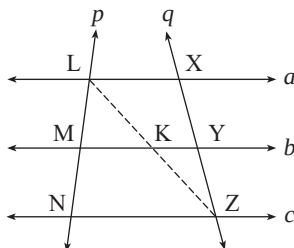


$\therefore \text{seg } QR \cong \text{seg } QM \dots$ (Converse of isosceles triangle theorem) ... (5)

$$\therefore \frac{PQ}{QR} = \frac{PS}{SR} \dots \text{[From (1) and (5)]}$$

Theorem 1.3 : Three parallel lines and its transversal

The ratio of the intercepts made on a transversal by three parallel lines is equal to the ratio of the corresponding intercepts made on any other transversal by the same parallel lines. (Nov. '20) (3 marks)



Given : Line $a \parallel$ line $b \parallel$ line c . Transversal p intersects lines a , b and c in points L , M and N respectively. Transversal q intersects lines a , b and c in points X , Y and Z respectively.

To prove : $\frac{LM}{MN} = \frac{XY}{YZ}$.

Construction : Draw seg LZ . Let seg LZ intersect seg MY at point K such that $M-K-Y$ and $L-K-Z$.

Proof : In $\triangle LNZ$,

seg $MK \parallel$ side NZ ... (Given)

$\therefore$ by basic proportionality theorem,

$$\frac{LM}{MN} = \frac{LK}{KZ} \dots (1)$$

In $\triangle LZX$,

seg $KY \parallel$ side LX ... (Given)

$\therefore$ by basic proportionality theorem,

$$\frac{LK}{KZ} = \frac{XY}{YZ} \dots (2)$$

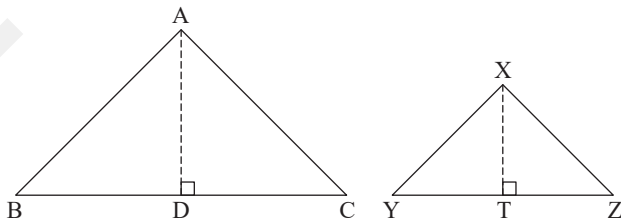
$\therefore$ from (1) and (2),

$$\frac{LM}{MN} = \frac{XY}{YZ}.$$



Theorem 1.4 : Theorem of Areas of Similar Triangles

When two triangles are similar, the ratio of the areas of those triangles is equal to the ratio of the squares of their corresponding sides.



Given : $\triangle ABC \sim \triangle XYZ$.

To prove : $\frac{A(\triangle ABC)}{A(\triangle XYZ)} = \frac{AB^2}{XY^2} = \frac{BC^2}{YZ^2} = \frac{AC^2}{XZ^2}$.

Construction : Draw seg $AD \perp$ side BC such that $B-D-C$ and seg $XT \perp$ side YZ such that $Y-T-Z$.

Proof : The ratio of areas of two triangles is equal to the ratio of the products of their bases and corresponding heights.

$$\therefore \frac{A(\triangle ABC)}{A(\triangle XYZ)} = \frac{BC \times AD}{YZ \times XT}$$

$$\therefore \frac{A(\triangle ABC)}{A(\triangle XYZ)} = \frac{BC}{YZ} \times \frac{AD}{XT} \quad \dots (1)$$

$$\triangle ABC \sim \triangle XYZ \quad \dots (\text{Given})$$

$$\therefore \frac{AB}{XY} = \frac{BC}{YZ} = \frac{AC}{XZ} \quad \dots (\text{Corresponding sides of similar triangles are in proportion}) \quad \dots (2)$$

In $\triangle ABD$ and $\triangle XYT$,

$$\angle ABD \cong \angle XYT \quad \dots (\text{Given})$$

$$\angle ADB \cong \angle XTY \quad \dots (\text{Each of measure } 90^\circ)$$

$$\therefore \triangle ABD \sim \triangle XYT \quad \dots (\text{AA test of similarity})$$

$$\therefore \frac{AB}{XY} = \frac{AD}{XT} \quad \dots (\text{Corresponding sides of similar triangles are in proportion}) \quad \dots (3)$$

$\therefore$ from (2) and (3), we get,

$$\frac{AB}{XY} = \frac{BC}{YZ} = \frac{AD}{XT}.$$

$$\therefore \frac{BC}{YZ} = \frac{AD}{XT} \quad \dots (4)$$

$$\frac{A(\triangle ABC)}{A(\triangle XYZ)} = \frac{BC}{YZ} \times \frac{BC}{YZ} \quad \dots [\text{From (1) and (4)}]$$

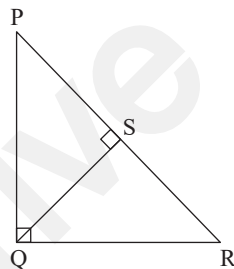
$$\therefore \frac{A(\triangle ABC)}{A(\triangle XYZ)} = \frac{BC^2}{YZ^2} \quad \dots (5)$$

$$\therefore \frac{A(\triangle ABC)}{A(\triangle XYZ)} = \frac{AB^2}{XY^2} = \frac{BC^2}{YZ^2} = \frac{AC^2}{XZ^2} \quad \dots [\text{From (1) and (5)}]$$

CHAPTER 2 PYTHAGORAS THEOREM

Theorem 2.1 : Similarity and Right Angled Triangle

In a right angled triangle, if the altitude is drawn to the hypotenuse, then the two triangles formed are similar to the original triangle and to each other.



Given : (i) In $\triangle PQR$, $\angle PQR = 90^\circ$.

(ii) Seg $QS \perp$ hypotenuse PR such that $P-S-R$.

To prove : (i) $\triangle PSQ \sim \triangle PQR$

(ii) $\triangle QSR \sim \triangle PQR$ (iii) $\triangle PSQ \sim \triangle QSR$.

Proof : In $\triangle PSQ$ and $\triangle PQR$,

$\angle QPS \cong \angle RPQ$... (Common angle)

$\angle PSQ \cong \angle PQR$... (Each of measure 90°)

$\therefore \triangle PSQ \sim \triangle PQR$... (AA test of similarity) ... (1)

In $\triangle QSR$ and $\triangle PQR$,

$\angle SRQ \cong \angle QRP$... (Common angle)

$\angle QSR \cong \angle PQR$... (Each of measure 90°)

$\therefore \triangle QSR \sim \triangle PQR$... (AA test of similarity) ... (2)

From (1) and (2), we get,

$\triangle PSQ \sim \triangle QSR$.

Theorem 2.2 : Theorem of Geometric Mean

In a right angled triangle, the perpendicular segment to the hypotenuse from the opposite vertex, is the geometric mean of the segments into which the hypotenuse is divided. (Sept. '21) (3 marks)

Given : In $\triangle ABC$, $\angle ABC = 90^\circ$,

seg $BD \perp$ hypotenuse AC such that $A-D-C$.

To prove : $BD^2 = AD \times DC$

Proof : In $\triangle ABC$, $\angle ABC = 90^\circ$

seg $BD \perp$ hypotenuse AC

$\therefore \triangle ADB \sim \triangle BDC$

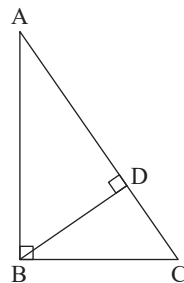
... (Similarity of right angled triangles)

$$\therefore \frac{AD}{BD} = \frac{BD}{CD}$$

... (Corresponding sides of similar triangles are in proportion)

$$\therefore BD^2 = AD \times DC$$

$$\text{i.e. } BD^2 = AD \times DC.$$



... (Given)

Theorem 2.3 : Pythagoras Theorem

In a right angled triangle, the square of the hypotenuse is equal to the sum of the squares of remaining two sides. (July '23) (3 marks)

Given : In $\triangle PQR$, $\angle PQR = 90^\circ$.

To prove : $PR^2 = PQ^2 + QR^2$.

Construction : Draw seg $QS \perp$ side PR such that $P-S-R$.

Proof : In $\triangle PQR$,

$$\angle PQR = 90^\circ$$

... (Given)

Seg $QS \perp$ hypotenuse PR

... (Construction)

$$\therefore \triangle PSQ \sim \triangle QSR \sim \triangle PQR$$

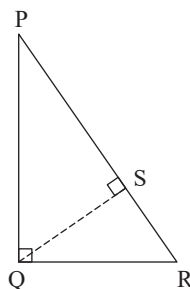
... (Similarity of right angled triangles) ... (1)

$$\triangle PSQ \sim \triangle PQR$$

... [From (1)]

$$\therefore \frac{PQ}{PR} = \frac{PS}{PQ}$$

... (Corresponding sides of similar triangles are in proportion)



$$\therefore PQ^2 = PS \times PR \quad \dots (2)$$

$$\triangle QSR \sim \triangle PQR \quad \dots [\text{From (1)}]$$

$$\therefore \frac{QR}{PR} = \frac{SR}{QR} \quad \dots (\text{Corresponding sides of similar triangles are in proportion})$$

$$\therefore QR^2 = SR \times PR \quad \dots (3)$$

Adding (2) and (3), we get,

$$PQ^2 + QR^2 = PS \times PR + SR \times PR$$

$$\therefore PQ^2 + QR^2 = PR(PS + SR)$$

$$\therefore PQ^2 + QR^2 = PR \times PR \quad \dots (P-S-R)$$

$$\therefore PQ^2 + QR^2 = PR^2 \quad \text{OR} \quad PR^2 = PQ^2 + QR^2.$$

Theorem 2.4 : Apollonius Theorem

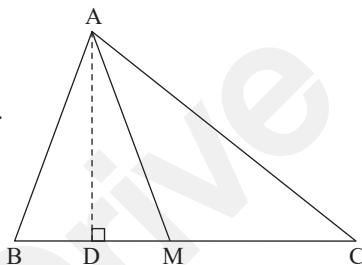
In $\triangle ABC$, if M is the midpoint of side BC , then $AB^2 + AC^2 = 2AM^2 + 2BM^2$.

Given : In $\triangle ABC$, M is the midpoint of seg BC .

To prove :

$$AB^2 + AC^2 = 2AM^2 + 2BM^2.$$

Construction : Draw seg $AD \perp$ side BC such that $B-D-C$.



Proof : Let's consider the case, where AM is not perpendicular to seg BC , then out of $\angle AMB$ and $\angle AMC$, one is acute angle and other is obtuse angle.

Let's consider $\angle AMB$ as an acute angle and $\angle AMC$ as an obtuse angle. $\triangle ABM$ is an acute angled triangle.

$\therefore$ by application of Pythagoras theorem, we get,

$$AB^2 = AM^2 + BM^2 - 2BM \cdot DM \quad \dots (1)$$

$\triangle AMC$ is an obtuse angled triangle.

By application of Pythagoras theorem, we get,

$$AC^2 = AM^2 + MC^2 + 2MC \cdot DM \quad \dots (2)$$

$$\text{But } MC = BM \quad \dots (M \text{ is the midpoint of side } BC) \quad \dots (3)$$

Substituting (3) in (2), we get,

$$AC^2 = AM^2 + BM^2 + 2BM \cdot DM \quad \dots (4)$$



Adding (1) and (4), we get,

$$AB^2 + AC^2 = AM^2 + BM^2 - 2BM \cdot DM + AM^2 + BM^2 + 2BM \cdot DM$$

$$\therefore AB^2 + AC^2 = 2AM^2 + 2BM^2.$$

Let's consider the case, where seg $AM \perp$ seg BC .

In $\triangle AMB$,

$$\angle AMB = 90^\circ.$$

$\therefore$ by Pythagoras theorem,

$$AB^2 = AM^2 + BM^2$$

... (5)

In $\triangle AMC$,

$$\angle AMC = 90^\circ.$$

$\therefore$ by Pythagoras theorem,

$$AC^2 = AM^2 + MC^2$$

... (6)

But $MC = BM$

... (M is the midpoint of side BC) ... (7)

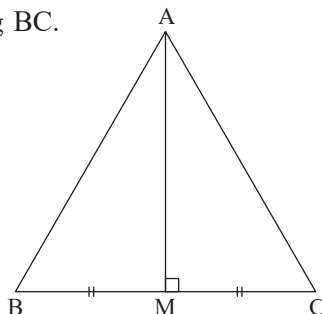
$$\therefore AC^2 = AM^2 + BM^2$$

... [From (6) and (7)] ... (8)

Adding (5) and (8), we get,

$$AB^2 + AC^2 = AM^2 + BM^2 + AM^2 + BM^2$$

$$\therefore AB^2 + AC^2 = 2AM^2 + 2BM^2.$$



CHAPTER 3

CIRCLE

Theorem 3.1 : Tangent Theorem

Statement : A tangent at any point of a circle is perpendicular to the radius at the point of contact.

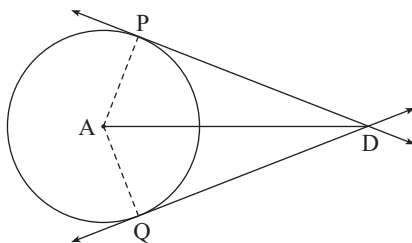
[Note : For the proof of the above theorem, please refer to Mathematics Digest (Part-II) Std. X.]

Theorem 3.2 : Tangent Segment Theorem

Tangent segments drawn from an external point to a circle are congruent. (March '20) (3 marks)

Given : (1) A circle with centre A.

(2) DP and DQ are tangents from an external point D.



To prove : $\text{seg DP} \cong \text{seg DQ}$.

Construction :

Draw radius AP and radius AQ.

Proof : In $\triangle PAD$ and $\triangle QAD$, $\angle APD = \angle AQD = 90^\circ$

$\text{seg PA} \cong \text{seg QA}$... (Radii of the same circle)

$\text{seg AD} \cong \text{seg AD}$... (Common side)

$\therefore \triangle PAD \cong \triangle QAD$... (Hypotenuse side theorem)

$\therefore \text{seg DP} \cong \text{seg DQ}$... (c.s.c.t.)

Theorem 3.3 : Theorem of Touching Circles

If two circles touch each other, their point of contact lies on the line joining the centres.

[Note : For the proof of the above theorem, please refer to Mathematics Digest (Part-II) Std. X.]

Theorem 3.4 : The chords corresponding to congruent arcs of a circle (or congruent circles) are congruent.

[Note : For the proof of the above theorem, please refer to Mathematics Digest (Part-II) Std. X.]

Theorem 3.5 : Corresponding arcs of congruent chords of a circle (or congruent circles) are congruent.

[Note : For the proof of the above theorem, please refer to Mathematics Digest (Part-II) Std. X.]

Theorem 3.6 : Inscribed Angle Theorem

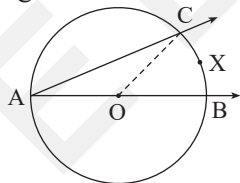
The measure of an inscribed angle is half of the measure of the arc intercepted by it.

Given : In a circle with centre O, $\angle BAC$ is inscribed in an arc BAC. $\angle BAC$ intercepts arc BXC of the circle.

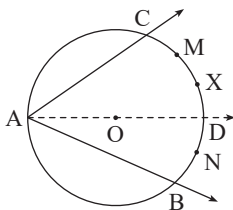
To prove : $m\angle BAC = \frac{1}{2} m(\text{arc BXC})$



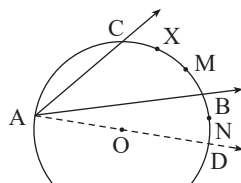
Proof : There arise three cases as shown in parts (a), (b), (c) of the figure.



(a)



(b)



(c)

Case 1 : The centre lies on the arm of the angle [as in figure (a)].

In this case, join C to O.

$\triangle OAC$ is isosceles with $OA = OC$.

Let $\angle OAC = \angle OCA = p$.

By remote interior angles theorem,

$\angle BOC = \angle OAC + \angle OCA$

$$\angle BOC = p + p = 2p = 2 \angle BAC. \quad \therefore \angle BAC = \frac{1}{2} \angle BOC$$

But $\angle BOC = m(\text{arc BXC})$... (Definition of measure of minor arc)

$$\therefore \angle BAC = \frac{1}{2} m(\text{arc BXC}).$$

Case 2 : The centre lies in the interior of the angle [as in figure (b)].

In this case, let D be the other endpoint of the diameter drawn through A.

Let arc CMD be the one intercepted by $\angle CAD$, and let arc BND be the one intercepted by $\angle DAB$.

Then as proved in the first situation, we have,

$$\angle CAD = \frac{1}{2} m(\text{arc CMD}) \text{ and } \angle DAB = \frac{1}{2} m(\text{arc BND})$$

$$\therefore \angle CAD + \angle DAB = \frac{1}{2} m(\text{arc CMD}) + \frac{1}{2} m(\text{arc BND})$$

$$\therefore m \angle BAC = \frac{1}{2} [m(\text{arc CMD}) + m(\text{arc BND})] = \frac{1}{2} m(\text{arc BXC}).$$

Case 3 : The centre lies in the exterior of the angle [as in figure (c)].

Again let D be the other endpoint of the diameter drawn through A.

Let arc CMD be the one intercepted by $\angle CAD$ and let arc BND be the one intercepted by $\angle DAB$.

Then as proved in the first situation, we have,

$$\angle CAD = \frac{1}{2} m(\text{arc CMD}) \text{ and } \angle DAB = \frac{1}{2} m(\text{arc BND})$$

$$\therefore \angle CAD - \angle DAB = \frac{1}{2} m(\text{arc CMD}) - \frac{1}{2} m(\text{arc BND})$$

$$\therefore \angle BAC = \frac{1}{2} [m(\text{arc CMD}) - m(\text{arc BND})] = \frac{1}{2} m(\text{arc BXC}).$$

Corollaries of inscribed angle theorem :

(1) Angles inscribed in the same arc are congruent.

(March '19) (Activity for 2 marks); (March '23) (3 marks)

Given : (1) A circle with centre O.

(2) $\angle ABD$ and $\angle ACD$ are inscribed in arc APD and intercepts arc APD.

To prove : $\angle ABD \cong \angle ACD$.

Proof :

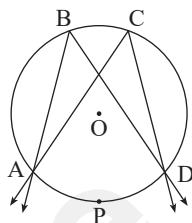
$$\angle ABD = \frac{1}{2} m(\text{arc APD}) \quad \dots (1)$$

$$\angle ACD = \frac{1}{2} m(\text{arc APD}) \quad \dots (2)$$

... (Inscribed angle theorem)

From (1) and (2), we get, $\angle ABD = \angle ACD$.

$\therefore \angle ABD \cong \angle ACD$.



(2) Angle inscribed in a semicircle is a right angle.

Given : (1) A circle with centre O.

(July '22) (As a activity for 2 marks)

(2) Seg AC is the diameter.

(3) $\angle ABC$ is inscribed in arc ABC and intercepts arc AMC.

To prove : $\angle ABC = 90^\circ$.

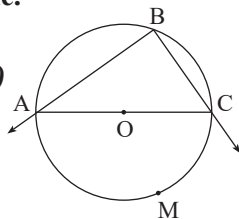
Proof :

$$m(\text{arc AMC}) = 180^\circ \quad \dots (\text{Measure of a semicircle is } 180^\circ) \quad \dots (1)$$

$$\angle ABC = \frac{1}{2} m(\text{arc AMC}) \quad \dots (\text{Inscribed angle theorem})$$

$$\therefore \angle ABC = \frac{1}{2} \times 180^\circ \quad \dots [\text{From (1)}]$$

$$\therefore \angle ABC = 90^\circ.$$



Theorem 3.7 : Theorem of Cyclic Quadrilateral

Opposite angles of a cyclic quadrilateral are supplementary.

(March '22; July '22–'24) (3 marks)

[Note : For proof refer to section 1, July 2024 Board Paper Solution to Q. 3 (B) (i).]

Corollary of cyclic quadrilateral theorem :

An exterior angle of a cyclic quadrilateral is congruent to the angle opposite to its adjacent interior angle.

Given : □ ABCD is cyclic. $\angle DCE$ is an exterior angle of □ ABCD.

To prove : $\angle DCE \cong \angle BAD$

Proof : $\angle DCE + \angle BCD = 180^\circ$

... (Linear pair of angles) ... (1)

□ ABCD is cyclic.

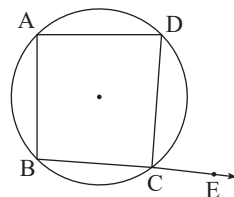
$\therefore \angle BAD + \angle BCD = 180^\circ$... (Theorem of cyclic quadrilateral) ... (2)

$\therefore$ from (1) and (2), we get,

$\angle DCE + \angle BCD = \angle BAD + \angle BCD$

$\therefore \angle DCE = \angle BAD$

$\therefore \angle DCE \cong \angle BAD$.



Theorem 3.8 : If two points on a given line subtend equal angles at two distinct points which lie on the same side of the line, then the four points are concyclic.

[Note : For the proof of the above theorem, please refer to Mathematics Digest (Part–II) Std. X.]

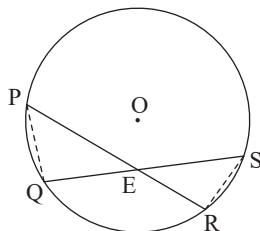
Theorem 3.9 : Theorem of Angle Between Tangent and Secant

If an angle has its vertex on the circle, its one side touches the circle and the other intersects the circle in one more point, then the measure of the angle is half the measure of its intercepted arc.

[Note : For the proof of the above theorem, please refer to Mathematics Digest (Part–II) Std. X.]

Theorem 3.10 : Theorem of Internal Division of Chords

Suppose two chords of a circle intersect each other in the interior of the circle, then the product of the lengths of the two segments of one chord is equal to the product of the lengths of the two segments of the other chord.



Given : (1) A circle with centre O.

(2) Chords PR and QS intersect at point E inside the circle.

To prove : $PE \times ER = QE \times ES$.

Construction : Draw seg PQ and seg RS.

Proof : $\angle PQS \cong \angle SRP$

... (Angles inscribed in the same arc are congruent)

i.e. $\angle PQE \cong \angle SRE$

... (P-E-R and Q-E-S) ... (1)

In $\triangle PQE$ and $\triangle SRE$,

$\angle PQE \cong \angle SRE$

... [From (1)]

$\angle PEQ \cong \angle SER$

... (Vertically opposite angles)

$\therefore \triangle PQE \sim \triangle SRE$

... (AA test of similarity)

$\therefore \frac{PE}{SE} = \frac{QE}{RE}$

... (Corresponding sides of similar triangles are in proportion)

$\therefore PE \times ER = QE \times ES$.

Theorem 3.11 : Theorem of External Division of Chords

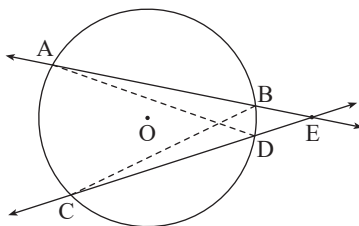
If secants containing chords AB and CD of a circle intersect outside the circle in point E, then $AE \times EB = CE \times ED$.

Given : (1) A circle with centre O.

(2) Secants AB and CD intersect at point E outside the circle.

To prove : $AE \times EB = CE \times ED$.

Construction : Draw seg AD and seg BC.



Proof : $\angle DAB \cong \angle BCD$

... (Angles inscribed in the same arc are congruent)

i.e. $\angle DAE \cong \angle BCE$

... (A-B-E and C-D-E) ... (1)



In $\triangle ADE$ and $\triangle CBE$,

$$\angle DAE \cong \angle BCE$$

... [From (1)]

$$\angle AED \cong \angle CEB$$

... (Common angle)

$$\therefore \triangle ADE \sim \triangle CBE$$

... (AA test of similarity)

$$\therefore \frac{AE}{CE} = \frac{DE}{BE}$$

... (Corresponding sides of similar triangles are in proportion)

$$\therefore AE \times BE = CE \times ED.$$

Theorem 3.12 : Tangent Secant Theorem

Point E is in the exterior of a circle. A secant through E intersects the circle at points A and B, and a tangent through E touches the circle at point T, then $EA \times EB = ET^2$.

Given : (1) A circle with centre O.

(2) Tangent ET touches the circle at point T.

(3) Secant EAB intersects the circle at points A and B.

To prove : $EA \times EB = ET^2$.

Construction : Draw seg TA and seg TB.

Proof : In $\triangle ETA$ and $\triangle EBT$,

$$\angle AET \cong \angle TEB$$

... (Common angle)

$$\angle ETA \cong \angle EBT$$

... (Tangent secant theorem)

$$\therefore \triangle ETA \sim \triangle EBT$$

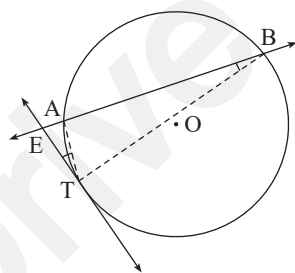
... (AA test of similarity)

$$\therefore \frac{ET}{EB} = \frac{EA}{ET}$$

... (Corresponding sides of similar triangles are in proportion)

$$\therefore ET \times ET = EA \times EB$$

$$\therefore ET^2 = EA \times EB, \text{ i.e. } EA \times EB = ET^2.$$



**Question
Set
2**

SIMILARITY

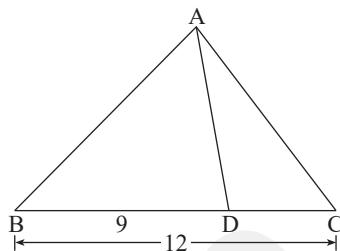
2.1 1 mark each

Q. Choose the correct alternative from those given below each question :

1. In the figure, $BD = 9$, $BC = 12$
and $B-D-C$, then

$$\frac{A(\triangle ABD)}{A(\triangle ADC)} = \dots\dots\dots$$

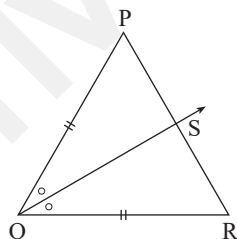
- (A) 1 : 3 (B) 3 : 1
(C) 3 : 4 (D) 4 : 3



Ans. (B) [Hint : The ratio of areas of two triangles with equal heights is equal to the ratio of their corresponding bases.]

2. In the figure, ray QS is the bisector of $\angle PQR$
and $PQ = QR$, then $\frac{PS}{SR} = \dots\dots\dots$

- (A) 1 : 1 (B) 1 : 2
(C) 2 : 1 (D) cannot be determined

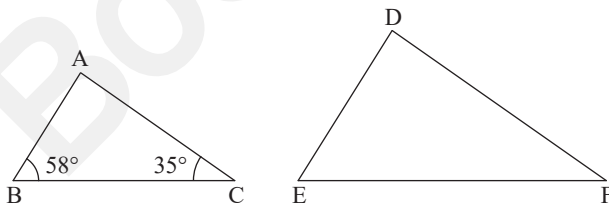


Ans. (A)

[Hint : By theorem of angle bisector of triangle.

$$\frac{PQ}{QR} = \frac{PS}{SR}, \frac{PQ}{QR} = 1 \quad \therefore \frac{PS}{SR} = 1]$$

3.



In the figure, $\triangle ABC \sim \triangle DEF$, $\angle B = 58^\circ$, $\angle C = 35^\circ$ then the measure of $\angle D = \dots\dots\dots$

- (A) 89° (B) 78° (C) 87° (D) 85°



Ans. (C) [Hint : $\angle A + \angle B + \angle C = 180^\circ \therefore \angle A = 87^\circ$,
 $\angle D = \angle A$, corresponding angles of similar triangles.]

4. $\triangle ABC \sim \triangle PQR$. If $AB = 4$ cm, $PQ = 6$ cm and $QR = 9$ cm then $BC = ?$ **(Sept. '21)**

(A) 7 cm (B) 6 cm (C) 8 cm (D) 9 cm

Ans. (B) [Hint : $\frac{AB}{PQ} = \frac{BC}{QR}$, corresponding sides of similar triangles.]

5. $\triangle ABC$ and $\triangle DEF$ are equilateral triangles, $A(\triangle ABC) : A(\triangle DEF) = 1 : 2$. If $AB = 4$, then what is the length of DE ? **(March '19)**

(A) $\sqrt{2}$ (B) 4 (C) 8 (D) $4\sqrt{2}$

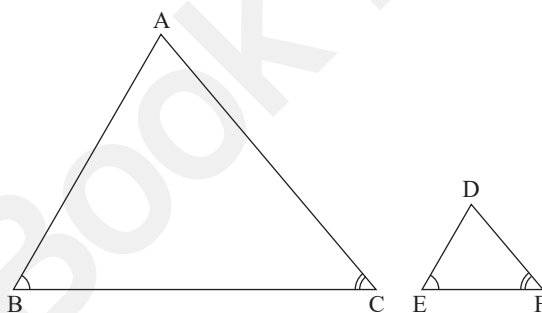
Ans. (D) [Hint : $\triangle ABC \sim \triangle DEF$, hence $\frac{A(\triangle ABC)}{A(\triangle DEF)} = \frac{AB^2}{DE^2}$.]

6. If $\triangle ABC \sim \triangle PQR$ and $4A(\triangle ABC) = 25A(\triangle PQR)$, then $AB : PQ = ?$ **(July '19)**

(A) 4 : 25 (B) 2 : 5 (C) 5 : 2 (D) 25 : 4

Ans. (C) [Hint : $\frac{A(\triangle ABC)}{A(\triangle PQR)} = \frac{AB^2}{PQ^2} \therefore \frac{AB^2}{PQ^2} = \frac{25}{4}$.]

7. In $\triangle ABC$ and $\triangle DEF$, $\angle B = \angle E$, $\angle F = \angle C$ and $AB = 3DE$, then which of the statements regarding the two triangles is true?



- (A) The triangles are not congruent and not similar.
 (B) The triangles are similar but not congruent.
 (C) The triangles are congruent and similar.
 (D) None of the statements above is true.

Ans. (B) [**Hint :** $\triangle ABC$ is larger than $\triangle DEF$, therefore, they are not congruent. $\triangle ABC \sim \triangle DEF$... (AA test of similarity)]

8. The shadow of a 40 m high pole falls 20 m long on the ground, at the same time the shadow of a building falls 50 m long. What is the height of the building?

(A) 100 m (B) 50 m (C) 25 m (D) 150 m

Ans. (A) [**Hint :** The triangle obtained by the pole with its shadow will be similar to triangle obtained by the building with

its shadow. So, $\frac{40}{20} = \frac{x}{50} \therefore x = 100 \text{ m}$]

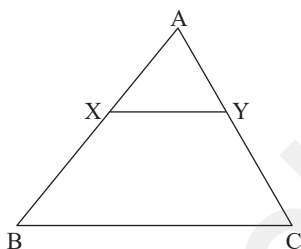
[**Note :** Hints are given only for understanding the solution. It is not expected to write in the examination.]

Assignment	2.1
-------------------	------------

1. If $\triangle ABC \sim \triangle DEF$ and $\angle A = 48^\circ$, then $\angle D = \dots\dots\dots$ (**March '22**)

(A) 48° (B) 83° (C) 49° (D) 132°

2.

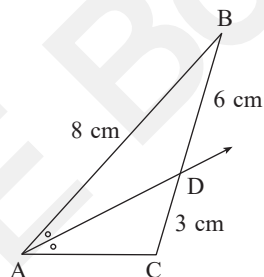


In the figure, $\text{seg } XY \parallel \text{seg } BC$, then which of the following statements is true?

(A) $\frac{AB}{AC} = \frac{AX}{AY}$ (B) $\frac{AX}{AB} = \frac{AY}{AC}$

(C) $\frac{AX}{YC} = \frac{AY}{XB}$ (D) $\frac{AB}{YC} = \frac{AC}{XB}$

3.



In $\triangle ABC$, $\text{seg } AD$ is a bisector of $\angle BAC$.

$B-D-C$. $AB = 8 \text{ cm}$, $BD = 6 \text{ cm}$ and

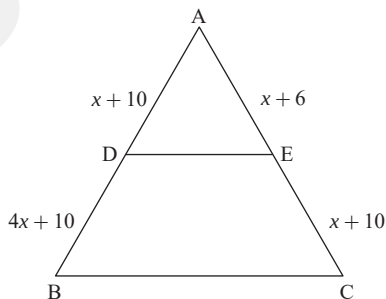
$DC = 3 \text{ cm}$, then length of $AC = \dots\dots\dots$

(A) 4 cm (B) 3 cm

(C) 6 cm (D) 8 cm



4. In the figure, for what value of x will seg DE be parallel to BC?

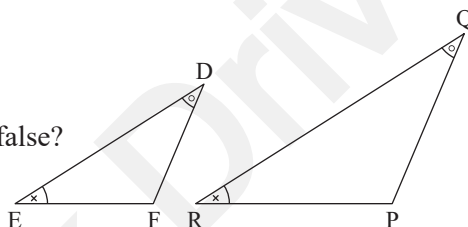


- (A) 2 (B) 3 (C) 20 (D) 2 and 20
5. $\triangle PQR \sim \triangle STU$ and $A(\triangle PQR) : A(\triangle STU) = 64 : 81$, then what is the ratio of corresponding sides? (July '22)
- (A) 8 : 9 (B) 64 : 81 (C) 9 : 8 (D) 16 : 27

6. $\triangle ABC \sim \triangle DEF$, then $\frac{AB}{DE} = \frac{\dots\dots\dots}{EF} = \frac{CA}{\dots\dots\dots}$

(A) AD, BE (B) AC, BD (C) BC, FD (D) BC, DE

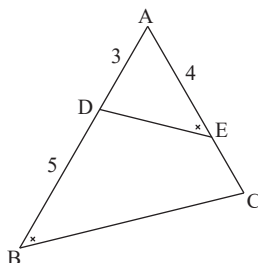
7. If in $\triangle DEF$ and $\triangle PQR$,
 $\angle D \cong \angle Q$, $\angle R \cong \angle E$,
 then which of the following statements is false?



- (A) $\frac{EF}{PR} = \frac{DF}{PQ}$ (B) $\frac{DE}{PQ} = \frac{EF}{RP}$ (C) $\frac{DE}{QR} = \frac{DF}{PQ}$ (D) $\frac{EF}{RP} = \frac{DE}{QR}$

8. In the figure,
 $\angle AED = \angle ABC$,
 $AD = 3$, $DB = 5$, $AE = 4$,
 then length of AC is

- (A) 2 (B) 6
 (C) 8 (D) 12



Answers :

1. (A)

[$\angle A \cong \angle D \therefore \angle D = 48^\circ$.]

2. (B) $\left[\begin{array}{l} \triangle AXY \sim \triangle ABC \quad \dots \text{ (AA test of similarity)} \\ \therefore \frac{AX}{AB} = \frac{AY}{AC} \quad \dots \text{ (Corresponding sides)} \end{array} \right]$
3. (A) $\left[\frac{AB}{AC} = \frac{BD}{DC} \quad \dots \text{ (Theorem of an angle bisector of a triangle)} \right]$
4. (A) [seg DE $\parallel$ seg BC, if $\frac{AD}{DB} = \frac{AE}{EC}$
 $\therefore \frac{x+10}{4x+10} = \frac{x+6}{x+10}$ On solving, $x = 2$.]
5. (A) [By theorem on areas of similar triangles.]
6. (C) [ABC $\longleftrightarrow$ DEF is the one-one correspondence.]
7. (B) [$\triangle DEF \sim \triangle QRP$ $\dots$ (By AA test of similarity)]
8. (B) $\left[\triangle AED \sim \triangle ABC \therefore \frac{AE}{AB} = \frac{AD}{AC}, \text{ corresponding sides of similar triangles.} \right]$

2.2 1 mark each

1. If $\triangle ABC \sim \triangle DEF$, then write the corresponding congruent angles and also write the ratios of corresponding sides. (July '19)

Ans.

$$\triangle ABC \sim \triangle DEF$$

$$\left. \begin{array}{l} \therefore \angle A \cong \angle D \\ \angle B \cong \angle E \\ \angle C \cong \angle F \end{array} \right\} \dots \text{ (Corresponding angles of similar triangles)}$$

$$\frac{AB}{DE} = \frac{BC}{EF} = \frac{AC}{DF} \quad \dots \text{ (Corresponding sides of similar triangles)}$$

2. If $\triangle ABC \sim \triangle PQR$ and $AB : PQ = 2 : 3$, then find the value of $\frac{A(\triangle ABC)}{A(\triangle PQR)}$. (March '24)

Ans.

$$\triangle ABC \sim \triangle PQR$$



$$\frac{A(\triangle ABC)}{A(\triangle PQR)} = \frac{AB^2}{PQ^2}$$

... (Theorem of areas of similar triangles)

$$\therefore \frac{A(\triangle ABC)}{A(\triangle PQR)} = \left(\frac{AB}{PQ}\right)^2$$

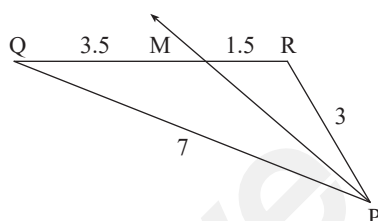
$$\therefore \frac{A(\triangle ABC)}{A(\triangle PQR)} = \left(\frac{2}{3}\right)^2$$

$$\therefore \frac{A(\triangle ABC)}{A(\triangle PQR)} = \frac{4}{9}$$

The value of $\frac{A(\triangle ABC)}{A(\triangle PQR)}$ is $\frac{4}{9}$.

3. In the figure, $PQ = 7$, $PR = 3$, $QM = 3.5$ and $MR = 1.5$.

State with reason whether PM is the bisector of $\angle QPR$ or not.



Ans. $\frac{PQ}{PR} = \frac{7}{3}$ and $\frac{QM}{MR} = \frac{3.5}{1.5} = \frac{7}{3}$

$$\therefore \frac{PQ}{PR} = \frac{QM}{MR}$$

$\therefore$ ray PM bisects $\angle QPR$... (Converse of theorem of angle bisector of a triangle.)

4. For $\triangle ABC$ and $\triangle XYZ$, if $\frac{AB}{YZ} = \frac{BC}{ZX} = \frac{AC}{XY}$, then state the correspondence under which $\triangle ABC$ and $\triangle XYZ$ are similar.

Ans. $ABC \longleftrightarrow YZX$.

5. If $\triangle ABC \sim \triangle PQR$ and $\frac{A(\triangle ABC)}{A(\triangle PQR)} = \frac{16}{25}$ then find $AB : PQ$.

(March '23)

Ans. $\triangle ABC \sim \triangle PQR$

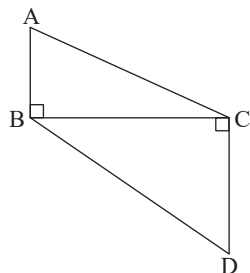
... (Given)

$$\therefore \frac{A(\triangle ABC)}{A(\triangle PQR)} = \frac{AB^2}{PQ^2} \quad \dots \text{(Theorem of areas of similar triangles)}$$

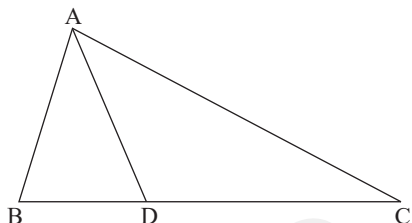
$$\therefore \frac{16}{25} = \frac{AB^2}{PQ^2} \quad \therefore \frac{AB}{PQ} = \frac{4}{5} \quad \dots \text{(Taking square roots of both the sides)}$$

$$AB : PQ = 4 : 5.$$

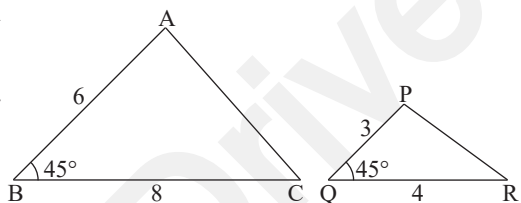
1. In the figure, $\text{seg } AB \perp \text{seg } BC$ and $\text{seg } DC \perp \text{seg } BC$. If $AB = 3 \text{ cm}$ and $CD = 4 \text{ cm}$, then find $\frac{A(\triangle ABC)}{A(\triangle DCB)}$. (Nov. 20; Sept. '21)



2. In the given figure, in $\triangle ABC$, $B-D-C$ and $BD = 7$, $BC = 20$, then find $\frac{A(\triangle ABD)}{A(\triangle ABC)}$. (March '25)

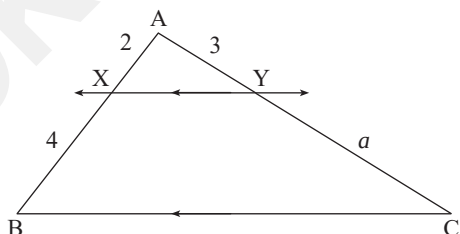


3. Observe the figure and state whether $\triangle ABC$ is similar to $\triangle PQR$ or not? Why?



4. The ratio of corresponding sides of two similar triangles is $3 : 5$, then write the ratio of their areas. (March '20)

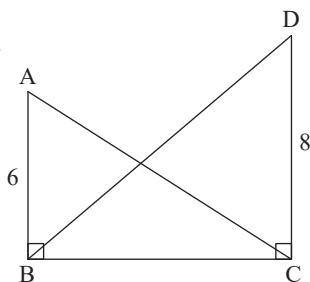
5. In the figure, line $XY \parallel \text{side } BC$. $AX = 2$, $XB = 4$, $AY = 3$ and $YC = a$, then write the value of a .



6. $\triangle ABC \sim \triangle DEF$. If $AB = \frac{1}{3} DE$, then find the value of $\frac{A(\triangle ABC)}{A(\triangle DEF)}$.

7. In the figure, $\angle ABC = \angle BCD = 90^\circ$,
 $AB = 6$, $DC = 8$, then find the value of
 $\frac{A(\triangle ABC)}{A(\triangle BCD)}$.

(July '23)



Answers :

1. $\frac{3}{4}$ 2. $\frac{7}{20}$ 3. $\triangle ABC \sim \triangle PQR$ by SAS test
 4. $9 : 25$ 5. $a = 6$. $1 : 9$ 7. $\frac{3}{4}$.

2.3 2 marks each

1. In $\triangle ABC$, seg $DE \parallel$ side BC .

If $AD = 6$ cm, $DB = 9$ cm

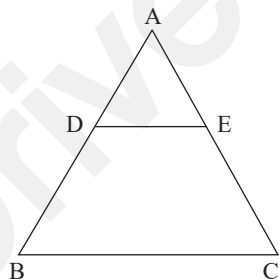
$EC = 7.5$ cm, then complete the following
 activity to find AE .

(Nov. '20)

Activity :

In $\triangle ABC$,

seg $DE \parallel$ side BC



... (Given)

$$\therefore \frac{AD}{DB} = \frac{AE}{EC} \quad \dots \left(\boxed{\hspace{2cm}} \right)$$

$$\therefore \frac{6}{9} = \frac{AE}{\boxed{\hspace{1cm}}} \quad \therefore AE = \frac{6 \times 7.5}{\boxed{\hspace{1cm}}}$$

$$\therefore AE = \boxed{\hspace{1cm}}.$$

Activity :

In $\triangle ABC$,

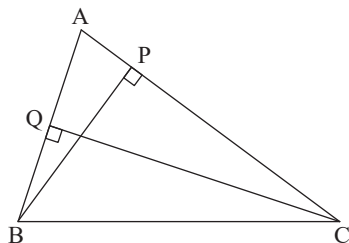
seg $DE \parallel$ side BC

... (Given)

$$\therefore \frac{AD}{DB} = \frac{AE}{EC} \quad \dots \left(\boxed{\text{By basic proportionality theorem}} \right)$$

$$\therefore \frac{6}{9} = \frac{AE}{\boxed{7.5}} \quad \therefore AE = \frac{6 \times 7.5}{9} \quad \therefore AE = \boxed{5 \text{ cm}}.$$

2. In the adjoining figure, $BP \perp AC$, $CQ \perp AB$, $A-P-C$, $A-Q-B$, then prove that $\triangle APB$ and $\triangle AQC$ are similar, by completing the following activity.



Activity :

In $\triangle APB$ and $\triangle AQC$,

$$\angle APB = \boxed{}^\circ \quad \dots (1)$$

$$\angle AQC = \boxed{}^\circ \quad \dots (2)$$

$$\therefore \angle APB \cong \angle AQC \quad \dots [\text{From (1) and (2)}]$$

$$\angle PAB \cong \angle QAC \quad \dots \left(\boxed{} \right)$$

$$\therefore \triangle APB \sim \triangle AQC. \quad \dots \left(\boxed{} \text{ test of similarity} \right)$$

Activity : In $\triangle APB$ and $\triangle AQC$,

$$\angle APB = \boxed{90}^\circ \quad \dots (1)$$

$$\angle AQC = \boxed{90}^\circ \quad \dots (2)$$

$$\therefore \angle APB \cong \angle AQC \quad \dots [\text{From (1) and (2)}]$$

$$\angle PAB \cong \angle QAC \quad \dots \left(\boxed{\text{Common angle}} \right)$$

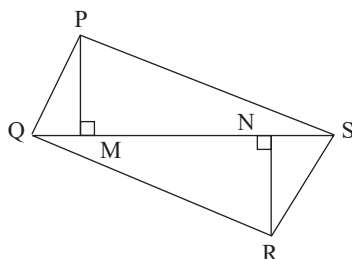
$$\therefore \triangle APB \sim \triangle AQC. \quad \dots \left(\boxed{\text{AA}} \text{ test of similarity} \right)$$

3. In the figure, $PM = 10 \text{ cm}$,

$$A(\triangle PQS) = 100 \text{ sq cm},$$

$$A(\triangle QRS) = 110 \text{ sq cm};$$

then find NR .



Solution :

$\triangle PQS$ and $\triangle QRS$ have same base QS.

Areas of triangles with the same base are proportional to their corresponding heights.

$$\therefore \frac{A(\triangle PQS)}{A(\triangle QRS)} = \frac{PM}{NR}$$

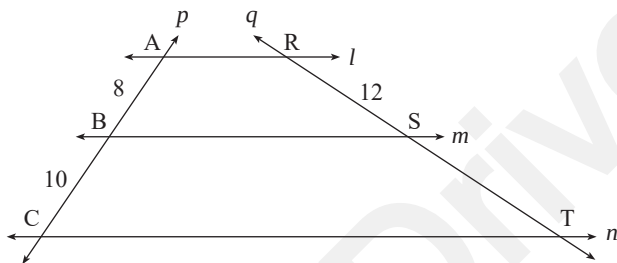
$$\therefore \frac{100}{110} = \frac{10}{NR}$$

$$\therefore NR = \frac{110 \times 10}{100}$$

$$\therefore NR = 11 \text{ cm}$$

Ans. NR = 11 cm.

4.



In the figure, line $l \parallel$ line $m \parallel$ line n . Line p and line q are transversals. With respect to given information find the length of segment ST.

Solution : line $l \parallel$ line $m \parallel$ line n and line p and line q are transversals.
... (Given)

$$\therefore \frac{AB}{BC} = \frac{RS}{ST} \quad \dots \text{(By the property of three parallel lines and their transversals)}$$

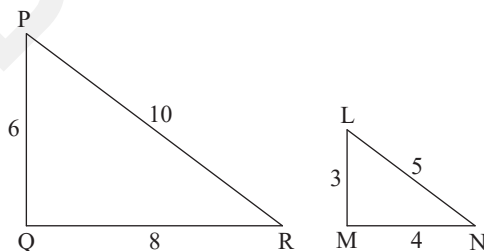
$$\therefore \frac{8}{10} = \frac{12}{ST}$$

$$\therefore ST = \frac{12 \times 10}{8}$$

$$\therefore ST = 15$$

Ans. ST = 15.

5. Are the triangles in following figures similar? If yes, by which test?



Solution : $\frac{PQ}{LM} = \frac{6}{3} = \frac{2}{1}$... (1)

$\frac{QR}{MN} = \frac{8}{4} = \frac{2}{1}$... (2)

$\frac{PR}{LN} = \frac{10}{5} = \frac{2}{1}$... (3)

In $\triangle PQR$ and $\triangle LMN$,

$\frac{PQ}{LM} = \frac{QR}{MN} = \frac{PR}{LN}$... [From (1), (2) and (3)]

$\therefore \triangle PQR \sim \triangle LMN$... (SSS test of similarity)

Ans. The triangles are similar by the SSS test of similarity.

6. In the figure, seg AC and seg BD intersect each other in point P

and $\frac{AP}{CP} = \frac{BP}{DP}$. Prove that $\triangle ABP \sim \triangle CDP$.

(Sept. '21) [(Activity) (March '22)]

Proof :

In $\triangle ABP$ and $\triangle CDP$,

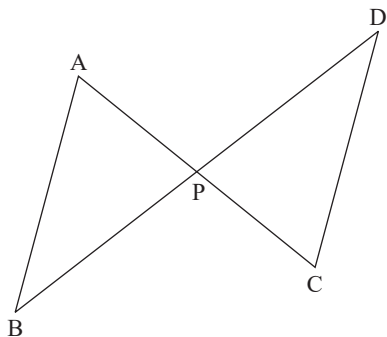
$\frac{AP}{CP} = \frac{BP}{DP}$... (Given)

$\angle APB \cong \angle CPD$

... (Vertically opposite angles)

$\therefore \triangle ABP \sim \triangle CDP$

... (SAS test of similarity)



7. $\triangle ABC \sim \triangle PQR$. $A(\triangle ABC) = 81 \text{ cm}^2$, $A(\triangle PQR) = 121 \text{ cm}^2$.

If $BC = 6.3 \text{ cm}$, then find QR .

(Nov. '20)

Solution :

$$\triangle ABC \sim \triangle PQR$$

$$\therefore \frac{A(\triangle ABC)}{A(\triangle PQR)} = \frac{BC^2}{QR^2} \quad \dots \text{ (Theorem of areas of similar triangles)}$$

$$\therefore \frac{81}{121} = \frac{(6.3)^2}{QR^2}$$

$$\therefore QR^2 = \frac{121 \times (6.3)^2}{81} \quad \therefore QR = \frac{11 \times 6.3}{9}$$

$$\therefore QR = 7.7 \text{ cm}$$

Ans. $QR = 7.7 \text{ cm}$.

8. In $\triangle ABC$, $DE \parallel BC$. If $DB = 5.4 \text{ cm}$, $AD = 1.8 \text{ cm}$, $EC = 7.2 \text{ cm}$, then find AE .
(March '25)

Solution :

In $\triangle ABC$,

seg $DE \parallel$ side BC ... (Given)

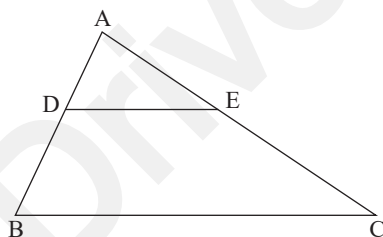
$\therefore$ by the basic proportionality theorem,

$$\frac{AD}{DB} = \frac{AE}{EC}$$

$$\therefore \frac{1.8}{5.4} = \frac{AE}{7.2}$$

$$\therefore AE = \frac{1.8 \times 7.2}{5.4} \quad \therefore AE = 2.4 \text{ cm}$$

Ans. $AE = 2.4 \text{ cm}$.



Assignment

2.3

1. In $\triangle PQR$, if $PN = 12$, $NR = 8$, $PM = 15$, $MQ = 12$, then complete the following activity to justify whether seg NM is parallel to side RQ or not.
(July '22)

[Note : The question has been modified, $MQ = 12$ instead of 10.]

Activity : In $\triangle PQR$,

$$\frac{PN}{NR} = \frac{12}{\boxed{}} = \frac{3}{2} \quad \dots \text{ (I)}$$

$$\text{and } \frac{PM}{MQ} = \frac{15}{12} = \frac{\boxed{}}{4} \quad \dots \text{ (II)}$$

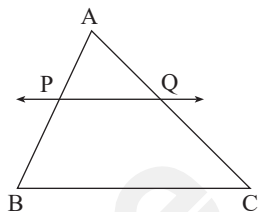
$$\therefore \frac{PN}{NR} \neq \frac{PM}{MQ} \quad \dots \text{ [From (I) and (II)]}$$

$$\therefore \text{ by } \boxed{}$$

seg NM is $\boxed{}$ to side RQ.

2. In $\triangle ABC$, line $PQ \parallel$ side BC. If $AP=10$, $PB=12$, $AQ=15$, then complete the following activity to find the value of QC.

(Sept. '21)



... (Given)

Activity : In $\triangle ABC$,
line $PQ \parallel$ side BC

$$\therefore \frac{AP}{PB} = \frac{AQ}{QC} \quad \dots \left(\boxed{} \right)$$

$$\therefore \frac{10}{12} = \frac{\boxed{}}{QC} \quad \therefore QC = \frac{\boxed{} \times 12}{10} \quad \therefore QC = \boxed{}.$$

3. In the figure, $\angle ABC = 75^\circ$, $\angle EDC = 75$. Complete the following activity to prove $\triangle ABC \sim \triangle EDC$.

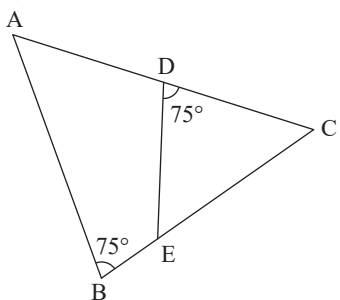
Activity : In $\triangle ABC$ and $\triangle EDC$,

$$\angle ABC \cong \boxed{}$$

... (Each measures 75°)

$$\angle C \cong \angle C \quad \dots \left(\boxed{} \right)$$

$$\therefore \triangle ABC \sim \triangle \boxed{} \quad \dots \left(\boxed{} \text{ test of similarity} \right)$$

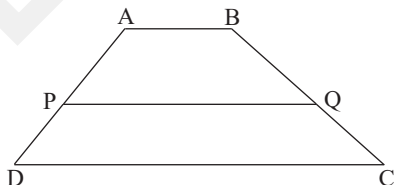


4. 'The bisector of an angle of a triangle divides the side opposite to the angle in the ratio of the remaining sides'.

(i) Draw a neat labelled figure for the above statement.

(ii) Write 'Given' and 'To prove' for the above statement.

5.



In trapezium ABCD,

side $AB \parallel PQ \parallel DC$.

$AP = 15$, $PD = 12$, $QC = 14$,

find BQ.

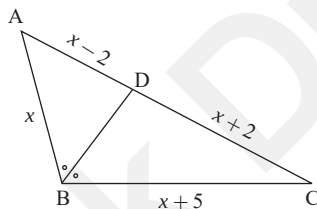
(March '23)

6. Base of a triangle is 9 and height is 5. Base of another triangle is 10 and height is 6. Find the ratio of areas of these triangles.

7. $\triangle ABC$ and $\triangle DEF$ are equilateral triangles.

If $A(\triangle ABC) : A(\triangle DEF) = 1 : 2$ and $AB = 4$, find DE.

8. In $\triangle ABC$, seg BD bisects $\angle ABC$. If $AB = x$, $BC = x + 5$, $AD = x - 2$, $DC = x + 2$, then find the value of x .

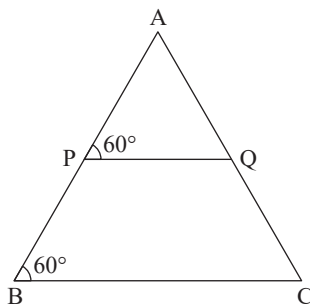


9. In $\triangle ABC \sim \triangle PQR$, $AB : PQ = 4 : 5$ and $A(\triangle PQR) = 125 \text{ cm}^2$, then find $A(\triangle ABC)$.

(March '22)

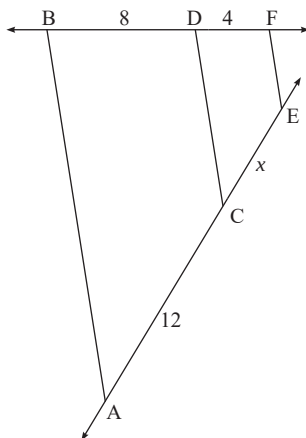
10. Measures of some angles in the figure are given.

Prove that $\frac{AP}{PB} = \frac{AQ}{QC}$.

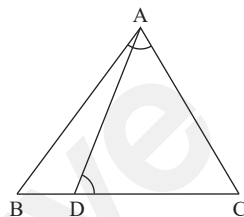


11. In the figure, if $AB \parallel CD \parallel EF$, then find x and AE .

(July '23)



12. In the figure, in $\triangle ABC$, point D on side BC is such that $\angle BAC = \angle ADC$. Prove that $CA^2 = CB \times CD$.



Answers :

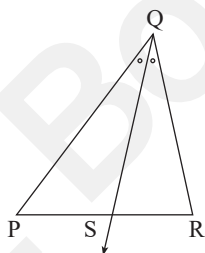
1. 8, 5, Converse of basic proportionality theorem,

not parallel

2. By basic proportionality theorem, 15, 15, 18

3. $\angle EDC$, Common angle, $\angle EDC$, AA

4. (i)



(ii) Given : In $\triangle PQR$,
ray QS bisects $\angle PQR$.

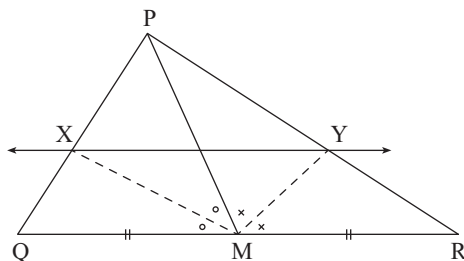
To prove : $\frac{PS}{SR} = \frac{PQ}{QR}$.

5. $BQ = 17.5$ 6. $3 : 4$ 7. $4\sqrt{2}$ 8. $x = 10$

9. 80 cm^2 11. $x = 6$ and $AE = 18$.

2.4 3 marks each

1. In $\triangle PQR$, seg PM is a median. Angle bisectors of $\angle PMQ$ and $\angle PMR$ intersect side PQ and side PR in points X and Y respectively. Prove that $XY \parallel QR$. Complete the proof by filling in the boxes.



(March '23)

Activity :

In $\triangle PMQ$,

Ray MX is the bisector of $\angle PMQ$.

$$\therefore \frac{MP}{MQ} = \frac{\boxed{}}{\boxed{}} \quad \dots \text{ (Theorem of angle bisector) } \dots (1)$$

Similarly, in $\triangle PMR$, ray MY is the bisector of $\angle PMR$.

$$\therefore \frac{MP}{MR} = \frac{\boxed{}}{\boxed{}} \quad \dots \text{ (Theorem of angle bisector) } \dots (2)$$

$$\text{But } \frac{MP}{MQ} = \frac{MP}{MR} \quad \dots \text{ (As } M \text{ is the midpoint of } QR \therefore MQ = MR) \dots (3)$$

$$\therefore \frac{PX}{\boxed{}} = \frac{\boxed{}}{YR} \quad \dots \text{ [From (1), (2) and (3)]}$$

$$\therefore XY \parallel QR \quad \dots \text{ (Converse of basic proportionality theorem)}$$

Activity : In $\triangle PMQ$,

Ray MX is the bisector of $\angle PMQ$.

$$\therefore \frac{MP}{MQ} = \frac{\boxed{PX}}{\boxed{XQ}} \quad \dots \text{ (Theorem of angle bisector) } \dots (1)$$

Similarly, in $\triangle PMR$, ray MY is the bisector of $\angle PMR$.

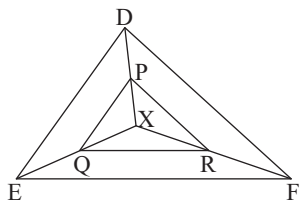
$$\therefore \frac{MP}{MR} = \frac{\boxed{PY}}{\boxed{YR}} \quad \dots \text{ (Theorem of angle bisector) } \dots (2)$$

$$\text{But } \frac{MP}{MQ} = \frac{MP}{MR} \quad \dots (\text{As } M \text{ is the midpoint of } QR \therefore MQ = MR) \quad \dots (3)$$

$$\therefore \frac{PX}{XQ} = \frac{PY}{YR} \quad \dots [\text{From (1), (2) and (3)}]$$

$$\therefore XY \parallel QR \quad \dots (\text{Converse of basic proportionality theorem})$$

2. In the figure, X is any point in the interior of the triangle. Point X is joined to the vertices of the triangle. Seg PQ $\parallel$ seg DE, seg QR $\parallel$ seg EF. Complete the following activity to prove seg PR $\parallel$ seg DF.



(March '20; July '23)

Activity :

$$\text{In } \triangle XDE, PQ \parallel DE \quad \dots (\text{Given})$$

$$\therefore \frac{XP}{\boxed{}} = \frac{\boxed{}}{QE} \quad \dots (\text{Basic proportionality theorem}) \quad \dots (1)$$

$$\text{In } \triangle XEF, QR \parallel EF \quad \dots (\text{Given})$$

$$\therefore \frac{XQ}{QE} = \frac{\boxed{}}{RF} \quad \dots \left(\boxed{} \right) \quad \dots (2)$$

$$\therefore \frac{XP}{PD} = \frac{\boxed{}}{\boxed{}} \quad \dots [\text{From (1) and (2)}]$$

$$\therefore \text{seg } PR \parallel \text{seg } DF \quad \dots (\text{Converse of basic proportionality theorem})$$

Activity :

$$\text{In } \triangle XDE, PQ \parallel DE \quad \dots (\text{Given})$$

$$\therefore \frac{XP}{\boxed{PD}} = \frac{\boxed{XQ}}{QE} \quad \dots (\text{Basic proportionality theorem}) \quad \dots (1)$$

$$\text{In } \triangle XEF, QR \parallel EF \quad \dots (\text{Given})$$

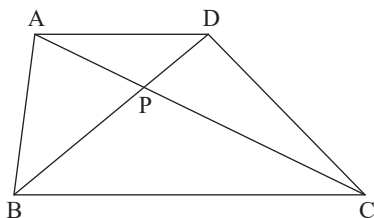
$$\therefore \frac{XQ}{QE} = \frac{\boxed{XR}}{RF} \quad \dots \left(\boxed{\text{Basic proportionality theorem}} \right) \quad \dots (2)$$

$$\therefore \frac{XP}{PD} = \frac{\boxed{XR}}{\boxed{RF}}$$

... [From (1) and (2)]

$\therefore$ seg PR $\parallel$ seg DF ... (Converse of basic proportionality theorem)

3. In $\square$ ABCD, seg AD $\parallel$ seg BC.
Diagonal AC and diagonal BD intersect each other in point P.



Show that $\frac{AP}{PD} = \frac{PC}{BP}$.

Proof : Seg AD $\parallel$ seg BC and line DB is transversal,

$\angle ADP \cong \angle CBP$... (Alternate angles) ... (1)

In $\triangle ADP$ and $\triangle CBP$,

$\angle ADP \cong \angle CBP$... [From (1)]

$\angle APD \cong \angle CPB$... (Vertically opposite angles)

$\therefore \triangle ADP \sim \triangle CBP$... (AA test of similarity)

$\therefore \frac{AP}{CP} = \frac{PD}{PB}$... (Corresponding sides of similar triangles are in proportion)

$\therefore \frac{AP}{PD} = \frac{PC}{PB}$... (By alternendo)

i.e. $\frac{AP}{PD} = \frac{PC}{BP}$.

4. In $\triangle ABC$, ray BD bisects $\angle ABC$
 and ray CE bisects $\angle ACB$.

If seg AB $\cong$ seg AC, then prove that
 ED $\parallel$ BC.

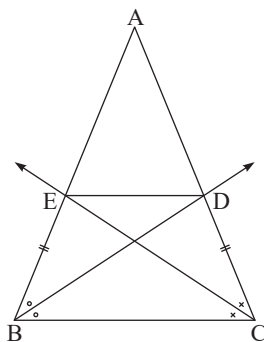
Proof : In $\triangle ABC$,

ray BD is the bisector of $\angle ABC$

$\therefore$ by the theorem of an angle bisector of a triangle,

$$\frac{AB}{BC} = \frac{AD}{DC} \quad \dots (1)$$

In $\triangle ABC$,



ray CE is the bisector of $\angle ACB$

$\therefore$ by the theorem of an angle bisector of a triangle,

$$\frac{AC}{BC} = \frac{AE}{EB} \quad \dots (2)$$

$$\text{seg AB} \cong \text{seg AC} \quad \dots (\text{Given}) \dots (3)$$

$$\therefore \frac{AE}{EB} = \frac{AD}{DC} \quad \dots [\text{From (1), (2) and (3)}]$$

$\therefore$ by converse of basic proportionality theorem,

seg ED $\parallel$ side BC

i.e. ED $\parallel$ BC.

5. In trapezium ABCD,

side AB $\parallel$ side DC, diagonals

AC and BD intersect in point O.

If AB = 20, DC = 6, OB = 15,

then find OD.

Solution : Side AB $\parallel$ side DC

and line DB is the transversal,

$$\angle CDB \cong \angle ABD$$

... (Alternate angles)

$$\text{i.e. } \angle CDO \cong \angle ABO$$

... (B-O-D) ... (1)

In $\triangle COD$ and $\triangle AOB$,

$$\angle CDO \cong \angle ABO$$

... [From (1)]

$$\angle COD \cong \angle AOB$$

... (Vertically opposite angles)

$$\therefore \triangle COD \sim \triangle AOB$$

... (AA test of similarity)

$$\therefore \frac{OD}{OB} = \frac{DC}{BA}$$

... (Corresponding sides of similar triangles are in proportion)

$$\therefore \frac{OD}{15} = \frac{6}{20}$$

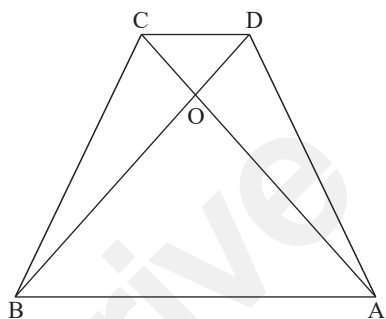
$$\therefore OD \times 20 = 15 \times 6$$

$$\therefore OD = \frac{6 \times 15}{20}$$

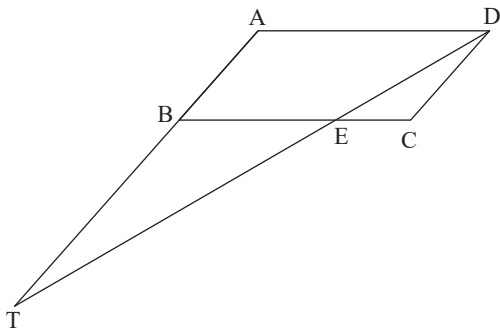
$$\therefore OD = \frac{9}{2}$$

$$\therefore OD = 4.5$$

Ans. OD = 4.5.



6. $\square$ ABCD is a parallelogram.
 Point E is on side BC.
 Line DE intersects ray AB in point T.
 Prove that
 $DE \times BE = CE \times TE$.



Proof :

Seg AB $\parallel$ seg DC ... (Opposite sides of parallelogram are parallel)
 i.e. seg AT $\parallel$ seg DC ... (A–B–T)

Line TD is the transversal,

$\therefore \angle ATD \cong \angle CDT$... (Alternate angles)

i.e. $\angle BTE \cong \angle CDE$... (A–B–T, T–E–D) ... (1)

In $\triangle BET$ and $\triangle CED$,

$\angle BTE \cong \angle CDE$... [From (1)]

$\angle BET \cong \angle CED$... (Vertically opposite angles)

$\therefore \triangle BET \sim \triangle CED$... (AA test of similarity)

$\therefore \frac{BE}{CE} = \frac{TE}{DE}$... (Corresponding sides of similar triangles are in proportion)

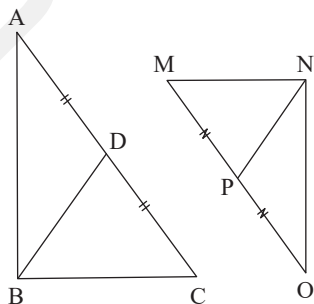
$\therefore BE \times DE = CE \times TE$.

7. In the figure, $\triangle ABC \sim \triangle MNO$,
 D is the midpoint of side AC and
 P is the midpoint of side MO.

Prove :

(i) $\triangle ABD \sim \triangle MNP$.

(ii) $\frac{BD}{NP} = \frac{AB}{MN}$.



(iii) Write your conclusion about the result obtained in (ii).

Proof :

$\triangle ABC \sim \triangle MNO$... (Given)

$\therefore \frac{AB}{MN} = \frac{BC}{NO} = \frac{AC}{MO}$... (Corresponding sides of similar triangles are in proportion.)

$$\therefore \frac{AB}{MN} = \frac{AC}{MO}$$

$$\therefore \frac{AB}{MN} = \frac{2AD}{2MP}$$

... (D is the midpoint of side AC and P is the midpoint of side MO)

$$\therefore \frac{AB}{MN} = \frac{AD}{MP} \quad \dots (1)$$

Also $\angle A = \angle M$... (2) ... (Corresponding angles of similar triangles are congruent)

In $\triangle ABD$ and $\triangle MNP$,

$$\frac{AB}{MN} = \frac{AD}{MP} \quad \dots [\text{From (1)}]$$

$$\angle A = \angle M \quad \dots [\text{From (2)}]$$

$$\therefore \triangle ABD \sim \triangle MNP \quad \dots (\text{SAS test of similarity})$$

$$\therefore \frac{BD}{NP} = \frac{AB}{MN} \quad \dots (\text{Corresponding sides of similar triangles are in proportion})$$

Conclusion : In similar triangles, the corresponding sides are in proportion to the corresponding medians.

Assignment	2.4
-------------------	------------

1. In the figure,

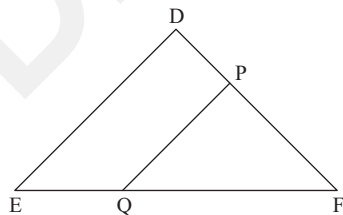
seg PQ $\parallel$ seg DE,

$A(\triangle PQF) = 20$ sq units,

$PF = 2DP$, then find

$A(\square DPQE)$

by completing the following activity.



Activity : $A(\triangle PQF) = 20$ sq units, $PF = 2DP$.

Let us assume that DP be x .

Then $PF = 2x$

$$DF = DP + \square = x + 2x = 3x$$

In $\triangle FDE$ and $\triangle FPQ$,

$$\angle FDE \cong \angle \square \quad \dots (\text{Corresponding angles})$$

$$\angle FED \cong \angle FQP \quad \dots (\text{Corresponding angles})$$

$$\therefore \triangle FDE \sim \triangle FPQ$$

... (AA test)

$$\therefore \frac{A(\triangle FDE)}{A(\triangle FPQ)} = \frac{\square}{PF^2} = \frac{(3x)^2}{(2x)^2} = \frac{9}{4}$$

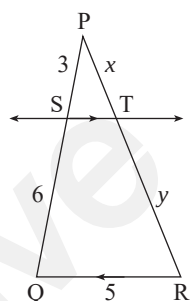
$$A(\triangle FDE) = \frac{9}{4}A(\triangle FPQ) = \frac{9}{4} \times \square = 45 \text{ sq units}$$

$$A(\square DPQE) = A(\triangle FDE) - A(\triangle FPQ)$$

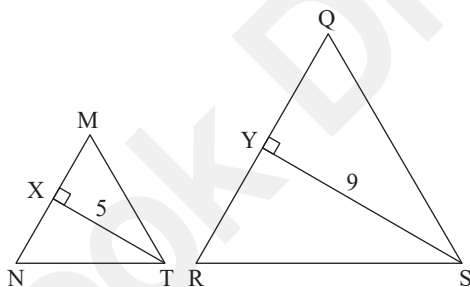
$$= 45 - \square$$

$$= \square.$$

2. In the figure, $PS = 3$, $SQ = 6$, $QR = 5$, $PT = x$ and $TR = y$. Suggest any two pairs of values of x and y such that $ST \parallel$ side QR . (Open ended question)

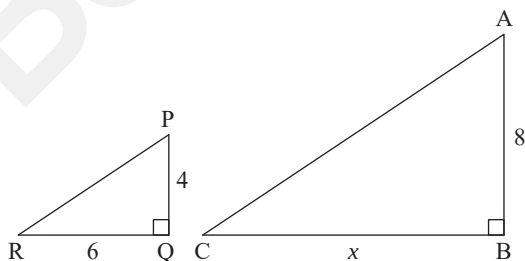


3.



$\triangle MNT \sim \triangle QRS$. Length of altitude drawn from point T is 5 and length of altitude drawn from point S is 9. Find the ratio $\frac{A(\triangle MNT)}{A(\triangle QRS)}$.

4. As shown in the figure, two poles of height 8 m and 4 m are perpendicular to the ground. If the length of shadow of smaller pole due to sunlight is 6 m, then how long will be the shadow of the bigger pole at the same time?

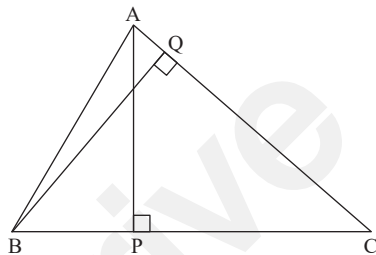


5. The sides of the smaller triangle, out of the two similar triangles are 4 cm, 5 cm and 6 cm respectively. If the perimeter of the larger triangle is 90 cm, then what are the lengths of the sides of the larger triangle?

6. In $\triangle ABC$, $AP \perp BC$,
 $BQ \perp AC$. $B-P-C$, $A-Q-C$, then
 prove that

$$\triangle CPA \sim \triangle CQB.$$

If $AP = 7$, $BQ = 8$, $BC = 12$, then
 find AC .

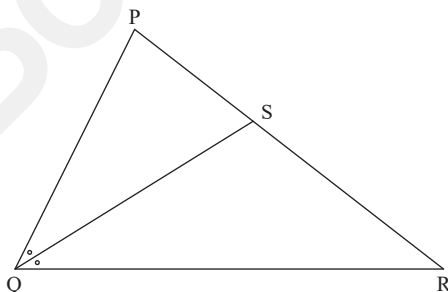


7. $\triangle ABC$ and $\triangle PQR$ are similar. seg $AD \perp$ side BC and $B-D-C$.
 seg $PM \perp$ side QR such that $Q-M-R$. If $AD : PM = 4 : 9$. Then find

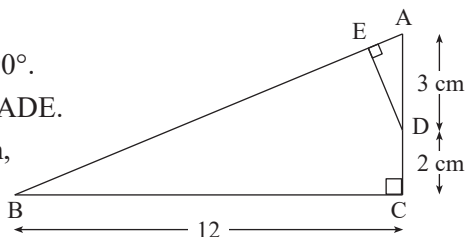
$$\frac{A(\triangle ABC)}{A(\triangle PQR)}.$$

8. In $\triangle PQR$, $\angle Q = 2\angle R$. If seg QS bisects $\angle PQR$ and $P-S-R$, then

prove that $\frac{QS}{SP} = \frac{QR}{QP}$.



9. In the figure,
 $\angle BED = 90^\circ$ and $\angle ACB = 90^\circ$.
 Then (a) Prove : $\triangle ABC \sim \triangle ADE$.
 (b) If $AD = 3$ cm, $CD = 2$ cm,
 $BC = 12$ cm, then find the
 lengths of AE and DE .



10. $\square ABCD$ is trapezium $AB \parallel CD$. Diagonals of trapezium intersect in point P . Write the answers to the following questions :
- Draw the figure using given information.
 - Write any one pair of alternate angles and opposite angles.
 - Write the names of similar triangles with test of similarity.

(March '24)

Answers :

1. $\square{PF}$, $\square{\angle FPQ}$, $\square{DF^2}$, $\square{20}$, $\square{20}$, $\square{25 \text{ sq units}}$

2. $(x = 2 \text{ and } y = 4)$ and $(x = 2.5 \text{ and } y = 5)$ are two pairs of values of x and y

[**Note :** The student can write any two pairs of values of x and y which satisfies the condition $y = 2x$ and sum of two sides of a triangle is greater than the third side.]

3. $\frac{A(\triangle MNT)}{A(\triangle QRS)} = \frac{25}{81}$

4. Shadow of the bigger pole = 12 m

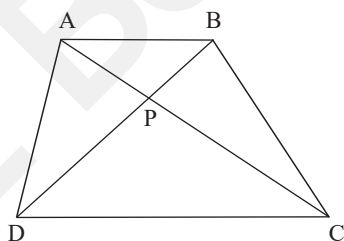
5. 24 cm, 30 cm, 36 cm

6. $AC = 10.5$

7. $16 : 81$

9. (b) $AE \approx 1.15$ cm and $DE \approx 2.77$ cm

10. (a)



(b) Pairs of alternate angles :

(i) $\angle ABP$ and $\angle CDP$ (ii) $\angle BAP$ and $\angle DCP$

Pairs of opposite angles :

(i) $\angle APB$ and $\angle CPD$ (ii) $\angle APD$ and $\angle CPB$

(c) $\triangle APB \sim \triangle CPD$... (AA test of similarity)

CHALLENGING QUESTIONS

2.5 4 marks each

1. In the figure, sides AB , BC , CA of $\triangle ABC$ are produced up to points R , P , F respectively, such that $AB = BR$, $BC = CP$ and $CA = AF$.

Prove that : $A(\triangle PFR) = 7A(\triangle ABC)$.

Construction : Draw seg PA , seg FB and seg RC .

Proof : In $\triangle APB$, seg AC is the median. The median of a triangle divides the triangle into two triangles of equal area.

$$\therefore A(\triangle ABC) = A(\triangle APC) \quad \dots (1)$$

Similarly, seg PA is the median of $\triangle PCF$.

$$\therefore A(\triangle APC) = A(\triangle PAF) \quad \dots (2)$$

$$\therefore A(\triangle APC) = \frac{1}{2} A(\triangle PCF) \quad \dots (3)$$

From (1) and (2), $A(\triangle PCF) = 2A(\triangle ABC)$

Similarly, we can prove that $A(\triangle FAR) = 2A(\triangle ABC)$

$$\text{and } A(\triangle RBP) = 2A(\triangle ABC) \quad \dots (4)$$

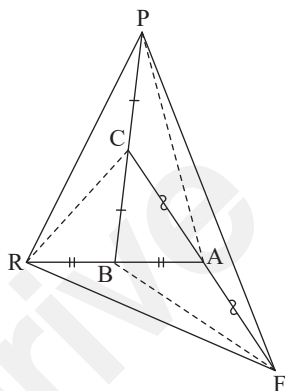
Adding results (3) and (4),

$$A(\triangle PCF) + A(\triangle FAR) + A(\triangle RBP) = 6A(\triangle ABC)$$

Adding $A(\triangle ABC)$ to both the sides,

$$A(\triangle PCF) + A(\triangle FAR) + A(\triangle RBP) + A(\triangle ABC) = 7A(\triangle ABC) \quad \dots (5)$$

By area addition postulate,



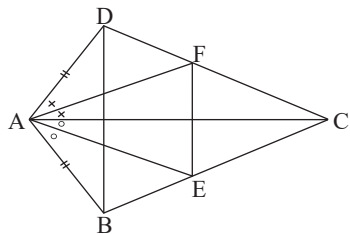
$$A(\triangle PCF) + A(\triangle FAR) + A(\triangle RBP) + A(\triangle ABC) = A(\triangle PFR) \quad \dots (6)$$

From (5) and (6), $A(\triangle PFR) = 7A(\triangle ABC)$.

2. In $\square ABCD$, side $AB \cong$ side AD .

Bisector of $\angle BAC$ cuts side BC at E and bisector of $\angle DAC$ cuts side CD at F .

Prove that $\text{seg } EF \parallel \text{seg } BD$.



Proof :

In $\triangle ABC$,

AE bisects $\angle BAC$

... (Given)

$\therefore$ by theorem of angle bisector of a triangle,

$$\frac{AC}{AB} = \frac{CE}{BE} \quad \dots (1)$$

In $\triangle ADC$,

AF bisects $\angle DAC$

... (Given)

$\therefore$ by theorem of angle bisector of a triangle,

$$\frac{AC}{AD} = \frac{CF}{DF} \quad \dots (2)$$

$AB = AD$

... (Given) ... (3)

From (1), (2) and (3),

$$\frac{CE}{BE} = \frac{CF}{DF}$$

$\therefore$ by converse of basic proportionality theorem,

$\text{seg } EF \parallel \text{seg } BD$.

3. $\square ABCD$ is a parallelogram.

Point P is the midpoint of side

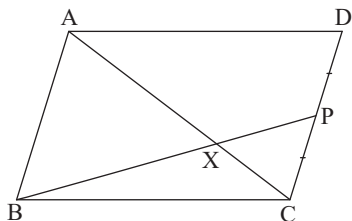
CD . $\text{Seg } BP$ intersects diagonal

AC at point X , then prove that

$3AX = 2AC$. (March '22)

Proof :

In $\triangle AXB$ and $\triangle CXP$,



$\angle AXB \cong \angle CXP$... (Vertically opposite angles)
 $\angle BAX \cong \angle PCX$... (Alternate interior angles)
 $\therefore \triangle AXB \sim \triangle CXP$... (AA test of similarity)

$\therefore \frac{AX}{CX} = \frac{XB}{XP} = \frac{AB}{CP}$... (Corresponding sides of similar triangles are in proportion)

$\therefore \frac{AX}{CX} = \frac{AB}{CP}$... (1)

$AB = CD$... (Opposite sides of parallelogram are equal)

$\therefore AB = 2CP$... (P is the midpoint of CD) ... (2)

$\therefore \frac{AX}{CX} = \frac{2CP}{CP}$... [From (1) and (2)]

$\therefore \frac{AX}{CX} = 2 \quad \therefore AX = 2CX$... (3)

Now, $\frac{AX}{AC} = \frac{AX}{AX + CX}$... (A-X-C)

$\therefore \frac{AX}{AC} = \frac{2CX}{2CX + CX}$... [From (3)]

$\therefore \frac{AX}{AC} = \frac{2CX}{3CX} \quad \therefore \frac{AX}{AC} = \frac{2}{3}$

$\therefore 3AX = 2AC$

- 4. In the figure, $\triangle FEC \cong \triangle GBD$ and also $\angle AED \cong \angle ADE$, then prove that $\triangle ADE \sim \triangle ABC$.**

Proof :

In $\triangle ADE$,

$\angle AED \cong \angle ADE$... (Given)

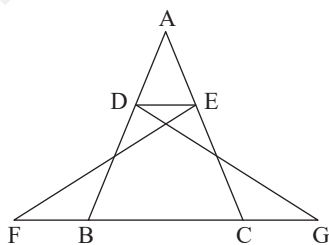
$\therefore AD = AE$... (Converse of isosceles triangle theorem) ... (1)

$\triangle GBD \cong \triangle FEC$... (Given)

$\therefore DB = CE$... (Corresponding sides of congruent triangles) ... (2)

Adding (1) and (2),

$AD + DB = AE + CE$



$$\therefore AB = AC$$

$$\dots (A-D-B \text{ and } A-E-C) \dots (3)$$

On dividing (1) and (3),

$$\frac{AD}{AB} = \frac{AE}{AC} \dots (4)$$

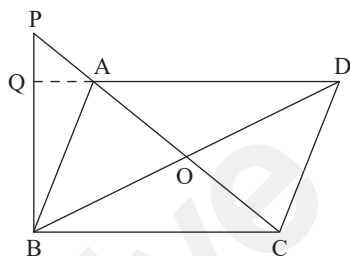
In $\triangle ADE$ and $\triangle ABC$,

$$\frac{AD}{AB} = \frac{AE}{AC} \dots [\text{From (4)}]$$

$$\angle DAE \cong \angle BAC \dots (\text{Common angle})$$

$$\therefore \triangle ADE \sim \triangle ABC \dots (\text{SAS test of similarity})$$

5. In the figure, $\square ABCD$ is a parallelogram. Diagonals AC and BD intersect at point O . Diagonal CA is extended to point P such that $PA : AO = 1 : 2$. Side DA is extended to intersect PB in point Q such that $D-A-Q$ and $P-Q-B$.



Find $\frac{A(\triangle PQA)}{A(\square QBCA)}$.

Solution :

In $\triangle PQA$ and $\triangle PBC$,

$$\angle PQA \cong \angle PBC \dots (\text{Corresponding angles})$$

$$\angle QPA \cong \angle BPC \dots (\text{Common angle})$$

$$\therefore \triangle PQA \sim \triangle PBC \dots (\text{AA test of similarity})$$

$$\therefore \frac{A(\triangle PQA)}{A(\triangle PBC)} = \frac{PA^2}{PC^2} \dots (\text{Theorem of areas of similar triangles})$$

$$\therefore \frac{A(\triangle PQA)}{A(\triangle PBC)} = \left(\frac{PA}{PC}\right)^2 \dots (1)$$

$$PA : AO = 1 : 2 \dots (\text{Given})$$

$$\therefore \frac{PA}{AO} = \frac{1}{2}$$

$$\therefore AO = 2PA \quad \dots (2)$$

$$\text{Now, } \frac{PA}{PC} = \frac{PA}{PA + AO + OC} \quad \dots (P-A-O-C)$$

$$\therefore \frac{PA}{PC} = \frac{PA}{PA + AO + AO} \quad \dots (AO = OC, \text{ diagonals of parallelogram bisect each other})$$

$$\therefore \frac{PA}{PC} = \frac{PA}{PA + 2PA + 2PA} \quad \dots [\text{From (2)}]$$

$$\therefore \frac{PA}{PC} = \frac{PA}{5PA}$$

$$\therefore \frac{PA}{PC} = \frac{1}{5} \quad \dots (3)$$

Substituting (3) in (1), we get,

$$\frac{A(\triangle PQA)}{A(\triangle PBC)} = \left(\frac{1}{5}\right)^2$$

$$\therefore \frac{A(\triangle PQA)}{A(\triangle PBC)} = \frac{1}{25}$$

Let $A(\triangle PQA)$ be x sq units, then $A(\triangle PBC) = 25$ sq units

$$\begin{aligned} A(\square QBCA) &= A(\triangle PBC) - A(\triangle PQA) \\ &= 25x - x = 24x \text{ sq units} \end{aligned}$$

$$\therefore \frac{A(\triangle PQA)}{A(\square QBCA)} = \frac{x}{24x} = \frac{1}{24}$$

$$\therefore A(\triangle PQA) : A(\square QBCA) = 1 : 24 .$$

- 6. In $\triangle ABC$, L, M, N are points on side AB, BC, AC respectively. Perpendiculars drawn at L, M, N form $\triangle PQR$.**

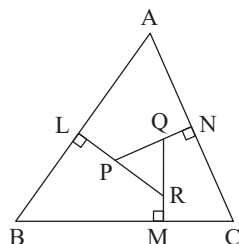
Prove that $\triangle ABC \sim \triangle PRQ$.

Proof :

In $\square MQNC$,

$$\angle QMC + \angle MCN + \angle QNC + \angle MQN = 360^\circ$$

... (Sum of all angles of quadrilateral is 360°)



$$\therefore 90^\circ + \angle MCN + 90^\circ + \angle MQN = 360^\circ$$

$$\therefore 180^\circ + \angle MCN + \angle MQN = 360^\circ$$

$$\therefore \angle MCN + \angle MQN = 360^\circ - 180^\circ$$

$$\therefore \angle MCN + \angle MQN = 180^\circ \quad \dots (1)$$

$$\text{also, } \angle MQN + \angle PQM = 180^\circ \quad \dots (\text{Angles in a linear pair}) \dots (2)$$

$\therefore$ from (1) and (2), we get,

$$\angle MCN = \angle PQM \quad \dots (\text{Eliminating } \angle MQN)$$

$$\text{i.e. } \angle ACB = \angle PQR \quad \dots (Q-R-M, A-N-C \text{ and } B-M-C) \quad \dots (3)$$

$$\text{Similarly, } \angle ABC = \angle PRQ \text{ can be proved.} \quad \dots (4)$$

In $\triangle ABC$ and $\triangle PRQ$,

$$\angle ACB \cong \angle PQR \quad \dots [\text{From (3)}]$$

$$\angle ABC \cong \angle PRQ \quad \dots [\text{From (4)}]$$

$$\therefore \triangle ABC \sim \triangle PRQ \quad \dots (\text{AA test of similarity})$$

Assignment

2.5

1. In the given figure, seg $DE \parallel$ side BC .

- (a) If $DE = 4$ cm, $BC = 8$ cm,

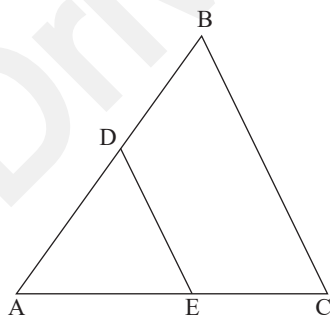
$$A(\triangle ADE) = 25 \text{ cm}^2, \text{ find } A(\triangle ABC).$$

- (b) If $DE : BC = 3 : 5$, then

$$\text{find } A(\triangle ADE) : A(\square DBCE).$$

(March '25)

$$[\text{Ans. (a) } 100 \text{ cm}^2 \quad (\text{b) } 9 : 16]$$



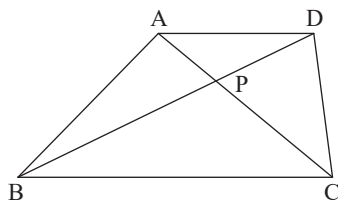
2. If the area of two similar triangles are equal, then prove that they are congruent.

3. In $\square ABCD$, side $BC \parallel$ side AD .

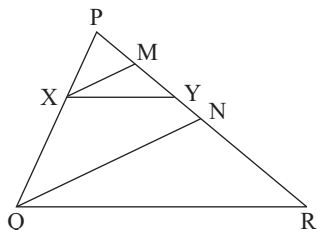
Diagonals AC and BD intersect

each other at P . If $AP = \frac{1}{3} AC$,

then prove that $DP = \frac{1}{2} BP$.



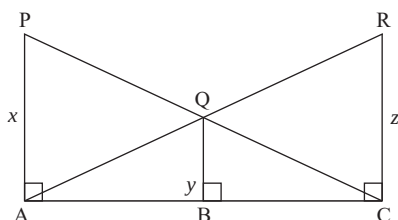
4. In $\triangle PQR$, seg $XY \parallel$ side QR , M and N are midpoints of seg PY and seg PR respectively. Prove that
- $\triangle PXM \sim \triangle PQN$.
 - seg $XM \parallel$ seg QN . **(Sept. '21)**



5. In the given figure, seg PA , seg QB and seg RC are perpendicular to seg AC .

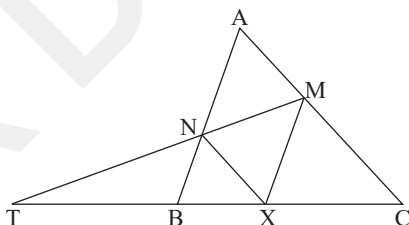
From the information given in the figure, prove that $\frac{1}{x} + \frac{1}{z} = \frac{1}{y}$.

(July '19)



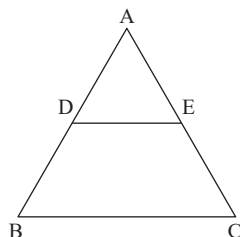
6. In $\triangle ABC$, PQ is a line segment intersecting AB at P and AC at Q such that seg $PQ \parallel$ seg BC . If PQ divides $\triangle ABC$ into two equal parts having equal areas, find $\frac{BP}{AB}$. **(March '20)** [Ans. $\frac{BP}{AB} = \frac{\sqrt{2}-1}{\sqrt{2}}$]

7. In $\triangle ABC$, point X is any point on side BC . Seg $XM \parallel$ seg AB and seg $XN \parallel$ seg AC . Extend seg MN such that it intersects extended side CB in point T . Prove that $TX^2 = TB \times TC$.



(July '22)

8. In $\triangle ABC$, seg $DE \parallel$ side BC .
If $2A(\triangle ADE) = A(\square DBCE)$,
find $AB : AD$ and show that $BC = \sqrt{3} \cdot DE$.
(March '23)
[Ans. $AB : AD = \sqrt{3} : 1$.]



9. In $\triangle PQR$, bisectors of $\angle Q$ and $\angle R$ intersect in point X. Line PX intersects side QR in point Y, then prove that $\frac{PQ + PR}{QR} = \frac{PX}{XY}$.

(July '23)

**Based on the Textbook And
Latest Activity Sheet Format**

MATHEMATICS WORKBOOK

**Part II
Standard X**

**Ideal for the Internal Evaluation
as well as Board's Examination**

Available from all booksellers

**Question
Set
3**

PYTHAGORAS THEOREM

3.1 1 mark each

Q. Choose the correct alternative from those given below each question :

1. Out of the following which is a Pythagorean triplet?

- (A) (1, 5, 10) (B) (3, 4, 5) (C) (2, 2, 2) (D) (5, 5, 2)

(March '25)

Ans. (B)

[Hint : $5^2 = 3^2 + 4^2$.]

2. In a right angled triangle, if the sum of the squares of the sides making right angle is 169, then what is the length of the hypotenuse?

(July '22)

- (A) 15 (B) 13 (C) 5 (D) 12

Ans. (B)

[Hint : Use Pythagoras theorem.]

3. Altitude on the hypotenuse of a right angled triangle divides it in two parts of lengths 4 cm and 9 cm. Find the length of the altitude.

- (A) 9 cm (B) 4 cm (C) 6 cm (D) $2\sqrt{6}$ cm

Ans. (C)

[Hint : Use Geometric mean theorem.]

4. Height and base of a right angled triangle are 24 cm and 18 cm; find the length of its hypotenuse.

- (A) 24 cm (B) 30 cm (C) 15 cm (D) 18 cm

Ans. (B)

[Hint : Use Pythagoras theorem.]

5. If a, b, c are sides of a triangle and $a^2 + b^2 = c^2$, then name the type of triangle.

(March '23)

- (A) Obtuse angled triangle (B) Acute angled triangle
(C) Right angled triangle (D) Equilateral triangle

Ans. (C) [Hint : By converse of Pythagoras theorem, the triangle is a right angled triangle.]



6. $\triangle PQR$ is an equilateral triangle of side 10 cm. What is the height of the triangle?

(A) 5 cm (B) $5\sqrt{3}$ cm (C) $5\sqrt{2}$ cm (D) 25 cm

Ans. (B) [Hint : Height of an equilateral triangle = $\frac{\sqrt{3}}{2} \times \text{side}$.]

7. Out of the given triplets, which is not a Pythagorean triplet?

(Sept. '21)

(A) (12, 9, 15) (B) (10, 24, 26)

(C) (12, 16, 25) (D) (15, 17, 8)

Ans. (C) [Hint : $25^2 \neq 12^2 + 16^2$.]

8. In an isosceles triangle ABC, $AC = BC$ and $AB^2 = 2AC^2$ then $\angle C = ?$

(A) 30° (B) 45° (C) 60° (D) 90°

Ans. (D) [Hint : $AB^2 = 2AC^2 \Rightarrow AB^2 = AC^2 + AC^2 \Rightarrow AB^2 = AC^2 + BC^2$, so AB is hypotenuse.]

Assignment

3.1

- Q. Choose the correct alternative from those given below each question :

1. Find the perimeter of a square if its diagonal is $10\sqrt{2}$ cm. (July '23)

(A) 10 cm (B) $40\sqrt{2}$ cm (C) 20 cm (D) 40 cm

2. Out of the dates given below which date constitutes a Pythagorean triplet? (March '24)

(A) 15/08/17 (B) 16/08/16 (C) 3/5/17 (D) 4/9/15

3. $\triangle PQR$ is an isosceles right angled triangle. $\angle Q = 90^\circ$.

If $PQ = 5$ cm, then $PR = ?$

(A) 5 cm (B) $5\sqrt{3}$ cm (C) $5\sqrt{2}$ cm (D) 25 cm

4. If two sides of the right angled triangle are 3 and 4, then the length of the third side is

(A) 5 (B) $\sqrt{7}$ (C) 5 or $\sqrt{7}$ (D) $7\sqrt{5}$

5. In $\triangle ABC$, if $AB = 10$ cm and $\angle ACB = 90^\circ$, $\angle ABC = 30^\circ$, then $BC = ?$

(A) $5\sqrt{2}$ cm (B) 5 cm (C) $10\sqrt{3}$ cm (D) $5\sqrt{3}$ cm

6. In right angled $\triangle PQR$, if hypotenuse $PR = 12$ and $PQ = 6$, then what is the measure of $\angle P = ?$ (July '19)
 (A) 30° (B) 60° (C) 90° (D) 45°
7. In $\triangle ABC$, seg AM is the median and $AB^2 + AC^2 = 410$ and $BC = 12$, then $AM = ?$
 (A) 12 (B) 13 (C) $\sqrt{12}$ (D) $\sqrt{13}$

Answers :

1. (D) [Diagonal of a square $= \sqrt{2} \times \text{side}$.]
 2. (A) [The square of the largest number is equal to the sum of the squares of the other two numbers.]
 3. (C) [Use Pythagoras theorem.]
 4. (C) [Use Pythagoras theorem.]
 5. (D) [Use $30^\circ - 60^\circ - 90^\circ$ triangle theorem.]
 6. (B) [By converse of $30^\circ - 60^\circ - 90^\circ$ triangle theorem, $\angle R = 30^\circ$,
 $\therefore \angle P = 60^\circ$.]
 7. (B) [Use Apollonius theorem.]

3.2 1 mark each

1. In the figure, find the length of RP using the information given in $\triangle PSR$. (July '23)

Ans.

In $\triangle PSR$,

$\angle S = 90^\circ$, $\angle P = 30^\circ$... (Given)

$\therefore \angle R = 60^\circ$... (Angle sum property of a triangle)

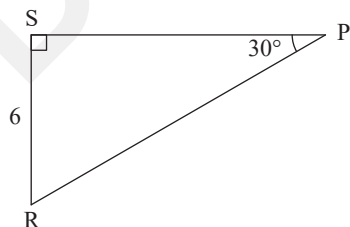
$\therefore$ by $30^\circ - 60^\circ - 90^\circ$ triangle theorem,

$SR = \frac{1}{2} PR$... (Side opposite to 30°)

$$\therefore 6 = \frac{1}{2} PR$$

$$\therefore PR = 12$$

The length of RP is **12**.



2. In the given figure, $\angle MNP = 90^\circ$, seg $NQ \perp$ seg MP , $MQ = 9$, $QP = 4$, find NQ .

(March '25)

Ans.

In $\triangle MNP$, $\angle MNP = 90^\circ$

seg $NQ \perp$ hypotenuse MP

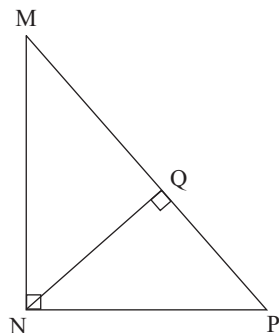
$\therefore$ by property of geometric mean,

$$NQ^2 = MQ \times PQ$$

$$\therefore NQ^2 = 9 \times 4$$

$$\therefore NQ^2 = 36$$

$$\therefore NQ = \sqrt{36} \quad \therefore NQ = 6$$



3. $\triangle PQR$ is an isosceles right angled triangle, $\angle PQR = 90^\circ$ and $PQ = 4$, then find the length of the side PR .

Ans. $\triangle PQR$ is a $45^\circ - 45^\circ - 90^\circ$ triangle.

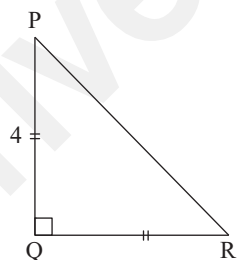
By $45^\circ - 45^\circ - 90^\circ$ triangle theorem,

$$PQ = \frac{1}{\sqrt{2}} \times PR$$

$$\therefore 4 = \frac{1}{\sqrt{2}} \times PR$$

$$\therefore PR = 4\sqrt{2}.$$

The length of side PR is $4\sqrt{2}$.



4. In the figure, $PQ = 5$ and $PR = 10$ then find the value of θ .

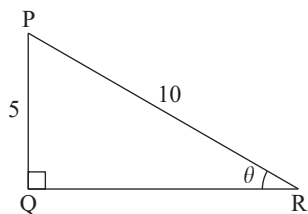
Ans.

$PQ = 5$ and $PR = 10$

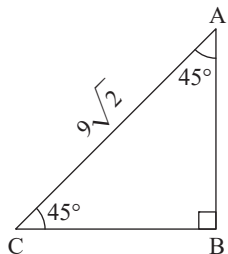
$$\therefore PQ = \frac{1}{2} PR$$

$\therefore \theta = 30^\circ \quad \dots$ [By converse of $30^\circ - 60^\circ - 90^\circ$ triangle theorem.]

The value of θ is 30° .



5. In the figure, $\angle ABC = 90^\circ$,
 $\angle BAC = \angle BCA = 45^\circ$. If $AC = 9\sqrt{2}$,
 then find AB. (March '22)



Ans.

$\triangle ABC$ is a $45^\circ-45^\circ-90^\circ$ triangle.

$\therefore$ by $45^\circ-45^\circ-90^\circ$ triangle theorem,

$$AB = \frac{1}{\sqrt{2}} \times AC$$

... (Side opposite to 45°)

$$\therefore AB = \frac{1}{\sqrt{2}} \times 9\sqrt{2}$$

$$\therefore AB = 9.$$

6. Find the diagonal of the square whose side is 10 cm.

(March '20)

Ans.

$$\begin{aligned} \text{Diagonal of the square} &= \sqrt{2} \times \text{side} \\ &= \sqrt{2} \times 10 \\ &= 10\sqrt{2} \text{ cm.} \end{aligned}$$

Diagonal of the square is $10\sqrt{2}$ cm.

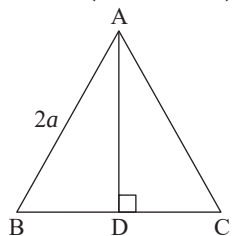
Assignment

3.2

- Is (6, 8, 10) a Pythagorean triplet or not?
- $\triangle PQR$ is a right angled triangle right angled at Q. State Pythagorean relation between its sides.
- In $\triangle ABC$, if $AB^2 = AC^2 + CB^2$, then which angle of $\triangle ABC$ is a right angle?
- In $\triangle RST$, $\angle S = 90^\circ$, $\angle T = 30^\circ$, $RT = 12$ cm, then find RS?

(March '23)

5. In the figure,
 $\triangle ABC$ is an equilateral triangle.
 $AB = 2a$, then $BD = ?$



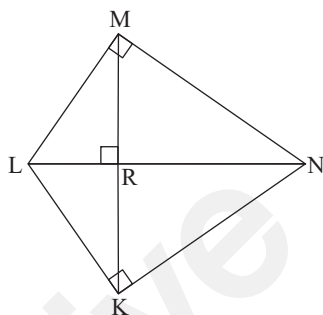
6. In $\triangle ABC$, $\angle B = 90^\circ$ and $AB^2 + BC^2 = 576$, then find the length of AC.

Answers :

1. (6, 8, 10) is a Pythagorean triplet. 2. $PR^2 = PQ^2 + QR^2$
 3. $\angle C = 90^\circ$ 4. RS = 6 cm 5. BD = a 6. AC = 24.

3.3 2 marks each

1. In the figure, $\angle LMN = \angle LKN = 90^\circ$,
 seg $MK \perp$ seg LN. Complete the
 following activity to prove R is the
 midpoint of seg MK.



Activity :

In $\triangle LMN$, $\angle LMN = 90^\circ$

seg $MR \perp$ hypotenuse LN.

$\therefore$ by theorem of geometric mean,

$$MR^2 = \boxed{} \times RN \quad \dots (1)$$

In $\triangle LKN$, $\angle LKN = 90^\circ$

seg $KR \perp$ hypotenuse LN

$\therefore$ by theorem of geometric mean,

$$KR^2 = LR \times \boxed{} \quad \dots (2)$$

From (1) and (2), we get,

$$MR^2 = \boxed{} \quad \therefore MR = \boxed{}$$

$\therefore$ R is the midpoint of seg MK.

Activity :

In $\triangle LMN$, $\angle LMN = 90^\circ$

seg $MR \perp$ hypotenuse LN

$\therefore$ by theorem of geometric mean,

$$MR^2 = \boxed{LR} \times RN \quad \dots (1)$$

In $\triangle LKN$, $\angle LKN = 90^\circ$

seg $KR \perp$ hypotenuse LN

∴ by theorem of geometric mean,

$$KR^2 = LR \times \boxed{RN} \quad \dots (2)$$

From (1) and (2), we get,

$$MR^2 = \boxed{KR^2}$$

$$\therefore MR = \boxed{KR}$$

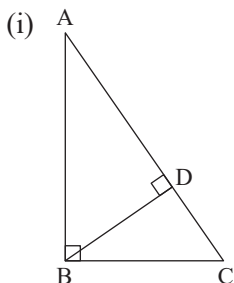
∴ R is the midpoint of seg MK.

2. For the statement, 'In a right angled triangle, the perpendicular segment to the hypotenuse from the opposite vertex, is the geometric mean of the segments into which the hypotenuse is divided,' then

(i) draw a neat labelled figure.

(ii) write 'Given' and 'To prove'.

Ans.



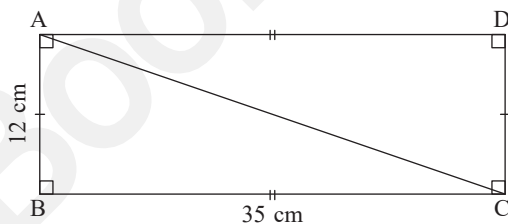
(ii) **Given :** In $\triangle ABC$, $\angle ABC = 90^\circ$.

seg $BD \perp$ hypotenuse AC such that
 $A-D-C$.

To prove : $BD^2 = AD \times DC$.

3. Find the length of the diagonal of a rectangle whose length is 35 cm and breadth is 12 cm. (March '23)

Solution :



Let $\square ABCD$ be required rectangle, $AB = 12$ cm, $BC = 35$ cm.

In right angled $\triangle ABC$,

by Pythagoras theorem,

$$AC^2 = AB^2 + BC^2$$

$$\therefore AC^2 = 12^2 + 35^2$$

$$\therefore AC^2 = 144 + 1225 \quad \therefore AC^2 = 1369$$

$$\therefore AC = 37 \text{ cm} \quad \dots \text{ (Taking square roots of both the sides)}$$

Ans. The length of the diagonal of the rectangle is **37 cm**.

- 4. In $\triangle ABC$, $AB = 9 \text{ cm}$, $BC = 40 \text{ cm}$, $AC = 41 \text{ cm}$. State whether $\triangle ABC$ is a right angled triangle or not. Write reason.**

(March '22)

Solution :

$$AC^2 = 41^2 = 1681$$

$$AB^2 + BC^2 = 9^2 + 40^2 = 81 + 1600 = 1681$$

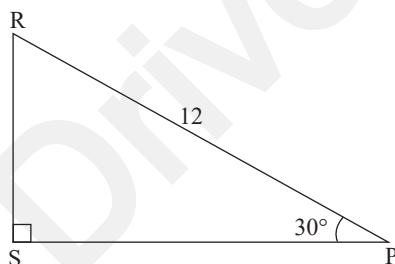
$$\therefore AC^2 = AB^2 + BC^2$$

$\therefore$ by converse of Pythagoras theorem,

$\triangle ABC$ is a right angled triangle.

Ans. $\triangle ABC$ is a right angled triangle.

- 5. In the given figure, find RS and PS using the information given in $\triangle PSR$. *(March '25)***



Solution :

In $\triangle RSP$, $\angle RSP = 90^\circ$, $\angle RPS = 30^\circ$,

$\therefore \angle SRP = 60^\circ \quad \dots \text{ (The remaining angle of } \triangle RSP \text{)}$

$\therefore \triangle RSP$ is a $30^\circ - 60^\circ - 90^\circ$ triangle.

$\therefore$ by $30^\circ - 60^\circ - 90^\circ$ triangle theorem,

$$RS = \frac{1}{2} \times RP \quad \dots \text{ (Side opposite to } 30^\circ \text{)}$$

$$\therefore RS = \frac{1}{2} \times 12$$

$$\therefore RS = 6$$

$$SP = \frac{\sqrt{3}}{2} \times RP \quad \dots \text{ (Side opposite to } 60^\circ \text{)}$$

$$\therefore SP = \frac{\sqrt{3}}{2} \times 12$$

$$\therefore SP = 6\sqrt{3}$$

Ans. RS = 6 and SP = $6\sqrt{3}$.

Assignment

3.3

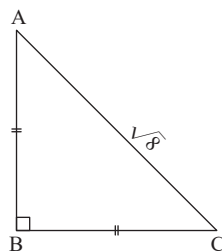
1. In the figure, $AB = BC$, $\angle ABC = 90^\circ$, $AC = \sqrt{8}$.

Complete the following activity to find the length of AB and BC.

Activity :

$AB = BC$ and $\angle ABC = 90^\circ$... (Given)

$\triangle ABC$ is an isosceles right angled triangle.



$$\therefore \angle BAC = \angle ACB = \boxed{}$$

$$\therefore AB = BC = \boxed{} \times AC$$

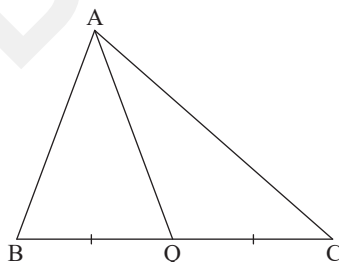
... (Side opposite to 45°)

$$= \boxed{} \times \sqrt{8}$$

$$= \frac{1}{\sqrt{2}} \times 2\sqrt{2} = \boxed{}$$

2. Complete the following activity to find the length of median AQ on side BC, if $AB^2 + AC^2 = 122$ and $BQ = 5$.

Activity : In $\triangle ABC$, seg AQ is the median.



... (Apollonius theorem)

$$\therefore AB^2 + AC^2 = 2AQ^2 + 2\boxed{}^2$$

$$\therefore 122 = 2AQ^2 + 2(5)^2$$

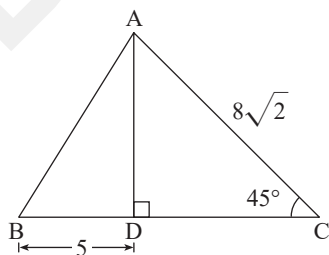
$$\therefore 2AQ^2 = 122 - 50$$

$$\therefore 2AQ^2 = \boxed{}$$

$$\therefore AQ^2 = \boxed{}$$

$$\therefore AQ = \boxed{} \quad \text{... (Taking square roots of both the sides)}$$

3. In order to prove, 'In a right angled triangle, the square of the hypotenuse is equal to the sum of the squares of remaining two sides.'
- (i) Draw a neat labelled figure.
- (ii) Write 'Given' and 'To prove' from the figure drawn by you.
4. Find the side and perimeter of a square whose diagonal is $13\sqrt{2}$ cm.
(July '19)
5. In $\triangle DEF$, $\angle E = 90^\circ$. If $DE = 33$ cm, $DF = 65$ cm, then find EF .
(Sept. '21)
6. Find the length of the hypotenuse of a right angled triangle, if remaining sides are 9 cm and 12 cm.
(March '24)
7. Determine whether (11, 12, 15) is a Pythagorean triplet or not. Justify.
8. Ramesh walks 30 m towards south and then walks 40 m due west. How far is he now from the starting point?
9. In $\triangle PQR$, $\angle P = 60^\circ$, $\angle Q = 90^\circ$ and $QR = 6\sqrt{3}$ cm, then find PR and PQ .
(Nov. '20)
10. Find the height of an equilateral triangle whose side is 12 cm.
11. Verify whether $(2a - 1)$ cm, $2\sqrt{2a}$ cm and $(2a + 1)$ cm are the sides of a right angled triangle or not?
12. In the figure, seg $AD \perp$ side BC , $\angle C = 45^\circ$, $BD = 5$, $AC = 8\sqrt{2}$. Find the values of AD and BC .



13. If the sides of the triangle are of the length 2, 1 and $\sqrt{3}$ then determine the type of triangle.

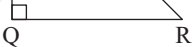
Answers :

1. 45° , $\frac{1}{\sqrt{2}}$, $\frac{1}{\sqrt{2}}$, 2 2. BQ, 72, 36, 6

P

(ii) **Given :** In $\triangle PQR$, $\angle PQR = 90^\circ$

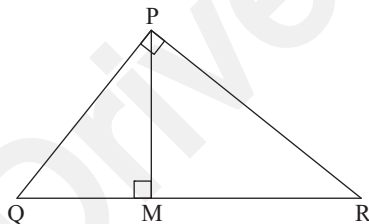
To prove : $PR^2 = PQ^2 + QR^2$.



4. side = 13 cm; Perimeter = 52 cm 5. EF = 56 cm 6. 15 cm
7. (11, 12, 15) is not a Pythagorean triplet
8. 50 m 9. PR = 12 and PQ = 6 10. $6\sqrt{3}$ cm
11. $(2a - 1)$ cm, $2\sqrt{2}a$ cm and $(2a + 1)$ cm are sides of a right angled triangle
12. AD = 8, BC = 13 13. The triangle is a $30^\circ - 60^\circ - 90^\circ$ triangle.

3.4 3 marks each

- In the figure, $\angle QPR = 90^\circ$,
seg $PM \perp$ seg QR and $Q-M-R$,
 $PM = 10$, $QM = 8$. Complete the
following activity to find the
value of QR . (July '22)**



Activity :

In $\triangle PQR$, $\angle QPR = 90^\circ$ and seg $PM \perp$ seg QR .

$$\therefore \text{PM}^2 = \boxed{} \times \text{MR} \quad \dots \left(\boxed{} \right)$$

$$\therefore \left(\square \right)^2 = 8 \times \text{MR}$$

$$\therefore \frac{100}{8} = \text{MR} \qquad \therefore \boxed{} = \text{MR}$$

Now, $QR = QM + MR$... (Q-M-R)

$$\therefore QR = 8 + \boxed{} \qquad \therefore QR = \boxed{}.$$

Activity :

In $\triangle PQR$, $\angle QPR = 90^\circ$ and seg $PM \perp$ seg QR .

$$\therefore \text{PM}^2 = \boxed{\text{QM}} \times \text{MR} \qquad \dots \quad \boxed{\text{Theorem of geometric mean}}$$



$$\therefore \left(\boxed{10} \right)^2 = 8 \times MR$$

$$\therefore \frac{100}{8} = MR$$

$$\therefore \boxed{12.5} = MR$$

Now, $QR = QM + MR$

... (Q-M-R)

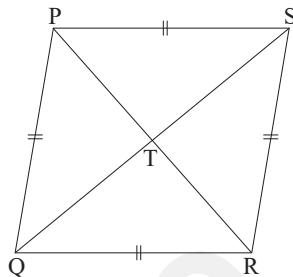
$$\therefore QR = 8 + \boxed{12.5}$$

$$\therefore QR = \boxed{20.5}.$$

2. Complete the following activity to prove that the sum of squares of the diagonals of a rhombus is equal to the sum of the squares of its sides.

(March '25)

Given : $\square PQRS$ is a rhombus.
Diagonals PR and SQ intersect each other at T.



To prove : $PS^2 + SR^2 + QR^2 + PQ^2 = PR^2 + QS^2$

Activity :

The diagonals of a rhombus bisect each other.

In $\triangle PQS$, PT is the median and in $\triangle QRS$, RT is the median.

$\therefore$ by Apollonius theorem,

$$PQ^2 + PS^2 = \boxed{} + 2QT^2 \quad \dots (1)$$

$$\text{and } QR^2 + SR^2 = \boxed{} + 2QT^2 \quad \dots (2)$$

Adding (1) and (2),

$$\begin{aligned} PQ^2 + PS^2 + QR^2 + SR^2 &= 2 \left(PT^2 + \boxed{} \right) + 4QT^2 \\ &= 2 \left(PT^2 + \boxed{} \right) + 4QT^2 \quad \dots (RT = PT) \\ &= 4PT^2 + 4QT^2 \\ &= \left(\boxed{} \right)^2 + (2QT)^2 \end{aligned}$$

$$\therefore PQ^2 + PS^2 + QR^2 + SR^2 = PR^2 + \boxed{}.$$

Activity :

The diagonals of a rhombus bisect each other.

In $\triangle PQS$, PT is the median and in $\triangle QRS$, RT is the median.

$\therefore$ by Apollonius theorem,

$$PQ^2 + PS^2 = \boxed{2PT^2} + 2QT^2 \quad \dots (1)$$

$$\text{and } QR^2 + SR^2 = \boxed{2RT^2} + 2QT^2 \quad \dots (2)$$

Adding (1) and (2),

$$\begin{aligned} PQ^2 + PS^2 + QR^2 + SR^2 &= 2 \left(PT^2 + \boxed{RT^2} \right) + 4QT^2 \\ &= 2 \left(PT^2 + \boxed{PT^2} \right) + 4QT^2 \quad \dots (RT = PT) \\ &= 4PT^2 + 4QT^2 \\ &= \left(\boxed{2PT} \right)^2 + (2QT)^2 \end{aligned}$$

$$\therefore PQ^2 + PS^2 + QR^2 + SR^2 = PR^2 + \boxed{QS^2}.$$

- 3. The perpendicular sides of a right angled triangle are $3x$ and $4x$. The length of its hypotenuse is 30 . Find the lengths of the perpendicular sides of the right angled triangle.**

Solution : Let $\triangle ABC$ be the given right angled triangle,

$$\angle ABC = 90^\circ.$$

$$AB = 4x, BC = 3x, AC = 30.$$

By Pythagoras theorem,

$$AC^2 = AB^2 + BC^2$$

$$\therefore 30^2 = (4x)^2 + (3x)^2$$

$$\therefore 900 = 16x^2 + 9x^2$$

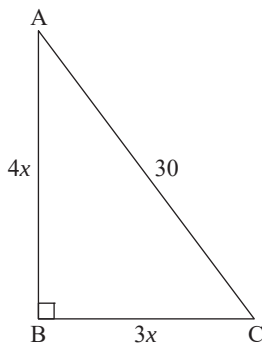
$$\therefore 25x^2 = 900$$

$$\therefore x^2 = \frac{900}{25} \quad \therefore x^2 = 36 \quad \therefore x = 6$$

$$AB = 4x = 4 \times 6 = 24$$

$$BC = 3x = 3 \times 6 = 18$$

Ans. The lengths of the perpendicular sides of the right angled triangle are **18** and **24**.



4. In $\triangle ABC$, seg $AD \perp$ side BC and $AD^2 = BD \cdot CD$. Prove that in $\triangle ABC$, $\angle A = 90^\circ$.

Proof : In $\triangle ADB$, $\angle ADB = 90^\circ$

$$\therefore AB^2 = AD^2 + BD^2$$

... (By Pythagoras theorem) ... (1)

In $\triangle ADC$, $\angle ADC = 90^\circ$.

$$\therefore AC^2 = AD^2 + CD^2$$

... (By Pythagoras theorem) ... (2)

Adding (1) and (2),

$$AB^2 + AC^2 = AD^2 + BD^2 + AD^2 + CD^2$$

$$= BD^2 + CD^2 + 2AD^2$$

$$= BD^2 + CD^2 + 2BD \cdot CD$$

... ($AD^2 = BD \cdot CD$)

$$= (BD + CD)^2$$

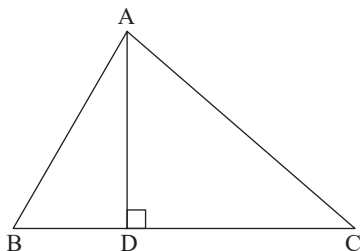
$$= BC^2$$

... (B-D-C)

$$\therefore AB^2 + AC^2 = BC^2$$

So by converse of Pythagoras theorem,

$$\angle A = 90^\circ.$$



5. Find the length of the side and perimeter of an equilateral triangle whose height is $\sqrt{3}$ cm.

Solution :

Let $\triangle ABC$ be given equilateral triangle and seg AD be its height.

$$AD = \sqrt{3} \text{ cm}$$

$$\angle B = 60^\circ$$

... (Angle of an equilateral triangle) ... (1)

In $\triangle ADB$,

$$\angle ADB = 90^\circ, \angle ABD = 60^\circ.$$

$$\therefore \angle BAD = 30^\circ$$

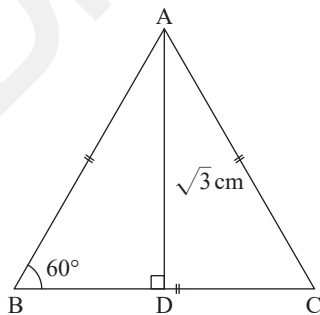
... (The remaining angle of $\triangle ADB$)

$\therefore \triangle ADB$ is a $30^\circ - 60^\circ - 90^\circ$ triangle.

$\therefore$ by $30^\circ - 60^\circ - 90^\circ$ triangle theorem,

$$AD = \frac{\sqrt{3}}{2} AB$$

... (Side opposite to 60°)



$$\therefore \sqrt{3} = \frac{\sqrt{3}}{2} AB \quad \therefore AB = \frac{2 \times \sqrt{3}}{\sqrt{3}}$$

$$\therefore AB = 2 \text{ cm}$$

$\therefore$ side of an equilateral triangle is 2 cm.

Perimeter of $\triangle ABC = 3 \times \text{side}$

$$= 3 \times AB$$

$$= 3 \times 2 = 6 \text{ cm}$$

Perimeter of equilateral triangle is 6 cm.

Ans. Side of an equilateral triangle is **2 cm** and its perimeter is **6 cm**.

- 6. In $\triangle ABC$, seg AP is a median. If $BC = 18$, $AB^2 + AC^2 = 260$. Find AP.** **(July '19; March '23)**

Solution :

$$BC = 18$$

... (Given)

$$BP = \frac{1}{2} BC \quad \dots \text{ (P is the midpoint of seg BC)}$$

$$\therefore BP = \frac{1}{2} \times 18$$

$$\therefore BP = 9$$

In $\triangle ABC$,

seg AP is the median ... (Given)

$\therefore$ by Apollonius theorem,

$$AB^2 + AC^2 = 2AP^2 + 2BP^2$$

$$\therefore 260 = 2AP^2 + 2(9)^2$$

$$\therefore 260 = 2AP^2 + 162$$

$$\therefore 2AP^2 = 260 - 162$$

$$\therefore 2AP^2 = 98$$

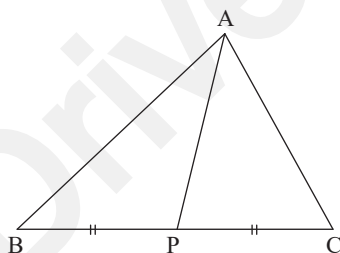
$$\therefore AP^2 = \frac{98}{2}$$

$$\therefore AP^2 = 49$$

$$\therefore AP = 7$$

... (Taking square roots of both the sides)

Ans. $AP = 7$.

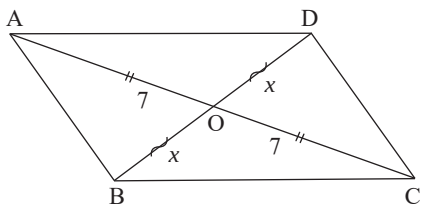


7. Sum of the squares of adjacent sides of a parallelogram is 130 cm^2 and length of one of its diagonals is 14 cm . Find the length of the other diagonal.

Solution :

Let $\square ABCD$ be the given parallelogram.

Let diagonals AC and BD intersect at point O .



$$AB^2 + BC^2 = 130 \text{ cm}^2 \text{ and } AC = 14 \text{ cm}$$

$$AO = \frac{1}{2}AC \quad \dots \text{ (Diagonals of parallelogram bisect each other)}$$

$$\therefore AO = \frac{1}{2} \times 14$$

$$\therefore AO = 7 \text{ cm}$$

In $\triangle ABC$,

seg BO is the median ... (By definition)

$\therefore$ by Apollonius theorem,

$$AB^2 + BC^2 = 2BO^2 + 2AO^2$$

$$\therefore 130 = 2BO^2 + 2(7)^2$$

$$\therefore 130 = 2BO^2 + 2 \times 49$$

$$\therefore 130 = 2BO^2 + 98$$

$$\therefore 2BO^2 = 130 - 98$$

$$\therefore 2BO^2 = 32$$

$$\therefore BO^2 = \frac{32}{2}$$

$$\therefore BO^2 = 16$$

$$\therefore BO = 4 \text{ cm} \quad \dots \text{ (Taking square roots of both the sides)}$$

$$BO = \frac{1}{2}BD \quad \dots \text{ (Diagonals of parallelogram bisect each other)}$$

$$\therefore 4 = \frac{1}{2} BD$$

$$\therefore BD = 8 \text{ cm}$$

Ans. Length of other diagonal of the parallelogram is **8 cm**.

- 8. Suppose m and n are any two numbers and $m > n$. If $m^2 - n^2$, $2mn$ and $m^2 + n^2$ are the three sides of a triangle. Show that it is a right angled triangle and hence write any two pairs of Pythagorean triplet.**

Solution :

$$\begin{aligned}(m^2 - n^2)^2 &= (m^2)^2 - 2(m^2)(n^2) + (n^2)^2 \\ &= m^4 - 2m^2 n^2 + n^4\end{aligned}\quad \dots (1)$$

$$(2mn)^2 = 4m^2 n^2 \quad \dots (2)$$

$$\begin{aligned}(m^2 + n^2)^2 &= (m^2)^2 + 2(m^2)(n^2) + (n^2)^2 \\ &= m^4 + 2m^2 n^2 + n^4\end{aligned}\quad \dots (3)$$

Adding (1) and (2), we get,

$$\begin{aligned}(m^2 - n^2)^2 + (2mn)^2 &= (m^4 - 2m^2 n^2 + n^4) + 4m^2 n^2 \\ &= m^4 - 2m^2 n^2 + n^4 + 4m^2 n^2 \\ &= m^4 + 2m^2 n^2 + n^4 \\ &= (m^2 + n^2)^2\end{aligned}\quad \dots [\text{From (3)}]$$

Thus, $(m^2 - n^2)$, $(2mn)$ and $(m^2 + n^2)$ are the three sides of a right angled triangle
... (By converse of Pythagoras theorem)

If $m = 5$ and $n = 2$ then

$$m^2 - n^2 = 5^2 - 2^2 = 25 - 4 = 21$$

$$2mn = 2 \times 5 \times 2 = 20$$

$$m^2 + n^2 = 5^2 + 2^2 = 25 + 4 = 29.$$

$\therefore$ **(21, 20, 29)** is a Pythagorean triplet.

If $m = 7$ and $n = 3$, then

$$m^2 - n^2 = 7^2 - 3^2 = 49 - 9 = 40$$

$$2mn = 2 \times 7 \times 3 = 42$$

$$m^2 + n^2 = 7^2 + 3^2 = 49 + 9 = 58.$$

$\therefore$ **(40, 42, 58)** is a Pythagorean triplet.

9. $\triangle ABC$ and $\triangle PQR$ are equilateral triangles with altitudes $2\sqrt{3}$ and $4\sqrt{3}$ respectively, then :

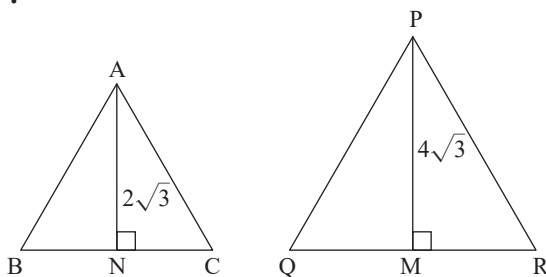
(a) Find the lengths of side AB and side PQ.

(b) Find $\frac{A(\triangle ABC)}{A(\triangle PQR)}$.

(c) Find the ratio of perimeter of $\triangle ABC$ to the perimeter of $\triangle PQR$.
(July '22)

Solution :

(a)



Let AN and PM be respective altitudes of $\triangle ABC$ and $\triangle PQR$.

Height of an equilateral triangle $= \frac{\sqrt{3}}{2} \times \text{side}$

$$\therefore AN = \frac{\sqrt{3}}{2} \times AB$$

$$\therefore 2\sqrt{3} = \frac{\sqrt{3}}{2} \times AB$$

$$\therefore AB = \frac{2\sqrt{3} \times 2}{\sqrt{3}}$$

$$\therefore AB = 4$$

$$PM = \frac{\sqrt{3}}{2} \times PQ$$

$$\therefore 4\sqrt{3} = \frac{\sqrt{3}}{2} \times PQ$$

$$\therefore PQ = \frac{4\sqrt{3} \times 2}{\sqrt{3}}$$

$$\therefore PQ = 8$$

(b) Two equilateral triangles are always similar.

$$\therefore \frac{A(\triangle ABC)}{A(\triangle PQR)} = \frac{AB^2}{PQ^2} \quad \dots \text{(Theorem on areas of similar triangles)}$$

$$= \frac{4^2}{8^2} = \left(\frac{4}{8}\right)^2$$

$$= \left(\frac{1}{2}\right)^2$$

$$\therefore \frac{A(\triangle ABC)}{A(\triangle PQR)} = \frac{1}{4}$$

(c) Perimeter of $\triangle ABC = 3 \times AB = 3 \times 4 = 12$

Perimeter of $\triangle PQR = 3 \times PQ = 3 \times 8 = 24$

$$\frac{\text{Perimeter of } \triangle ABC}{\text{Perimeter of } \triangle PQR} = \frac{12}{24} = \frac{1}{2}$$

Ans. (a) $AB = 4$, $PQ = 8$

(b) $\frac{A(\triangle ABC)}{A(\triangle PQR)} = \frac{1}{4}$

(c) $\frac{\text{Perimeter of } \triangle ABC}{\text{Perimeter of } \triangle PQR} = \frac{1}{2}$

Assignment	3.4
-------------------	------------

1. $\triangle PQR$ is an equilateral triangle.
seg $PS \perp$ side QR . Complete the following
activity to prove $PS^2 = 3QS^2$.

Activity :

In $\triangle PQS$,

$\angle PSQ = \boxed{}$

... (Given)

$\angle Q = \boxed{}$

... (Angle of an equilateral triangle)

$\therefore \angle QPS = 30^\circ$

... (Remaining angle of $\triangle PSQ$)

$\therefore \triangle PQS$ is a $30^\circ - 60^\circ - 90^\circ$ triangle

$\therefore PS = \boxed{} PQ$

... (Side opposite to 60°) ... (1)

$\therefore QS = \boxed{} PQ$

... (Side opposite to 30°)

$\therefore PQ = 2QS$

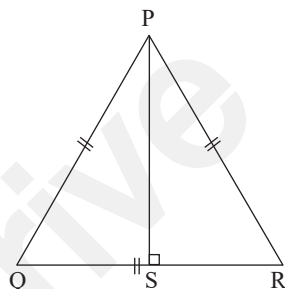
... (2)

Substituting (2) in (1), we get,

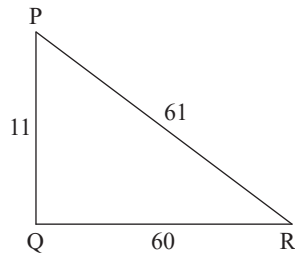
$PS = \boxed{} \times 2QS$

$\therefore PS = \boxed{} QS$

$\therefore PS^2 = 3QS^2$



2. In the figure, $PQ = 11$, $PR = 61$ and $QR = 60$. Complete the following activity to determine whether $\triangle PQR$ is a right angled triangle or not.



$$PQ = 11, QR = 60 \text{ and } PR = 61$$

$$PQ^2 = 11^2 = \boxed{}$$

$$QR^2 = 60^2 = 3600$$

$$PR^2 = 61^2 = \boxed{}$$

$$PQ^2 + QR^2 = \boxed{} + 3600 = \boxed{}$$

$$\therefore PQ^2 + QR^2 \boxed{} PR^2 \quad (\text{Put sign } = \text{ or } \neq)$$

$$\therefore \triangle PQR \boxed{} \text{ a right angled triangle.} \quad \dots (\text{is/is not})$$

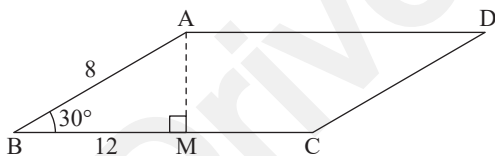
3. In the figure,

$\square ABCD$ is a

parallelogram. $AB = 8$,

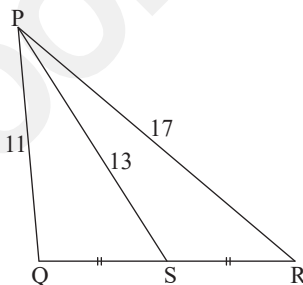
$BC = 12$ and $\angle B = 30^\circ$.

Find A ($\square ABCD$).



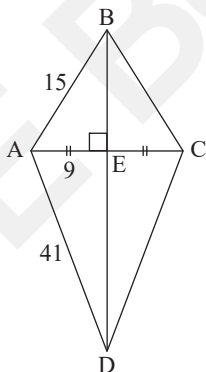
4. In $\triangle PQR$, seg PM is a median. $PM = 9$ and $PQ^2 + PR^2 = 290$. Find the length of QR .
(March '19-'24)

5.



In $\triangle PQR$, point S is the midpoint of side QR . If $PQ = 11$, $PR = 17$, $PS = 13$, find QR .
(March '20)

6.



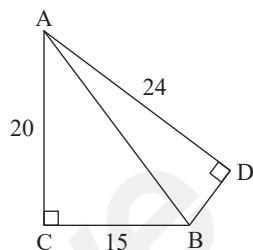
In the figure, smaller diagonal AC of a kite ABCD is 18 cm long. $AB = 15$ cm and $AD = 41$ cm. Find the length of

- (i) BE
- (ii) ED
- (iii) BD

7. In the figure,

$AC = 20$, $CB = 15$, $AD = 24$.

Find AB and BD.



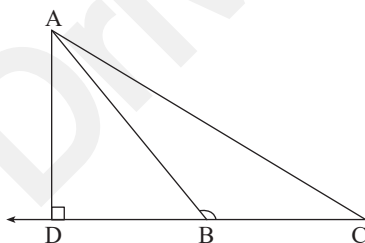
8. Construct a line segment of length $\sqrt{41}$ using Pythagoras theorem.

9. In obtuse angled $\triangle ABC$,

$\angle B > 90^\circ$. If seg $AD \perp$ side BC

and $D-B-C$, then prove that

$$AC^2 = AB^2 + BC^2 + 2BC \cdot DB.$$



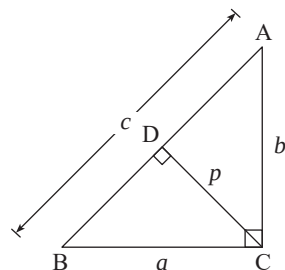
10. In $\triangle ABC$, $\angle C = 90^\circ$ and $CD \perp AB$

if $BC = a$, $CA = b$, $AB = c$ and $CD = p$ then

(a) Find $A(\triangle ABC)$ considering AB as the base.

(b) Find $A(\triangle ABC)$ considering BC as the base.

(c) Using (a) and (b) prove $cp = ab$.



11. Ramesh goes 24 km to south, then turns and goes to west for 10 km.

(i) From the given information, draw a proportionate figure.

(ii) If starting point and end points are joined, then which type of triangle do we get?

(iii) At last how far is he from the starting point?

Answers :

1. 90° , 60° , $\frac{\sqrt{3}}{2}$, $\frac{1}{2}$, $\frac{\sqrt{3}}{2}$, $\sqrt{3}$

2. 121, 3721, 121, 3721, =, is

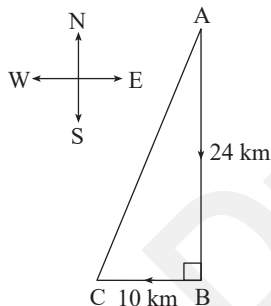
3. $A(\square ABCD) = 48$ sq units 4. $QR = 16$ 5. $QR = 12$

6. $BE = 12$ cm, $ED = 40$ cm, $BD = 52$ cm 7. $AB = 25$, $BD = 7$

10. (a) $A(\triangle ABC) = \frac{1}{2} \times c \times p$

(b) $A(\triangle ABC) = \frac{1}{2} \times a \times b$

11. (i)

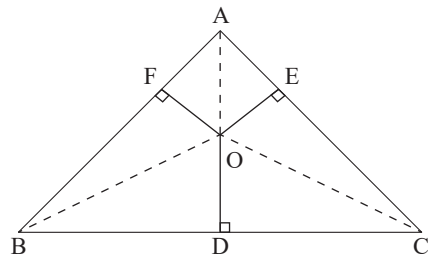


(ii) Right angled triangle

(iii) 26 km.

3.5 4 marks each

1. O is any point in $\triangle ABC$,
 seg OF, seg OD and
 seg OE $\perp$ sides AB, BC and
 AC respectively. Prove that
 $AF^2 + BD^2 + CE^2 =$
 $AE^2 + CD^2 + BF^2$.



Construction : Draw OA, OB and OC.

Proof : In $\triangle OFA$, $\angle OFA = 90^\circ$

∴ by Pythagoras theorem,

$$OA^2 = OF^2 + AF^2$$

$$\therefore AF^2 = OA^2 - OF^2 \quad \dots (1)$$

$$\text{Similarly, } BD^2 = OB^2 - OD^2 \quad \dots (2)$$

$$\text{and } CE^2 = OC^2 - OE^2 \quad \dots (3)$$

Adding (1), (2) and (3).

$$AF^2 + BD^2 + CE^2 = OA^2 - OF^2 + OB^2 - OD^2 + OC^2 - OE^2 \quad \dots (4)$$

In $\triangle OEA$, $\angle OEA = 90^\circ$

∴ by Pythagoras theorem,

$$OA^2 = OE^2 + AE^2$$

$$\therefore AE^2 = OA^2 - OE^2 \quad \dots (5)$$

$$\text{Similarly, } CD^2 = OC^2 - OD^2 \quad \dots (6)$$

$$BF^2 = OB^2 - OF^2 \quad \dots (7)$$

Adding (5), (6) and (7), we get,

$$AE^2 + CD^2 + BF^2 = OA^2 - OE^2 + OC^2 - OD^2 + OB^2 - OF^2 \quad \dots (8)$$

∴ from (4) and (8), we get,

$$AF^2 + BD^2 + CE^2 = AE^2 + CD^2 + BF^2.$$

2. In the figure, $\angle PQR = 90^\circ$.

Points S and T are the points of trisection of QR.

Prove that $8PT^2 = 3PR^2 + 5PS^2$.

Proof : Points S and T are the points of trisection.

Let $QS = ST = TR = x$

∴ $QT = 2x$ and $QR = 3x$

In $\triangle PQT$,

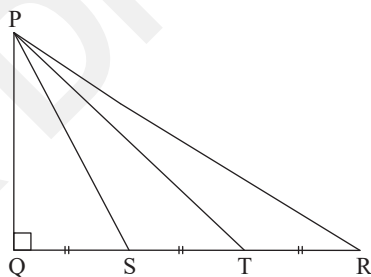
$\angle PQT = 90^\circ$

∴ by Pythagoras theorem,

$$PT^2 = PQ^2 + QT^2$$

$$\therefore PT^2 = PQ^2 + (2x)^2$$

$$\therefore PT^2 = PQ^2 + 4x^2$$



Multiplying each term by 8, we get

$$8PT^2 = 8PQ^2 + 32x^2 \quad \dots (1)$$

In $\triangle PQR$,

$$\angle PQR = 90^\circ$$

$\therefore$ by Pythagoras theorem,

$$PR^2 = PQ^2 + QR^2$$

$$\therefore PR^2 = PQ^2 + (3x)^2$$

$$\therefore PR^2 = PQ^2 + 9x^2$$

Multiplying each term by 3, we get

$$3PR^2 = 3PQ^2 + 27x^2 \quad \dots (2)$$

In $\triangle PQS$,

$$\angle PQS = 90^\circ$$

$\therefore$ by Pythagoras theorem,

$$PS^2 = PQ^2 + QS^2$$

$$PS^2 = PQ^2 + x^2$$

Multiplying each term by 5, we get

$$5PS^2 = 5PQ^2 + 5x^2 \quad \dots (3)$$

Adding (2) and (3), we get

$$\begin{aligned} 3PR^2 + 5PS^2 &= 3PQ^2 + 27x^2 + 5PQ^2 + 5x^2 \\ &= 8PQ^2 + 32x^2 \\ &= 8PT^2 \end{aligned} \quad \dots [\text{From (1)}]$$

$$\therefore 8PT^2 = 3PR^2 + 5PS^2.$$

- 3. In $\square ABCD$, M and N are the midpoints of diagonals AC and BD respectively.**

Prove that

$$AB^2 + BC^2 + CD^2 + DA^2 = AC^2 + BD^2 + 4MN^2.$$

Construction : Draw seg MB and seg MD.

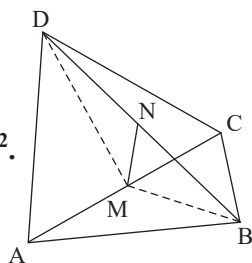
Proof :

In $\triangle ABC$, seg BM is the median.

$\therefore$ by Apollonius theorem,

$$AB^2 + BC^2 = 2BM^2 + 2AM^2 \quad \dots (1)$$

In $\triangle ADC$, seg DM is the median.



∴ by Apollonius theorem,

$$CD^2 + AD^2 = 2DM^2 + 2AM^2 \quad \dots (2)$$

On adding (1) and (2), we get,

$$AB^2 + BC^2 + CD^2 + AD^2 = 2BM^2 + 2AM^2 + 2DM^2 + 2AM^2$$

$$\therefore AB^2 + BC^2 + CD^2 + AD^2 = 2(BM^2 + DM^2) + 4AM^2$$

$$\therefore AB^2 + BC^2 + CD^2 + AD^2 = 2(BM^2 + DM^2) + 4 \left[\frac{1}{2} AC \right]^2$$
$$\dots \left[AM = \frac{1}{2} AC \right]$$

$$\therefore AB^2 + BC^2 + CD^2 + AD^2 = 2(BM^2 + DM^2) + AC^2 \quad \dots (3)$$

In $\triangle BMD$,

seg MN is the median.

∴ by Apollonius theorem,

$$BM^2 + DM^2 = 2MN^2 + 2BN^2$$

Multiplying both the sides by 2, we get,

$$2(BM^2 + DM^2) = 2(2MN^2 + 2BN^2)$$

$$\therefore 2(BM^2 + DM^2) = 4MN^2 + 4BN^2$$

$$\therefore 2(BM^2 + DM^2) = 4MN^2 + 4 \times \left(\frac{1}{2} BD \right)^2 \quad \dots \left[BN = \frac{1}{2} BD \right]$$

$$\therefore 2(BM^2 + DM^2) = 4MN^2 + 4 \times \frac{1}{4} BD^2$$

$$\therefore 2(BM^2 + DM^2) = 4MN^2 + BD^2 \quad \dots (4)$$

Substituting (4) in (3), we get,

$$AB^2 + BC^2 + CD^2 + AD^2 = 4MN^2 + BD^2 + AC^2,$$

$$\text{i.e. } AB^2 + BC^2 + CD^2 + AD^2 = AC^2 + BD^2 + 4MN^2.$$

-
4. $\triangle XYZ$ is a right angled triangle. $\angle XYZ = 90^\circ$, $YZ = b$ and $a = A (\triangle XYZ)$. If seg $YP \perp$ side XZ , then prove that :

$$YP = \frac{2ab}{\sqrt{b^4 + 4a^2}}.$$

Proof : $a = A (\triangle XYZ)$

... (Given) ... (1)

$$\text{and } A (\triangle XYZ) = \frac{1}{2} \times \text{base} \times \text{height}$$

$$\therefore A (\triangle XYZ) = \frac{1}{2} \times YZ \times XY \quad \dots (2)$$

From (1) and (2),

$$a = \frac{1}{2} \times b \times XY \quad \dots (YZ = b, \text{ given})$$

$$\therefore XY = \frac{2a}{b} \quad \dots (3)$$

$$\text{Similarly, } A (\triangle XYZ) = \frac{1}{2} \times XZ \times YP$$

... (XZ = base, YP = height) ... (4)

In right angled $\triangle XYZ$, by Pythagoras theorem,

$$XZ^2 = XY^2 + YZ^2 = \left(\frac{2a}{b}\right)^2 + (b)^2 \quad \dots [\text{From (3) and } YZ = b, \text{ given}]$$

$$= \frac{4a^2}{b^2} + b^2 = \frac{4a^2 + b^4}{b^2}$$

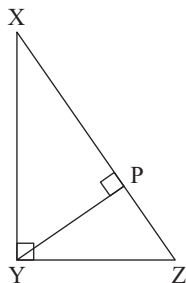
$$\therefore XZ = \frac{\sqrt{4a^2 + b^4}}{b} \quad \dots (\text{Taking square roots of both the sides})$$

Keeping value of XZ in (4), we get

$$A (\triangle XYZ) = \frac{1}{2} \times \frac{\sqrt{4a^2 + b^4}}{b} \times YP$$

$$\therefore YP = \frac{2A(\triangle XYZ) \times b}{\sqrt{4a^2 + b^4}} = \frac{2a \times b}{\sqrt{4a^2 + b^4}} \quad \dots [A (\triangle XYZ) = a]$$

$$\therefore YP = \frac{2ab}{\sqrt{b^4 + 4a^2}}$$



5. In $\triangle ABC$, $\angle BAC = 90^\circ$,

seg $AP \perp$ side BC , $B-P-C$.

D is the midpoint of side BC.

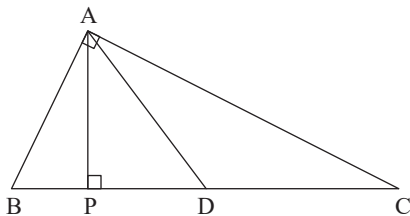
Prove that $2AD^2 = BD^2 + CD^2$.

(Nov. '20)

Proof :

D is the midpoint of side BC

... (Given)



$$\therefore BD = DC \quad \dots (1)$$

In $\triangle BAC$,

$$\angle BAC = 90^\circ \quad \dots (\text{Given})$$

$\therefore$ by Pythagoras theorem,

$$BC^2 = AB^2 + AC^2$$

$$\therefore (2BD)^2 = AB^2 + AC^2 \quad \dots (BC = 2BD)$$

$$\therefore AB^2 + AC^2 = 4BD^2 \quad \dots (2)$$

In $\triangle ABC$,

seg AD is the median.

$\therefore$ by Apollonius theorem,

$$AB^2 + AC^2 = 2AD^2 + 2BD^2$$

$$\therefore 4BD^2 = 2AD^2 + 2BD^2 \quad \dots [\text{From (2)}]$$

$$\therefore 2AD^2 = 4BD^2 - 2BD^2$$

$$\therefore 2AD^2 = 2BD^2$$

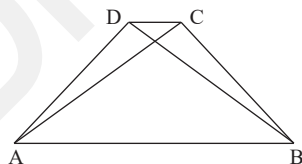
$$\therefore 2AD^2 = BD^2 + BD^2$$

$$\therefore 2AD^2 = BD^2 + DC^2 \quad \dots [\text{From (1)}]$$

6. In $\square ABCD$, $\angle A + \angle B = 90^\circ$. Prove that $AB^2 + CD^2 = AC^2 + BD^2$.

Construction :

Extend seg AD and seg BC to intersect each other at point P as shown.



Proof : In $\triangle PAB$,

$$\angle P + \angle A + \angle B = 180^\circ$$

$\dots$ (Sum of all angles of a triangle is 180°)

$$\therefore \angle P + 90^\circ = 180^\circ$$

$\dots (\angle A + \angle B = 90^\circ, \text{ given})$

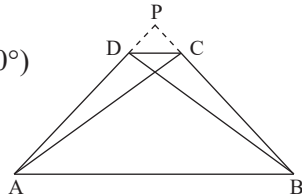
$$\therefore \angle P = 180 - 90^\circ$$

$$\therefore \angle P = 90^\circ$$

$\dots (1)$

In $\triangle APB$,

$$\angle APB = 90^\circ \quad \dots [\text{From (1)}]$$



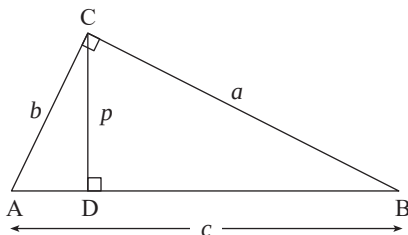
$$\therefore AB^2 + CD^2 = AC^2 + BD^2.$$

3.5

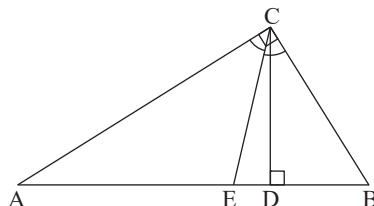
-

3. In $\triangle ABC$, $\angle C = 90^\circ$,
 seg $CD \perp$ side AB . $BC = a$,
 $AC = b$, $AB = c$ and $CD = p$.

Prove that $\frac{1}{p^2} = \frac{1}{a^2} + \frac{1}{b^2}$.



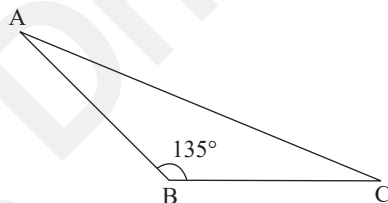
4. In a right angled triangle ACB,
 the perpendicular drawn from the
 point C on the hypotenuse meets
 the hypotenuse at D and the
 bisector of $\angle ACB$ meets the



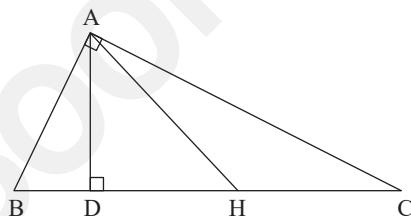
hypotenuse at E. Prove that $\frac{AD}{DB} = \frac{AE^2}{EB^2}$.

(March '19)

5. Prove that if the diagonals of a quadrilateral intersect each other in a right angle, then the sum of the squares of opposite sides are equal.
6. In $\triangle ABC$, $\angle ABC = 135^\circ$,
 prove that
 $AC^2 = AB^2 + BC^2 + 4A(\triangle ABC)$.



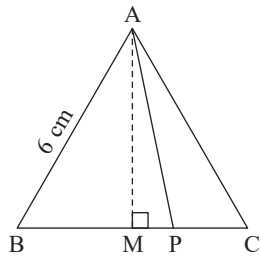
7.



In $\triangle ABC$, H is a point on side BC. Such that $BH : HC = 5 : 3$.

Prove that $3AB^2 + 5AC^2 = 8AH^2 + 3BH^2 + 5CH^2$.

8. $\triangle ABC$ is an equilateral triangle. Point P is on base BC such that $PC = \frac{1}{3} BC$, if $AB = 6$ cm, find AP.



(Ans. $AP = 2\sqrt{7}$ cm)

**Question
Set
4**

CIRCLE

4.1 1 mark each

Q. Choose the correct alternative from those given below each question :

1. $\angle ACB$ is an inscribed angle in a circle with centre O. If $\angle ACB = 65^\circ$, then what is the measure of its intercepted arc AXB? **(March '25)**
(A) 65° (B) 230° (C) 295° (D) 130°

Ans. (D) [Hint : Use inscribed angle theorem.]

2. A circle having radius 3 cm has the length of its largest chord
(March '24)

(A) 1.5 cm (B) 3 cm (C) 6 cm (D) 9 cm

Ans. (C) [Hint : In a circle diameter is the largest chord.]

3. If the points A, B, C are non-collinear points, then how many circles can be drawn which passes through points A, B and C? **(July '19)**
(A) Two (B) Three (C) One (D) Infinite

Ans. (C) [Hint : Use property of a circle.]

4. If P, Q, R are three points which lie on the circle with centre O such that $\angle POQ = 110^\circ$, $\angle QOR = 130^\circ$; then $\angle POR = \dots$
(A) 110° (B) 130° (C) 60° (D) 120°

Ans. (D) [Hint : $\angle POQ + \angle QOR + \angle POR = 360^\circ$.]

5. A tangent AB at a point A of a circle of radius 5 cm meets a line through the centre O at point B such that $OB = 12$ cm. Length AB is

(A) 5 cm (B) 12 cm (C) 13 cm (D) $\sqrt{119}$ cm

Ans. (D) [Hint : Use tangent theorem.]

6. The angle made by the tangent and the radius drawn at the point of contact is

(A) 0° (B) 45° (C) 90° (D) 180°

Ans. (C) [Hint : Use tangent theorem.]



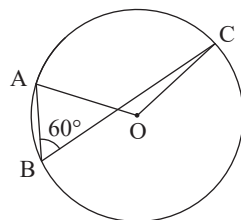
7. A circle touches all sides of a parallelogram. So the parallelogram must be a

- (A) rectangle (B) rhombus (C) square (D) trapezium

Ans. (B) [Hint : Using tangent segments theorem, conclude adjacent sides of parallelogram are equal. Hence the parallelogram is a rhombus.]

8. In the figure, if $\angle ABC = 60^\circ$, then $\angle AOC = \dots$

- (A) 120° (B) 60°
(C) 30° (D) 90°



Ans. (A) [Hint : Use inscribed angle theorem and definition of measure of an arc.]

9. In a cyclic $\square ABCD$, twice the measure of $\angle A$ is thrice the measure of $\angle C$. Find the measure of $\angle C$.

- (A) 36° (B) 72° (C) 90° (D) 108°

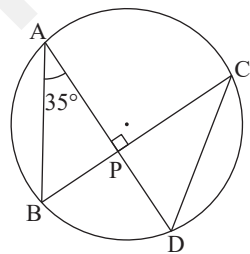
Ans. (B) [Hint : Use theorem of cyclic quadrilateral.]

10. In the figure, chords AD and BC intersect each other at right angles at a point P.

If $\angle DAB = 35^\circ$, then $\angle ADC = \dots$

- (A) 35° (B) 55° (C) 65° (D) 45°

Ans. (B) [Hint : $\angle ABP$ (i.e. $\angle ABC$) = 55° ,
 $\angle ADC = \angle ABC$.



Angles inscribed in the same arc are congruent.]

11. Length of a tangent segment drawn from a point which is at a distance 12.5 cm from the centre of a circle is 12 cm, find the diameter of the circle.

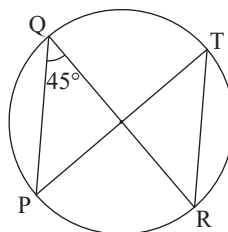
- (A) 25 cm (B) 24 cm (C) 7 cm (D) 14 cm

Ans. (C) [Hint : Use Tangent theorem and Pythagoras theorem.]

Q. Choose the correct alternative from those given below each question :

1. In the figure, $\angle PQR = 45^\circ$,
then $\angle PTR = ?$

(A) 90° (B) 45°
(C) 22.5° (D) 100°



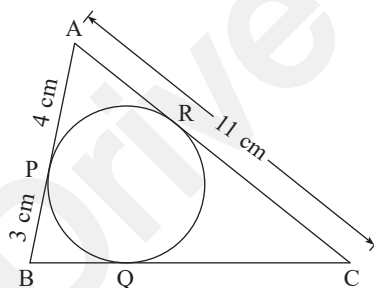
2. AP is a tangent at A drawn to the circle with centre O from an external point P. $OP = 12$ cm and $\angle OPA = 30^\circ$, then the radius of the circle is

(March '22)

(A) 12 cm (B) $6\sqrt{3}$ cm (C) 6 cm (D) $12\sqrt{3}$ cm

3. The sides AB, BC and CA of triangle ABC touch a circle at P, Q and R respectively. If $PA = 4$ cm, $BP = 3$ cm and $AC = 11$ cm; then length of BC is

(A) 7 cm (B) 10 cm
(C) 14 cm (D) 15 cm



4. Two circles of radii 5.5 cm and 3.3 cm respectively touch each other. What is the distance between their centres?

(March '20)

(A) 4.4 cm (B) 8.8 cm
(C) 2.2 cm (D) 8.8 or 2.2 cm

5. If two circles touch externally, how many common tangents of them can be drawn?

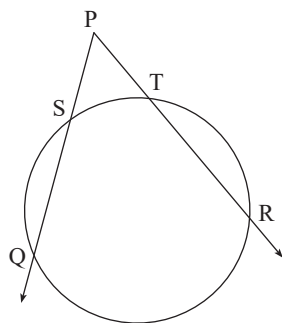
(A) One (B) Two (C) Three (D) Four

6. Two circles intersect each other such that each circle passes through the centre of the other. If the distance between their centres is 12 cm, what is the radius of each circle?

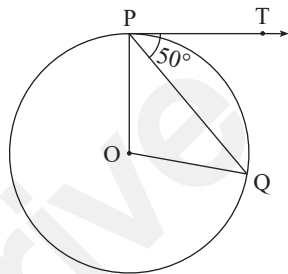
(A) 6 cm (B) 12 cm (C) 24 cm (D) can't say

7. $\angle PRQ$ is inscribed in the arc PRQ of a circle with centre O .
If $\angle PRQ = 75^\circ$, then $m(\text{arc } PRQ) = \dots\dots\dots$ (Nov. '20; Sept. '21)
(A) 75° (B) 150° (C) 285° (D) 210°

8. In the figure, $\angle QPR$ has its vertex outside the circle such that $m(\text{arc } QR) = 200^\circ$ and $m(\text{arc } ST) = 90^\circ$; then $\angle QPR = \dots\dots\dots$
(A) 45° (B) 100°
(C) 110° (D) 55°



9. In the figure, O is the centre of the circle.
 PT is the tangent. $\angle TPQ = 50^\circ$.
The measure of $\angle POQ = ?$
(A) 100° (B) 90°
(C) 80° (D) 75°



10. Chords AB and CD of a circle intersect inside the circle at point E .
If $AE = 4$, $EB = 10$, $CE = 8$, then find ED . (March '23)
(A) 7 (B) 5 (C) 8 (D) 9
11. How many circles can pass through a given point?
(A) One (B) Two (C) Three (D) Infinite
12. Points A, B, C are on a circle, such that
 $m(\text{arc } AB) = m(\text{arc } BC) = 120^\circ$. No point, except point B , is common to the arcs. Which is the type of $\triangle ABC$?
(A) Equilateral triangle (B) Scalene triangle
(C) Right angled triangle (D) Isosceles triangle

Answers :

1. (B) [Use Angles inscribed in the same arc are congruent.]
2. (C) [Use $30^\circ-60^\circ-90^\circ$ triangles theorem.]
3. (B) [Use Tangent segment theorem.]

4. (D) [Use Theorem of touching circles.]
5. (C)
6. (B) [Radius of the circle = Distance between the centres.]
7. (D) [Use Inscribed angle theorem.]
8. (D) $[\angle QPR = \frac{1}{2} [m(\text{arc QR}) - m(\text{arc ST})]]$
9. (A) $[\angle OPT = 90^\circ$. By tangent theorem.]
10. (B) [Use theorem of internal division of chords.]
11. (D)
12. (A) [Corresponding chords of equal arcs are equal.]

4.2 1 mark each

1. **What is the distance between two parallel tangents of a circle having radius 4.5 cm?** (July '23)

Ans. Radius of the circle (r) = 4.5 cm.

Distance between parallel tangents = Diameter of the circle

$$\begin{aligned}
 &= 2 \times r \\
 &= 2 \times 4.5 \\
 &= 9 \text{ cm}
 \end{aligned}$$

Distance between parallel tangents is **9 cm**.

2. **If the radius of a circle is 5 cm, then find the length of longest chord of the circle.** (March '23)

Ans. Radius of a circle (r) = 5 cm.

Diameter is the longest chord of the circle

$$d = 2r = 2 \times 5 = 10 \text{ cm.}$$

The length of the longest chord of the circle is **10 cm**.

3. **The radius of the circle with centre C is 6 cm. Line AB is a tangent at point A. What is the measure of $\angle CAB$?** (Sept. '21)

Ans. $\angle CAB = 90^\circ$.

... (Tangent theorem)



4. Two circles of radii 5 cm and 3 cm touch each other externally.
Find the distance between the centres. (March '24)

Ans. The two circles are touching each other externally.

$\therefore$ by theorem of touching circles, the distance between their centres is equal to the sum of their radii.

$\therefore$ distance between their centres = $5 + 3 = 8$ cm.

The distance between the centres is **8 cm**.

5. In a circle, the measure of a minor arc is 60° . Find the measure of its corresponding major arc.

Ans. Measure of minor arc = 60° .

Measure of its corresponding major arc = $360^\circ - 60^\circ = 300^\circ$.

Measure of corresponding major arc is **300°** .

6. Chord AB and chord CD of a circle with centre O are congruent.
If $m(\text{arc AB}) = 120^\circ$, then find $m(\text{arc CD})$. (March '22)

Ans. Chord $AB \cong$ chord CD

... (Given)

$\therefore m(\text{arc CD}) = m(\text{arc AB})$

... (Corresponding arcs of congruent chords)

$$m(\text{arc AB}) = 120^\circ$$

$\therefore m(\text{arc CD}) = 120^\circ$.

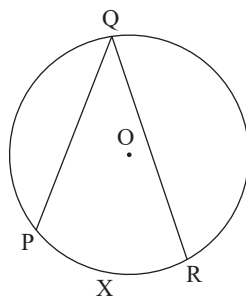
7. In the figure, O is the centre of the circle.
 $\angle PQR$ is inscribed in arc PQR and intercepts arc PXR. If $m(\text{arc PXR}) = 80^\circ$, then find the measure of $\angle PQR$.

Ans. $\angle PQR = \frac{1}{2} m(\text{arc PXR})$

... (Inscribed angle theorem)

$$= \frac{1}{2} \times 80^\circ = 40^\circ.$$

$$\angle PQR = 40^\circ.$$



8. In cyclic quadrilateral ABCD, $m\angle A = 100^\circ$, then find $m\angle C$.

(March '25)

Ans. $\square ABCD$ is a cyclic quadrilateral

$$\therefore \angle A + \angle C = 180^\circ$$

... (Opposite angles of a cyclic quadrilateral are supplementary)

$$\therefore 100^\circ + \angle C = 180^\circ$$

$$\therefore \angle C = 180^\circ - 100^\circ$$

$$\therefore \angle C = 80^\circ.$$

9. How many common tangents can be drawn to two internally touching circles?

Ans. Only one common tangent can be drawn to two internally touching circles.

10. In the figure, $m(\text{arc } ABC) = 50^\circ$ and $m(\text{arc } CDE) = 60^\circ$. Find $m(\text{arc } ACE)$.

Ans. $m(\text{arc } ABC) = 50^\circ$

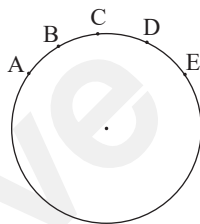
$$m(\text{arc } CDE) = 60^\circ$$

$$m(\text{arc } ACE) = m(\text{arc } ABC) + m(\text{arc } CDE)$$

... (Arcs addition property)

$$\therefore m(\text{arc } ACE) = 50^\circ + 60^\circ$$

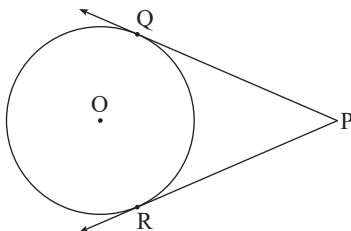
$$\therefore m(\text{arc } ACE) = 110^\circ.$$



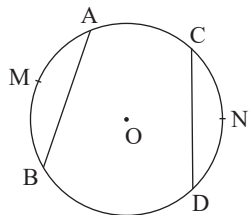
Assignment	4.2
------------	-----

- Two circles with radii 7 cm and 4 cm touch each other internally. Find the distance between their centres? (July '22)
- A and B are any two points such that $d(A, B) = 16$ cm. How many circles with radius 10 cm can we draw passing through A and B?
- O is the centre and AB is the diameter of a circle. P is any point on the circle other than A and B. Find measure of $\angle APB$.

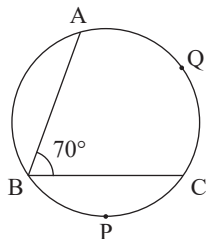
- In the figure, PQ and PR are tangents drawn from an external point P to the circle with centre O. Q and R are the points of contact. If $PQ = 5$ cm, then what is the length of segment PR? Why?



5. In the figure, arc $AMB \cong$ arc CND ,
if $AB = 5$ cm, then find CD .



6. In the figure, $\angle ABC = 70^\circ$,
then find the measure of arc AQC .



7. $\square ABCD$ is cyclic.

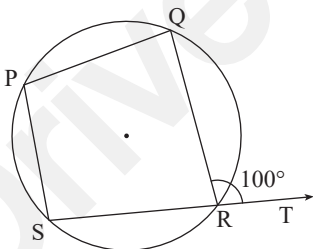
(i) If $\angle B = 110^\circ$, then find the measure of $\angle D$.

(March '20)

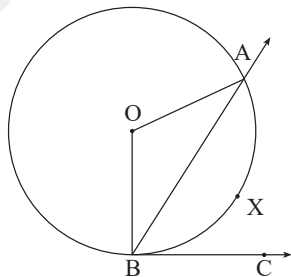
(ii) If $\angle B = 75^\circ$, then find $\angle D$.

(Nov. '20)

8. $\square PQRS$ is cyclic. T is a point of ray SR
such that $S-R-T$ and $\angle QRT = 100^\circ$,
then what is the measure of $\angle SPQ$?



9. In the figure, line BC is tangent at B , if
 $\angle AOB = 110^\circ$, then find the measure of
 $\angle ABC$.

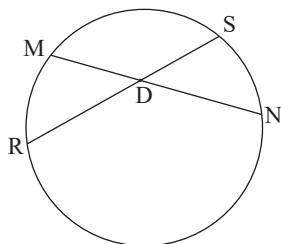


Answers :

1. 3 cm 2. Two 3. $\angle APB = 90^\circ$
4. $PR = 5$ cm, Tangent theorem 5. $CD = 5$ cm
6. $m(\text{arc } AQC) = 140^\circ$ 7. (i) $\angle D = 70^\circ$ (ii) $\angle D = 105^\circ$
8. $\angle SPQ = 100^\circ$ 9. $\angle ABC = 55^\circ$.

4.3 2 marks each

1. In the given figure, chord MN and chord RS intersect at point D. If $RD = 15$, $DS = 4$, $MD = 8$, find DN by completing the following activity. (March '25)



$$MD \times DN = \boxed{} \times DS$$

... (Theorem of internal division of chords)

$$\therefore \boxed{} \times DN = 15 \times 4$$

$$\therefore DN = \frac{\boxed{}}{8} \quad \therefore DN = \boxed{}.$$

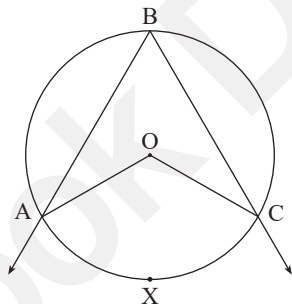
Activity :

$$MD \times DN = \boxed{RD} \times DS \quad \dots \text{(Theorem of internal division of chords)}$$

$$\therefore \boxed{8} \times DN = 15 \times 4$$

$$\therefore DN = \frac{\boxed{60}}{8} \quad \therefore DN = \boxed{7.5}.$$

2.



(March '24)

In the given figure, $\angle ABC$ is inscribed in arc AC.

If $\angle ABC = 60^\circ$, complete the following activity to find $\angle AOC$.

$$\angle ABC = \frac{1}{2} m(\text{arc } AC) \quad \dots \boxed{}$$

$$60^\circ = \frac{1}{2} m(\text{arc } AC)$$

$$\boxed{} = m(\text{arc } AC)$$

But $\angle AOC = \boxed{m(\text{arc })}$

... (Property of central angle)

$\angle AOC = \boxed{}$

Activity :

$\angle ABC = \frac{1}{2} m(\text{arc } AXC)$

... **Inscribed angle theorem**

$60^\circ = \frac{1}{2} m(\text{arc } AXC)$

$\boxed{120^\circ} = m(\text{arc } AXC)$

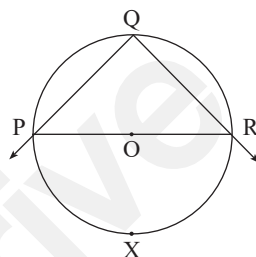
But $\angle AOC = \boxed{m(\text{arc } AXC)}$

... (Property of central angle)

$\angle AOC = \boxed{120^\circ}$

- 3. In the figure, $\angle PQR$ is inscribed in the semicircle PQR. Complete the following activity to find the measure of $\angle PQR$.**

(July '22)



$m(\text{arc } PQR) = 180^\circ$

... (Measure of a semicircle)

$\therefore m(\text{arc } PXR) = \boxed{}$

$\therefore \angle PQR = \frac{1}{2} m(\text{arc } \boxed{})$

... $\boxed{}$

$= \frac{1}{2} \times 180^\circ$

$\therefore \angle PQR = \boxed{}.$

Activity :

$m(\text{arc } PQR) = 180^\circ$

... (Measure of a semicircle)

$\therefore m(\text{arc } PXR) = \boxed{180^\circ}$

$\therefore \angle PQR = \frac{1}{2} m(\text{arc } \boxed{PXR})$

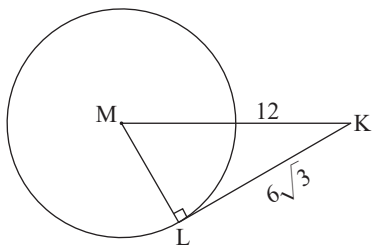
... **Inscribed angle theorem**

$= \frac{1}{2} \times 180^\circ$

$\therefore \angle PQR = \boxed{90^\circ}.$

4. In the figure, M is the centre of the circle and seg KL is a tangent segment. L is a point of contact. If $MK = 12$, $KL = 6\sqrt{3}$, then, find the radius of the circle.

(March '20)



Solution : In $\triangle MLK$,

$$\angle MLK = 90^\circ$$

... (Tangent theorem)

$\therefore$ by Pythagoras theorem,

$$MK^2 = ML^2 + LK^2$$

$$\therefore 12^2 = ML^2 + (6\sqrt{3})^2$$

$$\therefore 144 = ML^2 + 36 \times 3$$

$$\therefore 144 = ML^2 + 108$$

$$\therefore ML^2 = 144 - 108$$

$$\therefore ML^2 = 36$$

$$\therefore ML = 6$$

... (Taking square roots of both the sides)

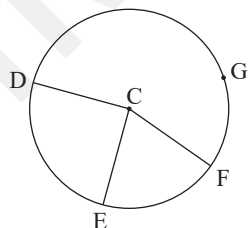
Ans. The radius of the circle is 6.

5. In the figure, points G, D, E, F are concyclic points of a circle with centre C.

$$\angle ECF = 70^\circ, m(\text{arc DGF}) = 200^\circ;$$

find $m(\text{arc DE})$ and $m(\text{arc DEF})$.

(March '23)



Solution : $m(\text{arc EF}) = \angle ECF$

... (Definition of measure of minor arc)

$$\angle ECF = 70^\circ$$

... (Given)

$$\therefore m(\text{arc EF}) = 70^\circ$$

$$m(\text{arc DGF}) + m(\text{arc EF}) + m(\text{arc DE}) = 360^\circ$$

... (Measure of a circle is 360°)

$$\therefore 200^\circ + 70^\circ + m(\text{arc DE}) = 360^\circ$$

$$\therefore 270^\circ + m(\text{arc DE}) = 360^\circ$$

$$\therefore m(\text{arc DE}) = 360^\circ - 270^\circ$$

$$\therefore m(\text{arc DE}) = 90^\circ$$

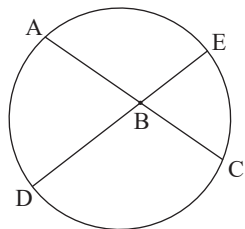
$$m(\text{arc DE}) + m(\text{arc EF}) = m(\text{arc DEF}) \quad \dots \text{ (Arcs addition postulate)}$$

$$\therefore 90^\circ + 70^\circ = m(\text{arc DEF})$$

$$\therefore m(\text{arc DEF}) = 160^\circ$$

Ans. $m(\text{arc DE}) = 90^\circ$ and $m(\text{arc DEF}) = 160^\circ$.

- 6. In the figure, chords AC and DE intersect at B. If $\angle ABE = 108^\circ$, $m(\text{arc AE}) = 95^\circ$, find $m(\text{arc DC})$. (July '22)**



Solution :

$$m(\text{arc AE}) = 95^\circ, \angle ABE = 108^\circ \dots (\text{Given})$$

$\angle ABE$ has its vertex inside the circle and

intercepts arc AE and its vertically opposite $\angle DBC$ intercepts arc DC.

$$\angle ABE = \frac{1}{2} [m(\text{arc AE}) + m(\text{arc DC})]$$

$$\therefore 108^\circ = \frac{1}{2} \times [95^\circ + m(\text{arc DC})]$$

$$\therefore 108^\circ \times 2 = 95^\circ + m(\text{arc DC})$$

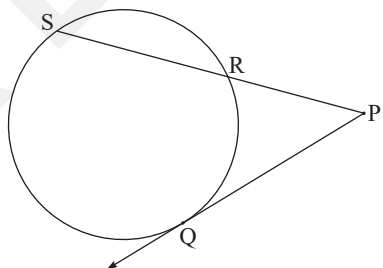
$$\therefore 216^\circ = 95^\circ + m(\text{arc DC})$$

$$\therefore m(\text{arc DC}) = 216^\circ - 95^\circ$$

$$\therefore m(\text{arc DC}) = 121^\circ$$

Ans. $m(\text{arc DC}) = 121^\circ$.

- 7. In the figure, ray PQ touches the circle at point Q. $PQ = 12$, $PR = 8$, find PS. (July '23)**



Solution : Ray PQ is tangent touching the circle at point Q and line PRS is secant intersecting the circle at points R and S.

$$\therefore PQ^2 = PR \times PS \dots (\text{By tangent secant segments theorem})$$

$$\therefore 12^2 = 8 \times PS$$

$$\therefore PS = \frac{12 \times 12}{8}$$

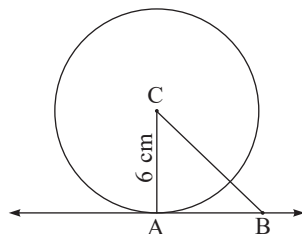
$$\therefore PS = 18$$

Ans. $PS = 18$.

8. In the figure, the radius of a circle with centre C is 6 cm, line AB is a tangent at A. Answer the following questions :

(a) What is the measure of $\angle CAB$? Why?

(b) What is the distance of point C from line AB? Why?



Ans.

(a) $\angle CAB = 90^\circ$... (Tangent theorem) ... (1)

(b) $\text{seg } CA \perp \text{line AB}$... [From (1)]

$CA = 6 \text{ cm}$... (Given)

$\therefore$ C is at a distance of **6 cm** from line AB.

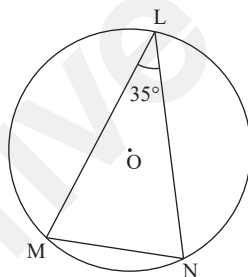
Assignment	4.3
------------	-----

1. In the figure, $\angle L = 35^\circ$. Complete the following activity to find

(i) $m(\text{arc } MN)$

(ii) $m(\text{arc } MLN)$.

(March '23)



Activity :

(i) $\angle L = \frac{1}{2} m(\text{arc } MN)$... (By inscribed angle theorem)

$$\therefore \square = \frac{1}{2} m(\text{arc } MN)$$

$$\therefore 2 \times 35^\circ = m(\text{arc } MN)$$

$$\therefore m(\text{arc } MN) = \square.$$

(ii) $m(\text{arc } MLN) = \square - m(\text{arc } MN)$

... (Definition of measure of a circle)

$$= 360^\circ - 70^\circ$$

$$\therefore m(\text{arc } MLN) = \square.$$

2. Two circles having radii 3.5 cm and 4.8 cm touch each other internally. Complete the following activity to find the distance between their centres.

Activity :

Let the radii of the two circles be r_1 and r_2 .

$$r_1 = 3.5 \text{ cm and } r_2 = 4.8 \text{ cm}$$

... (Given)

The two circles are touching each other internally.

$\therefore$ by theorem of touching circles, the centres and the point of contact are (collinear/ non-collinear)

The distance between the centres of the circles touching internally is equal to the difference of their radii.

$\therefore$ distance between their centres

$$= r_2 - \text{} = 4.8 - \text{} = \text{} \text{ cm.}$$

3. In the figure, chord $EF \parallel$ chord GH .

Prove that,

chord $EG \cong$ chord FH .

Fill in the blanks and

write the proof.

Proof : Draw seg GF .

$$\angle EFG = \angle FGH$$

$$\dots \left(\text{} \right) \dots (1)$$

$$\angle EFG = \text{}$$

... (Inscribed angle theorem) ... (2)

$$\angle FGH = \text{}$$

... (Inscribed angle theorem) ... (3)

$$\therefore m(\text{arc } EG) = \text{}$$

... [From (1), (2) and (3)]

$\therefore$ chord $EG \cong$ chord FH

... (Corresponding chords of congruent arcs)

4. In order to prove 'Opposite angles of a cyclic quadrilateral are supplementary'.

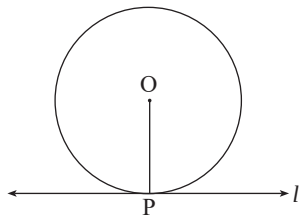
(1) Draw a neat labelled figure.

(2) Write 'Given' and 'To prove' from the figure drawn by you.

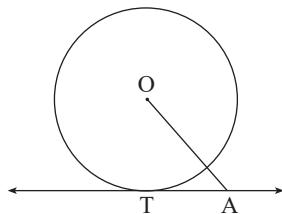
5. Line l touches a circle with centre O at point P . If radius of the circle is 9 cm, answer the following :

(i) What is $d(O, P)$ = ?

(ii) If $d(O, Q) = 8$ cm, then where does the point Q lie?



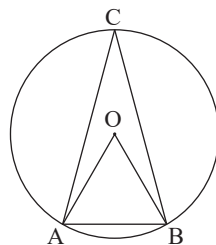
6. In the figure, O is the centre and line AT touches the circle at point T. If $OA = \frac{25}{3}$ cm and $AT = \frac{20}{3}$ cm, then find the radius of the circle.



7. In the figure, in a circle with centre O, length of chord AB is equal to the radius of the circle. Find measure of each of the following :

(i) $\angle AOB$ (ii) $\angle ACB$.

(Sept. '21)

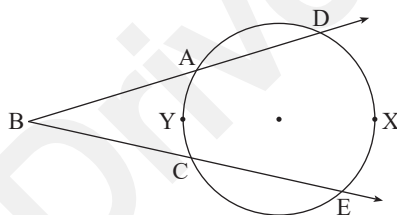


8. Measure of two arcs formed by a chord of a circle are $2x^\circ$ and $7x^\circ$. Find the measure of the minor arc.

(Sept. '21)

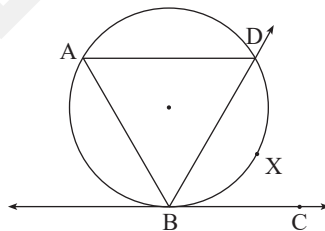
9. In the figure, $m(\text{arc } DXE) = 105^\circ$, $m(\text{arc } AYC) = 47^\circ$, then find the measure of $\angle DBE$.

(March '22)



10. In the figure, line BC is a tangent to the circle at point B. $\angle DBC = 50^\circ$, then find

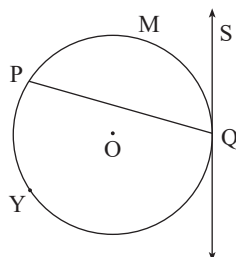
(i) $m(\text{arc } BXD)$ (ii) $\angle BAD$.



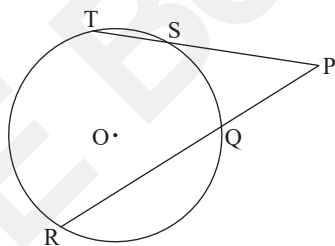
11. In the figure, O is the centre of the circle.

Line QS is the tangent at point Q, if $m(\text{arc } PMQ) = 160^\circ$.

Find (i) $\angle PQS$ (ii) $m(\text{arc } PYQ)$.

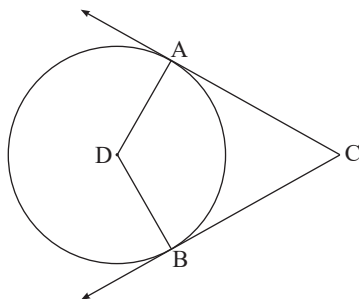


12.



In the figure,
if $PQ = 6$, $QR = 10$,
 $PS = 8$; then find TS .

13. In the figure, circle with centre D touches the sides of $\angle ACB$ at A and B. If $\angle ACB = 52^\circ$, find the measure of $\angle ADB$. (March '25)



Answers :

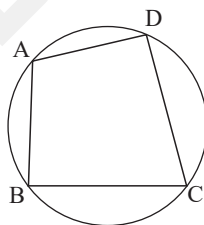
1. 35° , 70° , 360° , 290°

2. Collinear, r_1 , 3.5, 1.3.

3. Alternate angles, $\frac{1}{2} m(\text{arc EG})$, $\frac{1}{2} m(\text{arc FH})$, $m(\text{arc FH})$

4. Given : $\square ABCD$ is cyclic.

To prove : $\angle DAB + \angle DCB = 180^\circ$,
 $\angle ABC + \angle ADC = 180^\circ$.



5. (i) $d(O, P) = 9$ cm (ii) Q lies in the interior of the circle.

6. 5 cm 7. (i) $\angle AOB = 60^\circ$ (ii) $\angle ACB = 30^\circ$

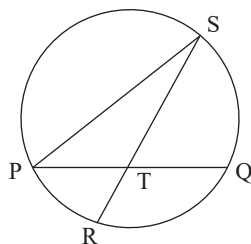
8. 80° 9. 29° 10. $m(\text{arc BXD}) = 100^\circ$ (ii) $\angle BAD = 50^\circ$

11. (i) $\angle PQS = 80^\circ$ (ii) $m(\text{arc PYQ}) = 200^\circ$

12. $TS = 4$ 13. $\angle ADB = 128^\circ$.

4.4 3 marks each

1. In the figure, chord PQ and chord RS intersect each other in point T. If $\angle STQ = 58^\circ$ and $\angle PSR = 24^\circ$, then complete the following activity to verify $\angle STQ = \frac{1}{2} [m(\text{arc PR}) + m(\text{arc SQ})]$.



(March '22)

In $\triangle PTS$,

$$\angle SPQ = \angle STQ - \boxed{} \quad \dots \text{ (Exterior angle theorem)}$$

$$\therefore \angle SPQ = 34^\circ$$

$$\therefore m(\text{arc QS}) = 2 \times \boxed{}^\circ = 68^\circ \quad \dots \left(\boxed{} \right)$$

$$\text{Similarly, } m(\text{arc PR}) = 2 \angle PSR = \boxed{}^\circ$$

$$\therefore \frac{1}{2} [m(\text{arc QS}) + m(\text{arc PR})] = \frac{1}{2} \times \boxed{}^\circ = 58^\circ \quad \dots (1)$$

$$\text{but } \angle STQ = 58^\circ \quad \dots \text{ (Given)} \quad \dots (2)$$

$$\therefore \frac{1}{2} [m(\text{arc PR}) + m(\text{arc QS})] = \boxed{\angle } \quad \dots \text{ [From (1) and (2)]}$$

Activity :

In $\triangle PTS$,

$$\angle SPQ = \angle STQ - \boxed{\angle PSR} \quad \dots \text{ (Exterior angle theorem)}$$

$$\therefore \angle SPQ = 34^\circ$$

$$\therefore m(\text{arc QS}) = 2 \times \boxed{34}^\circ = 68^\circ \quad \dots \left(\boxed{\text{Inscribed angle theorem}} \right)$$

$$\text{Similarly, } m(\text{arc PR}) = 2 \angle PSR = \boxed{48}^\circ$$

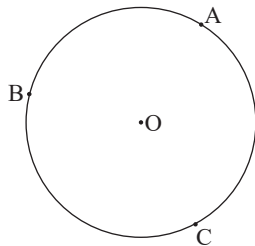
$$\therefore \frac{1}{2} [m(\text{arc QS}) + m(\text{arc PR})] = \frac{1}{2} \times \boxed{116}^\circ = 58^\circ \quad \dots (1)$$

$$\text{but } \angle STQ = 58^\circ \quad \dots \text{ (Given)} \quad \dots (2)$$

$$\therefore \frac{1}{2} [m(\text{arc PR}) + m(\text{arc QS})] = \boxed{\angle STQ} \quad \dots \text{ [From (1) and (2)]}$$



2. A, B, C are any points on the circle with centre O. If $m(\text{arc BC}) = 110^\circ$ and $m(\text{arc AB}) = 125^\circ$, complete the following activity to find $m(\text{arc ABC})$, $m(\text{arc AC})$, $m(\text{arc ACB})$ and $m(\text{arc BAC})$. (July '23)



$$\begin{aligned} m(\text{arc ABC}) &= m(\text{arc AB}) + \boxed{} \\ &= \boxed{}^\circ + 110^\circ \\ &= 235^\circ \end{aligned}$$

$$\begin{aligned} m(\text{arc AC}) &= 360^\circ - m(\text{arc } \boxed{}) \\ &= 360^\circ - \boxed{}^\circ \\ &= 125^\circ \end{aligned}$$

$$\begin{aligned} \text{Similarly, } m(\text{arc ACB}) &= 360^\circ - \boxed{} \\ &= 235^\circ \end{aligned}$$

$$\begin{aligned} \text{and } m(\text{arc BAC}) &= 360^\circ - \boxed{} \\ &= 250^\circ \end{aligned}$$

Activity :

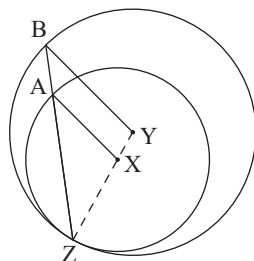
$$\begin{aligned} m(\text{arc ABC}) &= m(\text{arc AB}) + \boxed{m(\text{arc BC})} \\ &= \boxed{125}^\circ + 110^\circ \\ &= 235^\circ \end{aligned}$$

$$\begin{aligned} m(\text{arc AC}) &= 360^\circ - m(\text{arc } \boxed{ABC}) \\ &= 360^\circ - \boxed{235}^\circ \\ &= 125^\circ \end{aligned}$$

$$\begin{aligned} \text{Similarly, } m(\text{arc ACB}) &= 360^\circ - \boxed{125^\circ} \\ &= 235^\circ \end{aligned}$$

$$\begin{aligned} \text{and } m(\text{arc BAC}) &= 360^\circ - \boxed{110^\circ} \\ &= 250^\circ \end{aligned}$$

3. In the figure, circles with centres X and Y touch internally at point Z. Seg BZ is a chord of bigger circle and it intersects smaller circle at point A.



Prove that $\text{seg AX} \parallel \text{seg BY}$.

Construction : Draw YZ and XZ.

Proof : $Y-X-Z$

... (Theorem of touching circles)

In $\triangle YBZ$,

$\text{seg YB} \cong \text{seg YZ}$

... (Radii of the same circle)

$\therefore \angle YBZ \cong \angle YZB$

... (Isosceles triangle theorem)

i.e. $\angle YBZ \cong \angle XZA$

... ($Y-X-Z$ and $B-A-Z$) ... (1)

In $\triangle XAZ$,

$\text{seg XA} \cong \text{seg XZ}$

... (Radii of the same circle)

$\therefore \angle XAZ \cong \angle XZA$

... (Isosceles triangle theorem) ... (2)

$\therefore \angle YBZ \cong \angle XAZ$

... [From (1) and (2)]

$\therefore \text{seg AX} \parallel \text{seg BY}$... (Corresponding angles test for parallel lines)

4. $\square \text{MRPN}$ is cyclic, $\angle R = (5x - 13)^\circ$, $\angle N = (4x + 4)^\circ$. Find measures of $\angle R$ and $\angle N$.

Solution : $\square \text{MRPN}$ is cyclic

... (Given)

$\therefore$ by theorem of cyclic quadrilateral,

$\angle R + \angle N = 180^\circ$

$\therefore (5x - 13)^\circ + (4x + 4)^\circ = 180^\circ$

$\therefore 5x - 13 + 4x + 4 = 180$

$\therefore 9x - 9 = 180$

$\therefore x - 1 = 20$

... (Dividing both the sides by 9)

$\therefore x = 20 + 1$

$\therefore x = 21$

$\angle R = 5x - 13$

$\therefore \angle R = 5(21) - 13$

$\therefore \angle R = 105 - 13$

$\therefore \angle R = 92^\circ$

$\angle N = (4x + 4)$

$\therefore \angle N = 4(21) + 4$

$$\therefore \angle N = 84 + 4$$

$$\therefore \angle N = 88^\circ$$

Ans. $\angle R = 92^\circ$ and $\angle N = 88^\circ$.

- 5. In the figure, seg EF is a diameter and seg DF is a tangent segment. The radius of the circle is r .**

Prove that $DE \times GE = 4r^2$.

Proof :

Line DF is tangent to the circle touching the circle at point F and line DE is secant intersecting the circle at points G and E.

$\therefore$ by tangent secant segments theorem,

$$DF^2 = DG \times DE \quad \dots (1)$$

In $\triangle DFE$,

$$\angle DFE = 90^\circ \quad \dots \text{(By tangent theorem)}$$

$\therefore$ by Pythagoras theorem,

$$DE^2 = DF^2 + EF^2$$

$$\therefore DE^2 = DF^2 + (2r)^2 \quad \dots \text{(Diameter is twice the radius)}$$

$$\therefore DE^2 = DF^2 + 4r^2$$

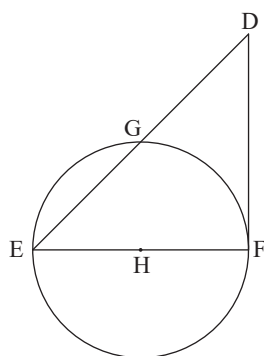
$$\therefore 4r^2 = DE^2 - DF^2$$

$$\therefore 4r^2 = DE^2 - DG \times DE \quad \dots [\text{From (1)}]$$

$$\therefore 4r^2 = DE (DE - DG)$$

$$\therefore 4r^2 = DE \times GE \quad \dots (D - G = E)$$

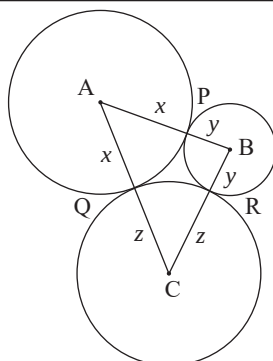
$$\therefore DE \times GE = 4r^2.$$



- 6. Circles with centres A, B and C touch each other externally. If $AB = 3$ cm, $BC = 3$ cm, $CA = 4$ cm, then find the radii of each circle. (March '20)**

Solution :

Let the circles with centres A, B and C touch each other in the points P, R and Q as shown.



By theorem of touching circles,
A–P–B, B–R–C and A–Q–C.

$$AB = x + y = 3 \quad \dots (1)$$

$$\text{Similarly, } BC = y + z = 3 \quad \dots (2)$$

$$\text{and } CA = z + x = 4 \quad \dots (3)$$

Adding (1), (2) and (3), we get,

$$x + y + y + z + z + x = 3 + 3 + 4$$

$$2x + 2y + 2z = 10$$

$$\therefore x + y + z = 5 \quad \dots (4)$$

Substituting equation (2) in equation (4),

$$x + y + z = 5$$

$$\therefore x + 3 = 5$$

$$\therefore x = 5 - 3$$

$$\therefore x = 2$$

Put $x = 2$ in equation (1),

$$x + y = 3$$

$$2 + y = 3$$

$$\therefore y = 1$$

Put $y = 1$ in equation (2),

$$y + z = 3$$

$$1 + z = 3$$

$$\therefore z = 2$$

Ans. The radius of circle with centre A is **2 cm**, the radius of the circle with centre B is **1 cm** and the radius of the circle with centre C is **2 cm**.

-
7. Let M be a point of contact of two internally touching circles. Let line AMB be their common tangent. The chord CD of the bigger circle touches the smaller circle at point N.

The chord CM and chord DM of bigger circle intersect the smaller circle at point P and R respectively.

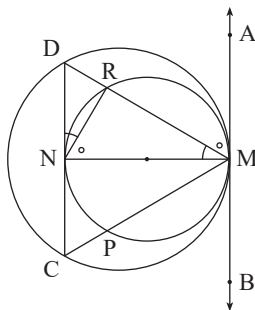
- (a) From the above information draw the suitable figure.
(b) Draw seg NR and seg NM and write the two pairs of congruent angles in smaller circle considering tangent and chord.



- (c) By using the property which is used in (b), write the two pairs of congruent angles in bigger circle. (Nov. '20)

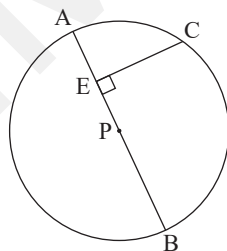
Ans.

(a)



- (b) In the smaller circle,
 (i) $\angle DNR \cong \angle NMR$
 (ii) $\angle RNM \cong \angle RMA$ } ... (Tangent secant angle theorem)
- (c) In the bigger circle,
 (i) $\angle DCM \cong \angle DMA$
 (ii) $\angle CDM \cong \angle CMB$ } ... (Tangent secant angle theorem)

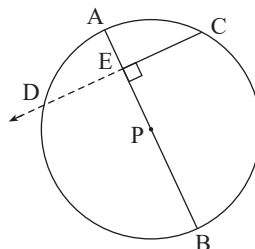
- 8. In the figure, seg AB is a diameter of a circle with centre P. C is any point on the circle. Seg $CE \perp$ seg AB. Prove that CE is the geometric mean of AE and EB. Write the proof with the help of the following steps.**



- (a) Draw ray CE. It intersects the circle at D.
 (b) Show that $CE = ED$.
 (c) Write the result using theorem of intersection of chords inside the circle.
 (d) Using $CE = ED$, complete the proof. (July '19)

Proof :

(a)



(b) seg $PE \perp$ chord CD .

$\therefore CE = ED \dots$ (Perpendicular drawn from the centre of the circle to the chord bisects the chord) $\dots (1)$

(c) Chords AB and CD intersect at point E inside the circle.

$\therefore$ by theorem of intersection of chords inside the circle,

$$AE \times EB = CE \times ED \dots (2)$$

(d) Now,

$$AE \times EB = CE \times ED \dots \text{[From (2)]}$$

$$\therefore AE \times EB = CE \times CE \dots \text{[From (1)]}$$

$$\therefore CE^2 = AE \times EB$$

$\therefore CE$ is the geometric mean of AE and EB .

Assignment

4.4

1. Given :

Chords AB and CD of a circle with centre P intersect at point E .

To prove :

$$AE \times EB = CE \times ED$$

Construction :

Draw seg AC and seg BD .

Fill in the blanks and complete the proof.

(March '24)

Proof :

In $\triangle CAE$ and $\triangle BDE$,

$$\angle AEC \cong \angle DEB$$

$$\dots \left(\boxed{} \right)$$

$$\boxed{} \cong \angle BDE$$

$\dots$ (Angles inscribed in the same arc)

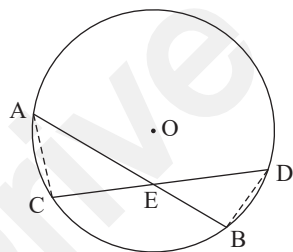
$$\therefore \triangle CAE \sim \triangle BDE$$

$$\dots \left(\boxed{} \right)$$

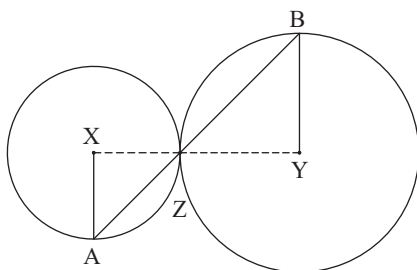
$$\therefore \frac{\boxed{}}{DE} = \frac{CE}{\boxed{}}$$

$$\dots \left(\boxed{} \right)$$

$$\therefore AE \times EB = CE \times ED.$$



2. In the figure, circles with centres X and Y touch each other at point Z. A secant passing through Z intersects the circles at points A and B respectively. Complete the following activity to prove that radius $XA \parallel$ radius YB .



(Sept. '21)

Activity : Draw segments XZ and YZ .

By theorem of touching circles, points X, Z and Y are .

$\angle XZA \cong$... (Vertically opposite angles) ... (1)

seg $XA \cong$ seg XZ ... ()

$\therefore \angle XAZ \cong$... (Isosceles triangle theorem) ... (2)

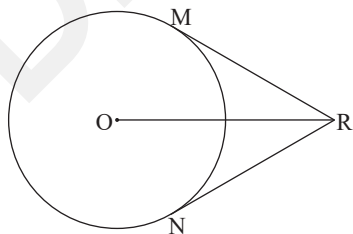
Similarly, seg $YB \cong$ seg YZ

$\therefore \angle BZY \cong \angle YBZ$... (3)

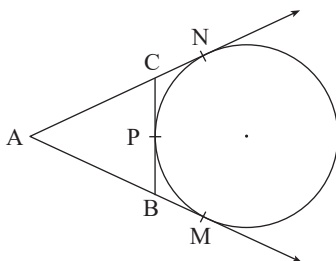
$\therefore \angle XAZ \cong$... [From (1), (2) and (3)]

$\therefore$ seg $XA \parallel$ seg YB ... ()

3. In the figure, O is the centre of the circle. From point R, seg RM and seg RN are tangent segments touching the circle at M and N. If $OR = 10$ cm and radius of the circle = 5 cm, then



- (i) What is the length of each tangent segment?
 (ii) What is the measure of $\angle MRO$?
 (iii) What is the measure of $\angle MRN$?
4. A circle touches side BC at point P of $\triangle ABC$ from outside of the triangle. Further extended lines AC and AB are tangents to the circle at N and M respectively.



Prove :

$$AM = \frac{1}{2} (\text{Perimeter of } \triangle ABC)$$

(March '23)

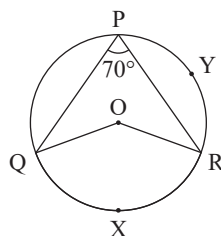
5. In the given figure, O is centre of circle, $\angle QPR = 70^\circ$ and $m(\text{arc PYR}) = 160^\circ$, then find the value of each of the following :

(i) $m(\text{arc QXR})$

(ii) $\angle QOR$

(iii) $\angle PQR$.

(March '19)



6. Prove that any rectangle is a cyclic quadrilateral.

7. $\square ABCD$ is cyclic. $AB = AD$,

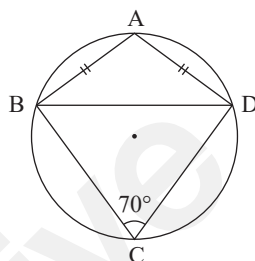
$\angle BCD = 70^\circ$, then find :

(i) $m(\text{arc BCD})$

(ii) $m(\text{arc BAD})$

(iii) $\angle ABD$.

(Sept. '21)



8. $\square ABCD$ is cyclic quadrilateral.

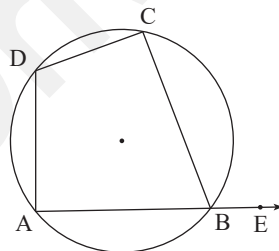
$m(\text{arc ABC}) = 220^\circ$

then find :

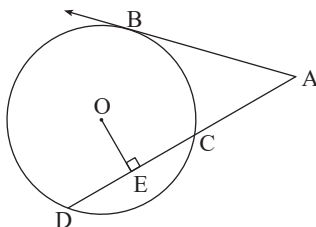
(i) $\angle ABC$

(ii) $\angle CDA$

(iii) $\angle CBE$.



9. In the figure, O is the centre of the circle and B is a point of contact. Seg $OE \perp$ seg AD, $AB = 12$, $AC = 8$, find (i) AD (ii) DC (iii) DE.



10. AB is a chord of a circle with centre O. AOC is diameter of circle, AT is a tangent at A.

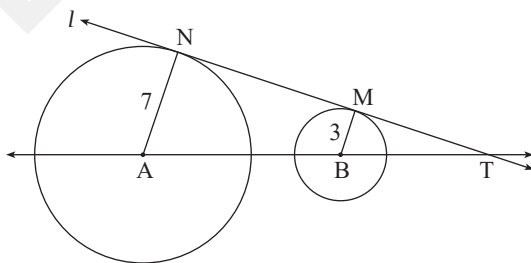
(March '24)

Write answers of the following questions :

(a) Draw the figure using given information.

- (b) Find the measures of $\angle CAT$ and $\angle ABC$ with reasons.
 (c) Whether $\angle CAT$ and $\angle ABC$ are congruent? Justify your answer.

11.



In the figure, line l touches the circle with centre A having radius 7 and circle with centre B having radius 3 at points N and M respectively.

- (i) Prove that $\triangle ATN \sim \triangle BTM$.
 - (ii) Write the corresponding sides of similar triangles ATN and BTM in proportion.
 - (iii) Determine TA : TB.
12. $\square ABCD$ is a rectangle. Taking AD as a diameter, a semicircle AXD is drawn which intersects the diagonal BD at X. If AB = 12 cm, AD = 9 cm, then find the values of BD and BX. **(March '25)**
13. In a circle with centre O, PA and PB are tangents from an external point P. E is the point on the circle such that O–E–P. Tangent drawn at E intersects PA and PB in point C and D respectively. If PA = 10, then write the answers to the following questions :
- (a) Draw the suitable figure using given information.
 - (b) Write the relation between seg PA and seg PB.
 - (c) Find the perimeter of $\triangle PCD$. **(July '22)**

Answers :

1. Vertically opposite angles , $\angle CAE$, AA test of similarity , AE , BE , Corresponding sides of similar triangles
2. collinear , $\angle YZB$, Radii of the same circle , $\angle XZA$, $\angle YBZ$, Alternate angles test
3. (i) $5\sqrt{3}$ cm (ii) $\angle MRO = 30^\circ$ (iii) $\angle MRN = 60^\circ$
5. (i) $m(\text{arc QXR}) = 140^\circ$ (ii) $\angle QOR = 140^\circ$ (iii) $\angle PQR = 80^\circ$

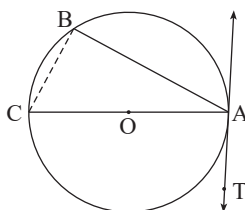
7. (i) $m(\text{arc BCD}) = 220^\circ$ (ii) $m(\text{arc BAD}) = 140^\circ$

(iii) $\angle ABD = 35^\circ$

8. (i) $\angle ABC = 70^\circ$ (ii) $\angle CDA = 110^\circ$ (iii) $\angle CBE = 110^\circ$

9. (i) $AD = 18$ (ii) $DC = 10$ (iii) $DE = 5$

10. (a)



(b) $\angle CAT = 90^\circ$

... (By tangent theorem)

$\angle ABC = 90^\circ$

... (Angle inscribed in a semicircle)

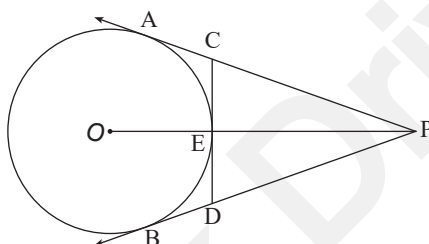
(c) $\angle CAT \cong \angle ABC$

... (90° each)

11. (b) $\frac{TA}{TB} = \frac{TN}{TM} = \frac{AN}{BM}$ (c) $TA : TB = 7 : 3$

12. $BD = 15$ cm, $BX = 9.6$ cm

13. (a)



(b) $PA = PB$

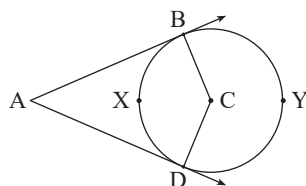
(c) Perimeter of $\triangle PCD = 20$.

CHALLENGING QUESTIONS

4.5 4 marks each

1. In the figure, AB and AD are tangent segments drawn to a circle with centre C from exterior point A, then prove that

$$\angle A = \frac{1}{2} [m(\text{arc BYD}) - m(\text{arc BXD})].$$



(March '22)



Proof :

$$\angle ABC = \angle ADC = 90^\circ \quad \dots \text{ (Tangent theorem)}$$

$$\angle BAD + \angle ABC + \angle BCD + \angle ADC = 360^\circ$$

... (Sum of all angles of a quadrilateral is 360°)

$$\angle A + 90^\circ + \angle BCD + 90^\circ = 360^\circ$$

$$\angle A + \angle BCD + 180^\circ = 360^\circ$$

$$\therefore \angle A + \angle BCD = 360^\circ - 180^\circ$$

$$\therefore \angle A + \angle BCD = 180^\circ$$

$$\therefore \angle A = 180^\circ - \angle BCD \quad \dots (1)$$

$$m(\text{arc BXD}) = m\angle BCD$$

... (Definition of measure of minor arc) ... (2)

$$\therefore \angle A = 180^\circ - m(\text{arc BXD}) \quad \dots \text{ [From (1) and (2)]} \quad \dots (3)$$

$$m(\text{arc BXD}) + m(\text{arc BYD}) = 360^\circ \quad \dots \text{ (Measure of a circle is } 360^\circ)$$

Multiplying each term by $\frac{1}{2}$, we get,

$$\frac{1}{2} m(\text{arc BXD}) + \frac{1}{2} m(\text{arc BYD}) = 180^\circ \quad \dots (4)$$

$$\therefore \angle A = \left[\frac{1}{2} m(\text{arc BXD}) + \frac{1}{2} m(\text{arc BYD}) \right] - m(\text{arc BXD})$$

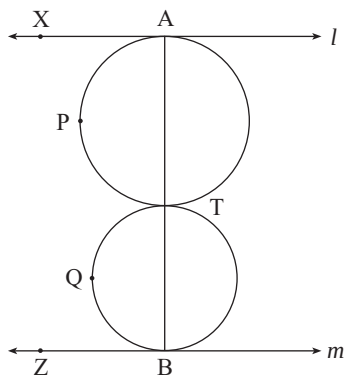
$$\therefore \angle A = \frac{1}{2} m(\text{arc BXD}) + \frac{1}{2} m(\text{arc BYD}) - m(\text{arc BXD})$$

$$\therefore \angle A = \frac{1}{2} m(\text{arc BYD}) - \frac{1}{2} m(\text{arc BXD})$$

$$\therefore \angle A = \frac{1}{2} [m(\text{arc BYD}) - m(\text{arc BXD})].$$

2. In the figure, two circles touch each other externally in point T. Through T a line is drawn intersecting the circles in points A and B as shown. Lines l and m are tangents to the circles at points A and B, then prove line $l \parallel$ line m .

Construction : Draw common tangent n at point T. Mark point Y on it.



Proof : $\angle XAT = \frac{1}{2} m(\text{arc APT})$... (Tangent secant theorem) ... (1)

$\angle ATY = \frac{1}{2} m(\text{arc APT})$... (Tangent secant theorem) ... (2)

$\therefore$ from (1) and (2), we get,

$\angle XAT = \angle ATY$... (3)

$\angle YTB = \frac{1}{2} m(\text{arc TQB})$
... (Tangent secant theorem) ... (4)

$\angle TBZ = \frac{1}{2} m(\text{arc TQB})$
... (Tangent secant theorem) ... (5)

$\therefore$ from (4) and (5), we get,

$\angle YTB = \angle TBZ$... (6)

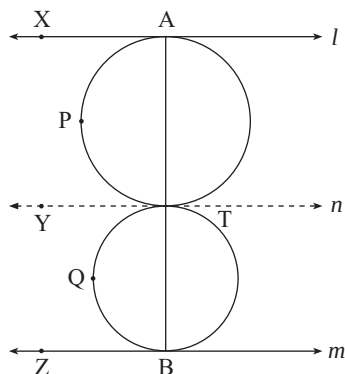
$\angle ATY + \angle YTB = 180^\circ$... (Angles in a linear pair)

$\therefore \angle XAT + \angle TBZ = 180^\circ$... [From (3) and (6)]

i.e. $\angle XAB + \angle ABZ = 180^\circ$... (A-T-B)

But they are pair of interior angles formed by lines l and m on transversal AB .

$\therefore$ line $l \parallel$ line m ... (Interior angles test)



3. In the figure, a circle touches the sides of a quadrilateral internally then prove that angles subtended by opposite sides of quadrilateral at the centre are supplementary.

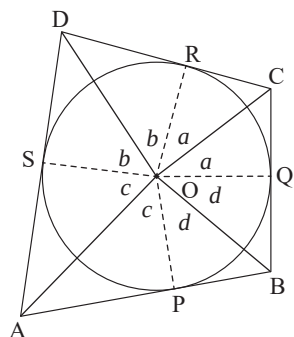
Given :

A circle touches the sides AB , BC , CD , AD of $\square ABCD$ in points P , Q , R and S respectively.

To prove : $\angle AOB + \angle COD = 180^\circ$

$\angle AOD + \angle BOC = 180^\circ$

Construction : Draw OP , OQ , OR and OS .



Proof :

In $\triangle OCR$ and $\triangle OCQ$

seg $OR \cong$ seg OQ ... (Radii of the same circle)

seg $CR \cong$ seg CQ ... (Tangent segment theorem)

seg $OC \cong$ seg OC ... (Common side)

$\therefore \triangle OCR \cong \triangle OCQ$... (SSS test)

$\therefore \angle COR = \angle COQ = a$... (c.a.c.t.) ... (1)

similarly, $\angle ROD = \angle SOD = b$... (2)

$\angle SOA = \angle POA = c$... (3)

$\angle POB = \angle QOB = d$... (4)

$\angle AOB = \angle BOC + \angle COD + \angle AOD = 360^\circ$

... (Sum of measures of all angles at a point is 360°)

$\therefore (\angle AOP + \angle POB) + (\angle BOQ + \angle COQ) + (\angle COR + \angle DOR) +$
 $(\angle AOS + \angle DOS) = 360^\circ$... (Angles addition axiom)

$\therefore (c + d) + (d + a) + (a + b) + (b + c) = 360^\circ$

... [From (1), (2) (3) and (4)]

$\therefore 2a + 2b + 2c + 2d = 360^\circ$

$\therefore a + b + c + d = 180^\circ$... (Dividing both the sides by 2) ... (5)

$\therefore (c + d) + (a + b) = 180^\circ$... [From (5)]

$\therefore \angle AOB + \angle COD = 180^\circ$

also, $(b + c) + (a + d) = 180^\circ$... [From (5)]

$\therefore \angle AOD + \angle BOC = 180^\circ$.

- 4. In $\triangle ABC$, $AB = AC$. A circle passing through point B touches side AC in its midpoint E and intersects side AB in point P.**

Prove : $BP = 3PA$.

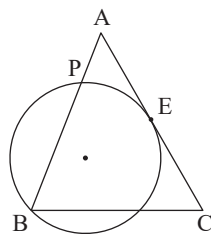
Proof :

Line AE is tangent to the circle at point E and line APB is secant intersecting the circle in points P and B.

$\therefore$ by tangent secant segments theorem,

$$AE^2 = AP \times AB$$

$\therefore \left(\frac{1}{2} AC\right)^2 = AP \times AB$... (E is the midpoint of AC)



$$\therefore \left(\frac{1}{2} AB\right)^2 = AP \times AB$$

... (AC = AB, given)

$$\therefore \frac{AB^2}{4} = AP \times AB$$

$$\therefore \frac{AB^2}{AB} = 4AP$$

$$\therefore AB = 4AP$$

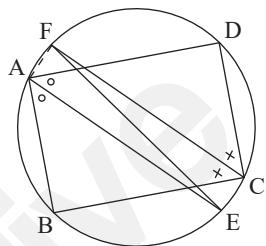
$$\therefore AP + BP = 4AP$$

... (A-P-B)

$$\therefore BP = 4AP - AP$$

$$\therefore BP = 3AP.$$

5. $\square ABCD$ is cyclic. The bisectors of $\angle BAD$ and $\angle BCD$ intersect the circle in points E and F respectively. Prove that EF is the diameter of the circle.



Construction : Draw seg AF.

Proof : Let $\angle BAE = \angle DAE = x$

... (AE bisects $\angle BAD$) ... (1)

$$\angle DCF = \angle BCF = y$$

... (CF bisects $\angle BCD$) ... (2)

$\square ABCD$ is cyclic

... (Given)

$$\therefore \angle BAD + \angle BCD = 180^\circ \quad \dots \text{(Theorem of cyclic quadrilateral)}$$

$$\angle BAE + \angle DAE + \angle BCF + \angle DCF = 180^\circ$$

... (Angles addition postulate)

$$\therefore x + x + y + y = 180^\circ$$

... [From (1) and (2)]

$$\therefore 2x + 2y = 180^\circ$$

$$\therefore x + y = 90^\circ$$

... (Dividing both the sides by 2) ... (3)

$$\angle DCF = \angle DAF = y$$

... (Angles inscribed in the same arc are congruent) ... (4)

$$\angle FAE = \angle DAE + \angle DAF$$

... (Angle addition postulate)

$$= x + y$$

... [From (1) and (4)]

$$= 90^\circ$$

... [From (3)]

$$\angle FAE = \frac{1}{2} m(\text{arc FDE})$$

... (Inscribed angle theorem)

$$90^\circ = \frac{1}{2} m(\text{arc FDE})$$

$$\therefore m(\text{arc FDE}) = 90^\circ \times 2$$

$\therefore$ arc FDE is a semicircle

$$\therefore m(\text{arc FDE}) = 180^\circ$$

$\therefore$ seg EF is the diameter.

- 6. Prove that the sum of the products of the lengths of opposite sides of a cyclic quadrilateral is equal to the product of the length of its diagonal.**

Given : $\square ABCD$ is cyclic.

To prove :

$$AB \times CD + AD \times BC = AC \times BD$$

Construction :

Draw seg AE such that $\angle DAC = \angle BAE$ and B-E-D.

Proof :

$$\angle DAC = \angle BAE$$

... (Construction)

Adding $\angle EAC$ to both the sides

$$\angle DAC + \angle EAC = \angle BAE + \angle EAC$$

$$\therefore \angle DAE = \angle CAB$$

... (Angles addition axiom) ... (1)

In $\triangle DAE$ and $\triangle CAB$

$$\angle DAE \cong \angle CAB$$

... [From (1)]

$$\angle ADE \cong \angle ACB$$

... (Angles inscribed in the same arc)

$$\therefore \triangle DAE \sim \triangle CAB$$

... (AA test of similarity)

$$\therefore \frac{AD}{AC} = \frac{ED}{BC}$$

... (Corresponding sides of similar triangles)

$$\therefore AD \times BC = ED \times AC$$

... (2)

In $\triangle BAE$ and $\triangle CAD$

$$\angle BAE \cong \angle CAD$$

... (Construction)

$$\angle ABE \cong \angle ACD$$

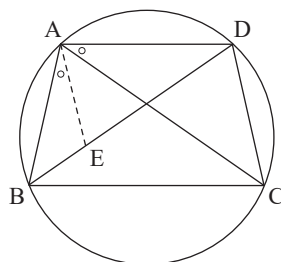
... (Angles inscribed in the same arc are congruent)

$$\therefore \triangle BAE \sim \triangle CAD$$

... (AA test of similarity)

$$\therefore \frac{AB}{AC} = \frac{BE}{CD}$$

... (Corresponding sides of similar triangles)



$$\therefore AB \times CD = AC \times BE \quad \dots (3)$$

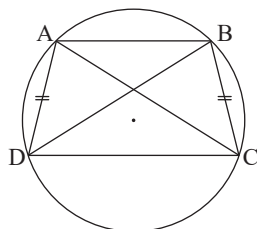
Adding (2) and (3), we get,

$$AD \times BC + AB \times CD = ED \times AC + AC \times BE$$

$$\begin{aligned} \therefore AD \times BC + AB \times CD &= AC(ED + BE) \\ &= AC \times BD \quad \dots (B-E-D) \end{aligned}$$

$$\therefore AD \times BC + AB \times CD = AC \times BD.$$

- 7. The diagonals of cyclic quadrilateral ABCD are congruent. Show that $AD = BC$ and $\text{seg } AB \parallel \text{seg } CD$.**



Proof :

$$AC = BD$$

$$\therefore \text{arc } ABC \cong \text{arc } DAB \quad \dots (\text{Corresponding arcs of congruent chords})$$

$$\begin{aligned} m(\text{arc } AD) + m(\text{arc } AB) &= m(\text{arc } AB) + m(\text{arc } BC) \\ &\dots (\text{Arcs addition postulate}) \end{aligned}$$

$$m(\text{arc } AD) = m(\text{arc } BC) \quad \dots (1)$$

$$\therefore AD = BC \quad \dots (\text{Congruent arcs have corresponding chords congruent})$$

$$\angle ABD = \frac{1}{2} m(\text{arc } AD) \quad \dots (\text{Inscribed angle theorem}) \quad \dots (2)$$

$$\angle BDC = \frac{1}{2} m(\text{arc } BC) \quad \dots (\text{Inscribed angle theorem}) \quad \dots (3)$$

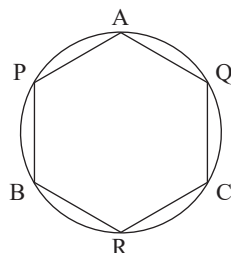
$$\therefore \angle ABD \cong \angle BDC \quad \dots [\text{From (1), (2), (3)}]$$

$$\therefore \text{seg } AB \parallel \text{seg } DC.$$

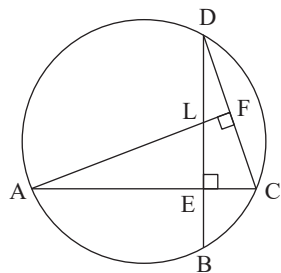
Assignment	4.5
-------------------	------------

- 1.** In the figure, points A, P, B, R, C, Q are on the circle. After joining the given points as shown in the figure it forms a hexagon, then prove that $\angle APB + \angle BRC = 360^\circ - \angle AQC$.

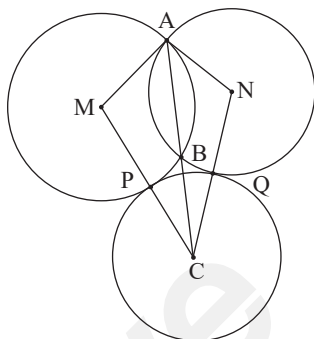
(July '22)



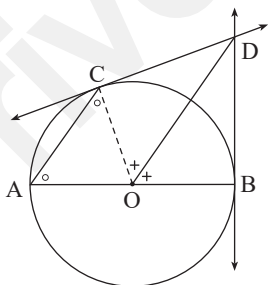
2. In the figure, chords AC and BD intersect at point E. $AC \perp BD$, $AF \perp CD$, $B-E-L-D$; then prove that $BE = EL$.



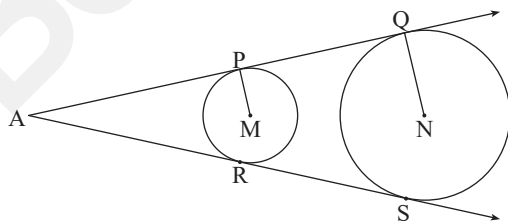
3. M and N are the centres of two congruent circles intersecting each other at points A and B. A circle with centre C touches the two circles at points P and Q as shown. Prove that points A, B and C are collinear.



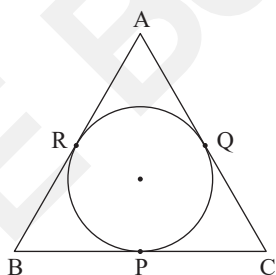
4. In the figure, O is the centre of the circle. Seg AB is the diameter at the point C on the circle, the tangent CD is drawn. Line BD is a tangent to the circle at point B. Show that $\text{seg } OD \parallel \text{chord } AC$.



5. In the figure, line APQ and line ARS are two tangents of the circles. M and N are centres of the circles. P, Q, R and S are points of contact, then prove that $\frac{AP}{AQ} = \frac{MP}{NQ} = \frac{PR}{QS}$.

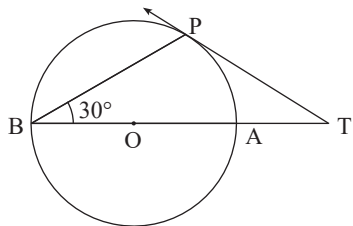


6.



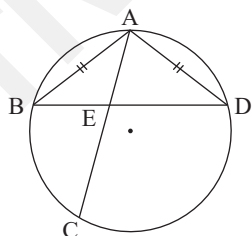
$\triangle ABC$ is an isosceles triangle in which $AB = AC$. A circle is inscribed in the triangle as shown in the figure. Prove that the base is bisected by the point of contact.

7. In the figure, O is the centre of the circle. Ray TP is tangent touching the circle at point P. $\angle PBT = 30^\circ$, then prove that $BA : AT = 2 : 1$.

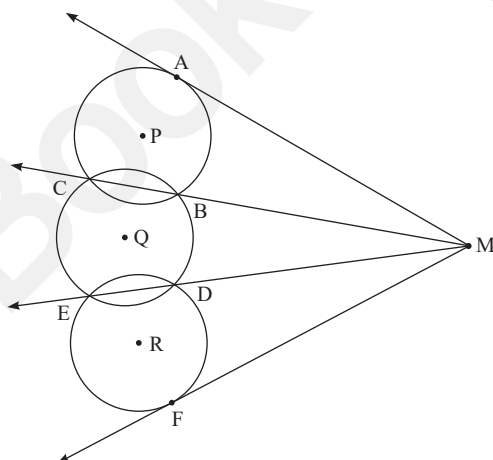


8. If two consecutive angles of a cyclic quadrilateral are congruent, then prove that one pair of opposite sides is congruent and the other is parallel. (Sept. '21)

9. In the figure, $AB = AD$, chord AC and chord BD intersect in point E. Prove that $AB^2 = AE \times AC$. (Nov. '20)



10.



In the figure, the circles with centre, P, Q and R intersect at points B, C, D and E as shown.

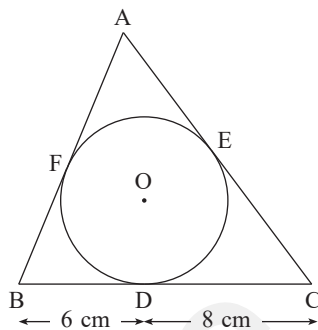
Lines CB and ED intersect at point M as shown. Line MA is tangent to the circle with centre P at point A and line MF is tangent to the circle with centre R at point F.

Prove that $MA = MF$.

(July '19)

11. In the given figure, O is the centre of the incircle of $\triangle ABC$. Sides BC, AC and AB touch the circle in points D, E and F respectively such that $BD = 6$ cm and $CD = 8$ cm. The radius of the circle is 4 cm. Find the length of seg AB and seg AC.

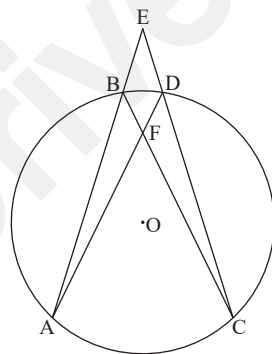
[Ans. $AB = 13$ cm and $AC = 15$ cm.]



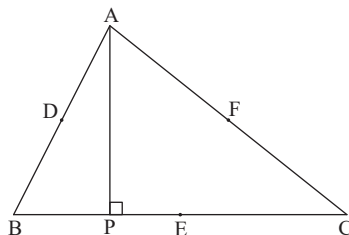
12. In the figure, O is the centre of the circle.

Prove that

$$\angle AOC = \angle AFC + \angle AEC.$$



13. In the figure, D, E and F are midpoints of sides AB, BC and AC respectively. P is the foot of the perpendicular from A to side BC. Show that points D, E, F and P are concyclic.



**Question
Set
5**

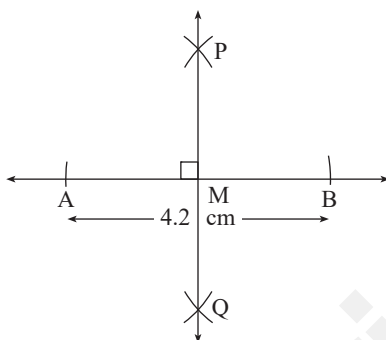
GEOMETRIC CONSTRUCTIONS

Basic Constructions

1. To draw a perpendicular bisector of a given line segment.

Draw seg AB of length 4.2 cm. Draw perpendicular bisector of seg AB.

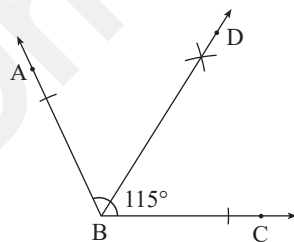
Ans.



2. To draw angle bisector of a given angle.

Draw $\angle ABC$ of measure 115° and bisect it.

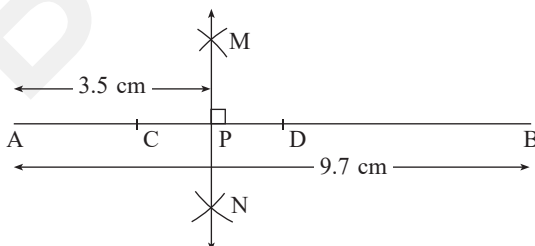
Ans.



3. To draw a perpendicular to a line segment at a given point on it.

Draw seg AB of length 9.7 cm. Take a point P on it, such that $A-P-B$, $AP=3.5$ cm. Construct a line MN perpendicular to seg AB through the point P.

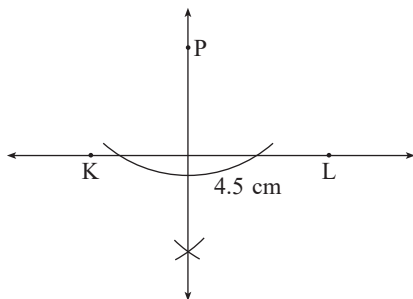
Ans.



4. To draw a perpendicular to a given line segment from a point outside it.

Draw seg KL of length 4.5 cm. Take a point P outside it. Construct a perpendicular to seg KL from the point P.

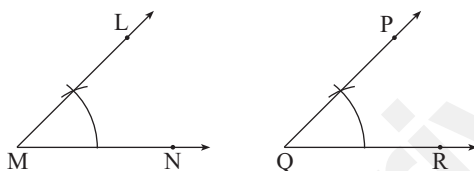
Ans.



5. To draw an angle congruent to a given angle.

$\angle LMN$ is given. Construct $\angle PQR$ such that $\angle PQR \cong \angle LMN$.

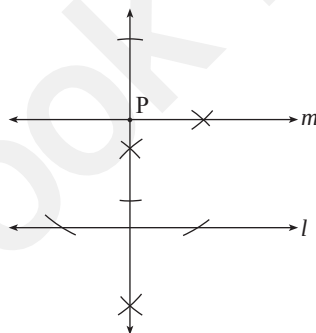
Ans.



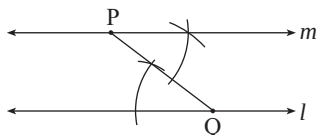
6. To draw a line parallel through a given point to given line.

Draw line l . P is any point not on line l . Draw another line parallel to line l through point P.

Ans. Method 1 :



Method 2 : Based on property if alternate angles are congruent, then lines are parallel to each other.



7. To divide a given line segment in a given ratio.

Draw seg PQ of length 7 cm. Divide it in the ratio 3 : 2.

Ans.

(1) Draw seg PQ of length 7 cm.

(2) Draw ray PR making an acute angle with seg PQ at its lower side.

(3) Draw five equal parts on ray PR from point P.

$PP_1, P_1P_2, P_2P_3, P_3P_4, P_4P_5$.

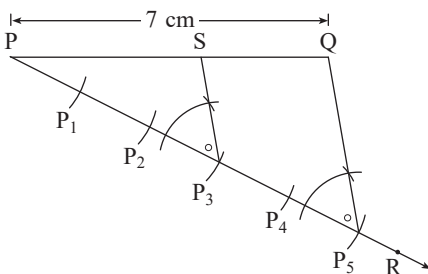
(4) Join QP_5 .

(5) Draw $\angle SP_3P$ such that $\angle SP_3P \cong \angle QP_5P$

... (Corresponding angles)

(6) Through point P_3 , draw $P_3S \parallel P_5Q$ and intersecting seg PQ in the point S.

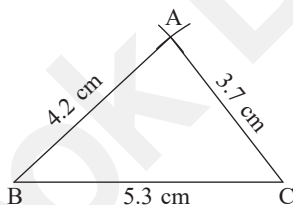
(7) $PS : SQ = 3 : 2$.



8. To draw a triangle when its sides are given.

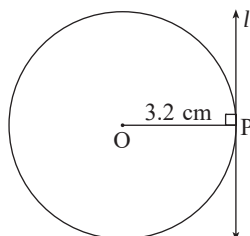
In $\triangle ABC$, $AB = 4.2$ cm, $BC = 5.3$ cm and $AC = 3.7$ cm, then draw $\triangle ABC$.

Ans.



5.1 2 marks each

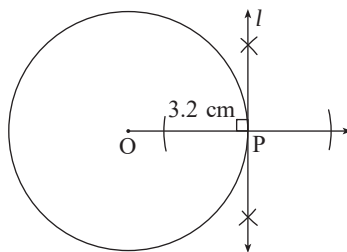
1. Draw a circle of radius 3.2 cm. Mark a point P on the circle. Draw tangent to the circle through point P using the centre of the circle. (Nov. '20; March '22; July '24)



Steps of construction :

- (1) Draw a circle with centre O and radius 3.2 cm.
- (2) Take any point P on it.
- (3) Draw ray OP.
- (4) Draw line $l \perp$ ray OP at point P.

Line l is the required tangent to the circle.

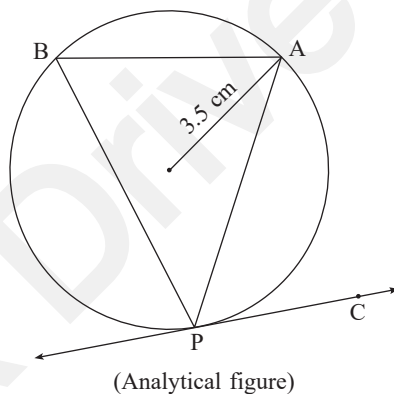
Construction :

[Note : Analysis and steps of construction are given to understand the construction. It is not necessary to write it in the examination.]

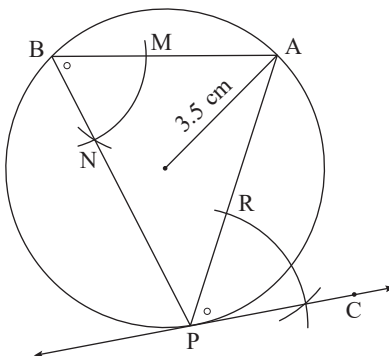
2. Draw a circle of radius 3.5 cm. Take any point P on it. Draw tangent to the circle through point P without using the centre of the circle.

Analysis :

Through P, a chord can be drawn. Let it be PA. Draw any $\angle PBA$ in the alternate segment. Now an $\angle APC$ can be constructed congruent to $\angle ABP$. Then by converse of tangent secant angle theorem, line PC is the required tangent.

**Steps of construction :**

- (1) Draw a circle of radius 3.5 cm. Take a point P on it.
- (2) Draw any chord AP and $\angle ABP$ in the alternate segment.

Construction :

- (3) With B as the centre and any convenient radius, draw an arc intersecting the sides BA and BP at points M and N respectively.
- (4) Using the same radius and centre P, draw an arc intersecting chord PA at point R.
- (5) Taking radius equal to $d(M, N)$ and centre R, draw an arc intersecting the arc drawn in the previous step. Let C be the point of intersection of these arcs. Draw line PC. Line PC is the required tangent to the circle.

Assignment	5.1
-------------------	------------

1. Complete the following activity to draw a tangent to a circle at a point on the circle.

Draw a circle of radius 3.2 cm with O as centre



Take a point P on the circle and draw ray OP



Draw a perpendicular line to ray OP at point P



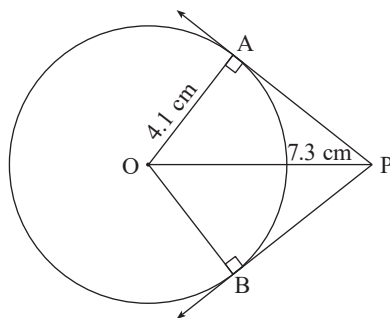
Name the perpendicular line as l , l is the tangent at point P

2. Construct a tangent to a circle with centre P and radius 3 cm at any point M on it without using the centre.
3. Draw a circle of radius 3.5 cm. Draw a tangent to the circle at any point M on it. **(July '19–'22)**
4. Construct tangent to a circle with centre A and radius 3.4 cm at any point P on it. **(March '19)**

5.2	3 marks each
------------	---------------------

1. Draw a circle with radius 4.1 cm. Construct tangents to the circle from a point at a distance of 7.3 cm from the centre.

(March '25)

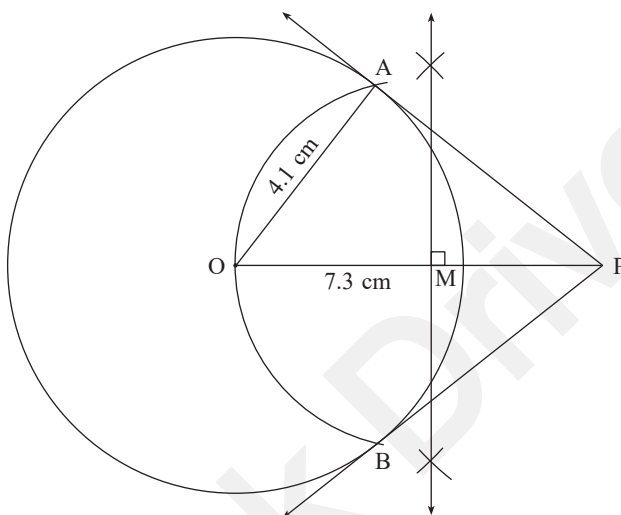


(Analytical figure)



Steps of construction :

- (1) Draw a circle of radius 4.1 cm.
- (2) Take a point P in the exterior of the circle such that $d(O, P) = 7.3$ cm.
- (3) Draw seg OP. Draw perpendicular bisector of seg OP to get its midpoint M.
- (4) Draw an arc with centre M and radius OM.
- (5) Mark the points of intersection of the arc with the circle as A and B.
- (6) Draw lines PA and PB.

Construction :

-
2. $\triangle ABC \sim \triangle PQR$. In $\triangle ABC$, $AB = 5.4$ cm, $BC = 4.2$ cm, $AC = 6$ cm. $AB : PQ = 3 : 2$ then construct $\triangle ABC$ and $\triangle PQR$.
(March '22)

Analysis :

$$\triangle ABC \sim \triangle PQR$$

... (Given)

$$\therefore \frac{AB}{PQ} = \frac{BC}{QR} = \frac{AC}{PR} \quad \dots \text{(Corresponding sides of similar triangles)}$$

$$\therefore \frac{5.4}{PQ} = \frac{4.2}{QR} = \frac{6}{PR} = \frac{3}{2}$$

$$\therefore \frac{5.4}{PQ} = \frac{3}{2}$$

$$\therefore PQ = \frac{5.4 \times 2}{3}$$

$$\therefore PQ = 3.6 \text{ cm}$$

$$\frac{4.2}{QR} = \frac{3}{2}$$

$$\therefore QR = \frac{4.2 \times 2}{3}$$

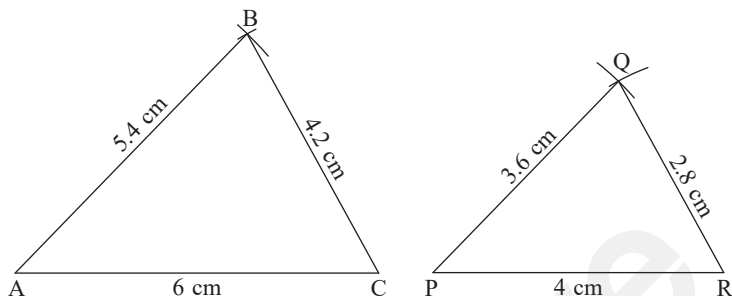
$$\therefore QR = 2.8 \text{ cm}$$

$$\frac{6}{PR} = \frac{3}{2}$$

$$\therefore PR = \frac{6 \times 2}{3}$$

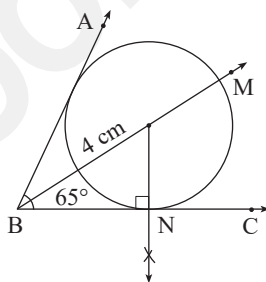
$$\therefore PR = 4 \text{ cm}$$

Construction :



3. Draw the $\angle ABC$ of measure 65° . Draw ray BM which is a bisector of $\angle B$. Take point P on ray BM such that $BP = 4 \text{ cm}$. Draw perpendicular on arm BC through the point P. Draw a circle with centre P and length of perpendicular as a radius. Write the measure of radius. Observe the figure and write the relation between circle and arms of the angle. (Sept. '21)

Analysis :

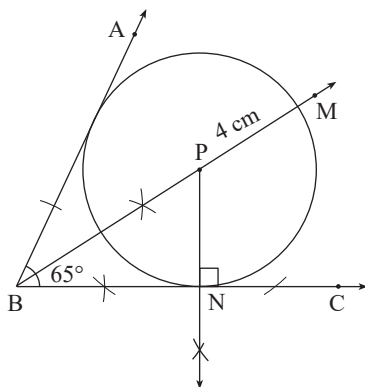


(Rough Figure)

Ans.

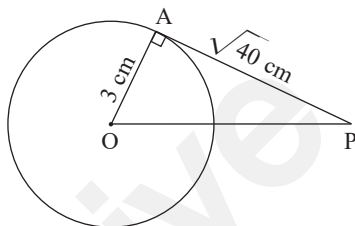
- (i) Radius of the circle = 2.1 cm.
- (ii) The arms of the angle, i.e. ray BA and ray BC are tangents to the circle.

Construction :



- 4. Draw a circle with centre 'O' and radius 3 cm. Draw a tangent segment PA having length $\sqrt{40}$ cm from an exterior point P.**

(Nov. '20)



Analysis :

$$\angle OAP = 90^\circ$$

... (By tangent radius theorem)

In $\triangle OAP$,

by Pythagoras theorem,

$$OP^2 = OA^2 + AP^2$$

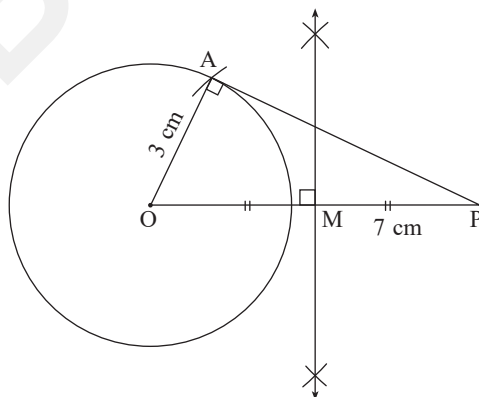
$$= (3)^2 + (\sqrt{40})^2 = 9 + 40 = 49$$

$$\therefore OP = \sqrt{49} = 7 \text{ cm.}$$

Steps of construction :

- (1) Draw a circle with centre O and radius 3 cm.
- (2) Draw segment OP of length 7 cm and construct its perpendicular bisector. Intersecting OP in point M.
- (3) Taking M as a centre and MO as a radius. Draw an arc intersecting the circle in point A.
- (4) Draw tangent segment PA.

Construction :



Assignment

5.2

1. $\triangle ABC \sim \triangle LMN$. In $\triangle ABC$, $AB = 5.5$ cm, $BC = 6$ cm, $CA = 4.5$ cm.
Construct $\triangle ABC$ and $\triangle LMN$ such that $\frac{BC}{MN} = \frac{5}{4}$. (March '24)
2. Draw circle with centre O and radius 3.5 cm. Take a point P at a distance of 7.5 cm from the centre. Draw tangents to the circle from point P. (July '22)
3. $\triangle PQR \sim \triangle LTR$. In $\triangle PQR$, $PQ = 4.2$ cm, $QR = 5.4$ cm, $PR = 4.8$ cm.
Construct $\triangle PQR$ and $\triangle LTR$ such that $\frac{PQ}{LT} = \frac{3}{4}$. (July '23)
4. Draw a circle with centre P and radius 3.5 cm. Draw an arc AB of 120° measure. Draw tangents to the circle at point A and point B. (Sept. '21)
5. Draw a circle of radius 3.3 cm. Draw a chord PQ of length 6.6 cm. Draw tangents to the circle at points P and Q. (March '23)
6. Construct two concentric circles with centre O with radii 3 cm and 5 cm. Construct tangent to a smaller circle from any point A on the larger circle. Measure and write the length of tangent segment. Calculate the length of tangent segment using Pythagoras theorem. (March '22) (Ans. 4 cm)



CHALLENGING QUESTIONS

5.3 4 marks each

1. $\triangle ABC \sim \triangle PQR$. In $\triangle ABC$, $AB = 3.6$ cm, $BC = 4$ cm and $AC = 4.2$ cm. The corresponding sides of $\triangle ABC$ and $\triangle PQR$ are in the ratio $2 : 3$, construct $\triangle ABC$ and $\triangle PQR$. (March '25)

Analysis :

$$\triangle ABC \sim \triangle PQR$$

$$\therefore \frac{AB}{PQ} = \frac{BC}{QR} = \frac{AC}{PR} = \frac{2}{3} \quad \dots \text{ (Corresponding sides of similar triangles)}$$

$$\therefore \frac{AB}{PQ} = \frac{2}{3} \quad \therefore \frac{3.6}{PQ} = \frac{2}{3}$$

$$\therefore PQ = \frac{3.6 \times 3}{2} = 5.4 \text{ cm}$$

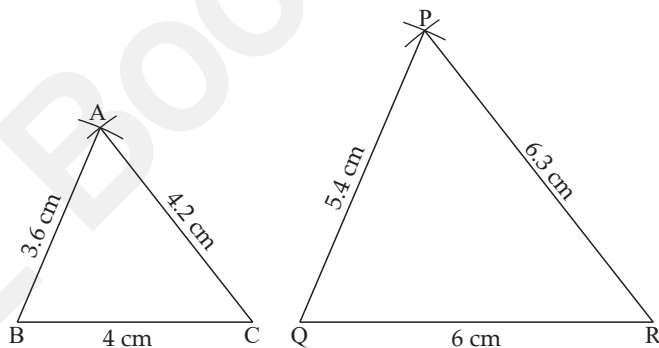
$$\therefore \frac{BC}{QR} = \frac{2}{3} \quad \therefore \frac{4}{QR} = \frac{2}{3}$$

$$\therefore QR = \frac{4 \times 3}{2} = 6 \text{ cm}$$

$$\therefore \frac{AC}{PR} = \frac{2}{3} \quad \therefore \frac{4.2}{PR} = \frac{2}{3}$$

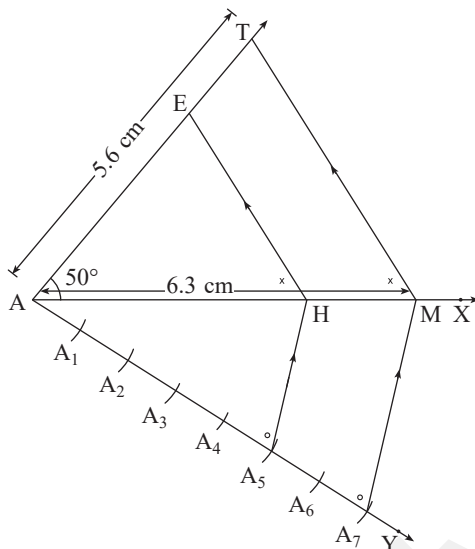
$$\therefore PR = \frac{4.2 \times 3}{2} = 6.3 \text{ cm}$$

Construction :



2. $\triangle AMT \sim \triangle AHE$. In $\triangle AMT$, $AM = 6.3$ cm, $\angle TAM = 50^\circ$,

$AT = 5.6$ cm and $\frac{AM}{AH} = \frac{7}{5}$. Construct $\triangle AHE$.

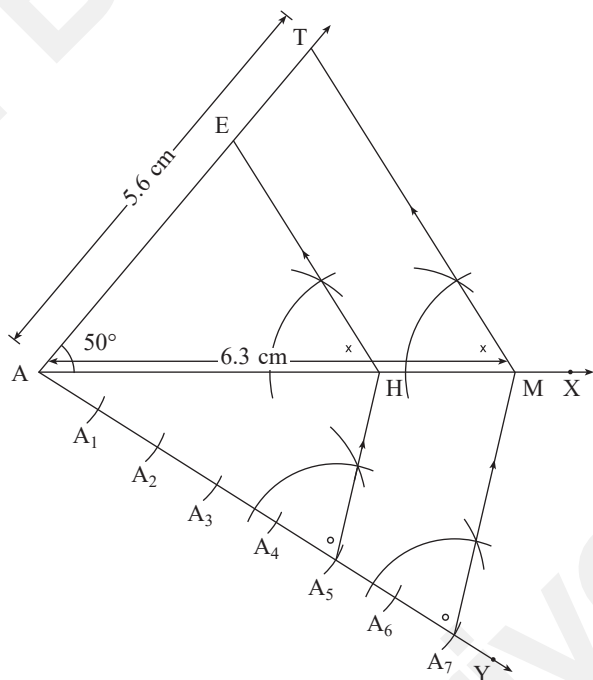


(Analytical figure)

Steps of construction :

- (1) Draw $\triangle AMT$ such that $AM = 6.3$ cm, $\angle TAM = 50^\circ$ and $AT = 5.6$ cm.
- (2) Draw a ray AY making a suitable angle with ray AX .
- (3) Make 7 equal parts AA_1 , A_1A_2 , A_2A_3 , A_3A_4 , A_4A_5 , A_5A_6 and A_6A_7 on ray AY .
- (4) Draw seg A_7M .
- (5) Draw seg $A_5H \parallel \text{seg } A_7M$. Intersecting seg AM at point H . Through point H draw seg $HE \parallel \text{seg } MT$. It intersecting AT at point E .
 $\triangle AHE$ is the required triangle.

Construction :



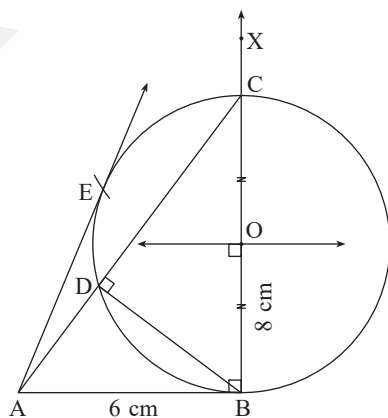
3. Draw a right angled $\triangle ABC$ such that $AB = 6$ cm, $BC = 8$ cm and $\angle B = 90^\circ$. From point B, draw a perpendicular segment BD to the hypotenuse AC. Draw a circle that passes through points B, C and D. From point A, draw a tangents to this circle.

Steps of construction :

- (1) Draw seg AB of length 6 cm.
- (2) At point B draw $\angle ABX = 90^\circ$.
- (3) On ray BX, take a point C such that $BC = 8$ cm.

Draw seg AC.

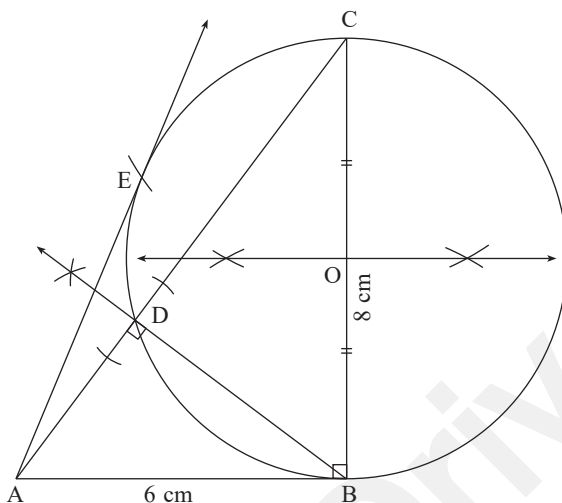
- (4) Construct $\text{seg BD} \perp \text{hypotenuse AC}$.
- (5) Construct perpendicular bisector of seg BC to intersect BC at point O .



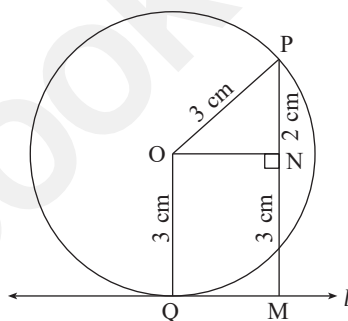
(Analytical figure)

- (6) With O as centre and OB as radius, draw a circle passing through B, D and C.
- (7) Since $OB \perp AB$; then AB is tangent to the circle. Thus with A as centre and radius equal to AB cut an arc intersecting the circle in point E.
8. AB and AE are tangents to the circle.

Construction :



4. Draw a line l . Take a point P at a distance of 5 cm from line l . Draw a circle with a radius of 3 cm, such that it touches line l and passes through point P.



(Analytical figure)

Analysis :

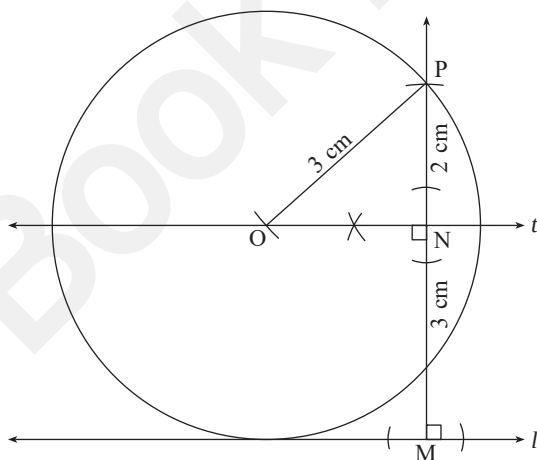
1. Let the required circle have its centre at point O.

- Line l can be drawn. A point M can be marked on line l such that $PM \perp$ line l and $PM = 5$ cm.
- If we draw $OQ \perp$ line l then $OQ = 3$ cm, since OQ is the radius at the point of contact.
- Let's draw seg $ON \perp$ seg PM . Therefore, $\square OQMN$ is a rectangle.
- Thus $MN = OQ = 3$ cm as opposite sides of rectangle are equal.
- Now, $PM = PM - MN = 5 - 3 = 2$ cm.
- Hence the centre of the required circle (point O) is on the line perpendicular to segment PM at point N and at a distance of 3 cm from point P .

Steps of construction :

- Draw a line l and take a point M on it.
- Draw perpendicular to line l through point M . Take points P and N on this perpendicular such that $PM = 5$ cm and $MN = 3$ cm.
- Through point N , draw a line parallel to line l . Name this line t .
- From point P , cut an arc with a radius 3 cm that intersects line t . Mark the point of intersection as O .
- Taking O as the centre and radius equal to OP , draw the circle that passes through P and touches line l .

Construction :



5. Draw a circle with centre O having radius 3 cm. Draw tangent segments PA and PB through the point P outside the circle such that $\angle APB = 70^\circ$. (March '24)

Analysis :

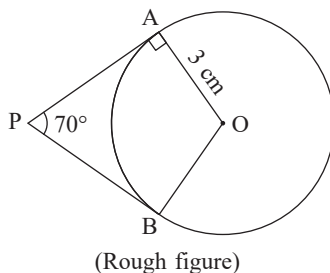
By tangent theorem,

$$\angle OAP = 90^\circ \text{ and } \angle OBP = 90^\circ$$

$$\angle APB = 70^\circ \quad \dots \text{ (Given)}$$

$$\therefore \angle AOB = 110^\circ$$

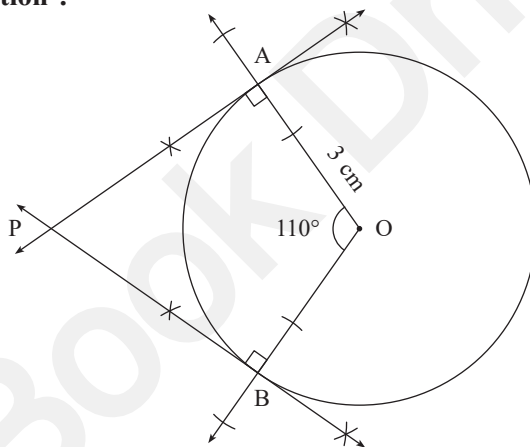
... (Remaining angle of $\square OAPB$)



Steps of construction :

- (1) Draw circle with centre O and radius 3 cm. Take any point A on the circle.
- (2) Draw $\angle AOB$ such that point B is on the circle and $\angle AOB = 110^\circ$.
- (3) Draw tangents to the circle at points A and B.
- (4) $\angle APB$ is required angle.

Construction :



Assignment

5.3

1. $\triangle LMN \sim \triangle LQP$. In $\triangle LMN$, $LM = 3.6$ cm, $\angle L = 50^\circ$, $LN = 4.2$ cm and $\frac{LM}{LQ} = \frac{4}{7}$, then construct $\triangle LQP$ and $\triangle LMN$. (Sept. '21)



2. Draw a circle with centre P and radius 3 cm. Draw a chord MN of length 4 cm. Draw tangents to the circle through points M and N which intersect in point Q. Measure the length of seg PQ.

(July '23) (Ans. PQ = 4 cm)

3. $\triangle SHR \sim \triangle SVU$. In $\triangle SHR$, SH = 4.5 cm, HR = 5.2 cm,

SR = 5.8 cm and $\frac{SH}{SV} = \frac{3}{5}$. Construct $\triangle SVU$. (March '23)

4. Draw $\angle ABC = 60^\circ$. Draw bisector of $\angle ABC$ and take a point Q on it. Such that $d(B, Q) = 8$ cm. With Q as centre, draw a circle which touches ray BA and ray BC. Measure the radius of the circle and length of tangent segment drawn from B.

(Ans. radius QP = 4 cm, length of tangent segment ≈ 6.9 cm.)

5. Construct an isosceles triangle such that its base is of length 8 cm and its height is 4 cm. Construct a triangle similar to the given isosceles triangle whose sides are $\frac{2}{3}$ the sides of the given isosceles triangle.
6. Draw a circle with centre O and radius 3.2 cm. Take points A and B on the circle such that $\angle AOB = 60^\circ$. Let the bisector of $\angle AOB$ intersect the circle in point K. Draw a circle passing through K such that ray OA and ray OB are tangents to it.
7. Draw a circle with centre M and radius 2.7 cm. Take a point P such that length of segment PM is 7.5 cm. Draw tangents to the circle through P. Draw a circle that touches the circle and the tangents.
8. Draw $\angle ABC = 40^\circ$. Take a point S in the interior of $\angle ABC$. Draw a circle passing through point S and touching the sides of $\angle ABC$.
9. Draw a circle of radius 2.7 cm and draw chord PQ of length 4.5 cm. Draw tangents at points P and Q without using the centre of the circle. (March '20)

POINTS TO REMEMBER

- Coordinates of origin are $(0, 0)$. Hence, if coordinates of point A are (x, y) , then $d(O, A) = \sqrt{x^2 + y^2}$.
- The distance between any two points $P(x_1, 0)$ and $Q(x_2, 0)$ on X-axis is find out by the formula : $d(P, Q) = x_2 - x_1 \dots (x_2 > x_1)$
- The distance between any two points $A(0, y_1)$ and $B(0, y_2)$ on Y-axis is find out by following formula :
 $d(A, B) = y_2 - y_1 \dots (y_2 > y_1)$
- If points $A(x_1, y_1)$, $B(x_2, y_2)$ lie in the XY-plane,
then $d(A, B) = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$ that is,
 $AB^2 = (x_2 - x_1)^2 + (y_2 - y_1)^2 = (x_1 - x_2)^2 + (y_1 - y_2)^2$.
This is known as distance formula.
- Section formula :** The coordinates of a point which divides the line segment joined by two distinct points (x_1, y_1) and (x_2, y_2) in the ratio $m : n$ are $\left(\frac{mx_2 + nx_1}{m+n}, \frac{my_2 + ny_1}{m+n} \right)$.
- Midpoint formula :** The coordinates of midpoint of a line segment joining two distinct points (x_1, y_1) and (x_2, y_2) are $\left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$.
- The medians of a triangle are concurrent. The point of concurrence of the medians is called centroid. The centroid divides the median in the ratio $2 : 1$.
- Centroid formula :** If (x_1, y_1) , (x_2, y_2) and (x_3, y_3) are the vertices of a triangle, then coordinates of the centroid are $\left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3} \right)$.

9. Slope of a line : If $P(x_1, y_1)$ and $Q(x_2, y_2)$ are any two points in the XY-plane, then

$$\text{slope of line PQ } (m) = \frac{y_2 - y_1}{x_2 - x_1}.$$

- 10.** Slope of a line making an angle θ with the positive direction of X-axis is $(m) = \tan \theta$.
- 11.** (i) Slope of X-axis and any line parallel to X-axis is zero.
(ii) Slope of Y-axis and any line parallel to Y-axis cannot be determined.
(iii) The parallel lines have equal slopes.
(iv) When two distinct lines have same slope, then the two lines are parallel.

6.1 1 mark each

Q. Choose the correct alternative from those given below each question :

- 1.** Distance of point $(3, 4)$ from the origin is **(March '25)**

(A) 7 (B) 1 (C) 5 (D) -5

Ans. (C)

[Hint : Use distance formula.]

- 2.** Point $P(3, a)$ is the midpoint of line segment joining $A(4, 7)$ and $B(2, 1)$. The value of a is

(A) 4 (B) 8 (C) 6 (D) 3

Ans. (A)

[Hint : Use midpoint formula.]

- 3.** From the following points, point lies to the right side of the origin and X-axis. **(July '22)**

(A) $(-2, 0)$ (B) $(0, 2)$ (C) $(2, 3)$ (D) $(2, 0)$

Ans. (D) **[Hint :** When the point lies on X-axis and right side of origin its y-coordinate is 0 and x-coordinate is positive.]

- 4.** The coordinates of the point on X-axis equidistant from $A(-1, 0)$ and $B(5, 0)$ are

(A) $(0, 2)$ (B) $(2, 0)$ (C) $(3, 0)$ (D) $(0, 3)$

Ans. (B) [**Hint :** Both A and B lie on X-axis, so the required coordinates can be found using midpoint formula.]

5. If the distance between points P(2, 2) and Q(5, x) is 5, then the value of x is

(A) 2 (B) 6 (C) 3 (D) 1

Ans. (B) [**Hint :** Use distance formula.]

6. The sum of the x-coordinates of the vertices of the triangle is 12 and that of y-coordinates is 9, then the coordinates of the centroid are

(A) (12, 9) (B) (9, 12) (C) (4, 3) (D) (3, 4)

Ans. (C) [**Hint :** Coordinates of centroid are

$$\left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3} \right).]$$

7. If the slope of a line joining points A(x, 0) and B(-3, -2) is $\frac{2}{7}$, then the value of x is

(A) 4 (B) 16 (C) -3 (D) -2

Ans. (A) [**Hint :** Use slope of line = $\frac{y_2 - y_1}{x_2 - x_1}$.]

8. A line makes an angle of 30° with the positive direction of X-axis, so the slope of the line is (July '23)

(A) $\frac{1}{2}$ (B) $\frac{\sqrt{3}}{2}$ (C) $\frac{1}{\sqrt{3}}$ (D) $\sqrt{3}$

Ans. (C) [**Hint :** Slope of line = $\tan \theta$.]

Assignment	6.1
-------------------	------------

1. The coordinates of the origin are (March '23)

(A) (0, 0) (B) (0, 1) (C) (1, 0) (D) (1, 1)

2. Seg AB is parallel to the X-axis and coordinates of the point A are (1, 3), then coordinates of the point B can be (March '22)

(A) (-3, 1) (B) (5, 1) (C) (3, 0) (D) (-5, 3)



3. The distance between the points $P(-1, 1)$ and $Q(5, -7)$ is
 (A) 11 (B) 10 (C) 5 (D) 7
4. The line segment joining the points $(-3, -4)$ and $(1, -2)$ is divided by Y-axis in the ratio
 (A) 2 : 3 (B) 3 : 2 (C) 3 : 1 (D) 1 : 3
5. The ratio in which X-axis divides the segment joining $(3, 6)$ and $(12, -3)$ is
 (A) 1 : 2 (B) 3 : 2 (C) 2 : 3 (D) 2 : 1
6. $P(-1, -1)$ is the midpoint of line segment joining $A(-3, b)$ and $B(1, b+4)$, then the value of b is
 (A) 1 (B) -3 (C) 2 (D) 0
7. For what value of x will the points $(-1, 4)$, $(2, x)$ and $(8, 1)$ will be collinear?
 (A) -1 (B) 1 (C) 3 (D) -3
8. A line makes an angle of 60° with the positive direction of X-axis, so the slope of the line is
 (A) $\frac{1}{2}$ (B) $\frac{\sqrt{3}}{2}$ (C) $\sqrt{3}$ (D) $\frac{1}{\sqrt{3}}$ (Nov. '20)
9. The slope of X-axis is
 (A) 1 (B) -1 (C) 0 (D) cannot be determined (March '24)
10. Distance of point $(-3, 4)$ from the origin is
 (A) 7 (B) 1 (C) 5 (D) -5 (March '20)

Answers :

1. (A)
2. (D) [The y-coordinates of the points on the line parallel to X-axis will be same.]
3. (B) [Use distance formula.]
4. (C) [Use section formula. Here, $x = 0$.]
5. (D) [Use section formula. Here, $y = 0$.]

6. (B) [Use midpoint formula.]
 7. (C) [Use slope formula.]
 8. (C) [$m = \tan \theta$, $\tan 60^\circ = \sqrt{3}$.]
 9. (C)
 10. (C) [Use distance formula.]

6.2 1 mark each

1. The angle made by a line with the positive direction of X-axis is 30° . Find slope of that line. (March '25)

Ans. Slope of the line = $\tan \theta$

$$= \tan 30^\circ = \frac{1}{\sqrt{3}}.$$

Slope of line is $\frac{1}{\sqrt{3}}$.

2. Find the coordinates of the midpoint of the segment joining the points A(4, 6) and B(-2, 2). (July '23)

Ans.

Let the coordinates of the midpoint joining the points

A(4, 6) $\equiv (x_1, y_1)$ and B(-2, 2) $\equiv (x_2, y_2)$ be (x, y).

$\therefore$ by midpoint formula,

$$x = \frac{x_1 + x_2}{2} \quad \text{and} \quad y = \frac{y_1 + y_2}{2}$$

$$\therefore x = \frac{4 + (-2)}{2} \quad \text{and} \quad y = \frac{6 + 2}{2}$$

$$\therefore x = \frac{4 - 2}{2} \quad \text{and} \quad y = \frac{8}{2}$$

$$\therefore x = 1 \quad \text{and} \quad y = 4$$

The coordinates of the midpoint joining the points A(4, 6) and B(-2, 2) are (1, 4).

3. Find the distance between the points O(0, 0) and P(3, 4).

(March '23)

Ans.

Let $O(0, 0) \equiv (x_1, y_1)$ and $P(3, 4) \equiv (x_2, y_2)$

$$\begin{aligned} d(O, P) &= \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2} && \dots \text{ (By distance formula)} \\ &= \sqrt{(3 - 0)^2 + (4 - 0)^2} \\ &= \sqrt{3^2 + 4^2} \\ &= \sqrt{9 + 16} \\ &= \sqrt{25} \\ &= 5 \end{aligned}$$

Distance between the points O and P is 5.

4. Find the distance of A(-3, 4) from X-axis.

Ans. Distance of A(-3, 4) from X-axis is 4.

5. If P(x, 6) is the midpoint of line segment joining A(6, 5) and B(4, y), then find the value of y.

Ans. P(x, 6), A(6, 5) and B(4, y)

By midpoint formula,

$$6 = \frac{5 + y}{2}$$

$$\therefore 5 + y = 6 \times 2$$

$$\therefore 5 + y = 12$$

$$\therefore y = 12 - 5$$

$$\therefore y = 7.$$

The value of y is 7.

6. Points A, B and C are collinear. If slope of line AB is $-\frac{1}{2}$, then find the slope of line BC. (Nov. '20)

Ans.

Points A, B and C are collinear.

$\therefore$ slope of line AB = slope of BC

Slope of line BC is $-\frac{1}{2}$.

7. Find the y-coordinate of the centroid of a triangle whose vertices are (4, -3), (7, 5) and (-2, 1). (March '22)

Ans.

Here, $y_1 = -3$, $y_2 = 5$, and $y_3 = 1$

By centroid formula,

$$y = \frac{y_1 + y_2 + y_3}{3}$$

$$y = \frac{-3 + 5 + 1}{3} \quad \therefore y = \frac{3}{3} \quad \therefore y = 1$$

The y-coordinate of the centroid of the triangle is 1.

8. Find the slope of the line making an angle of 45° with the positive direction of X-axis. (March '24)

Ans.

The inclination of the line with positive direction of X-axis (θ) = 45° .

Slope of line = $\tan \theta$

$$= \tan 45^\circ$$

$$= 1$$

The slope of line is 1.

Assignment	6.2
------------	-----

1. Find the coordinates of the midpoint of the segment joining the points (22, 20) and (0, 16). (July '22)
2. Find the distance between the points O(0, 0) and P(6, 8).
3. The coordinates of points A and B are (-6, 7) and (-1, -5) respectively, then find the distance 2AB.
4. The slope of a line is $\frac{1}{\sqrt{3}}$. Find the inclination of the line with the positive direction of X-axis.
5. What is the slope of Y-axis?
6. The slope of AB is -1. Line AB || line CD. Write the slope of line CD.



7. Write the abscissa and the ordinate of the point P(-5, 4).
8. Write the coordinates of the point of intersection of lines $x = 5$ and $y = -3$.

Answers :

1. (11, 18) 2. 10 3. $2AB = 26$ 4. $\theta = 30^\circ$
5. Slope of Y-axis cannot be determined 6. -1
7. Abscissa is -5 and ordinate is 4 8. (5, -3).

6.3 2 marks each

1. If C(3, 5) and D(-2, -3), then complete the following activity to find the distance between points C and D. (Nov. '20)

Activity :

Let $C(3, 5) \equiv (x_1, y_1)$, $D(-2, -3) \equiv (x_2, y_2)$

$$CD = \sqrt{(x_2 - \boxed{})^2 + (y_2 - y_1)^2} \quad \dots \text{ (Distance formula)}$$

$$\therefore CD = \sqrt{(-2 - \boxed{})^2 + (-3 - 5)^2}$$

$$\therefore CD = \sqrt{\boxed{} + 64} \quad \therefore CD = \sqrt{\boxed{}}.$$

Activity :

Let $C(3, 5) \equiv (x_1, y_1)$, $D(-2, -3) \equiv (x_2, y_2)$

$$CD = \sqrt{(x_2 - \boxed{x_1})^2 + (y_2 - y_1)^2} \quad \dots \text{ (Distance formula)}$$

$$\therefore CD = \sqrt{(-2 - \boxed{3})^2 + (-3 - 5)^2}$$

$$\therefore CD = \sqrt{\boxed{25} + 64} \quad \therefore CD = \sqrt{\boxed{89}}.$$

2. Find x, if distance between L(x, 7) and M(1, 15) is 10.

Solution :

Let $L(x, 7) \equiv (x_1, y_1)$ and $M(1, 15) \equiv (x_2, y_2)$

By distance formula,

$$LM^2 = (x_2 - x_1)^2 + (y_2 - y_1)^2$$

$$\therefore 10^2 = (1 - x)^2 + (15 - 7)^2 \quad \therefore 10^2 = (1 - x)^2 + (8)^2$$

$$\therefore 100 = (x - 1)^2 + 64 \quad \therefore (x - 1)^2 = 100 - 64$$

$$\therefore (x - 1)^2 = 36$$

$$\therefore x - 1 = \pm 6 \quad \dots \text{(Taking square roots of both the sides)}$$

$$\text{Either } x - 1 = 6 \quad \text{or } x - 1 = -6$$

$$\therefore x = 6 + 1 \quad \text{or } x = -6 + 1$$

$$\therefore x = 7 \quad \text{or } x = -5$$

Ans. $x = 7$ or $x = -5$.

-
- 3. Find the centroids of the triangles whose vertices are $(-7, 6)$, $(2, -2)$, $(8, 5)$. (Sept. '21)**

Solution :

Let $A(-7, 6)$, $B(2, -2)$ and $C(8, 5)$.

Suppose the coordinates of point A are (x_1, y_1) , coordinates of point B are (x_2, y_2) and coordinates of point C are (x_3, y_3) .

$$\therefore x_1 = -7, y_1 = 6, x_2 = 2, y_2 = -2, x_3 = 8 \text{ and } y_3 = 5.$$

Let $G(x, y)$ be the centroid.

By centroid formula,

$$x = \frac{x_1 + x_2 + x_3}{3}$$

$$\therefore x = \frac{-7 + 2 + 8}{3}$$

$$\therefore x = \frac{3}{3}$$

$$\therefore x = 1$$

$$y = \frac{y_1 + y_2 + y_3}{3}$$

$$\therefore y = \frac{6 + (-2) + 5}{3}$$

$$\therefore y = \frac{9}{3}$$

$$\therefore y = 3$$

Ans. The coordinates of the centroid are $(1, 3)$.

-
- 4. Show that points $A(-1, -1)$, $B(0, 1)$, $C(1, 3)$ are collinear.**

Proof :

(March '23)

Let $A(-1, -1) \equiv (x_1, y_1)$, $B(0, 1) \equiv (x_2, y_2)$, $C(1, 3) \equiv (x_3, y_3)$

$$\text{Slope of line AB} = \frac{y_2 - y_1}{x_2 - x_1} = \frac{1 - (-1)}{0 - (-1)} = \frac{2}{1} = 2$$



$$\text{Slope of line BC} = \frac{y_3 - y_2}{x_3 - x_2} = \frac{3 - 1}{1 - 0} = \frac{2}{1} = 2$$

Slope of line AB = Slope of line BC and point B is the common point.

$\therefore$ points A(-1, -1), B(0, 1) and C(1, 3) are collinear.

- 5. Write the equations of any two lines which are parallel to X-axis.**
(OEQ)

Solution :

$y = b$ is the form of the line which is parallel to X-axis.

$y = 5$ and $y = -3$ are the two lines which are parallel to X-axis.

[**Note :** Students can write the equations of any two lines of the form $y = b$ of their own choice.]

- 6. Find the slope of the line passing through the points A(2, 3) and B(4, 7).**
(March '20-'24)

Solution :

$$A(2, 3) \equiv (x_1, y_1), B(4, 7) \equiv (x_2, y_2)$$

$$\begin{aligned} \text{Slope of line AB} &= \frac{y_2 - y_1}{x_2 - x_1} \\ &= \frac{7 - 3}{4 - 2} = \frac{4}{2} = 2. \end{aligned}$$

Ans. Slope of line AB is 2.

Assignment

6.3

- 1.** To find the distance between the points P(6, -6) and Q(3, -7) complete the following activity.
(Sept. '21)

Activity :

$$\text{Let } P(6, -6) \equiv (x_1, y_1), Q(3, -7) \equiv (x_2, y_2)$$

By distance formula,

$$\begin{aligned} d(P, Q) &= \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2} \\ &= \sqrt{(3 - 6)^2 + \left(-7 - \boxed{}\right)^2} \end{aligned}$$

$$= \sqrt{\left(\boxed{}\right)^2 + (-1)^2}$$

$$= \sqrt{\boxed{} + 1}$$

$$\therefore d(P, Q) = \sqrt{\boxed{}}$$

2. Complete the following activity to show that the line joining the points A(4, 8) and B(5, 5) is parallel to the line joining the points C(2, 4) and D(1, 7).

Activity : A(4, 8), B(5, 5), C(2, 4) and D(1, 7).

$$\text{Slope of line AB} = \frac{5-8}{5-4} = \frac{\boxed{}}{1} = \boxed{}$$

$$\text{Slope of line CD} = \frac{7-4}{1-2} = \frac{3}{\boxed{}} = \boxed{}$$

$$\therefore \text{slope of line AB} = \text{slope of line CD}$$

Parallel lines have equal slopes.

$$\therefore \text{line AB} \parallel \text{line CD.}$$

3. If the slope of the line passing through B(k, -5) and C(1, 2) is 7, then find the value of k.
4. Find the distance between the points P(-1, 1) and Q(5, -7).
(July '22)
5. Find the point on X-axis equidistant from (-2, 5) and (2, -3).
6. Prove that point P(-4, 2) lies on the line joining A(-4, 6) and B(-4, -6).
7. Point P is the centre of the circle and AB is a diameter. Find the coordinates of point B, if coordinates of points A and P are (2, -3) and (-2, 0) respectively.
8. A line intersects the Y-axis and X-axis at the points P and Q respectively. If (2, -5) is the midpoint of PQ, then find the coordinates of P and Q.



9. Find the slope of (i) A(2, 5) and B(4, -1) (ii) E(-4, -2) and F(6, 3).
(Nov. '20; July '23)

10. Verify whether the points (i) A(1, -3), B(2, -5), C(-4, 7) and (ii) A(1, 3), B(2, 5), C(-4, 7) are collinear or not.

(March '19 - '25)

Answers :

1. $\boxed{-6}$, $\boxed{-3}$, $\boxed{9}$, $\boxed{10}$

2. $\boxed{-3}$, $\boxed{-3}$, $\boxed{-1}$, $\boxed{-3}$ 3. $k=0$ 4. 10 5. $(-2, 0)$

7. $(-6, 3)$ 8. P(0, -10) and Q(4, 0) 9. (i) $m = -3$ (ii) $m = \frac{1}{2}$

10. (i) Points are collinear (ii) Points are collinear.

6.4 3 marks each

1. Complete the following activity to find the coordinates of point P which divides seg AB in the ratio 3 : 1 where A(4, -3) and B(8, 5).
(March '22)

Activity :



By section formula,

$$x = \frac{mx_2 + nx_1}{\boxed{}}$$

$$y = \frac{\boxed{}}{m+n}$$

$$\therefore x = \frac{3 \times 8 + 1 \times 4}{3 + 1}$$

$$y = \frac{3 \times 5 + 1 \times (-3)}{3 + 1}$$

$$= \frac{\boxed{} + 4}{4}$$

$$= \frac{\boxed{} - 3}{4}$$

$$\therefore x = \boxed{}$$

$$y = \boxed{}.$$

Activity :

By section formula,

$$x = \frac{mx_2 + nx_1}{m+n}$$

$$\therefore x = \frac{3 \times 8 + 1 \times 4}{3+1}$$

$$= \frac{24 + 4}{4}$$

$$\therefore x = 7$$

$$y = \frac{my_2 + ny_1}{m+n}$$

$$y = \frac{3 \times 5 + 1 \times (-3)}{3+1}$$

$$= \frac{15 - 3}{4}$$

$$\therefore y = 3$$

2. A(h, -6), B(2, 3) and C(-6, k) are the coordinates of vertices of triangle whose centroid is G(1, 5). Complete the following activity to find the values of h and k.

A(h, -6), B(2, 3), C(-6, k) and G(1, 5).

By centroid formula,

$$\therefore 1 = \frac{h+2+(-6)}{3}$$

$$\therefore 3 = h - \square$$

$$\therefore h = 3 + \square$$

$$\therefore h = \square$$

$$\therefore 5 = \frac{-6+3+k}{3}$$

$$\therefore 15 = \square + k$$

$$\therefore k = 15 + \square$$

$$\therefore k = \square$$

Activity : A(h, -6), B(2, 3), C(-6, k) and G(1, 5).

By centroid formula,

$$\therefore 1 = \frac{h+2+(-6)}{3}$$

$$\therefore 3 = h - 4$$

$$\therefore h = 3 + 4$$

$$\therefore h = 7$$

$$\therefore 5 = \frac{-6+3+k}{3}$$

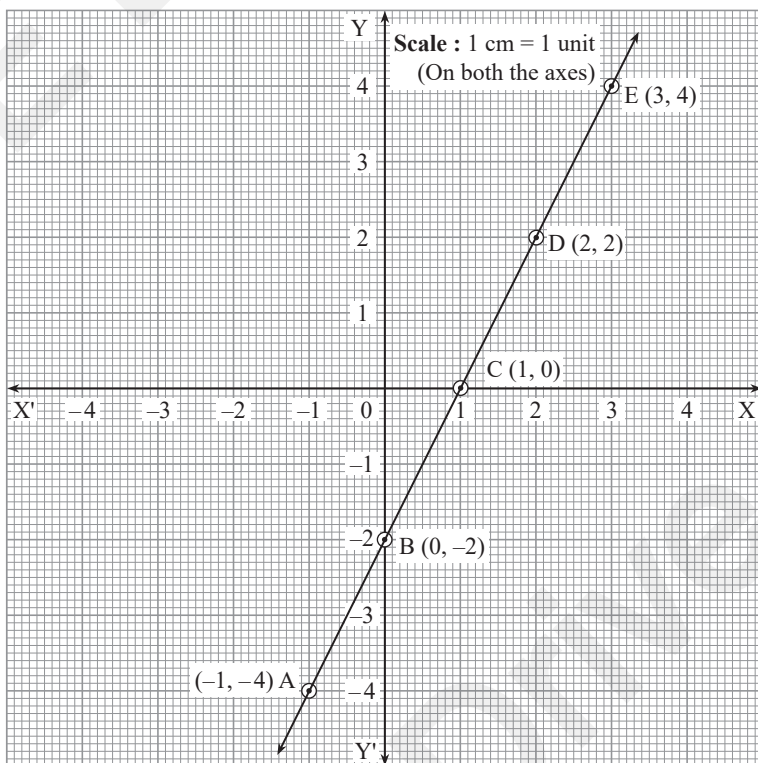
$$\therefore 15 = -3 + k$$

$$\therefore k = 15 + 3$$

$$\therefore k = 18$$



3. Complete the table below the graph with the help of the following graph.



Sr. No.	First point	Second point	Coordinates of first point (x_1, y_1)	Coordinates of second point (x_2, y_2)	$\frac{y_2 - y_1}{x_2 - x_1}$
1	C	E	(1, 0)	(3, 4)	=
2	A	B	(-1, -4)	(0, -2)	=
3	B	D	(0, -2)	(2, 2)	=

Write your observation from the table.

Solution :

Sr. No.	First point	Second point	Coordinates of first point (x_1, y_1)	Coordinates of second point (x_2, y_2)	$\frac{y_2 - y_1}{x_2 - x_1}$
1	C	E	(1, 0)	(3, 4)	$\frac{4}{2} = 2$
2	A	B	(-1, -4)	(0, -2)	$\frac{2}{1} = 2$
3	B	D	(0, -2)	(2, 2)	$\frac{4}{2} = 2$

The slopes of lines CE, AB and BD are the same.

4. Show that the points (2, 0), (-2, 0) and (0, 2) are the vertices of a triangle. Also state with reason the type of the triangle.

(March '19)

Proof : Let $A(2, 0) \equiv (x_1, y_1)$, $B(-2, 0) \equiv (x_2, y_2)$ and $C(0, 2) \equiv (x_3, y_3)$.

By distance formula,

$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2} = \sqrt{(-2 - 2)^2 + (0 - 0)^2} \\ = \sqrt{(-4)^2 + (0)^2} = \sqrt{16} = 4$$

Similarly, $BC = 2\sqrt{2}$ and $AC = 2\sqrt{2}$

For any triangle, sum of two sides will always be greater than the third side.

$$\therefore 2\sqrt{2} + 2\sqrt{2} > 4$$

$$\text{i.e. } 4\sqrt{2} > 4 \text{ and also } 4 + 2\sqrt{2} > 2\sqrt{2}$$

$\therefore$ A, B and C are the vertices of a triangle.

Now,

$$BC^2 + AC^2 = (2\sqrt{2})^2 + (2\sqrt{2})^2$$

$$\therefore BC^2 + AC^2 = 8 + 8$$



$$\therefore BC^2 + AC^2 = 16 \quad \dots (1)$$

$$AB^2 = 4^2$$

$$\therefore AB^2 = 16 \quad \dots (2)$$

$$\therefore BC^2 + AC^2 = AB^2 \quad \dots [\text{From (1) and (2)}]$$

$\therefore$ by converse of Pythagoras theorem,

$\triangle ABC$ is a right angled triangle and also $BC = AC$.

$\therefore \triangle ABC$ is an isosceles right angled triangle.

- 5. (0, 4); (0, 0) and (3, 0) are the vertices of the triangle. Find the perimeter of the triangle.**

Solution :

Let A(0, 4), B(0, 0), and C(3, 0).

By distance formula,

$$AB = \sqrt{(0-0)^2 + (0-4)^2} = \sqrt{0^2 + 4^2} = \sqrt{16} = 4$$

$$BC = \sqrt{(3-0)^2 + (0-0)^2} = \sqrt{3^2 + 0^2} = \sqrt{9} = 3$$

$$AC = \sqrt{(3-0)^2 + (0-4)^2} = \sqrt{3^2 + (-4)^2} = \sqrt{9 + 16} = \sqrt{25} = 5$$

$$\text{Perimeter of } \triangle ABC = AB + BC + AC$$

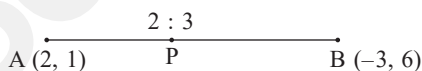
$$= 4 + 3 + 5$$

$$= 12$$

Ans. The perimeter of $\triangle ABC$ is **12**.

- 6. If point P divides the seg AB joining the points A(2, 1) and B(-3, 6) in the ratio 2 : 3, then determine whether the point P lies on the line $x - 5y + 15 = 0$ or not. (Sept. '21)**

Solution :



(Figure for reference only)

Let $A(2, 1) \equiv (x_1, y_1)$, $B(-3, 6) \equiv (x_2, y_2)$

$$m : n = 2 : 3$$

Let P(x, y) divide the join of A and B in the ratio 2 : 3

∴ by section formula,

$$\begin{aligned}x &= \frac{mx_2 + nx_1}{m+n} \\&= \frac{2(-3) + 3(2)}{2+3} \\&= \frac{-6+6}{5} \\&= \frac{0}{5} \\&= 0\end{aligned}$$

$$\begin{aligned}y &= \frac{my_2 + ny_1}{m+n} \\&= \frac{2(6) + 3(1)}{2+3} \\&= \frac{12+3}{5} \\&= \frac{15}{5} \\&= 3\end{aligned}$$

∴ P(0, 3)

Substituting $x=0$ and $y=3$ in LHS of the equation $x-5y+15=0$, we get,

$$\begin{aligned}\text{LHS} &= x - 5y + 15 \\&= 0 - 5(3) + 15 \\&= 0 - 15 + 15 \\&= 0 \\&= \text{RHS}\end{aligned}$$

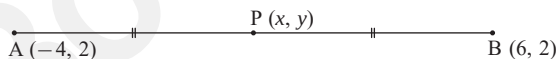
The coordinates of point P(0, 3) satisfy the equation $x-5y+15=0$. Hence point **P(0, 3)** lies on the line $x-5y+15=0$.

Assignment

6.4

- Find the coordinates of point P, where P is the midpoint of line segment AB with A(-4, 2) and B(6, 2). (March '23)

Activity :



Suppose $(-4, 2) = (x_1, y_1)$ and $(6, 2) = (x_2, y_2)$ and coordinates of P are (x, y) .

According to midpoint formula,

$$x = \frac{x_1 + x_2}{2} = \frac{\boxed{} + 6}{2} = \frac{\boxed{}}{2} = \boxed{}$$



$$y = \frac{y_1 + y_2}{2} = \frac{2 + \boxed{}}{2} = \frac{4}{2} = \boxed{}$$

∴ coordinates of midpoint P, are $\boxed{}$.

2. If A(-2, -1), B(p, 0), C(4, q) and D(1, 2) are the vertices of a parallelogram, then complete the following activity to find the values of p and q.

Diagonals of parallelogram bisect each other.

∴ midpoint of AC = Midpoint of BD

By midpoint formula,

$$\therefore \left(\frac{-2+4}{2}, \frac{-1+q}{2} \right) = \left(\frac{p+1}{2}, \frac{0+\boxed{}}{2} \right)$$

$$\therefore \left(\boxed{}, \frac{-1+q}{2} \right) = \left(\frac{p+1}{2}, 1 \right)$$

$$\therefore \frac{p+1}{2} = \boxed{} \text{ and } \frac{-1+q}{2} = \boxed{}$$

∴ on solving, we get,

$$p = \boxed{} \text{ and } q = \boxed{}.$$

3. A(3, 3), B(6, y), C(x, 7) and D(5, 6) are the vertices of a parallelogram. Find the values of x and y.
4. Find the coordinates of centroid of the triangle whose vertices are (4, 7), (8, 4), (7, 11). (July '22)
5. Find the ratio in which point P(k, 7) divides the segment joining A(8, 9) and B(1, 2). Also find k.
6. Point A lies on the line joining P(6, -6) and Q(-4, -1) such that $\frac{PA}{PQ} = \frac{2}{5}$. Line $3x + k(y+1) = 0$ passes through point A. Find the value of k.
7. Show that the point (0, 9) is equidistant from the points (-4, 1) and (4, 1).

8. If point $P(x, y)$ is equidistant from $A(5, 1)$ and $B(-1, 5)$, then prove $3x = 2y$.
9. Determine whether the points are collinear or not :
 $A(1, -3)$, $B(2, -5)$, $C(-4, 7)$. (March '24)
10. $A(3, 5)$ and $B(7, 9)$. Point Q divides seg AB in the ratio $2 : 3$. Find the coordinates of point Q . (July '23)
11. Using slope concept, determine whether $L(2, 5)$, $M(3, 3)$ and $N(5, 1)$ are collinear or not.
12. Points $A(1, 2)$, $B\left(\frac{1}{2}, 3\right)$ and $C(0, k)$ are collinear.
- Find the slope of AB .
 - Find the slope of BC .
 - Thus determine the value of k .

Answers :

1. -4 , 2 , 1 , 2 , 2 , $(1, 2)$

2. 2 , 1 , 1 , 1 , 1 , 3

3. $x = 8, y = 4$ 4. $\left(\frac{19}{3}, \frac{22}{3}\right)$ 5. $2 : 5, k = 6$ 6. $k = 2$

9. Points are collinear 10. $\left(\frac{23}{5}, \frac{33}{5}\right)$

11. Points are not collinear.

12. (i) -2 (ii) $2(3 - k)$ (iii) $k = 4$.

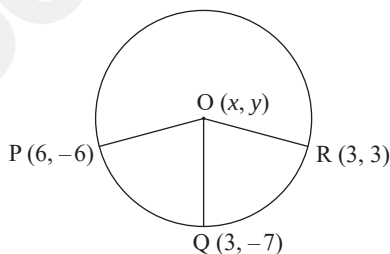
CHALLENGING QUESTIONS

6.5 4 marks each

1. Find the coordinates of the centre of the circle passing through the points $P(6, -6)$, $Q(3, -7)$ and $R(3, 3)$.



Solution :



(Fig. : Just for understanding)

$P(6, -6) \equiv (x_1, y_1)$, $Q(3, -7) \equiv (x_2, y_2)$ and $R(3, 3) \equiv (x_3, y_3)$.

Let $O(x, y)$ be the centre of the circle passing through the points P, Q and R.

$OP = OQ$... (Radii of the same circle)

$$\sqrt{(x - x_1)^2 + (y - y_1)^2} = \sqrt{(x - x_2)^2 + (y - y_2)^2}$$

$$\sqrt{(x - 6)^2 + [y - (-6)]^2} = \sqrt{(x - 3)^2 + [y - (-7)]^2}$$

Squaring both the sides, we get,

$$(x - 6)^2 + (y + 6)^2 = (x - 3)^2 + (y + 7)^2$$

$$\therefore x^2 - 12x + 36 + y^2 + 12y + 36 = x^2 - 6x + 9 + y^2 + 14y + 49$$

$$\therefore -12x + 12y + 72 = -6x + 14y + 58$$

$$\therefore -12x + 6x + 12y - 14y = 58 - 72$$

$$\therefore -6x - 2y = -14$$

$$\therefore -2(3x + y) = -14$$

$$\therefore 3x + y = \frac{-14}{-2}$$

$$\therefore 3x + y = 7 \quad \dots (1)$$

$OQ = OR$... (Radii of the same circle)

$$\sqrt{(x - x_2)^2 + (y - y_2)^2} = \sqrt{(x - x_3)^2 + (y - y_3)^2}$$

$$\sqrt{(x - 3)^2 + [y - (-7)]^2} = \sqrt{(x - 3)^2 + (y - 3)^2}$$

Squaring both the sides, we get,

$$(x - 3)^2 + (y + 7)^2 = (x - 3)^2 + (y - 3)^2$$

$$\therefore (y + 7)^2 = (y - 3)^2$$

$$\therefore y^2 + 14y + 49 = y^2 - 6y + 9$$

$$\therefore 14y + 6y = 9 - 49$$

$$\therefore 20y = -40$$

$$\therefore y = \frac{-40}{20}$$

$$\therefore y = -2$$

Substituting $y = -2$ in (1), we get,

$$3x + y = 7$$

$$\therefore 3x - 2 = 7$$

$$\therefore 3x = 7 + 2 \quad \therefore 3x = 9$$

$$\therefore x = \frac{9}{3} \quad \therefore x = 3$$

Ans. Coordinates of the centre of the circle are **(3, -2)**.

2. Points A(-1, y) and B(5, 7) lie on the circle having centre O(2, -3y). Find the value of y and hence find the radius of the circle.

Solution :

$$\text{Let } A(-1, y) \equiv (x_1, y_1)$$

$$B(5, 7) \equiv (x_2, y_2)$$

$$O(2, -3y) \equiv (x_3, y_3)$$

OA = OB ... (Radii of the same circle)

$\therefore$ by distance formula,

(Fig. : Just for understanding)

$$\sqrt{(x_3 - x_1)^2 + (y_3 - y_1)^2} = \sqrt{(x_3 - x_2)^2 + (y_3 - y_2)^2}$$

$$\therefore \sqrt{[2 - (-1)]^2 + [-3y - y]^2} = \sqrt{(2 - 5)^2 + (-3y - 7)^2}$$

Taking squares of both the sides, we get,

$$(2 + 1)^2 + (-4y)^2 = (-3)^2 + (-3y - 7)^2$$

$$\therefore 9 + 16y^2 = 9 + [(-3y)^2 - 2(-3y)(7) + (7)^2]$$

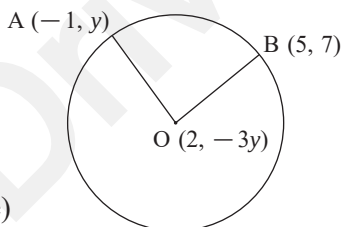
$$\therefore 9 + 16y^2 = 9 + 9y^2 + 42y + 49$$

$$\therefore 16y^2 - 9y^2 - 42y - 49 = 0$$

$$\therefore 7y^2 - 42y - 49 = 0$$

$$\therefore 7(y^2 - 6y - 7) = 0$$

$$\therefore y^2 - 6y - 7 = 0$$



$$\therefore y^2 - 7y + y - 7 = 0$$

$$\therefore y(y - 7) + y(y - 7) = 0$$

$$\therefore (y - 7)(y + 1) = 0$$

$$\text{Either } y - 7 = 0 \quad \text{or} \quad y + 1 = 0$$

$$\therefore y = 7 \quad \text{or} \quad y = -1$$

$$\text{Case 1 : if } y = 7 \quad \text{then} \quad O(2, -21)$$

$$\text{Case 2 : if } y = -1 \quad \text{then} \quad O(2, 3)$$

Considering case 1 :

$$\begin{aligned} \text{Radius OB} &= \sqrt{(x_3 - x_2)^2 + (y_3 - y_2)^2} \\ &= \sqrt{(2 - 5)^2 + (-21 - 7)^2} \\ &= \sqrt{(-3)^2 + (-28)^2} \\ &= \sqrt{9 + 784} \\ &= \sqrt{793} \\ &\approx 28.16 \\ &\approx 28.2 \text{ units} \end{aligned}$$

Considering case 2 :

$$\begin{aligned} \text{Radius OB} &= \sqrt{(x_3 - x_2)^2 + (y_3 - y_2)^2} \\ &= \sqrt{(2 - 5)^2 + (3 - 7)^2} \\ &= \sqrt{(-3)^2 + (-4)^2} \\ &= \sqrt{9 + 16} \\ &= \sqrt{25} \\ &= 5 \text{ units} \end{aligned}$$

Ans. $y = 7$ or $y = 1$ and radius = **28.2 units** or radius = **5 units**.

- 3. P is the midpoint of the line segment joining points A(-10, 4) and B(-2, 0). Point P also lies on line segment joining points C(-9, -4) and D(-4, y). In what ratio does P divide the segment CD? Also, find the value of y.**

Solution :

$$A(-10, 4) \equiv (x_1, y_1), B(-2, 0) \equiv (x_2, y_2)$$

Let $P(x, y)$

P is the midpoint of AB .

$\therefore$ by midpoint formula,

$$x = \frac{x_1 + x_2}{2} \text{ and } y = \frac{y_1 + y_2}{2}$$

$$\therefore x = \frac{-10 + (-2)}{2} \quad \text{and} \quad y = \frac{4 + 0}{2}$$

$$\therefore x = \frac{-12}{2} \quad y = \frac{4}{2}$$

$$\therefore x = -6 \quad \text{and} \quad y = 2$$

$$\therefore P(-6, 2)$$

Let $P(-6, 2)$ divide seg CD in the ratio $m : n$.

$$x = \frac{mx_2 + nx_1}{m + n} \quad \dots \text{ (Section formula)}$$

$$-6 = \frac{m(-4) + n(-9)}{m + n}$$

$$\therefore -6(m + n) = -4m - 9n$$

$$\therefore -6m - 6n = -4m - 9n$$

$$\therefore -6m + 4m = -9n + 6n$$

$$\therefore -2m = -3n$$

$$\therefore \frac{m}{n} = \frac{-3}{-2}$$

$$\therefore \frac{m}{n} = \frac{3}{2}$$

$$\therefore m : n = 3 : 2$$

$$\text{Now, } 2 = \frac{3y + 2(-4)}{3 + 2}$$

$$\therefore 2 = \frac{3y - 8}{5}$$

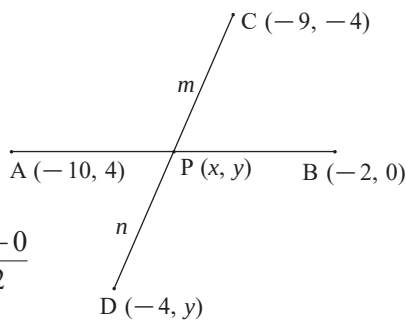
$$\therefore 10 = 3y - 8 \quad \therefore 3y = 10 + 8$$

$$\therefore 3y = 18$$

$$\therefore y = \frac{18}{3}$$

$$\therefore y = 6$$

Ans. P divides seg CD in the ratio **3 : 2** and $y = 6$.



(Fig. : Just for understanding)



4. Prove that quadrilateral formed by joining the points (1, 1), (−1, 5), (7, 9) and (9, 5) in order is a rectangle. Also, find the area of this rectangle.

Solution :

$$\text{Let } A(1, 1) \equiv (x_1, y_1),$$

$$B(-1, 5) \equiv (x_2, y_2),$$

$$C(7, 9) \equiv (x_3, y_3),$$

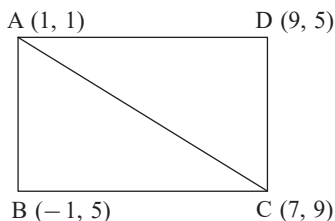
$$D(9, 5) \equiv (x_4, y_4).$$

by distance formula,

$$\begin{aligned} AB &= \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2} \\ &= \sqrt{(-1 - 1)^2 + (5 - 1)^2} \\ &= \sqrt{(-2)^2 + (4)^2} \\ &= \sqrt{4 + 16} \\ &= \sqrt{20} \\ &= 2\sqrt{5}. \end{aligned}$$

$$\begin{aligned} BC &= \sqrt{(x_3 - x_2)^2 + (y_3 - y_2)^2} \\ &= \sqrt{[7 - (-1)]^2 + (9 - 5)^2} \\ &= \sqrt{(7 + 1)^2 + (4)^2} \\ &= \sqrt{(8)^2 + (4)^2} \\ &= \sqrt{64 + 16} \\ &= \sqrt{80} \\ &= 4\sqrt{5}. \end{aligned}$$

$$\begin{aligned} CD &= \sqrt{(x_4 - x_3)^2 + (y_4 - y_3)^2} \\ &= \sqrt{(9 - 7)^2 + (5 - 9)^2} \\ &= \sqrt{(2)^2 + (-4)^2} \\ &= \sqrt{4 + 16} \\ &= \sqrt{20} \\ &= 2\sqrt{5}. \end{aligned}$$



(Fig. : Just for understanding)

$$\begin{aligned}
 AD &= \sqrt{(x_4 - x_1)^2 + (y_4 - y_1)^2} \\
 &= \sqrt{(9 - 1)^2 + (5 - 1)^2} \\
 &= \sqrt{(8)^2 + (4)^2} \\
 &= \sqrt{64 + 16} \\
 &= \sqrt{80} \\
 &= 4\sqrt{5}.
 \end{aligned}$$

In $\square ABCD$,

$$AB = CD$$

$$BC = AD$$

$\therefore \square ABCD$ is a parallelogram

... (A quadrilateral is a parallelogram if its opposite sides are equal)

$$\begin{aligned}
 AC &= \sqrt{(x_3 - x_1)^2 + (y_3 - y_1)^2} \\
 &= \sqrt{(7 - 1)^2 + (9 - 1)^2} \\
 &= \sqrt{(6)^2 + (8)^2} \\
 &= \sqrt{36 + 64} \\
 &= \sqrt{100} \\
 &= 10.
 \end{aligned}$$

$$(AC)^2 = (10)^2 = 100$$

$$\begin{aligned}
 (AB)^2 + (BC)^2 &= (2\sqrt{5})^2 + (4\sqrt{5})^2 \\
 &= 20 + 80 \\
 &= 100 \\
 &= AC^2
 \end{aligned}$$

$$\therefore AB^2 + BC^2 = AC^2$$

$\therefore$ by converse of Pythagoras theorem,

$$\angle ABC = 90^\circ.$$

$\therefore$ in parallelogram $ABCD$,

$$\angle ABC = 90^\circ$$

$\therefore \square ABCD$ is a rectangle

... (A parallelogram is a rectangle if one of its angles is a right angle)



$$\begin{aligned}
 \text{Area of rectangle} &= \text{length} \times \text{breadth} \\
 &= BC \times AB \\
 &= 4\sqrt{5} \times 2\sqrt{5} \\
 &= 40 \text{ sq units.}
 \end{aligned}$$

Ans. Area of rectangle is **40 sq units**.

Assignment

6.5

- Find the coordinates of the point Q on the X-axis, which lies on the perpendicular bisector of the line segment joining the points A(-5, -2) and B(4, -2). Name the type of triangle formed by the points Q, A and B.
- Prove that (a, a) , $(-a, -a)$ and $(-\sqrt{3}a, \sqrt{3}a)$ are the vertices of an equilateral triangle. Also find the area of the triangle.
- $(-4, 3)$ and $(4, 3)$ are the vertices of an equilateral triangle. Origin lies in the interior of the triangle. Find the coordinates of the third vertex.
- Show that $\triangle ABC$ with A(-2, 0), B(2, 0), C(0, 2) and $\triangle PQR$ with P(-4, 0), Q(4, 0), R(0, 4) are similar triangles.
- Point P divides the line segment joining the points A(2, 1) and B(5, -8) such that $\frac{AP}{AB} = \frac{1}{3}$. If P lies on the line $2x - y + k = 0$, find the value of k.
- Find the coordinates of the points which divides the segment joining A(-2, 2) and B(2, 8) into four equal parts.
- If P(9a - 2, -b) divides the line segment joining A(3a + 1, -3) and B(8a, 5) in the ratio 3 : 1, then find the values of a, b and $a^2 + b^2$.
- Find the coordinates of the centroid of the triangle, if points D(-7, 6), E(8, 5) and F(2, -2) are the midpoints of the sides of that triangle. **(March '22)**
- The coordinates of the points A and B are (-2, -2) and (2, -4) respectively. If P lies on segment AB such that A-P-B and $AP = \frac{3}{7} AB$, then find the coordinates of point P.

Answers :

1. $(-0.5, 0)$, Isosceles triangle 2. $2\sqrt{3}a^2$ sq units
3. $(0, 3 - 4\sqrt{3})$ 5. $k = -8$ 6. $\left(-1, \frac{7}{2}\right)\left(1, \frac{13}{2}\right)(0, 5)$
7. $a = 1, b = -3, a^2 + b^2 = 10$ 8. $(1, 3)$ 9. $\left(\frac{-2}{7}, \frac{-20}{7}\right)$.

Based on the Board's Evaluation Pattern

Vikas

MATHEMATICS PRACTICE BOOK

(Part I & II)

(Internal Evaluation Handbook)

With Solution on QR Code

Based on the Textbook And
Latest Activity Sheet Format

**MATHEMATICS
WORKBOOK**

Part II

Standard X

**Ideal for the Internal Evaluation
as well as Board's Examination**

Available from all booksellers



**Question
Set
7**

TRIGONOMETRY

POINTS TO REMEMBER

1. Trigonometric ratios of an acute angle θ in a right angled triangle :

$$(i) \sin \theta = \frac{\text{side opposite of angle } \theta}{\text{hypotenuse}}$$

$$(ii) \cos \theta = \frac{\text{side adjacent of angle } \theta}{\text{hypotenuse}}$$

$$(iii) \tan \theta = \frac{\text{side opposite of angle } \theta}{\text{side adjacent of angle } \theta}$$

$$(iv) \cot \theta = \frac{\text{side adjacent of angle } \theta}{\text{side opposite of angle } \theta}$$

$$(v) \sec \theta = \frac{\text{hypotenuse}}{\text{side adjacent of angle } \theta}$$

$$(vi) \operatorname{cosec} \theta = \frac{\text{hypotenuse}}{\text{side opposite of angle } \theta}.$$

2. Relations among trigonometric ratios :

$$(i) \tan \theta = \frac{\sin \theta}{\cos \theta} \quad (ii) \cot \theta = \frac{\cos \theta}{\sin \theta} \quad (iii) \sec \theta = \frac{1}{\cos \theta}$$

$$(iv) \operatorname{cosec} \theta = \frac{1}{\sin \theta} \quad (v) \cot \theta = \frac{1}{\tan \theta}.$$

3. Trigonometric identities : (*Important*)

$$(i) \sin^2 \theta + \cos^2 \theta = 1$$

$$(ii) 1 + \tan^2 \theta = \sec^2 \theta$$

$$(iii) 1 + \cot^2 \theta = \operatorname{cosec}^2 \theta.$$

4. Trigonometric ratios of complementary angles :

$$(i) \sin (90 - \theta) = \cos \theta$$

$$(ii) \cos (90 - \theta) = \sin \theta$$

$$(iii) \tan (90 - \theta) = \cot \theta$$

$$(iv) \cot (90 - \theta) = \tan \theta$$

$$(v) \sec (90 - \theta) = \operatorname{cosec} \theta$$

$$(vi) \operatorname{cosec} (90 - \theta) = \sec \theta.$$

5. The table for the values of trigonometric ratios of angles :

Trigonometric ratios	Angle (θ)				
	0°	30°	45°	60°	90°
$\sin \theta$	0	$\frac{1}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{\sqrt{3}}{2}$	1
$\cos \theta$	1	$\frac{\sqrt{3}}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{1}{2}$	0
$\tan \theta$	0	$\frac{1}{\sqrt{3}}$	1	$\sqrt{3}$	Not defined
$\cot \theta$	Not defined	$\sqrt{3}$	1	$\frac{1}{\sqrt{3}}$	0
$\sec \theta$	1	$\frac{2}{\sqrt{3}}$	$\sqrt{2}$	2	Not defined
$\operatorname{cosec} \theta$	Not defined	2	$\sqrt{2}$	$\frac{2}{\sqrt{3}}$	1

Application of trigonometry :

Trigonometric ratios can be used in estimating the distance between two inaccessible points or heights of points too high to determine by direct measurement.

For the purpose of computation we shall treat ‘trees’, ‘hills’ or ‘towers’ as ideal vertical objects. We shall represent them in diagram, by segments perpendicular to the ground. Unless there is a mention of observer’s height, we shall always regard observers eye to be in the plane of the ground.

Let us study a few related terms.

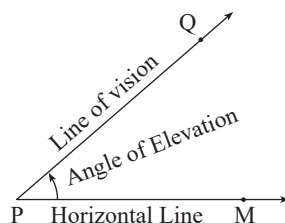
(i) Line of vision :



If the observer at the location P is looking at an object Q, then the line PQ is called **line of vision**.

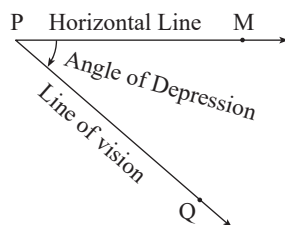
(ii) Angle of elevation :

If P and Q are two points such that Q is at higher level than P and PM is the horizontal line through P, then $\angle MPQ$ is the **angle of elevation** of Q with respect to line PM.



(iii) Angle of depression :

If P and Q are two points such that P is at higher level than Q and PM is the horizontal line through P, then $\angle MPQ$ is the **angle of depression** of Q with respect to line PM.



7.1 1 mark each

Q. Choose the correct alternative from those given below each question :

1. $\sin \theta \operatorname{cosec} \theta = ?$

(July '19; March '24)

- (A) 1 (B) 0 (C) $\frac{1}{2}$ (D) $\sqrt{2}$

Ans. (A)

[**Hint :** $\operatorname{cosec} \theta = \frac{1}{\sin \theta}$.]

2. If $\cos \theta = \frac{4}{5}$, then $\sin \theta = ?$

- (A) $\frac{5}{3}$ (B) $\frac{3}{5}$ (C) $\frac{4}{5}$ (D) $\frac{5}{4}$

Ans. (B)

[**Hint :** Use $\sin^2 \theta + \cos^2 \theta = 1$.]

3. $\sin 30^\circ - \cos 60^\circ = ?$

- (A) 1 (B) 0 (C) -1 (D) $\frac{1}{2}$

Ans. (B)

[**Hint :** $\sin 30^\circ = \frac{1}{2}$, $\cos 60^\circ = \frac{1}{2}$.]

4. If $\tan x = 3 \cot x$, then the value of x ?

- (A) 60° (B) 45° (C) 30° (D) 15°

Ans. (A)

[**Hint :** Substitute $\cot x = \frac{1}{\tan x}$ and simplify. We get,
 $\tan x = \sqrt{3} \quad \therefore x = 60^\circ$.]

5. The value of $2 \tan 45^\circ - 2 \sin 30^\circ$ is ...

- (A) 2 (B) 1 (C) $\frac{1}{2}$ (D) $\frac{3}{4}$

(March '22)

Ans. (B)

$$\left[\text{Hint : } \tan 45^\circ = 1, \sin 30^\circ = \frac{1}{2} \right]$$

6. If $\tan \theta = \sqrt{3}$, then the value of θ is ...

(July '22)

- (A) 60° (B) 30° (C) 90° (D) 45°

Ans. (A)

$$[\text{Hint : } \tan 60^\circ = \sqrt{3}.]$$

7. $\sec 45^\circ = \dots$

- (A) $\frac{1}{\sqrt{2}}$ (B) $\sqrt{2}$ (C) $\frac{\sqrt{3}}{2}$ (D) $\frac{2}{\sqrt{3}}$

Ans. (B)

$$[\text{Hint : } \sec 45^\circ = \sqrt{2}.]$$

8. The ratio of the height of a tower and the length of its shadow on the ground is $\sqrt{3} : 1$. What is the angle of elevation of the sun?

- (A) 60° (B) 30° (C) 90° (D) 45°

Ans. (A)

$$\left[\text{Hint : } \tan \theta = \frac{\sqrt{3}}{1} \text{ and } \tan 60^\circ = \frac{\sqrt{3}}{1} \right]$$

9. If $\cos \theta + \sec \theta = \frac{5}{2}$, then $\cos^2 \theta + \sec^2 \theta = ?$

- (A) $\frac{29}{4}$ (B) $\frac{21}{4}$ (C) $\frac{17}{4}$ (D) $\frac{33}{4}$

Ans. (C)

$$[\text{Hint : Use } (\cos \theta + \sec \theta)^2 = \cos^2 \theta + \sec^2 \theta + 2.]$$

Assignment

7.1

Q. Choose the correct alternative answer for each of the following question :

1. $1 + \tan^2 \theta = ?$

(March '19)

- (A) $\cot^2 \theta$ (B) $\operatorname{cosec}^2 \theta$ (C) $\sec^2 \theta$ (D) $\tan^2 \theta$

2. Which of the following options has a value equal to 1?

- (A) $\cot^2 \theta - \operatorname{cosec}^2 \theta$ (B) $\operatorname{cosec}^2 \theta - \cot^2 \theta$
(C) $\operatorname{cosec}^2 \theta + \cot^2 \theta$ (D) $\tan^2 \theta - \sec^2 \theta$



3. If $\cos \theta = \frac{11}{61}$, then $\sec \theta = ?$
 (A) $\frac{60}{61}$ (B) $\frac{60}{11}$ (C) $\frac{11}{61}$ (D) $\frac{61}{11}$
4. If $24 \sin A - 7 \cos A = 0$, then find the value of $\tan A$.
 (A) $\frac{25}{24}$ (B) $\frac{7}{24}$ (C) $\frac{25}{7}$ (D) $\frac{24}{7}$
5. $\frac{1 - \tan^2 45^\circ}{1 + \tan^2 45^\circ}$ is equal to
 (A) $\sin 0^\circ$ (B) $\sin 90^\circ$ (C) $\sin 45^\circ$ (D) $\sin 30^\circ$
6. $\sec^2 60^\circ + \sec 0^\circ = ?$
 (A) 5 (B) 1 (C) 0 (D) 2
7. $\sin 0^\circ \times \sin 45^\circ \times \sin 90^\circ = ?$
 (A) $\frac{1}{2}$ (B) -1 (C) 1 (D) 0
8. If $\tan \theta = \frac{1}{\sqrt{3}}$, then $\sin (90^\circ - \theta) = ?$
 (A) 0 (B) $\frac{1}{2}$ (C) $\frac{\sqrt{3}}{2}$ (D) 1
9. If $\cot^2 \theta = 3$, then $\sin \theta = ?$
 (A) 0 (B) 2 (C) $\frac{1}{2}$ (D) $\frac{3}{2}$
10. If $2 \cos^2 \theta = \frac{1}{2}$, then $\theta = ?$
 (A) 60° (B) 90° (C) 30° (D) 45°

Answers :

1. (C) [Trigonometric identity.] 2. (B) [Trigonometric identity.]
 3. (D) $\left[\sec \theta = \frac{1}{\cos \theta} \right]$ 4. (B) $\left[\tan A = \frac{\sin A}{\cos A} \right]$ 5. (A) [$\tan 45^\circ = 1$.]
 6. (A) [$\sec 60^\circ = 2$, $\sec 0^\circ = 1$.] 7. (D) [$\sin 0^\circ = 0$.]
 8. (C) [$\theta = 30^\circ$.] 9. (C) [$1 + \cot^2 \theta = \operatorname{cosec}^2 \theta$.]
 10. (A) $\left[\cos 60^\circ = \frac{1}{2} \right]$

7.2 1 mark each

1. If $\tan \theta = \sqrt{3}$, then find the value of θ .

(Sept. '21)

Ans.

$$\tan \theta = \sqrt{3}$$

... (Given)

$$\tan 60^\circ = \sqrt{3}$$

$$\therefore \theta = 60^\circ$$

The value of θ is 60° .

2. If $\cos^2 \theta = 0.36$, then find the value of $\sin^2 \theta$.

Ans.

$$\sin^2 \theta + \cos^2 \theta = 1$$

$$\therefore \sin^2 \theta + 0.36 = 1$$

$$\therefore \sin^2 \theta = 1 - 0.36$$

$$\therefore \sin^2 \theta = 0.64$$

The value of $\sin^2 \theta$ is **0.64**.

3. If $3 \sin \theta = 4 \cos \theta$, then find the value of $\tan \theta$.

(Nov. '20)

Ans. $3 \sin \theta = 4 \cos \theta$

$$\therefore \frac{\sin \theta}{\cos \theta} = \frac{4}{3} \quad \therefore \tan \theta = \frac{4}{3}$$

$$\tan \theta = \frac{4}{3}$$

4. If $\sin \theta = \cos \theta$, then find the measure of θ .

(March '22)

Ans.

$$\sin \theta = \cos \theta$$

$$\sin \theta = \cos \theta \quad \dots (1)$$

$$\therefore \frac{\sin \theta}{\cos \theta} = 1$$

$$\sin \theta = \cos (90 - \theta) \quad \dots (2)$$

From (1) and (2),

$$\therefore \tan \theta = 1$$

OR

$$\cos \theta = \cos (90 - \theta)$$

We know that,

$$\therefore \theta = 90 - \theta$$

$$\tan 45^\circ = 1$$

$$\therefore 2\theta = 90$$

$$\therefore \theta = 45^\circ$$

$$\therefore \theta = 45^\circ$$

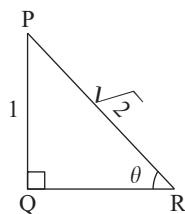


5. In the figure, find the measure of θ .

Ans. $\sin \theta = \frac{PQ}{PR} = \frac{1}{\sqrt{2}}$

We know that, $\sin 45^\circ = \frac{1}{\sqrt{2}}$

$\therefore \theta = 45^\circ$.



6. If $\sin \theta = \frac{1}{2}$, then find the value of θ .

(July '19)

Ans. $\sin \theta = \frac{1}{2}$

We know that, $\sin 30^\circ = \frac{1}{2}$

$\therefore \theta = 30^\circ$

The value of θ is 30° .

7. If $\operatorname{cosec} \theta = \sqrt{5}$, then find the value of $\cot^2 \theta$.

Ans.

$$1 + \cot^2 \theta = \operatorname{cosec}^2 \theta$$

$$\therefore 1 + \cot^2 \theta = (\sqrt{5})^2$$

$$\therefore 1 + \cot^2 \theta = 5$$

$$\therefore \cot^2 \theta = 5 - 1$$

$$\therefore \cot^2 \theta = 4.$$

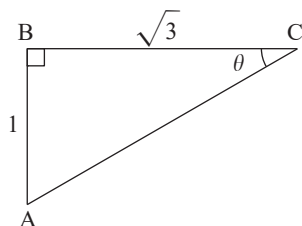
Assignment

7.2

1. In the figure, $\angle ABC = 90^\circ$,

$AB = 1$ and $BC = \sqrt{3}$.

$\angle ACB = \theta$, then what is the value of θ ?



- If $\cos (45^\circ + x) = \sin 30^\circ$, then what is the value of x ?
- Find the value of $\sin^2 90^\circ + \cos^2 90^\circ$.
- If $\sin \alpha = \cos \beta$, then find the value of $\alpha + \beta$.

5. Find the value of $9 \cot^2 \theta - 9 \operatorname{cosec}^2 \theta$.
6. If $\sin \theta = \frac{3}{5}$, then find the value of $\operatorname{cosec} \theta$.
7. What is the value of $2 \operatorname{cosec} 10^\circ \times \sin 10^\circ$?
8. Find the value of $\cot^2 \theta - \frac{1}{\sin^2 \theta}$.

Answers :

1. $\theta = 30^\circ$ 2. $x = 15$ 3. 1 4. $\alpha + \beta = 90^\circ$
 5. -9 6. $\frac{5}{3}$ 7. 2 8. -1.

7.3 2 marks each

1. An observer at a distance of 10 m from the tree, looks at the top of the tree. The angle of elevation is 60° . Complete the activity to find the height of the tree. ($\sqrt{3} = 1.73$) (March '25)

Activity :

In the given figure,

$AB = h$ = height of the tree, distance of the observer from the tree, $BC = 10$ m.

Angle of elevation (θ) = $\angle BCA = 60^\circ$.

$$\tan \theta = \frac{\boxed{}}{BC} \quad \dots (1)$$

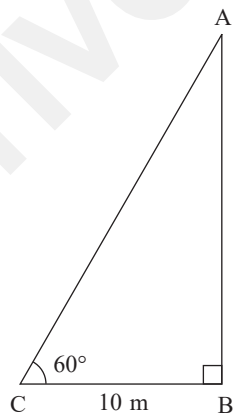
$$\tan 60^\circ = \frac{\boxed{}}{} \quad \dots (2)$$

$$\therefore \frac{AB}{BC} = \sqrt{3} \quad \dots [\text{From (1) and (2)}]$$

$$\therefore AB = \sqrt{3} \times BC = 10\sqrt{3}$$

$$\therefore AB = 10 \times 1.73 = \boxed{}$$

$\therefore$ the height of the tree is $\boxed{}$ m.



Activity :

In the given figure,

$AB = h$ = height of the tree, distance of the observer from the tree, $BC = 10$ m.

Angle of elevation (θ) = $\angle BCA = 60^\circ$.

$$\tan \theta = \frac{\boxed{AB}}{BC} \quad \dots (1)$$

$$\tan 60^\circ = \frac{\boxed{\sqrt{3}}}{BC} \quad \dots (2)$$

$$\therefore \frac{AB}{BC} = \sqrt{3} \quad \dots [\text{From (1) and (2)}]$$

$$\therefore AB = \sqrt{3} \times BC = 10\sqrt{3}$$

$$\therefore AB = 10 \times 1.73 = \boxed{17.3}$$

$$\therefore \text{the height of the tree is } \boxed{17.3} \text{ m.}$$

2. If $7 \sin^2 \theta + 3 \cos^2 \theta = 4$, then prove that $\sin \theta = \frac{1}{2}$.

Proof :

$$7 \sin^2 \theta + 3 \cos^2 \theta = 4$$

$$\therefore 7 \sin^2 \theta + 3(1 - \sin^2 \theta) = 4 \dots (\sin^2 \theta + \cos^2 \theta = 1 \therefore \cos^2 \theta = 1 - \sin^2 \theta)$$

$$\therefore 7 \sin^2 \theta + 3 - 3 \sin^2 \theta = 4$$

$$\therefore 4 \sin^2 \theta + 3 = 4$$

$$\therefore 4 \sin^2 \theta = 4 - 3$$

$$\therefore 4 \sin^2 \theta = 1$$

$$\therefore \sin^2 \theta = \frac{1}{4}$$

$$\therefore \sin \theta = \frac{1}{2} \quad \dots (\text{Taking square roots of both the sides})$$

3. If $\frac{\sin^2 \theta}{7} + \frac{\cos^2 \theta}{7} = \frac{x}{21}$, then find the value of x.

Solution :

$$\frac{\sin^2 \theta}{7} + \frac{\cos^2 \theta}{7} = \frac{x}{21}.$$

Multiplying both the sides term by 21,

$$\frac{\sin^2 \theta}{7} \times 21 + \frac{\cos^2 \theta}{7} \times 21 = \frac{x}{21} \times 21$$

$$\therefore 3 \sin^2 \theta + 3 \cos^2 \theta = x$$

$$\therefore 3 (\sin^2 \theta + \cos^2 \theta) = x$$

$$\therefore 3(1) = x$$

$$\dots (\sin^2 \theta + \cos^2 \theta = 1)$$

$$\therefore x = 3$$

Ans. The value of x is **3**.

4. If $\sec \theta = \frac{25}{7}$, find the value of $\tan \theta$.

[March '19; (July '24-Activity)]

Solution : $\sec^2 \theta = 1 + \tan^2 \theta$

$$\therefore \left(\frac{25}{7} \right)^2 = 1 + \tan^2 \theta$$

$$\therefore \tan^2 \theta = \left(\frac{25}{7} \right)^2 - 1 \quad \therefore \tan^2 \theta = \frac{625}{49} - 1$$

$$\therefore \tan^2 \theta = \frac{625 - 49}{49} \quad \therefore \tan^2 \theta = \frac{576}{49}$$

$$\therefore \tan \theta = \frac{24}{7} \quad \dots (\text{Taking square roots of both the sides})$$

Ans. $\tan \theta = \frac{24}{7}$.

5. If $\tan A + \frac{1}{\tan A} = 2$, then show that $\tan^2 A + \frac{1}{\tan^2 A} = 2$.

Proof : $\tan A + \frac{1}{\tan A} = 2$

$$\therefore \left(\tan A + \frac{1}{\tan A} \right)^2 = (2)^2$$

$$\therefore \tan^2 A + 2 \times \tan A \times \frac{1}{\tan A} + \frac{1}{\tan^2 A} = 4$$

$$\therefore \tan^2 A + 2 + \frac{1}{\tan^2 A} = 4$$

$$\therefore \tan^2 A + \frac{1}{\tan^2 A} = 4 - 2$$

$$\therefore \tan^2 A + \frac{1}{\tan^2 A} = 2.$$



6. A person is standing at a distance of 50 m from a temple looking at its top. The angle of elevation is of 45° . Find the height of the temple. (March '23)

Solution :

In the figure, AB represents the temple.

C is the position of the observer.

$\angle ACB$ is the angle of elevation.

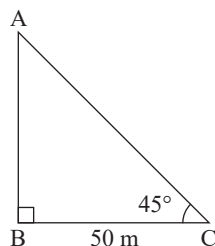
In $\triangle ABC$,

$$\angle ABC = 90^\circ$$

$$\tan \angle ACB = \tan 45^\circ = \frac{AB}{BC}$$

$$\therefore 1 = \frac{AB}{50} \quad \therefore AB = 50 \text{ m}$$

Ans. Height of the temple is **50 m**.



Assignment	7.3
-------------------	------------

1. Complete the following activity and find the value of $6 \tan^2 \theta - \frac{6}{\cos^2 \theta}$.

Activity :

$$\begin{aligned} 6 \tan^2 \theta - \frac{6}{\cos^2 \theta} &= 6 \left(\tan^2 \theta - \frac{1}{\boxed{}} \right) \\ &= 6 \left(\tan^2 \theta - \boxed{} \right) \quad \dots \text{(Writing the reciprocal)} \\ &= 6 \left(\boxed{} \right) = \boxed{}. \end{aligned}$$

2. Complete the following activity to prove that

$$\tan^4 \theta + \tan^2 \theta = \sec^4 \theta - \sec^2 \theta.$$

Activity :

$$\begin{aligned} \text{LHS} &= \tan^4 \theta + \tan^2 \theta \\ &= \tan^2 \theta \left(\boxed{} + 1 \right) \\ &= \tan^2 \theta \times \boxed{} \\ &= \left(\sec^2 \theta - \boxed{} \right) \times \sec^2 \theta \end{aligned}$$

$$= \sec^4 \theta - \boxed{}$$

$$= \text{RHS}$$

3. Complete the following activity to show that,

$$\cot \theta + \tan \theta = \operatorname{cosec} \theta \times \sec \theta.$$

(March '23)

Activity :

$$\text{LHS} = \cot \theta + \tan \theta$$

$$= \frac{\cos \theta}{\sin \theta} + \frac{\sin \theta}{\cos \theta} = \frac{\boxed{} + \boxed{}}{\sin \theta \times \cos \theta}$$

$$= \frac{1}{\sin \theta \times \cos \theta} \quad \dots \left(\boxed{} \right)$$

$$= \frac{1}{\sin \theta} \times \frac{1}{\boxed{}} = \operatorname{cosec} \theta \times \sec \theta$$

$$\text{LHS} = \text{RHS}$$

$$\therefore \cot \theta + \tan \theta = \operatorname{cosec} \theta \times \sec \theta.$$

4. If $\theta = 45^\circ$, then find the value of $\frac{\sin^3 \theta + \cos^3 \theta}{\sin \theta + \cos \theta}$.

5. Find the value of x : if $x \cdot \tan 45^\circ \cdot \cot 60^\circ = \sin 30^\circ \times \operatorname{cosec} 60^\circ$.

6. If $\tan \theta = 1$, then find the value of $\left(\frac{\cos \theta}{\sin \theta} + \frac{\sin \theta}{\cos \theta} \right)$.

7. In $\triangle ABC$, $\angle C = 90^\circ$, if $\tan A = \frac{1}{\sqrt{3}}$, then find the value of $\sin A \times \cos B + \cos A \times \sin B$.

8. If $\sin \theta = \frac{7}{25}$, then find the value of $\cos \theta$ and $\tan \theta$. (July '22)

9. Prove that $\cos^2 \theta (1 + \tan^2 \theta) = 1$.

10. Prove that $\frac{\sin^2 \theta}{\cos \theta} + \cos \theta = \sec \theta$.

11. If $\cos \theta = \frac{3}{5}$, then find $\sin \theta$. (July '23)

12. If $\sin \theta = \frac{11}{61}$, find the value of $\cos \theta$ using trigonometric identity.

(March '25)



13. A person is standing at a distance of 80 m from a church and looking at its top. The angle of elevation is of 45° . Find the height of the church. (March '20)

14. Find the value of $\frac{\tan 49^\circ}{\cot 41^\circ}$.

Answers :

1. $\cos^2\theta$, $\sec^2\theta$, -1 , -6

2. $\tan^2\theta$, $\sec^2\theta$, 1 , $\sec^2\theta$

3. $\cos^2\theta$, $\sin^2\theta$, $\sin^2\theta + \cos^2\theta = 1$, $\cos\theta$

4. $\frac{1}{2}$ 5. $x = 1$ 6. 2 7. 1

8. $\cos\theta = \frac{24}{25}$ and $\tan\theta = \frac{7}{24}$

11. $\sin\theta = \frac{4}{5}$ 12. $\cos\theta = \frac{60}{61}$ 13. 80 m 14. 1.

7.4 3 marks each

1. If $\tan\theta = \frac{3}{4}$, then complete the following activity to find the value of $\sec\theta$ and $\cos\theta$.

Activity :

$\tan\theta = \frac{3}{4}$... (Given)

$1 + \boxed{} = \sec^2\theta \quad \therefore 1 + \left(\boxed{-}\right)^2 = \sec^2\theta \quad \therefore \sec^2\theta = \boxed{-}$

$\therefore \sec\theta = \boxed{-}$... (Taking square roots of both the sides)

$\cos\theta = \frac{1}{\boxed{-}} \quad \therefore \cos\theta = \frac{1}{\boxed{-}} \quad \therefore \cos\theta = \frac{4}{5}$

Activity : $\tan\theta = \frac{3}{4}$... (Given)

$1 + \boxed{\tan^2\theta} = \sec^2\theta \quad \therefore 1 + \left(\boxed{\frac{3}{4}}\right)^2 = \sec^2\theta \quad \therefore \sec^2\theta = \boxed{\frac{25}{16}}$

$$\therefore \sec \theta = \boxed{\frac{5}{4}} \quad \dots \text{ (Taking square roots of both the sides)}$$

$$\cos \theta = \frac{1}{\boxed{\sec \theta}} \quad \therefore \cos \theta = \frac{1}{\boxed{\frac{5}{4}}} \quad \therefore \cos \theta = \frac{4}{5}.$$

2. Prove that $\frac{1}{\sec \theta - \tan \theta} = \sec \theta + \tan \theta$. (Nov. '20)

Proof :

$$\begin{aligned} \text{LHS} &= \frac{1}{\sec \theta - \tan \theta} \\ &= \frac{1}{(\sec \theta - \tan \theta)} \times \frac{(\sec \theta + \tan \theta)}{(\sec \theta + \tan \theta)} \\ &= \frac{(\sec \theta + \tan \theta)}{(\sec \theta - \tan \theta)(\sec \theta + \tan \theta)} \\ &= \frac{\sec \theta + \tan \theta}{\sec^2 \theta - \tan^2 \theta} \\ &= \frac{\sec \theta + \tan \theta}{1} \quad \dots \left[\begin{array}{l} 1 + \tan^2 \theta = \sec^2 \theta, \\ \therefore \sec^2 \theta - \tan^2 \theta = 1 \end{array} \right] \\ &= \text{RHS} \end{aligned}$$

$$\therefore \text{LHS} = \text{RHS}$$

$$\therefore \frac{1}{\sec \theta - \tan \theta} = \sec \theta + \tan \theta.$$

3. A boy standing at a distance of 48 metres from a building observes the top of the building and makes an angle of elevation of 30° . Find the height of the building.

Solution :

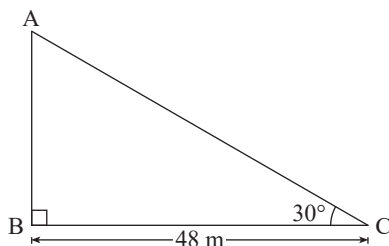
Let AB represent the building.

Point C is the position of the boy.

$\angle ACB$ is the angle of elevation.

$BC = 48$ m and $\angle ACB = 30^\circ$

... (Given)



In right angled $\triangle ABC$,

$$\tan \angle ACB = \tan 30^\circ = \frac{AB}{BC} \quad \dots \text{ (By definition)}$$

$$\therefore \frac{1}{\sqrt{3}} = \frac{AB}{48} \quad \dots \left[\because \tan 30^\circ = \frac{1}{\sqrt{3}} \right]$$

$$\therefore AB = \frac{48}{\sqrt{3}}$$

$$\therefore AB = \frac{48}{\sqrt{3}} \times \frac{\sqrt{3}}{\sqrt{3}}$$

$$\therefore AB = \frac{48\sqrt{3}}{3} = 16\sqrt{3}$$

$$\therefore AB = 16\sqrt{3} \text{ m}$$

Ans. The height of the building is $16\sqrt{3} \text{ m}$.

-
4. If $\sin \theta + \sin^2 \theta = 1$, then prove $\cos^2 \theta + \cos^4 \theta = 1$. (March '20)

Proof :

$$\sin \theta + \sin^2 \theta = 1 \quad \dots \text{ (Given)} \quad \dots (1)$$

$$\therefore \sin \theta = 1 - \sin^2 \theta$$

$$\therefore \sin \theta = \cos^2 \theta \quad \dots \left[\begin{array}{l} \sin^2 \theta + \cos^2 \theta = 1 \\ \therefore \cos^2 \theta = 1 - \sin^2 \theta \end{array} \right] \quad \dots (2)$$

Substituting (1) in (2), we get,

$$\cos^2 \theta + (\cos^2 \theta)^2 = 1$$

$$\therefore \cos^2 \theta + \cos^4 \theta = 1.$$

-
5. From top of the building, Ramesh is looking at a bicycle parked at some distance away from the building on the road.

If $AB \rightarrow$ Height of building is 40 m.

$C \rightarrow$ Position of bicycle, $A \rightarrow$ Position of Ramesh on top of the building.

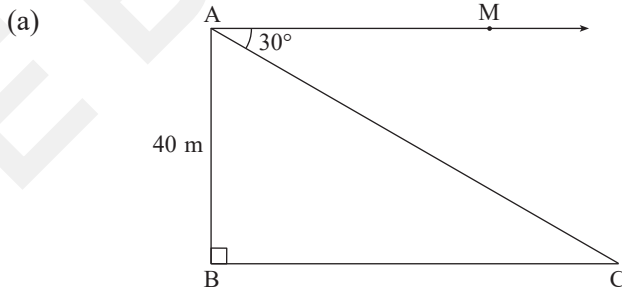
$\angle MAC$ is the angle of depression and $\angle MAC = 30^\circ$, then

(a) Draw a figure with the given information.

(b) Find the distance between the building and the bicycle.

$$(\sqrt{3} = 1.73) \quad \text{(July '23)}$$

Solution :



(b) $\angle MAC = 30^\circ$... (Given)

$\angle ACB = \angle MAC = 30^\circ$... (Alternate angles)

$\triangle ABC$ is a right angled triangle.

$$\tan \angle ACB = \tan 30^\circ = \frac{AB}{BC}$$

$$\therefore \frac{1}{\sqrt{3}} = \frac{40}{BC}$$

$$\begin{aligned} \therefore BC &= 40\sqrt{3} \\ &= 40 \times 1.73 \\ &= 69.20 \text{ m} \end{aligned}$$

Ans. The distance between the building and the bicycle is **69.20 m**.

Assignment

7.4

1. If $\cos \theta + \sin \theta = \sqrt{2} \cos \theta$, then complete the following activity to prove that $\cos \theta - \sin \theta = \sqrt{2} \sin \theta$.

Activity :

$$\cos \theta + \sin \theta = \sqrt{2} \cos \theta.$$

Taking squares of both the sides, we get,

$$(\cos \theta + \sin \theta)^2 = \boxed{} \cos^2 \theta$$

$$\therefore \cos^2 \theta + 2 \cos \theta \boxed{} + \sin^2 \theta = 2 \cos^2 \theta$$

$$\therefore \cos^2 \theta - 2 \cos \theta \sin \theta = \boxed{}$$

Adding $\sin^2\theta$ to both the sides, we get,

$$\square - 2 \cos \theta \cdot \sin \theta + \sin^2 \theta = \sin^2 \theta + \sin^2 \theta$$

$$\therefore (\cos \theta - \sin \theta)^2 = 2 \square$$

$$\therefore \cos \theta - \sin \theta = \square \sin \theta.$$

2. Prove the following :

$$(i) \cos^2 A \cdot \cos^2 B - \sin^2 A \cdot \sin^2 B = \cos^2 A - \sin^2 B.$$

$$(ii) \sqrt{\frac{1 + \sin A}{1 - \sin A}} + \sqrt{\frac{1 - \sin A}{1 + \sin A}} = 2 \sec A.$$

$$(iii) \sec^6 x - \tan^6 x = 1 + 3 \sec^2 x \times \tan^2 x.$$

$$(iv) \sqrt{\frac{1 - \cos A}{1 + \cos A}} = \operatorname{cosec} A - \cot A. \quad (\text{Sept. '21})$$

$$(v) \frac{\tan A}{(1 + \tan^2 A)^2} + \frac{\cot A}{(1 + \cot^2 A)^2} = \sin A \cdot \cos A. \quad (\text{March '22})$$

$$(vi) \sqrt{\frac{1 - \sin \theta}{1 + \sin \theta}} = \sec \theta - \tan \theta. \quad (\text{July '19})$$

$$(vii) \sec \theta + \tan \theta = \frac{\cos \theta}{1 - \sin \theta}. \quad (\text{July '22})$$

$$3. \text{ If } 7 \sin^2 \theta + 3 \cos^2 \theta = 4, \text{ then prove that } \tan \theta = \frac{1}{\sqrt{3}}.$$

$$4. \text{ If } 5x = \sec \theta \text{ and } \frac{5}{x} = \tan \theta, \text{ then find the value of } 5 \left(x^2 - \frac{1}{x^2} \right).$$

$$5. \text{ If } \tan \theta + \sin \theta = m \text{ and } \tan \theta - \sin \theta = n, \text{ then show that } m^2 - n^2 = 4\sqrt{mn}.$$

$$6. \text{ If } 8 \sin x - \cos x = 4, \text{ then find the possible values of } \sin x.$$

$$7. \text{ Find the angle of elevation of the sun if the length of the shadow of a building having height 30 m is } 10\sqrt{3} \text{ m.}$$

$$8. \text{ When an observer at a distance of 12 m from a tree looks at the top of the tree, the angle of elevation is } 60^\circ. \text{ What is the height of the tree? } (\sqrt{3} = 1.73) \quad (\text{March '19})$$

9. Eliminate θ , if $x = r \cos \theta$ and $y = r \sin \theta$. (March '23)
10. Two buildings are facing each other on a road of width 12 metres. From the top of the first building which is 10 metres high, the angle of elevation of the top of the second is found to be 60° . What is the height of the second building?
11. A storm broke a tree and treetop rested 20 m from the base of the tree, making an angle of 60° with the horizontal. Find the height of the tree.
12. A kite is flying at a height of 60 m above the ground. The string attached to the kite is tied at the ground. It makes an angle of 60° with the ground. Assuming that the string is straight, find the length of the string. ($\sqrt{3} = 1.73$)
13. Taking $\theta = 30^\circ$ to verify the following Trigonometric identities :
- (a) $\sin^2 \theta + \cos^2 \theta = 1$
 (b) $1 + \tan^2 \theta = \sec^2 \theta$
 (c) $1 + \cot^2 \theta = \operatorname{cosec}^2 \theta$. (March '25)

Answers :

1. 2 , $\sin \theta$, $\sin^2 \theta$, $\cos^2 \theta$, $\sin^2 \theta$, $\sqrt{2}$
 4. $\frac{1}{5}$ 6. $\frac{3}{5}$ or $\frac{5}{13}$ 7. 60° 8. 20.76 m 9. $x^2 + y^2 = r^2$
 10. $(12\sqrt{3} + 10)$ m 11. $(40 + 20\sqrt{3})$ m 12. 69.20 m.

CHALLENGING QUESTIONS

7.5 4 marks each

1. The angle of depression from the top of a tower of a point A on the ground is 30° . On moving a distance of 20 m from the point A towards the foot of the tower to a point B, the angle of elevation of the top of the tower from the point B is 60° . Find the height of the tower and its distance from point A.



Solution :

Let seg CD represent the tower.

$\angle ECA$ is the angle of depression of point A from point C.

$\angle CBD$ is the angle of elevation of point C from point B.

$AB = 20$ cm, $\angle ECA = 30^\circ$ and

$\angle CBD = 60^\circ$

$\angle CAD = \angle ECA = 30^\circ$

... (Alternate angles)

In $\triangle CDA$,

$\angle CDA = 90^\circ$

$$\tan \angle CAD = \tan 30^\circ = \frac{CD}{AD}$$

... (By definition)

$$\therefore \frac{1}{\sqrt{3}} = \frac{CD}{AD}$$

$$\dots \left(\because \tan 30^\circ = \frac{1}{\sqrt{3}} \right)$$

$$\therefore CD = \frac{AD}{\sqrt{3}}$$

... (1)

In $\triangle CDB$,

$\angle CDB = 90^\circ$

$$\tan \angle CBD = \tan 60^\circ = \frac{CD}{BD}$$

... (By definition)

$$\therefore \sqrt{3} = \frac{CD}{BD}$$

$$\therefore CD = \sqrt{3} \times BD$$

... (2)

$\therefore$ from (1) and (2), we get

$$\frac{AD}{\sqrt{3}} = \sqrt{3} \times BD$$

$$\therefore AD = 3BD$$

$$\therefore AB + BD = 3BD$$

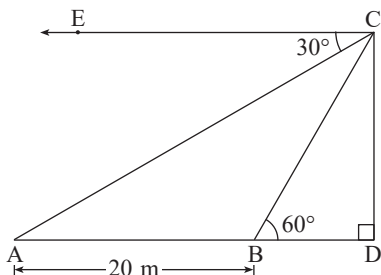
... (A + B = D)

$$\therefore 20 + BD = 3BD$$

$$\therefore 3BD - BD = 20$$

$$\therefore 2BD = 20$$

$$\therefore BD = \frac{20}{2}$$



$$\therefore BD = 10$$

$$CD = \sqrt{3} \times BD$$

... [From (2)]

$$\therefore CD = \sqrt{3} \times 10$$

$$\therefore CD = 10\sqrt{3} \text{ m}$$

$$AD = AB + BD$$

... (A-B-D)

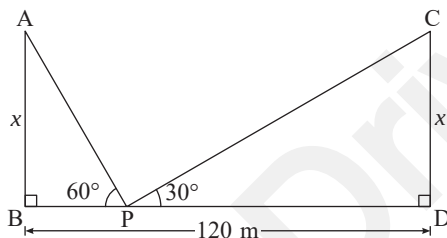
$$\therefore AD = 20 + 10$$

$$\therefore AD = 30 \text{ m}$$

Ans. Height of the tower is $10\sqrt{3} \text{ m}$ and its distance from point A is **30 m**.

2. Two pillars of equal height stand on either side of a roadway which is 120 m wide. At a point in the road between the pillars, the angles of elevation of the pillars are 60° and 30° . Find the height of each pillar to the nearest metre and position of the point from both the pillars. ($\sqrt{3} = 1.732$)

Solution :



AB and CD represent two pillars and P is the point on roadway.

$\angle APB$ and $\angle CPD$ are the angles of elevation.

$$\angle APB = 60^\circ \text{ and } \angle CPD = 30^\circ$$

$$\text{Let } AB = CD = x$$

$$BD = 120 \text{ m}$$

In $\triangle ABP$,

$$\angle ABP = 90^\circ$$

$$\tan \angle APB = \tan 60^\circ = \frac{x}{BP}$$

$$\therefore \sqrt{3} = \frac{x}{BP}$$

$$\therefore BP = \frac{x}{\sqrt{3}}$$

In $\triangle CDP$,

$$\angle CDP = 90^\circ$$

$$\tan \angle CPD = \tan 30^\circ = \frac{x}{PD}$$

$$\therefore \frac{1}{\sqrt{3}} = \frac{x}{PD}$$

$$\therefore PD = x\sqrt{3}$$



$$BP + PD = BD \quad \dots (B - P - D)$$

$$\therefore \frac{x}{\sqrt{3}} + x\sqrt{3} = 120$$

$$\therefore \frac{x + 3x}{\sqrt{3}} = 120$$

$$\therefore 4x = 120\sqrt{3}$$

$$\therefore x = \frac{120\sqrt{3}}{4}$$

$$\therefore x = 30\sqrt{3}$$

$$\therefore x = 30 \times 1.732$$

$$\therefore x = 51.96$$

$$\therefore x \approx 52 \text{ m}$$

$$BP = \frac{x}{\sqrt{3}} = \frac{30\sqrt{3}}{\sqrt{3}} = 30 \text{ m}$$

$$PD = x\sqrt{3} = 30\sqrt{3} \times \sqrt{3} = 90 \text{ m.}$$

Ans. Height of pillar is **52 m** and distance of point from pillars is **30 m** and **90 m**.

3. The angle of elevation of an aeroplane from the point A on the ground is 60° . After a flight of 15 seconds the angle of elevation changes to 30° . If the aeroplane is flying at a speed of 720 km/h, then find the constant height at which the aeroplane is flying.

$$(\sqrt{3} = 1.732)$$

Solution :

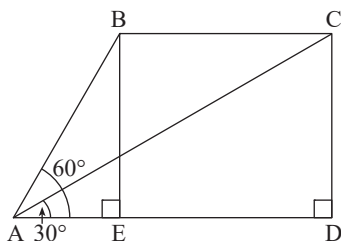
In the figure,

A is a point on the ground.

B is the first position of the aeroplane.

C is the second position of the aeroplane.

seg BE and seg CD represent the constant height of the aeroplane at points B and C.



$\angle BAE$ and $\angle CAD$ are the angles of elevation.

The speed of the aeroplane is 720 km/hr.

$\therefore$ distance covered by aeroplane in 1 hour, i.e. 3600 seconds is 720 km.

$\therefore$ distance covered by aeroplane in 15 seconds = BC

$$= \frac{720 \times 15}{3600} = 3 \text{ km}$$

$\square BCDE$ is a rectangle.

$\therefore ED = BC = 3 \text{ km}$... (Opposite sides of rectangle are equal)

Let the constant height of the aeroplane be x .

$\therefore CD = BE = x$

In $\triangle BEA$,

$\angle BEA = 90^\circ$

$$\tan \angle BAE = \tan 60^\circ = \frac{BE}{AE} \quad \dots \text{ (By definition)}$$

$$\therefore \sqrt{3} = \frac{x}{AE}$$

$$\therefore AE = \frac{x}{\sqrt{3}} \text{ km} \quad \dots (1)$$

In $\triangle CDA$,

$\angle CDA = 90^\circ$

$$\tan \angle CAD = \tan 30^\circ = \frac{CD}{AD} \quad \dots \text{ (By definition)}$$

$$\therefore \frac{1}{\sqrt{3}} = \frac{x}{AD}$$

$$\therefore AD = \sqrt{3} \times x \text{ km} \quad \dots (2)$$

$$AE + ED = AD \quad (\because A-E-D)$$

$$\therefore \frac{x}{\sqrt{3}} + 3 = \sqrt{3} \times x$$

$$\therefore \frac{x + 3\sqrt{3}}{\sqrt{3}} = \sqrt{3} \times x$$

$$\therefore x + 3\sqrt{3} = 3x$$

$$\therefore 3x - x = 3\sqrt{3}$$

$$\therefore 2x = 3\sqrt{3}$$

$$\therefore x = \frac{3}{2}\sqrt{3}$$

$$\therefore x = 1.5 \times 1.732$$

$$\therefore x = 2.598 \text{ km}$$

Ans. The aeroplane is flying at a constant height of **2.598 km**.

4. Find the value of θ , if

$$\frac{1}{\sin^2\theta} - \frac{1}{\cos^2\theta} - \frac{1}{\tan^2\theta} - \frac{1}{\cot^2\theta} - \frac{1}{\sec^2\theta} - \frac{1}{\operatorname{cosec}^2\theta} = -3. \text{ (March '24)}$$

Solution :

$$\begin{aligned} \text{LHS} &= \frac{1}{\sin^2\theta} - \frac{1}{\cos^2\theta} - \frac{1}{\tan^2\theta} - \frac{1}{\cot^2\theta} - \frac{1}{\sec^2\theta} - \frac{1}{\operatorname{cosec}^2\theta} \\ &= \operatorname{cosec}^2\theta - \sec^2\theta - \cot^2\theta - \tan^2\theta - \cos^2\theta - \sin^2\theta \\ &= (1 + \cot^2\theta) - (1 + \tan^2\theta) - \cot^2\theta - \tan^2\theta - (\cos^2\theta + \sin^2\theta) \\ &= 1 + \cot^2\theta - 1 - \tan^2\theta - \cot^2\theta - \tan^2\theta - 1 \\ &= -1 - 2 \tan^2\theta \end{aligned}$$

$$\therefore -1 - 2 \tan^2\theta = -3$$

$$\therefore -2 \tan^2\theta = -3 + 1$$

$$\therefore -2 \tan^2\theta = -2 \quad \therefore \tan^2\theta = 1 \quad \therefore \tan \theta = 1$$

$$\text{But } \tan 45^\circ = 1 \quad \therefore \theta = 45^\circ$$

Ans. The value of θ is **45°** .

Assignment

7.5

1. If $\sec \theta + \tan \theta = p$, then prove that $\frac{p^2 - 1}{p^2 + 1} = \sin \theta$.
2. Prove that $(1 - \cos^2 A) \cdot \sec^2 B + \tan^2 B (1 - \sin^2 A) = \sin^2 A + \tan^2 B$.
3. Prove that $\frac{1 + \sin x - \cos x}{1 + \sin x + \cos x} + \frac{1 + \sin x + \cos x}{1 + \sin x - \cos x} = 2 \operatorname{cosec} x$.
4. Prove that $\frac{\tan \theta}{1 - \cot \theta} + \frac{\cot \theta}{1 - \tan \theta} = 1 + \tan \theta + \cot \theta$.

5. Show that $\sin^6\theta + \cos^6\theta = 1 - 3 \sin^2\theta \cdot \cos^2\theta$.
6. A person, standing on the bank of a river, observes that the angle subtended by a tree on the opposite bank is 60° . When he retreats 40 m from the bank, he finds the angle to be 30° . Find the height of the tree and breadth of the river. ($\sqrt{3} = 1.732$)
7. An aeroplane flying horizontally 1200 m above the ground is observed at an angle of elevation of 60° . After 6 seconds, its angle of elevation is observed to be 30° . Find the speed of the aeroplane in km/h. ($\sqrt{3} = 1.73$)
8. A straight road leads to the foot of the tower of height 48 m. From the top of the tower the angles of depression of two cars standing on the road are 30° and 60° respectively. Find the distance between the two cars. ($\sqrt{3} = 1.73$) (Nov. '20)
9. Two men on the either side of the cliff of height 75 m observes the top of the cliff. The angles of the elevation of the top of the cliff was found to be 30° and 60° respectively. Find the distance between the two men.
10. The angle of elevation of a cloud from a point 60 m above a lake is 30° and the angle of depression of the reflection of cloud in the lake is 60° . Find the height of the cloud.

Answers :

6. (i) Height of the tree 34.6 m (ii) breadth of the river = 20 m
 7. Speed of the aeroplane = 830 km/h 8. 55.36 m
 9. $100\sqrt{3}$ m 10. 120 m.



**Question
Set
8**

MENSURATION

FORMULAE

1. Length of the arc (l) = $\frac{\theta}{360} \times 2\pi r$

... (θ is the measure of the central angle.)

2. Area of the sector = $\frac{\theta}{360} \times \pi r^2$.

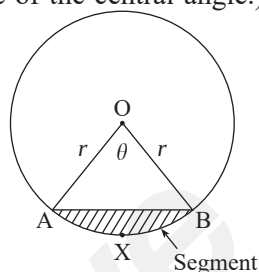
3. Area of the sector = $\frac{r}{2} \times l$.

4. Area of the segment

= Area of minor sector (O—AXB) — Area of $\triangle OAB$

= $r^2 \left[\frac{\pi\theta}{360} - \frac{\sin \theta}{2} \right]$.

[Note : A($\triangle OAB$) = $\frac{1}{2} r^2 \sin \theta$]



5. Vertical surface area of a cuboid

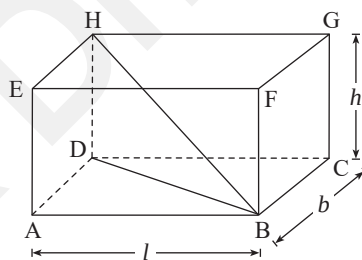
= $2(l + b)h$.

6. Total surface area of a cuboid

= $2(lb + bh + lh)$.

7. The diagonal of a cuboid

= $\sqrt{l^2 + b^2 + h^2}$.



Segment HB is the diagonal of the cuboid.

[Note : $(l + b + h)^2 = l^2 + b^2 + h^2 + 2(lb + bh + lh)$]

8. The total surface area of a cube = $6l^2$.

9. The volume of a cube = l^3 .

10. The diagonal of a cube = $\sqrt{3}l$.

11. The curved surface area (S_c) of a cylinder = $2\pi rh$.

12. The total surface area (S_t) of a cylinder = $2\pi r(h + r)$.

13. The volume of a cylinder = $\pi r^2 h$.

14. The slant height (l) of a cone $= \sqrt{h^2 + r^2}$.
15. The curved surface area (S_c) of a cone $= \pi r l$.
16. The total surface area (S_t) of a cone $= \pi r (l + r)$.
17. The volume of a cone $= \frac{1}{3} \pi r^2 h$.
18. The slant height (l) of a frustum $= \sqrt{h^2 + (r_1 - r_2)^2}$.
19. The curved surface area of a frustum $= \pi (r_1 + r_2) l$.
20. The total surface area of a frustum $= \pi (r_1 + r_2) l + \pi r_1^2 + \pi r_2^2$.
21. The volume of a frustum $= \frac{1}{3} \pi (r_1^2 + r_2^2 + r_1 r_2) h$.
22. The surface area (i.e. total surface area) of a sphere $= 4\pi r^2$.
23. The volume of a sphere $= \frac{4}{3} \pi r^3$.
24. The curved surface area of a hemisphere $= 2\pi r^2$.
25. The total surface area of a hemisphere $= 3\pi r^2$.
26. The volume of a hemisphere $= \frac{2}{3} \pi r^3$.

8.1 1 mark each

Q. Choose the correct alternative from those given below each question :

1. The volume of a cube of side 10 cm is
 (A) 1 cm^3 (B) 10 cm^3
 (C) 100 cm^3 (D) 1000 cm^3 (July '23-'24)

Ans. (D) [Hint : Use Volume of cube $= \text{side}^3$.]

2. If the radius of the base of cone is 7 cm and height is 24 cm, then find its slant height. (March '23)
 (A) 23 cm (B) 26 cm (C) 31 cm (D) 25 cm

Ans. (D) [Hint : Use $l^2 = r^2 + h^2$.]

3. If the radius of the sphere is triple, then the volume of the sphere increases times.
 (A) 3 (B) 6 (C) 9 (D) 27

Ans. (D) [Hint : New radius $r_1 = 3r$. Volume of sphere $= \frac{4}{3} \pi r^3$.]



4. A circle is inscribed in a square. The radius of the circle is 4 cm, then find the perimeter of the square.

(A) 32 cm (B) 16 cm (C) 64 cm (D) 8 cm

Ans. (A) [**Hint :** Side of the square = Diameter of the circle.]

5. If the radius of the sector is 5 cm and the length of its corresponding arc is 10 cm, then area of the sector is (**Nov. '20**)

(A) 50 cm² (B) 25 cm² (C) 25 m² (D) 10 cm²

Ans. (B) [**Hint :** Area of sector = $\frac{1}{2} \times \text{Length of arc} \times \text{Radius}$.]

6. Find the perimeter of a sector of a circle if its measure is 90° and radius is 7 cm.

(A) 44 cm (B) 25 cm (C) 36 cm (D) 56 cm

Ans. (B) [**Hint :** Use length of an arc = $\frac{\theta}{360} \times 2\pi r$.]

7. If the total surface area of a hemisphere is 300 cm², then what is the area of its base?

(A) 100 cm² (B) 150 cm² (C) 400 cm² (D) 200 cm²

Ans. (A) [**Hint :** Total surface area of hemisphere = $3\pi r^2$ and area of its base = πr^2 .]

8. If the area of the circle is 314 cm² and area of the major segment is 214 cm², then what is the area of its minor segment?

(A) 314 cm² (B) 100 cm² (C) 114 cm² (D) 214 cm²

Ans. (B) [**Hint :** A(Minor segment) = A(Circle) – A(Major segment).]

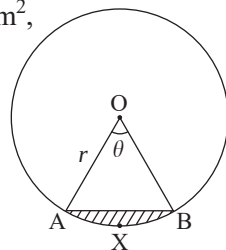
9. The radii of two circles are 4 cm and 3 cm respectively. The radius of a circle having an area equal to the sum of the areas of these two circles is

(A) 5 cm (B) 7 cm (C) 10 cm (D) 14 cm

Ans. (A) [**Hint :** Area of the circle = πr^2 .]

Q. Choose the correct alternative from those given below each question :

- If the radius of a cone is 5 cm and its perpendicular height is 12 cm, then the slant height is (March '25)
(A) 17 cm (B) 4 cm (C) 13 cm (D) 60 cm
- Find the volume of a cube of side 3 cm. (March '20)
(A) 27 cm^3 (B) 9 cm^3 (C) 81 cm^3 (D) 3 cm^3
- Find the side of a cube of volume 1 m^3 .
(A) 1 cm (B) 10 cm (C) 100 cm (D) 1000 cm
- The curved surface area of a cylinder is 440 cm^2 and its radius is 5 cm. Find its height.
(A) $\frac{44}{\pi} \text{ cm}$ (B) $22\pi \text{ cm}$ (C) $44\pi \text{ cm}$ (D) $\frac{22}{\pi} \text{ cm}$
- The radius and height of the right circular cylinder and right circular cone are equal. The volume of the cylinder is 600 cm^3 . The volume of the cone is
(A) 600 cm^3 (B) 1200 cm^3 (C) 300 cm^3 (D) 200 cm^3
- The numbers denoting the ratio of the circumference and the area of a circle is 2 : 7. Find its circumference.
(A) 14π (B) $\frac{7}{\pi}$ (C) 7π (D) $\frac{14}{\pi}$
- Area of the sector is times the length of its corresponding arc.
(A) $\frac{1}{2}$ (B) $\frac{\text{diameter}}{2}$ (C) $\frac{\text{radius}}{2}$ (D) $\frac{\text{radius}}{4}$
- $A(\text{O}-\text{AXB}) = 113.04 \text{ cm}^2$ and $A(\triangle \text{OAB}) = 100 \text{ cm}^2$, then find area of segment AXB.
(A) 13.04 cm^2 (B) 213.04 cm^2
(C) 110 cm^2 (D) 103.04 cm^2
- What will be the ratio of the volumes of a cylinder and a cone having the same radius and the same height?
(A) 1 : 2 (B) 2 : 1 (C) 1 : 3 (D) 3 : 1



Answers :

1. (C) [$l^2 = r^2 + h^2$.]
2. (A) [Volume of the cube = l^3 .]
3. (C) [Volume of the cube = l^3 .]
4. (A) [Curved surface area of cylinder = $2\pi rh$.]
5. (D) $\left[\text{Volume of cone} = \frac{1}{3} \times \text{Volume of cylinder.} \right]$
6. (A) [Circumference of the circle = $2\pi r$, Area of the circle = πr^2 .]
7. (C) $\left[l = \frac{\theta}{360} \times 2\pi r. \text{ Area of the sector} = \frac{l \times r}{2}. \right]$
8. (A) [$A(\text{Segment AXB}) = A(\text{O-AXB}) - A(\triangle OAB)$.]
9. (D) $\left[\text{Volume of cylinder} = \pi r^2 h \text{ and volume of the cone} = \frac{1}{3} \pi r^2 h. \right]$

8.2 1 mark each

1. Find the area of the circle whose radius is 7 cm. (March '19)

Ans. The radius of the circle (r) = 7 cm.

Area of the circle = πr^2

$$= \frac{22}{7} \times 7 \times 7$$

$$= 154 \text{ cm}^2$$

Area of the circle is **154 cm²**.

2. A sphere touches all the faces of a cube of side 12 cm. What is the radius of that sphere?

Ans.

Diameter of sphere = The length of the side of the cube = 12 cm.

$$\text{Its radius} = \frac{1}{2} \times \text{Diameter} = \frac{1}{2} \times 12 = 6 \text{ cm.}$$

The radius of the sphere is **6 cm**.

3. The volume of a cone is 300 cm^3 . The cone is melted and 6 smaller cones of the same size are made. Find the volume of each cone.

Ans. The volume of each small cone

$$= \frac{\text{The volume of the cone}}{\text{The number of smaller cones}} = \frac{300}{6} = 50 \text{ cm}^3$$

The volume of each smaller cone is 50 cm^3 .

4. The volume of a frustum-shaped bucket is 10780 cm^3 . What is its volume in litres?

Ans. Volume of frustum-shaped bucket = 10780 cm^3

$$= \frac{10780}{1000} \text{ litres} = 10.78 \text{ litres.}$$

Volume of frustum-shaped bucket in litres is **10.78 litres**.

5. If the area of the sector is $\frac{1}{4}$ th the area of circle, then what is the angular measure of arc of that sector?

Ans. Area of the sector = $\frac{1}{4} \times \text{Area of a circle}$... (Given)

$$\therefore \frac{\theta}{360} \times (\pi r^2) = \frac{1}{4} \times (\pi r^2)$$

$$\therefore \theta = \frac{1}{4} \times 360^\circ$$

$$\therefore \theta = 90^\circ.$$

The angular measure of arc of that sector is **90°** .

6. The circumference of a wheel is 44 m. Find the distance covered by the wheel in 10 revolutions.

Ans. The distance covered by the wheel in 10 revolutions

= The number of revolutions $\times$ The circumference of the wheel

$$= 10 \times 44$$

$$= 440 \text{ m}$$

The distance covered by the wheel is **440 m**.



7. Find the length of the arc whose radius is 7 cm and the angular measure of the arc is 90° .

Ans.

$$\begin{aligned}\text{Length of the arc} &= \frac{\theta}{360} \times 2\pi r \\ &= \frac{90}{360} \times 2 \times \frac{22}{7} \times 7 \\ &= 11 \text{ cm}\end{aligned}$$

Length of the arc is **11 cm**.

Assignment

8.2

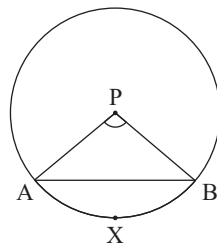
1. A solid sphere is cut into two equal halves. If the total surface area of the two halves is 90 cm^2 , what is the surface area of the sphere?
2. Find the volume of a cube of side 5 cm. (July '19)
3. Volume of a godown is 5000 m^3 . How many boxes of volume 10 m^3 each, placed one over the other, can be kept in the godown?
4. Area of the square is equal to the area of the circle. Find the ratio of their perimeters.
5. The area of a circle is 100 cm^2 . The area of a sector of the same circle is 25 cm^2 . Find the measure of the arc of that sector.
6. Find the capacity of a cube in litres if its side is of length 1 m.
7. The ratio of the areas of two circles is 9 : 4. What is the ratio of their circumferences?
8. The length of the wire is 6600 cm. How many circles with a circumference of 66 cm can be made from this wire?

Answers :

1. 60 cm^2 2. 125 cm^3 3. 500 boxes 4. $2\sqrt{\pi} : \pi$ 5. 90°
6. 1000 litres 7. 3 : 2 8. 100.

8.3 2 marks each

1. The radius of a circle with centre P is 10 cm. If chord AB of the circle subtends a right angle at P, find area of minor sector by using the following activity. ($\pi = 3.14$) (March '25)



Activity :

$$r = 10 \text{ cm}, \theta = 90^\circ, \pi = 3.14$$

$$\begin{aligned} A(P-AXB) &= \frac{\theta}{360} \times \boxed{} \\ &= \frac{\boxed{}}{360} \times 3.14 \times 10^2 \\ &= \frac{1}{4} \times \boxed{} \end{aligned}$$

$$\therefore A(P-AXB) = \boxed{} \text{ sq. cm.}$$

Activity :

$$r = 10 \text{ cm}, \theta = 90^\circ, \pi = 3.14$$

$$\begin{aligned} A(P-AXB) &= \frac{\theta}{360} \times \boxed{\pi r^2} \\ &= \frac{\boxed{90}}{360} \times 3.14 \times 10^2 \\ &= \frac{1}{4} \times \boxed{314} \end{aligned}$$

$$\therefore A(P-AXB) = \boxed{78.5} \text{ sq cm.}$$

2. The radius of the sphere is 7 cm. Complete the following activity to find its surface area. (March '23)

Activity :

$$\text{Surface area of the sphere} = 4\pi r^2$$

$$= 4 \times \frac{22}{7} \times \boxed{}^2$$

$$= 4 \times \frac{22}{7} \times \boxed{}$$

$$= \boxed{} \times 7$$

$$= \boxed{} \text{ cm}^2.$$

Activity :

Surface area of the sphere = $4\pi r^2$

$$= 4 \times \frac{22}{7} \times \boxed{7}^2$$

$$= 4 \times \frac{22}{7} \times \boxed{49}$$

$$= \boxed{88} \times 7$$

$$= \boxed{616} \text{ cm}^2.$$

3. If the radius of the base of the cone is 2.1 cm and its perpendicular height is 6 cm, then find the volume of the cone.

Solution : For cone,

radius (r) = 2.1 cm, height (h) = 6 cm.

$$\text{Volume of the cone} = \frac{1}{3} \pi r^2 h$$

$$= \frac{1}{3} \times \frac{22}{7} \times (2.1)^2 \times 6$$

$$= \frac{1}{3} \times \frac{22}{7} \times 2.1 \times 2.1 \times 6$$

$$= 27.72 \text{ cm}^3$$

Ans. Volume of the cone is **27.72 cm³**.

4. In the figure, AB and CD are the diameters of the circle with centre O. The radius of the circle is 2.1 cm; $\angle COA = 30^\circ$. Find the area of the shaded portion.

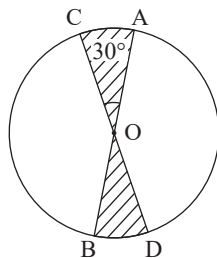
Solution :

The radius of the circle (r) = 2.1 cm.

$$\angle COA = \theta = 30^\circ$$

$$A(\text{Sector O-AC}) = \frac{\theta}{360} \times \pi r^2$$

$$= \frac{30}{360} \times \frac{22}{7} \times \frac{21}{10} \times \frac{21}{10}$$



$$= \frac{231}{200} \text{ cm}^2$$

$$\text{Similarly, } A(\text{Sector O-BD}) = \frac{231}{200} \text{ cm}^2$$

$$A(\text{Shaded portion}) = A(\text{Sector O-AC}) + A(\text{Sector O-BD})$$

$$= \frac{231}{200} + \frac{231}{200}$$

$$= \frac{462}{200} = 2.31 \text{ cm}^2.$$

Ans. Area of the shaded portion is **2.31 cm²**.

- 5. The radius of the base of a right circular cylinder is 5 cm and its height is 40 cm, then find its total surface area. ($\pi = 3.14$)**

Solution : For right circular cylinder,

radius (r) = 5 cm, height (h) = 40 cm.

Total surface area of cylinder = $2\pi r (r + h)$

$$= 2 \times 3.14 \times 5 \times (5 + 40)$$

$$= 3.14 \times 10 \times 45$$

$$= 1413 \text{ cm}^2$$

Ans. Total surface area of cylinder is **1413 cm²**.

- 6. The angular measure of an arc of the circle is 80° and its radius is 18 cm. Find the length of the arc. ($\pi = 3.14$)**

Solution : For an arc,

radius (r) = 18 cm, angular measure (θ) = 80°.

$$\text{Length of an arc} = \frac{\theta}{360} \times 2\pi r$$

$$= \frac{80}{360} \times 2 \times 3.14 \times 18$$

$$= 25.12 \text{ cm}$$

Ans. Length of the arc is **25.12 cm**.

1. A circle is inscribed in square ABCD of side 14 cm. Complete the following activity to find the area of shaded portion.

(July '19; March '24)

Activity :

$$\text{Area of square ABCD} = \left(\boxed{} \right)^2 \dots (\text{Formula})$$

$$= 14^2$$

$$= \boxed{} \text{ cm}^2$$

$$\text{Area of circle} = \boxed{}$$

... (Formula)

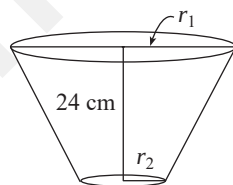
$$= \frac{22}{7} \times 7 \times 7 = 154 \text{ cm}^2$$

Area of the shaded portion = Area of square ABCD – Area of circle

$$= 196 - 154 = \boxed{} \text{ cm}^2$$

2. The circumferences of circular faces of a frustum are 132 cm and 88 cm and its height is 24 cm. Complete the following activity to find the slant height of the frustum.

$$\left(\pi = \frac{22}{7} \right)$$



Activity : Circumference₁ = $2\pi r_1 = 132$

$$\therefore r_1 = \frac{132}{2\pi} = \boxed{}$$

Circumference₂ = $2\pi r_2 = 88$

$$\therefore r_2 = \frac{88}{2\pi} = \boxed{}$$

Slant height of frustum, $l = \sqrt{h^2 + (r_1 - r_2)^2}$

$$= \sqrt{24^2 + \boxed{}^2} = \boxed{} \text{ cm}$$

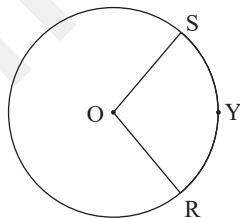
3. The length, breadth and height of a cuboid are 12 cm, 10 cm and 8 cm respectively. Complete the following activity to find its volume.

Activity :

$$l = 12 \text{ cm}, b = 10 \text{ cm}, h = 8 \text{ cm}.$$

$$\begin{aligned} \text{Volume of the cuboid} &= \boxed{} \times b \times h \\ &= 12 \times \boxed{} \times \boxed{} \\ &= \boxed{} \text{ cm}^3 \end{aligned}$$

4. How many lead spheres of radius 1 cm can be made from a lead sphere of radius 8 cm?
5. How much metal sheet will be required to make a pipe 10 cm long and with a radius of 7 cm?
6. The total surface area of an iron cube is 600 cm^2 . Find the length of its side.
7. The curved surface area of a sphere is 50.24 cm^2 . Find its radius.
($\pi = 3.14$)
8. If the length of an arc of sector of a circle is 20 cm and its radius is 7 cm. Find the area of the sector. **(July '19)**
9. In the figure, O is the centre of the circle with radius 7 cm. If $m(\text{arc RYS}) = 60^\circ$, then find $A(\text{O-RYS})$.



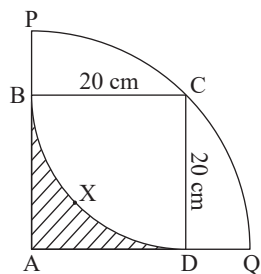
10. The radius of a sector of a circle is 3.5 cm and length of its arc is 2.2 cm. Find the area of the sector. **(March '24)**
11. The area of a minor sector of a circle is 3.85 cm^2 and the measure of its central angle is 36° . Find the radius of the circle.

Answers :

1. (side), 196, πr^2 , 42 2. 21 cm, 14 cm, 7, 25
 3. l, 10, 8, 960 4. 512 5. 440 cm^2 6. 10 cm 7. 2 cm
 8. 70 cm^2 9. 25.67 cm^2 10. 3.85 cm^2 11. 3.5 cm.

8.4 3 marks each

1. In the figure, square ABCD is inscribed in the sector A-PCQ. The radius of sector C-BXD is 20 cm. Complete the following activity to find the area of shaded region. Side of square ABCD = radius of sector C-BXD = 20 cm.



$$\text{Area of square} = (\text{side})^2 = \boxed{}^2 = \boxed{} \quad \dots (1)$$

Area of shaded region inside the square
= Area of square ABCD – Area of sector C-BXD

$$\begin{aligned} &= \boxed{} - \frac{\theta}{360} \times \pi r^2 \\ &= \boxed{} - \frac{90}{360} \times \frac{3.14}{1} \times \frac{400}{1} \\ &= \boxed{} - 314 \\ &= \boxed{} \end{aligned}$$

Activity :

Side of square ABCD = radius of sector C-BXD = 20 cm.

$$\text{Area of square} = (\text{side})^2 = \boxed{20}^2 = \boxed{400 \text{ cm}^2} \quad \dots (1)$$

Area of shaded region inside the square
= Area of square ABCD – Area of sector C-BXD

$$\begin{aligned} &= \boxed{BC^2} - \frac{\theta}{360} \times \pi r^2 \\ &= \boxed{20^2} - \frac{90}{360} \times \frac{3.14}{1} \times \frac{400}{1} \\ &= \boxed{400} - 314 \\ &= \boxed{86 \text{ cm}^2} \end{aligned}$$

3. The perimeter of a sector of a circle with radius 3.4 cm is 12.8 cm. Complete the following activity to find the area of the sector P–ABC.

Perimeter of sector = PA + PC + length of arc ABC.

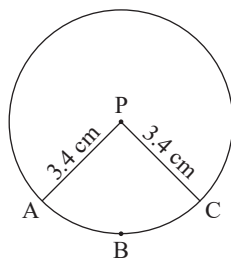
$$\therefore \boxed{} = 3.4 + \boxed{} + \text{Length of arc ABC}$$

$$\therefore \text{length of arc ABC} = \boxed{}$$

$$\text{Area of sector P–ABC} = \frac{l \times \boxed{}}{2}$$

$$= \frac{6 \times \boxed{}}{2}$$

$$= \boxed{} \text{ cm}^2$$



Activity :

Perimeter of sector = PA + PC + length of arc ABC.

$$\therefore \boxed{12.8} = 3.4 + \boxed{3.4} + \text{Length of arc ABC}$$

$$\therefore \text{length of arc ABC} = \boxed{6 \text{ cm}}$$

$$\text{Area of sector P–ABC} = \frac{l \times \boxed{r}}{2}$$

$$= \frac{6 \times \boxed{3.4}}{2}$$

$$= \boxed{10.2} \text{ cm}^2.$$

3. A metal cuboid of measures 16 cm × 11 cm × 10 cm was melted to make coins. How many coins were made if the thickness and diameter of each coin was 2 mm and 2 cm respectively?

(March '20)

Solution :

The length of a metal cuboid (l) = 16 cm,

its breadth (b) = 11 cm, its height (h) = 10 cm.

The diameter of coin = 2 cm.



$$\therefore \text{its radius } (r) = \frac{2}{2} = 1 \text{ cm}$$

$$\text{its thickness } (h_1) = 2 \text{ mm} = \frac{2}{10} \text{ cm}$$

Number of coins that were made

$$= \frac{\text{Volume of metal cuboid}}{\text{Volume of each coin}}$$

$$= \frac{l \times b \times h}{\pi r^2 h_1}$$

$$= \frac{16 \times 11 \times 10}{\frac{22}{7} \times 1 \times 1 \times \frac{2}{10}}$$

$$= \frac{16 \times 11 \times 10 \times 7 \times 10}{22 \times 2} = 2800$$

Ans. 2800 coins were made from the given metal cuboid.

- 4. The radii of circular ends of a frustum are 14 cm and 6 cm respectively and its height is 6 cm. Find its curved surface area.**

($\pi = 3.14$) (*March '23*)

Solution : The radii of ends of a frustum are 14 cm and 6 cm.

$\therefore r_1 = 14 \text{ cm}$ and $r_2 = 6 \text{ cm}$, its height (h) = 6 cm.

Let the slant height of the frustum be l .

$$l = \sqrt{h^2 + (r_1 - r_2)^2}$$

$$\therefore l = \sqrt{6^2 + (14 - 6)^2}$$

$$\therefore l = \sqrt{36 + 8^2} \quad \therefore l = \sqrt{36 + 64}$$

$$\therefore l = \sqrt{100} \quad \therefore l = 10 \text{ cm}$$

Curved surface area of a frustum = $\pi l (r_1 + r_2)$

$$= 3.14 \times 10 \times (14 + 6)$$

$$= 3.14 \times 10 \times 20$$

$$= 628 \text{ cm}^2$$

Ans. The curved surface area of the frustum is **628 cm²**.

5. The radius of a metallic sphere is 9 cm. It was melted to make a wire of diameter 4 mm. Find the length of the wire.

Solution :

The radius (r) of a metallic sphere = 9 cm.

The diameter of a wire = 4 mm.

$$\therefore \text{its radius } (r_1) = 2 \text{ mm} = \frac{2}{10} \text{ cm} \quad \dots (1 \text{ cm} = 10 \text{ mm})$$

Let the length of the wire be h .

Sphere is melted to make a wire.

$\therefore$ volume of wire = Volume of sphere

$$\therefore \pi r_1^2 h = \frac{4}{3} \pi r^3$$

$$\therefore \frac{2}{10} \times \frac{2}{10} \times h = \frac{4}{3} \times 9 \times 9 \times 9$$

$$\therefore h = \frac{4 \times 9 \times 9 \times 9 \times 10 \times 10}{3 \times 2 \times 2}$$

$$\therefore h = 24300 \text{ cm or } 243 \text{ m} \quad \dots (1 \text{ m} = 100 \text{ cm})$$

Ans. The length of the wire is **243 m**.

-
6. How many solid cylinders of radius 10 cm and height 6 cm can be made by melting a solid sphere of radius 30 cm? (March '25)

Solution : For cylinder,

radius (r) = 10 cm, height (h) = 6 cm.

For sphere,

radius (r_1) = 30 cm.

Number of solid cylinders that can be made

$$\begin{aligned} &= \frac{\text{Volume of sphere}}{\text{Volume of each cylinder}} \\ &= \frac{\frac{4}{3} \pi r_1^3}{\pi r^2 h} = \frac{\frac{4}{3} \times 30 \times 30 \times 30}{10 \times 10 \times 6} = 60 \end{aligned}$$

Ans. 60 solid cylinders can be made.

7. The figure shows a toy. Its lower part is a hemisphere and the upper part is a cone. Find the volume of the toy from the measures shown in the figure.

$$(\pi = 3.14)$$

Solution : A toy is made up of a cone and a hemisphere.

They have equal bases.

Let the radius of the base be r .

$$\therefore r = 3 \text{ cm.}$$

Height of the conical part (h) = 4 cm.

Volume of the toy = Volume of conical part +

Volume of hemispherical part

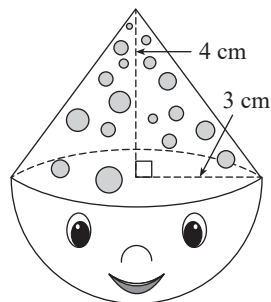
$$= \frac{1}{3} \pi r^2 h + \frac{2}{3} \pi r^3$$

$$= \frac{1}{3} \pi r^2 (h + 2r)$$

$$= \frac{1}{3} \times 3.14 \times 3 \times 3 (4 + 2 \times 3)$$

$$= 3.14 \times 3 \times 10 = 94.2 \text{ cm}^3$$

Ans. The volume of the toy is **94.2 cm³**.



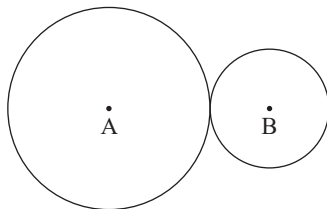
8. In the figure, two discs with centres A and B touch each other externally. The radius of the smaller disc is $\frac{1}{3}$ rd the radius of the

larger disc. If the radius of the smaller circle is 7 cm; then

(a) Find the circumference of the larger disc.

(b) Find the circumference of the smaller disc.

(c) How many rotations will the smaller disc make to cover a distance equal to the circumference of the larger disc?



Solution :

- (a) Let the radius of the smaller disc be r_1 and the radius of the larger disc be r_2 .

$$r_1 = 7 \text{ cm}$$

$$r_1 = \frac{1}{3} r_2 \quad \dots \text{ (Given)}$$

$$\therefore 7 = \frac{1}{3} r_2$$

$$\therefore r_2 = 7 \times 3$$

$$\therefore r_2 = 21 \text{ cm}$$

$$\text{Circumference of the larger disc} = 2\pi r_2$$

$$= 2 \times \frac{22}{7} \times 21 = 132 \text{ cm.}$$

- (b) Circumference of the smaller disc $= 2\pi r_1$

$$= 2 \times \frac{22}{7} \times 7 = 44 \text{ cm.}$$

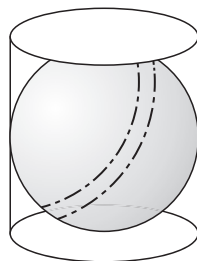
- (c) Number of rotations smaller disc makes to cover the distance equal to circumference of larger disc,

$$\begin{aligned} &= \frac{\text{Circumference of larger disc}}{\text{Circumference of the smaller disc}} \\ &= \frac{132}{44} = 3. \end{aligned}$$

Ans. (a) 132 cm (b) 44 cm (c) 3 rotations.

- 9. In the figure, a sphere is placed in a cylinder.**

It touches the top, bottom and the curved surface of the cylinder. If radius of the base of the cylinder is ' r ', write the answers of the following questions :



- (a) What is the height of the cylinder in terms of ' r '?
- (b) What is the ratio of the curved surface area of the cylinder and the surface area of the sphere?
- (c) What is the ratio of volumes of the cylinder and of the sphere?

(July '19)



Solution :

(a) Height of the cylinder = Diameter of the sphere

$$h = 2r$$

$$\begin{aligned} \text{(b) } \frac{\text{Curved surface area of the cylinder}}{\text{Surface area of the sphere}} &= \frac{2\pi rh}{4\pi r^2} \\ &= \frac{2 \times \pi \times r \times 2r}{4\pi r^2} = \frac{4\pi r^2}{4\pi r^2} = 1 \end{aligned}$$

$$\begin{aligned} \text{(c) } \frac{\text{Volume of the cylinder}}{\text{Volume of the sphere}} &= \frac{\pi r^2 h}{\frac{4}{3}\pi r^3} \\ &= \frac{3\pi r^2 h}{4\pi r^3} \\ &= \frac{3h}{4r} \\ &= \frac{3(2r)}{4r} = \frac{6r}{4r} = \frac{3}{2} \end{aligned}$$

Ans.

(a) $h = 2r$.

$$\text{(b) } \frac{\text{Curved surface area of the cylinder}}{\text{Surface area of the sphere}} = \frac{1}{1} = 1 : 1.$$

$$\text{(c) } \frac{\text{Volume of the cylinder}}{\text{Volume of the sphere}} = \frac{3}{2} = 3 : 2.$$

Assignment

8.4

1. The diameter and length of a roller is 120 cm and 84 cm respectively. To level the ground, 200 rotations of the roller are required. Complete the following activity to find the expenditure to level the ground at the rate of ₹ 10 per m^2 .

Activity : The diameter of the roller = 120 cm.

$$\therefore \text{its radius } (r) = \frac{120}{2} = \boxed{} \text{ cm}$$

its length $(h) = 84 \text{ cm}$

$$\begin{aligned}
 \text{Curved surface area of the roller} &= \boxed{} \\
 &= 2 \times \frac{22}{7} \times 60 \times 84 = \boxed{} \text{ cm}^2 \\
 &= \frac{31680}{\boxed{}} \text{ m}^2 \dots (1 \text{ m}^2 = 10000 \text{ cm}^2) \\
 &= 3.168 \text{ m}^2
 \end{aligned}$$

Area of ground levelled in 1 rotation is 3.168 m^2

$\therefore$ area of ground levelled in 200 rotations

$$= 200 \times \boxed{} = 633.6 \text{ m}^2$$

Cost of levelling the ground per $\text{m}^2 = ₹ 10$

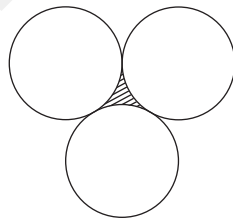
$$\therefore \text{total cost of levelling the ground} = 633.6 \times 10 = ₹ \boxed{}.$$

2. In the figure, a cylindrical wrapper of flat tablets is shown. The radius of a tablet is 7 mm and its thickness is 5 mm. How many such tablets are wrapped in the wrapper?



3. In the given figure, three circles each of radius 3.5 cm are drawn in such a way that each of them touches the other two. Find the area enclosed between these three circles (shaded region).

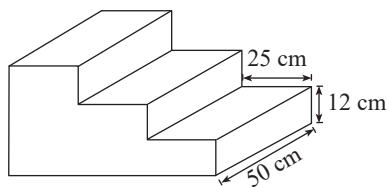
$$\left(\pi = \frac{22}{7} \right)$$



4. A hemispherical tank, of diameter 3 m is full of water. It is being emptied by a tap at the rate of $3\frac{4}{7}$ litres per second. How much time will it take to make the tank half empty?
5. The $\frac{3}{4}$ th part of a conical vessel of internal radius 5 cm and height 24 cm is full of water. The water is emptied into a cylindrical vessel with internal radius 10 cm. Find the height of the water level in cylindrical vessel.

6. The radius of a circle is 6 cm, the area of a sector of this circle is $15\pi \text{ cm}^2$. Find the measure of the arc and the length of the arc corresponding to that sector. (July '23)

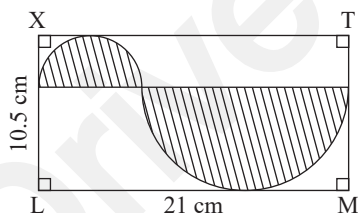
7. There are 3 stair steps as shown in the figure. Each stair step has width 25 cm, height 12 cm and length 50 cm. How many bricks have been used in it, if each brick is $12.5 \text{ cm} \times 6.25 \text{ cm} \times 4 \text{ cm}$?



8. Two types of water tanks are available in a shop at the same price. One is in a cubical form of dimensions $1 \text{ m} \times 1 \text{ m} \times 1 \text{ m}$ and the other is in a cylindrical form of diameter 1 m and height 1 m.

Calculate the volume of both tanks. Why does the shopkeeper advise to purchase cubical tank instead of cylindrical tank? ($\pi = 3.14$)

9. In the figure, $\square XLMT$ is a rectangle, $LM = 21 \text{ cm}$, $XL = 10.5 \text{ cm}$. Diameter of the smaller semicircle is half the diameter of the larger semicircle. Find the area of nonshaded region.



(March '19)

10. Radius of circular base of an ear of corn is 6.6 cm and its length is 11.2 cm. If on an average 1 sq cm area contains 2 corn kernels, find the total number of kernels on a corn.

Answers :

1. 60 , $2\pi rh$, 31680 , 10000 , 3.168 , 6336

2. 20 3. 1.967 cm^2 4. 16.5 minutes 5. 1.5 cm

6. Measure of the arc is 150° and length of the arc is $5\pi \text{ cm}$ 7. 288

8. (i) For cubical tank : Volume 1 m^3 .

For cylindrical tank : Volume $= 0.785 \text{ m}^3$. The shopkeeper advises to purchase cubical tank, because its volume is more than that of cylindrical tank

9. 124.25 cm^2 10. 539.

CHALLENGING QUESTIONS

8.5 *4 marks each*

1. A right circular cone has perpendicular height of 10 cm. This cone is cut exactly at the midpoint of the height parallel to its base such that two parts are formed. Find the ratio of the volume of these two parts.

Solution :

Consider points A, B, C, D and E as shown.

$$AD = 10 \text{ cm} \quad \dots \text{ (Given)}$$

$$AB = BD = h = \frac{1}{2} \times AD \quad \dots \text{ (Given)}$$

$$\therefore AB = BD = \frac{1}{2} \times 10 = 5 \text{ cm.}$$

$$\therefore h = 5 \text{ cm}$$

Let $BC = r_1$ and $DE = r_2$.

In $\triangle ABC$ and $\triangle ADE$,

$$\angle ABC \cong \angle ADE$$

... (Each measures 90°)

$$\angle BAC \cong \angle DAE$$

... (Common angle)

$$\therefore \triangle ABC \sim \triangle ADE$$

... (AA test of similarity)

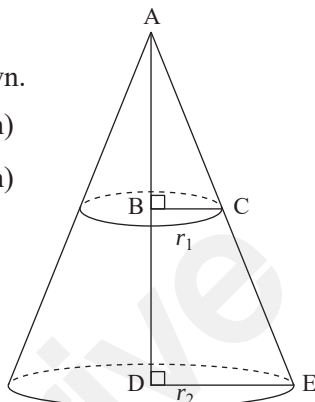
$$\therefore \frac{AB}{AD} = \frac{BC}{DE} \quad \dots \text{ (Corresponding sides of similar triangles)}$$

$$\therefore \frac{5}{10} = \frac{r_1}{r_2}$$

$$\therefore \frac{1}{2} = \frac{r_1}{r_2}$$

$$\therefore r_2 = 2 \times r_1 \quad \dots (1)$$

$$\begin{aligned} \frac{\text{Volume of the smaller cone}}{\text{Volume of the frustum}} &= \frac{\frac{1}{3} \pi r_1^2 h}{\frac{1}{3} \pi (r_1^2 + r_2^2 + r_1 \times r_2) h} \\ &= \frac{r_1^2}{r_1^2 + r_2^2 + r_1 \times r_2} \end{aligned}$$



$$= \frac{r_1^2}{r_1^2 + (2r_1)^2 + r_1 \times (2r_1)} \dots [\text{From (1)}]$$

$$= \frac{r_1^2}{r_1^2 + 4r_1^2 + 2r_1^2}$$

$$= \frac{r_1^2}{7r_1^2} = \frac{1}{7}$$

Ans. The ratio of volume of two parts of the cone is **1 : 7**.

- 2. The radii of the circular ends of a frustum of a cone are 14 cm and 8 cm. If the height of the frustum is 8 cm. Find**

(a) Curved surface area of the frustum.

(b) Total surface area of the frustum.

(c) Volume of the frustum.

$(\pi = 3.14)$ **(March '25)**

Solution : For frustum of the cone,

$$\therefore r_1 = 14 \text{ cm}, r_2 = 8 \text{ cm}, h = 8 \text{ cm}$$

Let the slant height be l .

$$l = \sqrt{h^2 + (r_1 - r_2)^2}$$

$$\therefore l = \sqrt{8^2 + (14 - 8)^2}$$

$$\therefore l = \sqrt{64 + 36}$$

$$\therefore l = \sqrt{100}$$

$$\therefore l = 10 \text{ cm}$$

$$(a) \text{ Curved surface area of the frustum} = \pi(r_1 + r_2)l$$

$$= 3.14(14 + 8) \times 10$$

$$= 3.14 \times 22 \times 10$$

$$= 690.8 \text{ cm}^2$$

$$(b) \text{ Total surface area of the frustum} = \pi(r_1 + r_2)l + \pi r_1^2 + \pi r_2^2$$

$$= 690.8 + \pi(r_1^2 + r_2^2)$$

$$= 690.8 + 3.14(14^2 + 8^2)$$

$$= 690.8 + 3.14(196 + 64)$$

$$\begin{aligned}
 &= 690.8 + 3.14 \times 260 \\
 &= 690.8 + 816.4 \\
 &= 1507.2 \text{ cm}^2
 \end{aligned}$$

$$\begin{aligned}
 \text{(c) Volume of the cone} &= \frac{1}{3} \pi h (r_1^2 + r_2^2 + r_1 \times r_2) \\
 &= \frac{1}{3} \times 3.14 \times 8 (14^2 + 8^2 + 14 \times 8) \\
 &= \frac{1}{3} \times 3.14 \times 8 \times (196 + 64 + 112) \\
 &= \frac{1}{3} \times 3.14 \times 8 \times 372 = 3114.88 \text{ cm}^3
 \end{aligned}$$

Ans. (a) Curved surface area of the frustum of the cone is **690.8 cm²**.

(b) Total surface area of the frustum of the cone is **1507.2 cm²**.

(c) Volume of the frustum of the cone is **3114.88 cm³**.

- 3. Water is flowing from a cylindrical pipe at a speed of 2.52 km/h. This water falls into a cylindrical tank with base radius of 40 cm. If the water level in the tank rises by 3.15 metres in half an hour, then find the internal diameter of the pipe.**

Solution :

$$\begin{aligned}
 \text{Rate of water flowing in the cylindrical pipe} &= 2.52 \text{ km/h} \\
 &= 2520 \text{ m/h} \\
 &= 1260 \text{ m/30 minutes}
 \end{aligned}$$

$\therefore$ water flows 1260 m in 30 minutes (i.e. half an hour)

$$\therefore h_1 = 1260 \text{ m}$$

Let the internal radius of the pipe be r_1

$$\begin{aligned}
 \therefore \text{volume of water collected in pipe in 30 minutes} &= \pi r_1^2 h_1 \\
 &= (\pi r_1^2 \times 1260) \text{ cm}^3
 \end{aligned}$$

Base radius of cylindrical tank (r) = 40 cm = 0.4 m

Height to which water rises (h) = 3.15 m

Volume of water that flows in pipe for 30 minutes

= Volume of water that gets collected in cylindrical tank for 30 minutes

$$\therefore \pi r_1^2 \times 1260 = \pi r^2 h$$



$$\therefore r_1^2 \times 1260 = \frac{4}{10} \times \frac{4}{10} \times \frac{315}{100}$$

$$\therefore r_1^2 = \frac{4}{10} \times \frac{4}{10} \times \frac{315}{100} \times \frac{1}{1260}$$

$$\therefore r_1^2 = \frac{4 \times 4}{100 \times 100 \times 4}$$

$$\therefore r_1 = \frac{4}{10 \times 10 \times 2}$$

$$\therefore r_1 = \frac{2}{100}$$

$$\therefore r_1 = 0.02 \text{ m or } 2 \text{ cm}$$

$$\text{Internal diameter of the pipe} = 2 \times r_1$$

$$= 2 \times 2$$

$$= 4 \text{ cm}$$

Ans. Internal diameter of the pipe is **4 cm**.

- 4. A tinmaker converts a cubical metallic box into 10 cylindrical tins. Side of a cube is 50 cm and radius of the cylinder is 7 cm. Find the height of each cylinder so made if there is a wastage of 12% in the process.**

Solution :

For cube $l = 50 \text{ cm}$,

$$\text{total surface area of box} = 6l^2 = 6 \times 50^2 = 15000 \text{ cm}^2$$

12% wasted

$\therefore$ 88% area of total surface area of box is used.

88% of 15000

$$= \frac{88}{100} \times 15000 = 13200 \text{ cm}^2.$$

$\therefore$ 88% of 15000 = 13200 cm² is used to make 10 cylindrical tins.

$$\text{Area of tin required for 1 box} = \frac{13200}{10} = 1320 \text{ cm}^2.$$

$\therefore$ total surface area of 1 cylindrical tin = 1320 cm².

$$\therefore 2\pi r(r+h) = 1320$$

$$\therefore 2 \times \frac{22}{7} \times 7(7+h) = 1320$$

$$\therefore 2 \times 22(7+h) = 1320$$

$$\therefore 7+h = \frac{1320}{2 \times 22} \quad \therefore 7+h = 30$$

$$h = 23 \text{ cm}$$

Ans. Height of each cylindrical tin is **23 cm**.

- 5. The diameter of each wheel of a car is 80 cm. The car is travelling at a speed of 66 km per hour, then how many revolutions will each wheel make in 10 minutes?**

Solution : The car is travelling at a speed of 66 km per hour

$\therefore$ distance covered by car in 1 hour, i.e. 60 minutes = 66 km

$$\begin{aligned} \therefore \text{distance covered by car in 10 minutes} &= \frac{66 \times 10}{60} \\ &= 11 \text{ km} \\ &= 1100000 \text{ cm.} \end{aligned}$$

Diameter of the wheel = 80 cm

Number of revolutions each wheel makes in 10 minutes

$$= \frac{\text{Distance covered by wheel in 10 minutes}}{\text{Circumference of the wheel}}$$

$$= \frac{1100000}{\pi \times d}$$

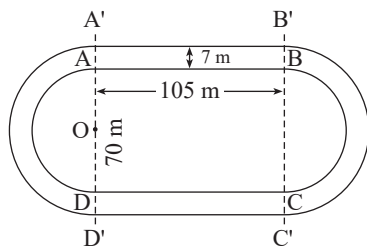
$$= \frac{1100000}{\frac{22}{7} \times 80}$$

$$\begin{aligned} &= \frac{1100000 \times 7}{22 \times 80} = 4375 \end{aligned}$$

Ans. The wheel makes **4375** revolutions in 10 minutes.



6. An athletic track 7 m wide consists of two straight sections 105 m long joining the semicircular ends. The distance between two straight sections is 70 m. Find the difference between the length of edges of track.



Solution : From figure,

$$AB = CD = A'B' = C'D' = 105 \text{ m}$$

The addition of semicircular tracks on both the sides will be a complete circle.

Inner edge diameter of semicircular track = $AD = BC = 70 \text{ m}$.

$$\therefore \text{radius } (r) \text{ of this track} = \frac{70}{2} = 35 \text{ m.}$$

$$\text{Outer edge diameter} = A'D' = B'C' = 70 + 7 + 7 = 84 \text{ m}$$

$$\therefore \text{radius } (R) \text{ of this track} = 42 \text{ m}$$

$AB, A'B', CD$ and $C'D'$ are the lengths of inner and outer edges of the track which are equal in lengths.

$\therefore$ difference between outer and inner edges of straight path is zero.

In case of semicircular track, length of outer semicircular edge will be greater than inner semicircular edge.

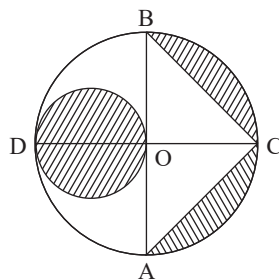
$\therefore$ difference between the length of outer and inner edges of the track = outer circumference of the circle – inner circumference of the circle

$$= 2\pi R - 2\pi r = 2\pi(R - r) = 2 \times \frac{22}{7} \times (42 - 35)$$

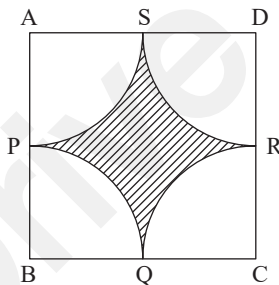
$$= 2 \times \frac{22}{7} \times 7 = 44 \text{ m}$$

Ans. The difference between the length of outer and inner edges of the track is **44 m**.

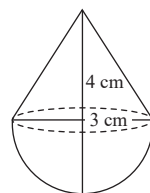
1. O is the centre of the circle. AB and CD are the diameters of the circle. Seg OD is the diameter of the smaller circle. If $OA = 7$ cm, then find the area of the shaded portion.



2. Solid plastic spheres each of radius 1 cm are melted and cast into a hollow cylindrical drum having height 90 cm and thickness 2 cm. Outer radius of the drum is 30 cm. Find how many plastic spheres are required to form the drum.
3. In the figure, $\square ABCD$ is a square of side 50 m. Points P, Q, R and S are midpoints of sides AB, BC, CD and AD respectively. Find the area of shaded region. ($\pi = 3.14$) (March '20)



4. Oil tins of cuboidal shape are made from metallic sheet with length 8 m and breadth 4 m. Each tin has dimensions $60 \times 40 \times 20$ in cm and is open from top. Find the number of such tins that can be made.
5. How many silver coins, 1.75 cm in diameter and of thickness 2 mm, must be melted and cast to form a cuboid of dimensions $5.5 \text{ cm} \times 10 \text{ cm} \times 3.5 \text{ cm}$?
6. Lower part of a solid toy is hemisphere and its upper part is conical. The height of the conical part is 4 cm and diameter of the base is 6 cm, then find (i) volume of the toy (ii) curved surface area of the toy. ($\pi = 3.14$)



7. An ice-cream pot has a right circular cylindrical shape. The radius of the base is 12 cm and height is 7 cm. This pot is completely

filled with ice cream. The entire ice cream is given to the students in the form of right circular ice-cream cones, having diameter 4 cm and height is 3.5 cm. If each student is given one cone, how many students can be served? **(March '23)**

8. A cylinder of radius 12 cm contains water up to the height of 20 cm. A spherical iron ball is dropped into the cylinder and thus water level is raised by 6.75 cm. What is the radius of iron ball? **(March '24)**

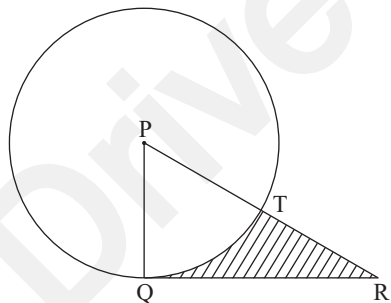
9. A bucket is in the form of a frustum of a cone. It holds 28.490 litres of water. The radii of the top and the bottom are 28 cm and 21 cm respectively. Find the height of the bucket. $\left(\pi = \frac{22}{7}\right)$

(July '23)

10. The radii of three solid metallic spheres are 6 cm, 8 cm and 10 cm respectively. These spheres are melted to form a single solid sphere, then find the radius of this new sphere.

11. In the figure, segment QR is a tangent to the circle with centre P. $PR = 12$ cm and $PQ = 6$ cm. Find the area of shaded region.

$$(\sqrt{3} = 1.73, \pi = 3.14)$$



Answers :

1. 66.5 cm^2 2. 9006 3. 535.71 cm^2 4. 50 5. 400
6. (i) 94.2 cm^3 (ii) 103.62 cm^2 7. 216 8. 9 cm 9. $h = 15 \text{ cm}$
10. 12 cm 11. 12.30 cm^2 .

Time : 2 Hours]**Solution on QR Code****[Max. Marks : 40**

- Note :** (i) All questions are compulsory.
(ii) Use of calculator is not allowed.
(iii) The numbers to the right of the questions indicate full marks.
(iv) In case of MCQ's [Q. No. 1 (A)], only the first attempt will be evaluated and will be given credit.
(v) Draw proper figures wherever necessary.
(vi) The marks of construction should be clear. Do not erase them.
(vii) Diagram is essential for writing the proof of the theorem.

Q. 1. (A) Choose the correct alternative from given : 4

- (i) If $\triangle ABC \sim \triangle DEF$, $m\angle B = 60^\circ$, then $m\angle E = \dots\dots\dots$
(A) 30° (B) 60° (C) 90° (D) 45°
- (ii) Two circles of radii 5.5 cm and 4.2 cm touch each other externally then distance between their centres is $\dots\dots\dots$
(A) 9.7 cm (B) 1.3 cm (C) 5.5 cm (D) 4.2 cm
- (iii) A line makes an angle of 45° with the positive direction of X-axis. So the slope of the line is $\dots\dots\dots$
(A) $\frac{1}{2}$ (B) $\frac{\sqrt{3}}{2}$ (C) 1 (D) $\sqrt{3}$
- (iv) The volume of a cube of side 2 cm is $\dots\dots\dots$
(A) 4 cm^3 (B) 2 cm^3 (C) 6 cm^3 (D) 8 cm^3

Q. 1. (B) Solve the following subquestions : 4

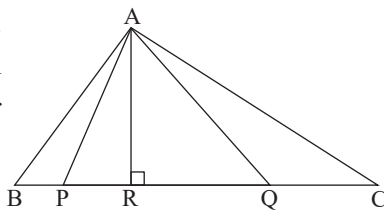
- (i) Find the diagonal of a square whose side is 10 cm.
(ii) The ratio of corresponding sides of similar triangles is 3 : 5, then find the ratio of their areas.

- (iii) Find the slope of the line passing through the points A(2, 3) and B(4, 7).
- (iv) If $\sin \theta = \frac{7}{25}$, then find the value of $\operatorname{cosec} \theta$.

Q. 2. (A) Complete the following activities and rewrite them :

(Any two) 4

- (i) In the given figure, $AR \perp BC$, $AR \perp PQ$, then complete the activity for finding $\frac{A(\triangle ABC)}{A(\triangle APQ)}$.

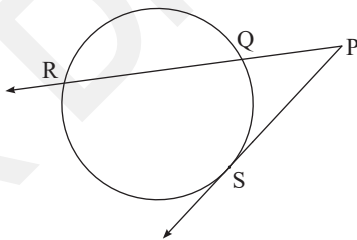


Activity :

$$\frac{A(\triangle ABC)}{A(\triangle APQ)} = \frac{\boxed{} \times AR}{PQ \times \boxed{}}$$

$$\therefore \frac{A(\triangle ABC)}{A(\triangle APQ)} = \frac{\boxed{}}{\boxed{}}$$

- (ii) In the given figure, seg PS is a tangent segment, line PR is a secant. If $PQ = 3.6$, $QR = 6.4$, then find PS by completing the following activity.



Activity :

$$PS^2 = PQ \times \boxed{} \dots (\text{Tangent secant segments theorem})$$

$$\therefore PS^2 = PQ \times (PQ + \boxed{})$$

$$\therefore PS^2 = 3.6 \times (3.6 + \boxed{})$$

$$\therefore PS^2 = 3.6 \times 10$$

$$\therefore PS^2 = 36$$

$$\therefore PS = \boxed{}$$

- (iii) Measure of an arc of a circle is 90° and its radius is 14 cm. Complete the following activity to find the length of an arc.

Activity :

$$\text{Length of an arc} = \frac{\theta}{360} \times \boxed{} \quad \dots \text{ (Formula)}$$

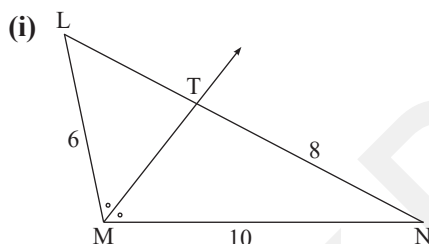
$$= \frac{90}{360} \times 2 \times \frac{22}{7} \times \boxed{}$$

$$= \frac{1}{4} \times \boxed{}$$

$$\text{Length of an arc} = \boxed{} \text{ cm}$$

Q. 2. (B) Solve the following subquestions : (Any four)

8



In $\triangle LMN$, ray MT bisects $\angle LMN$.

If $LM = 6$, $MN = 10$, $TN = 8$, then find LT .

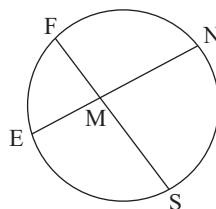
- (ii) Find surface area of a sphere of radius 7 cm.

- (iii) In the following figure,

$$m(\text{arc } NS) = 125^\circ,$$

$$m(\text{arc } EF) = 37^\circ.$$

Find $m\angle NMS$.



- (iv) Find the coordinates of midpoint of the segment joining the points $P(22, 20)$ and $Q(0, 16)$.

- (v) Find the volume of a cone if radius of its base is 7 cm and its perpendicular height is 15 cm.

Q. 3. (A) Complete the following activities and rewrite it :

(Any one) 3

- (i) If $\tan \theta = 1$, then find the value of $\frac{\sin \theta + \cos \theta}{\sec \theta + \operatorname{cosec} \theta}$ by completing the following activity.

Activity :

$$\tan \theta = 1$$

... (Given)

$$\text{but } \tan \boxed{} = 1$$

$$\therefore \theta = \boxed{}$$

$$\therefore \frac{\sin \theta + \cos \theta}{\sec \theta + \operatorname{cosec} \theta} = \frac{\sin 45^\circ + \cos 45^\circ}{\sec 45^\circ + \operatorname{cosec} 45^\circ}$$

$$= \frac{\frac{1}{\boxed{}} + \frac{1}{\sqrt{2}}}{\sqrt{2} + \boxed{}}$$

$$= \frac{\frac{2}{\sqrt{2}}}{\boxed{}}$$

$$\frac{\sin \theta + \cos \theta}{\sec \theta + \operatorname{cosec} \theta} = \frac{1}{\boxed{}}$$

- (ii) In the following figure, point O is the centre of the circle and length of chord AB is equal to the radius of the circle.

Find the measures of :

(i) $\angle AOB$

(ii) arc AB

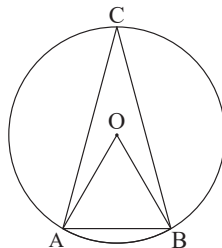
(iii) $\angle ACB$

by completing the activity.

Activity :

In $\triangle AOB$,

$$AO = OB = AB$$



$\therefore \triangle AOB$ is an triangle.

$\therefore m\angle AOB =$

$m\angle AOB = m(\text{arc } AB) =$

... (Definition of measure of an arc)

$m\angle ACB = \frac{1}{2} \times$... $\left(\text{} \right)$

$$= \frac{1}{2} \times 60^\circ$$

$m\angle ACB =$

Q. 3. (B) Solve the following subquestions : (Any two)

6

- (i) Find coordinates of point P, if P divides the line segment joining the points A(−1, 7) and B(4, −3) in the ratio 2 : 3.
- (ii) Draw a circle with centre O of radius 3.4 cm. Draw a chord MN of length 5.7 cm in it. Construct tangents at point M and N to the circle.
- (iii) A storm broke a tree and the treetop rested 20 m from the base of the tree, making an angle of 60° with the horizontal. Find the height of the tree.
- (iv) Prove that, 'In a right angled triangle, the square of the hypotenuse is equal to the sum of the squares of remaining two sides.'

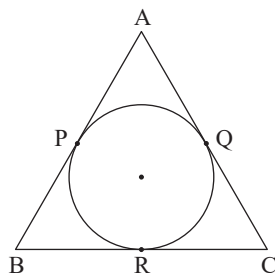
Q. 4. Solve the following subquestions : (Any two)

8

- (i) $\triangle ABC$ has sides of length 4 cm, 5 cm and 6 cm while $\triangle PQR$ has perimeter of 90 cm. If $\triangle ABC$ is similar to $\triangle PQR$, then find the length of corresponding sides of $\triangle PQR$.
- (ii) $\triangle ABC \sim \triangle PBR$, $BC = 8$ cm, $AC = 10$ cm, $\angle B = 90^\circ$, $\frac{BC}{BR} = \frac{5}{4}$, then construct $\triangle PBR$.



- (iii) In the following figure, $\triangle ABC$ is an isosceles triangle with perimeter 44 cm. The base BC is of length 12 cm. Side AB and AC are congruent. A circle touches the three sides of triangle as shown.



Find the length of tangent segment from A to circle.

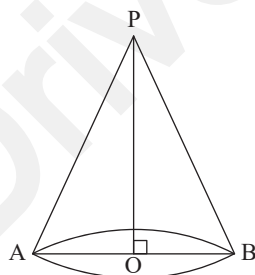
Q. 5. Solve the following subquestions : (Any one)

3

- (i) Draw a right angled $\triangle ABC$ with sides of lengths 3 cm, 4 cm and 5 cm. Draw median on the hypotenuse of $\triangle ABC$. Then :
- Measure the length of median and write it.
 - By observing lengths of median and hypotenuse write your observations.

- (ii) Observe the given figure and answer the following questions :

- How many surfaces does a solid cone have?
- What are the names of slant height and perpendicular height in the given figure?
- If slant height of solid cone is 10 cm and perpendicular height is 8 cm, then find diameter of base of solid cone?



ALL THE BEST

Scan  this **QR Code** for the
Model Answers to
Board's July 2025 Question Paper.



NOTES

Handwriting practice lines consisting of 20 horizontal dashed lines.

Ramesh D0060

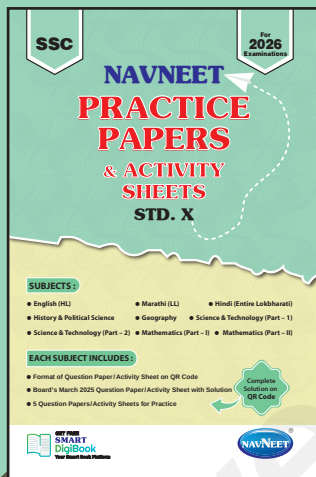


Perfect preparation for 2026 Board's
Examination

NAVNEET PRACTICE PAPERS & ACTIVITY SHEETS STD. X



GET FREE
**SMART
DigiBook**
Your Smart Book Platform



Question Papers /
Activity Sheets
with complete
solution through
QR Codes

SUBJECTS:

- English (HL)
- History & Political Science
- Science & Technology (Part 2)
- Marathi (LL)
- Geography
- Mathematics (Part-I)
- Hindi (Entire Lokbharati)
- Science & Technology (Part 1)
- Mathematics (Part-II)

Salient Features :

- Updated format of Question Papers/Activity Sheets on QR Code
- Board's March 2025 Question Paper/Activity Sheet in each subject with complete solution and detailed marking scheme
- 5 Question Papers/Activity Sheets for practice in each subject prepared by expert and experienced team of authors



NAVNEET EDUCATION LIMITED

WIN EXCITING REWARDS WITH NAVNEET 21 MOST LIKELY QUESTION SETS



UNLOCK REWARDS AND SCHOLARSHIP OPPORTUNITIES



Step-01
Buy Navneet
'21 Most Likely Question Sets'



Step-03
Enter the unique access code
from 21 MLQ sets. Get assured
21% off on Portronics products



Step-02
Register on
reader.smartdigibook.com



Step-04
Win Youva Voucher, Smart Watch,
Scholarship*



SCAN
THE
QR CODE
NOW!

Join the Winners' List



* T&C Apply. The Management reserves the right to modify/terminate the program and rewards at any point in time.



PORTRONICS

Making Learning Smarter



Scan to ace your goals now

ISBN 978-93-7354-457-1



D0060